Kubernetes architecture

04

Kubernetes architecture

API server

etcd

Controller manager

Scheduler

Kubelet

Kubernetes architecture

API server

etcd

Controller manager

Scheduler

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etcd

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Kubernetes architecture

API server

etcd

Controller manager

Scheduler

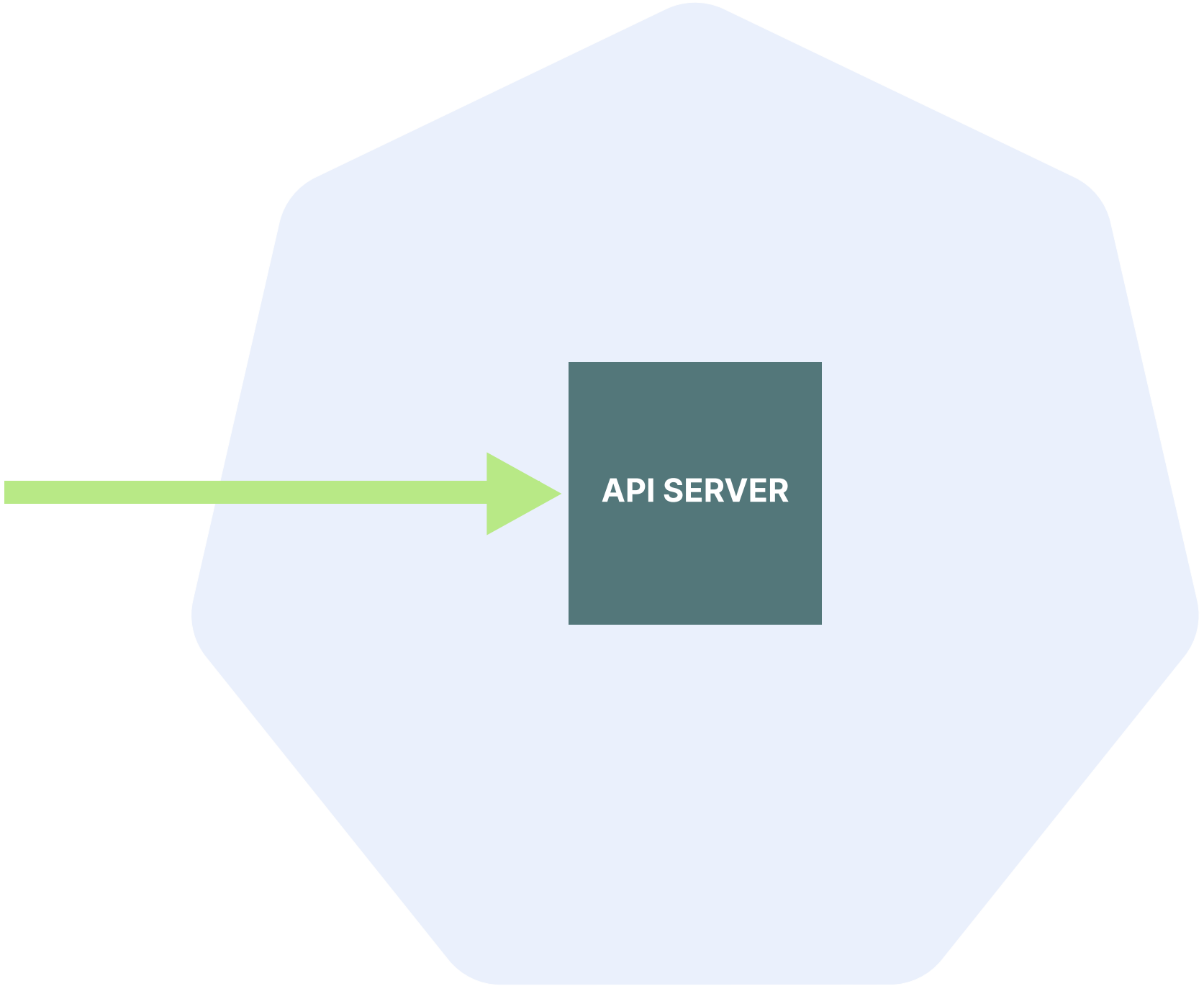
Kubelet

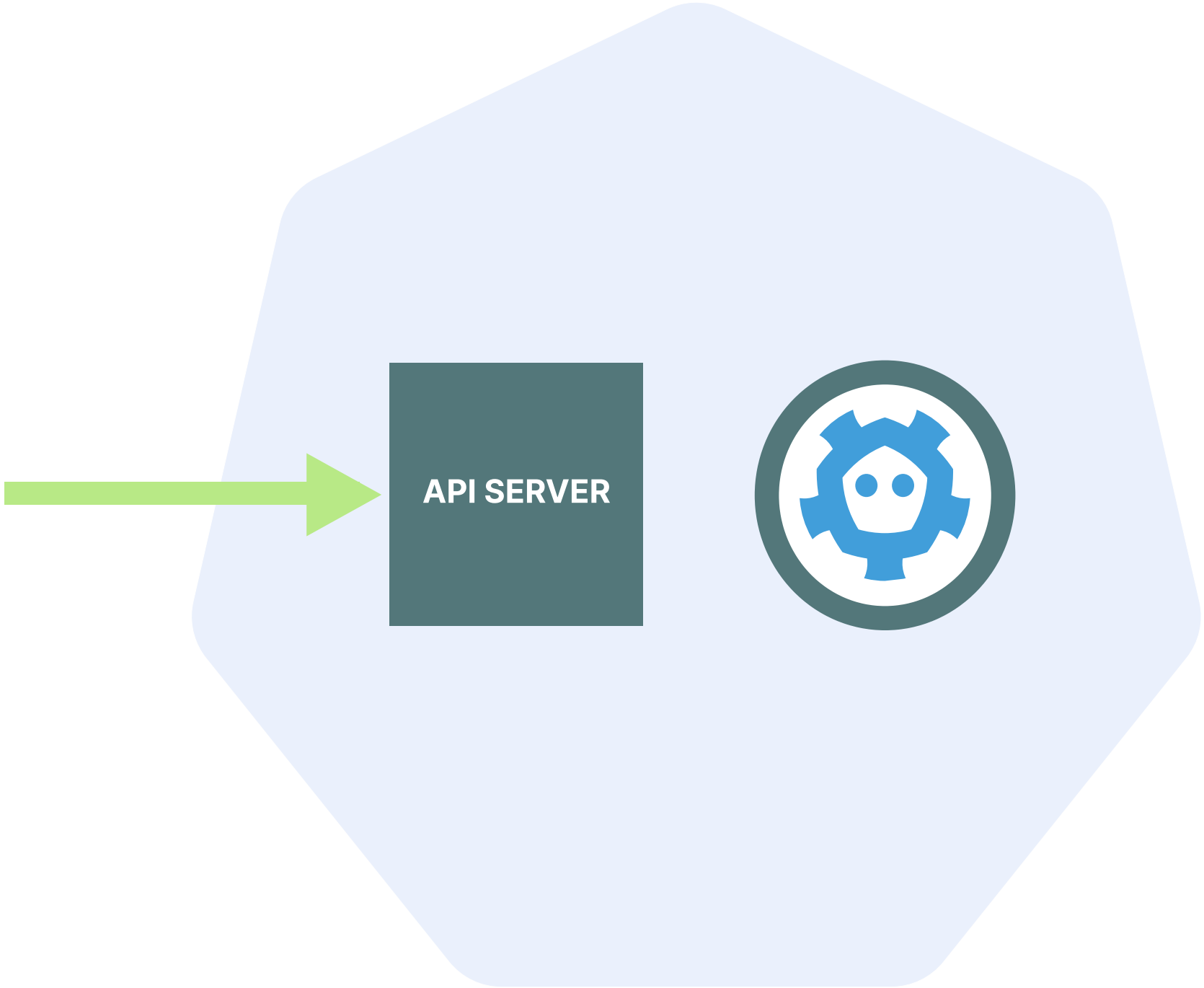
The lifecycle of a request

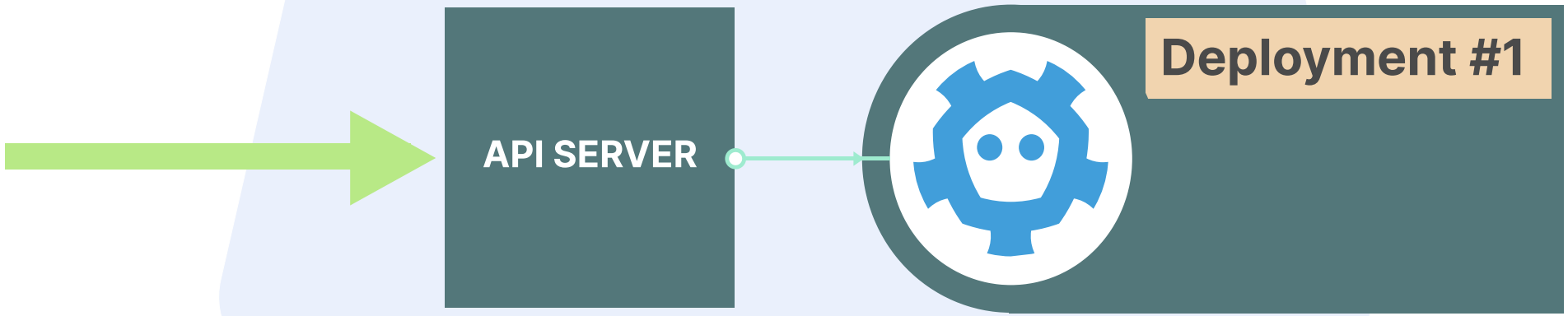
kubectl apply -f my-pod.yaml






API SERVER


API SERVER


API SERVER
Deployment #1

Etcd

etcd

Key-value store

Distributed

Change notification

etcd

Key-value store

Distributed

Change notification

Key-value store

Distributed

Change notification

Leaders and followers

RAFT

Follower

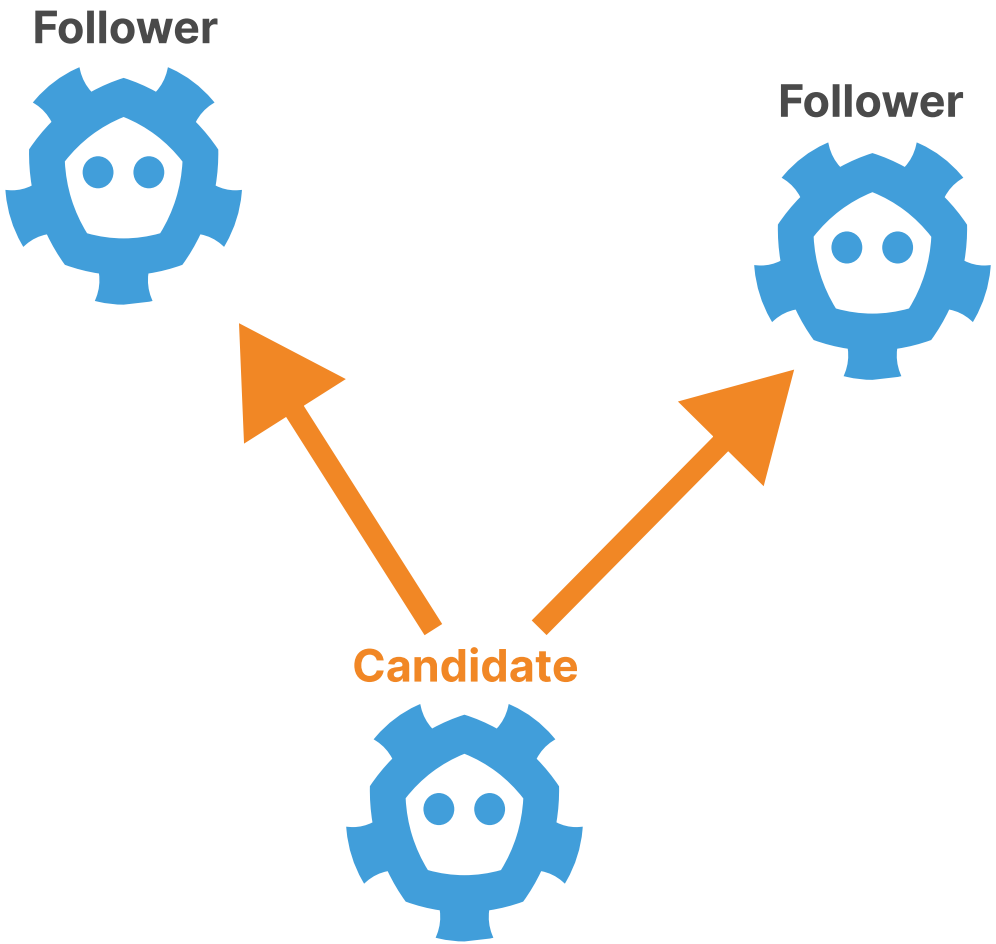


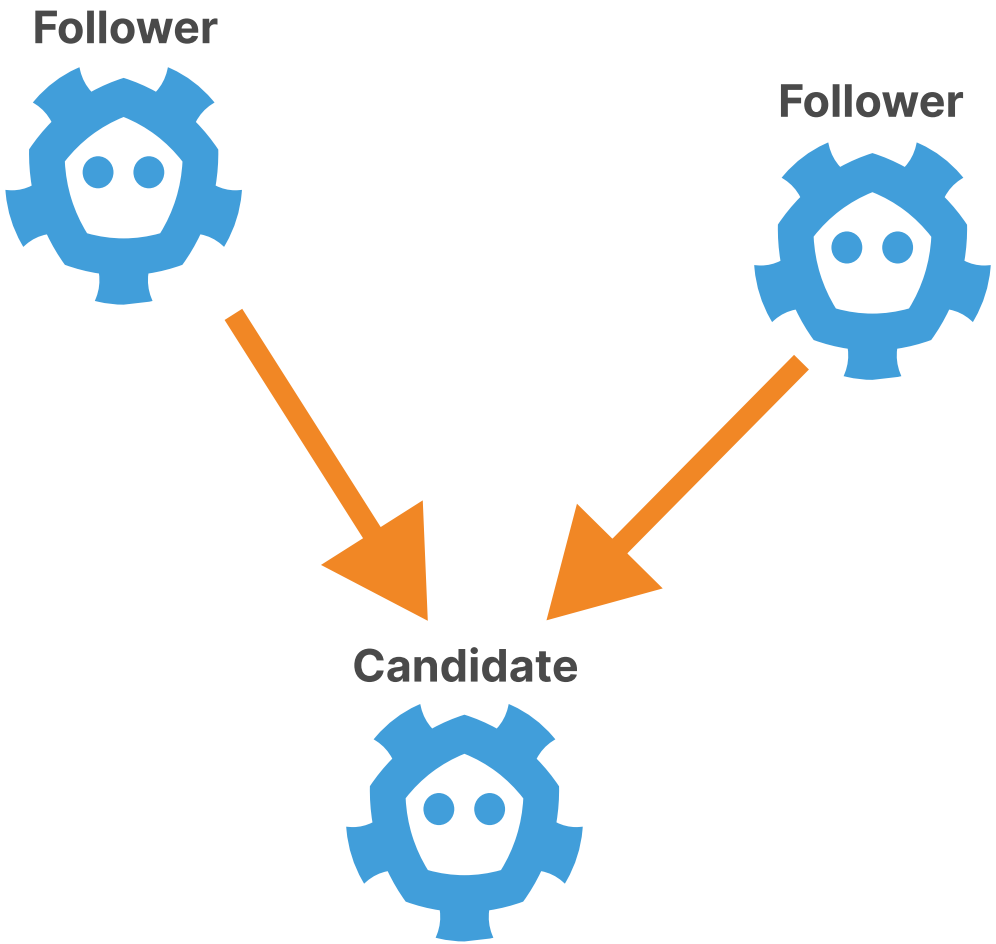
Follower



Follower




Follower
Follower
Candidate


Follower
Follower
Candidate

Follower



Follower

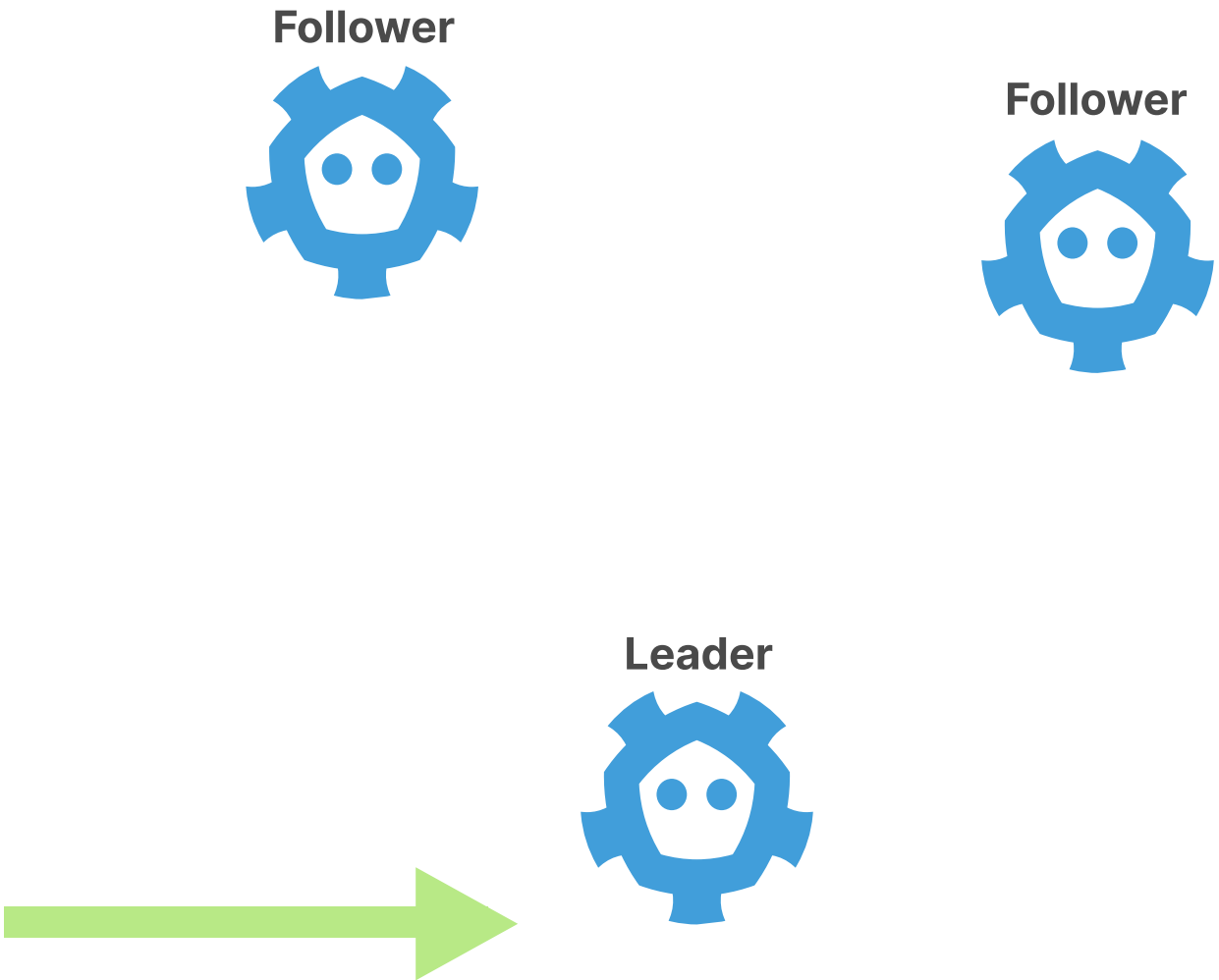


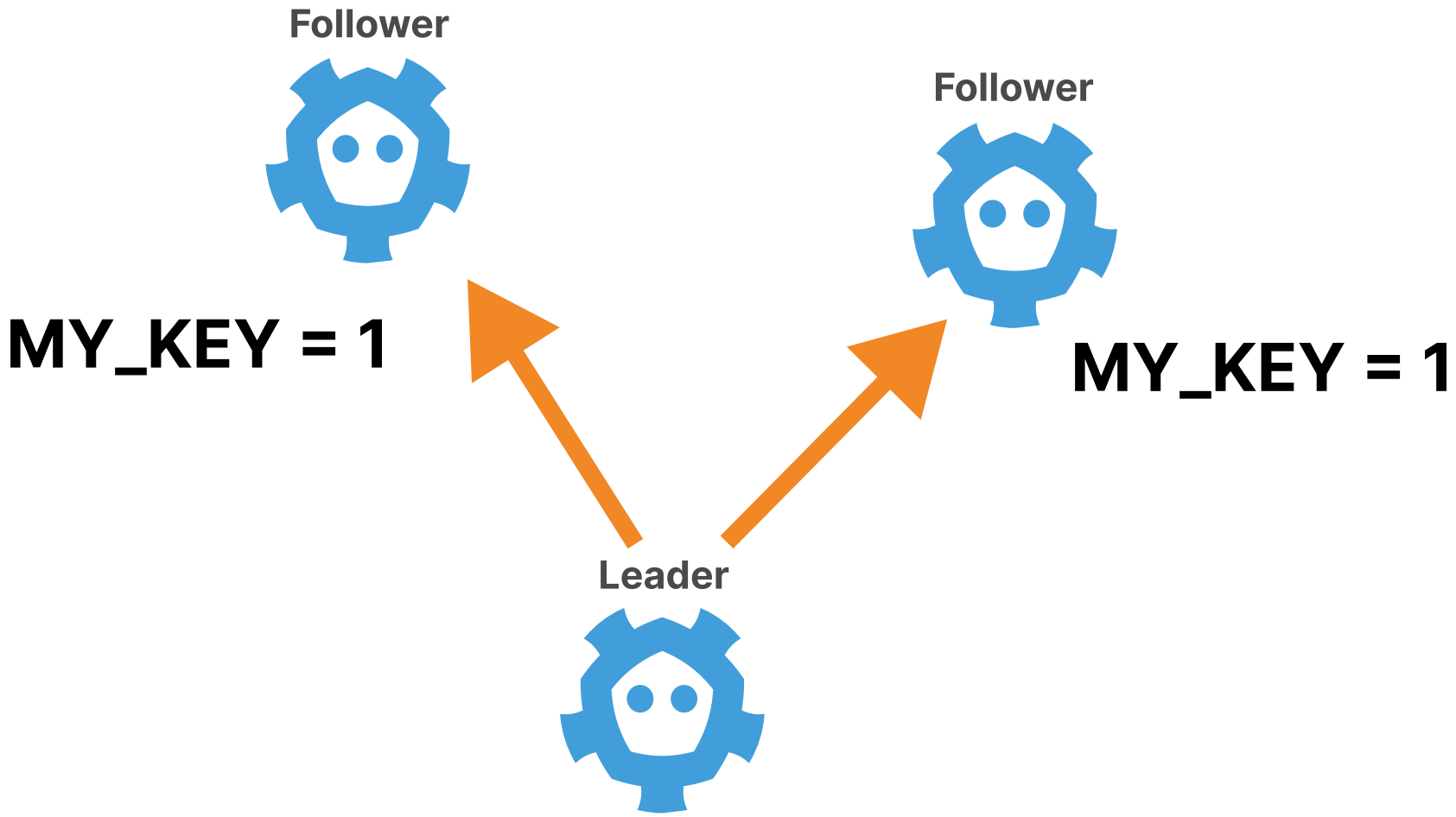
Leader

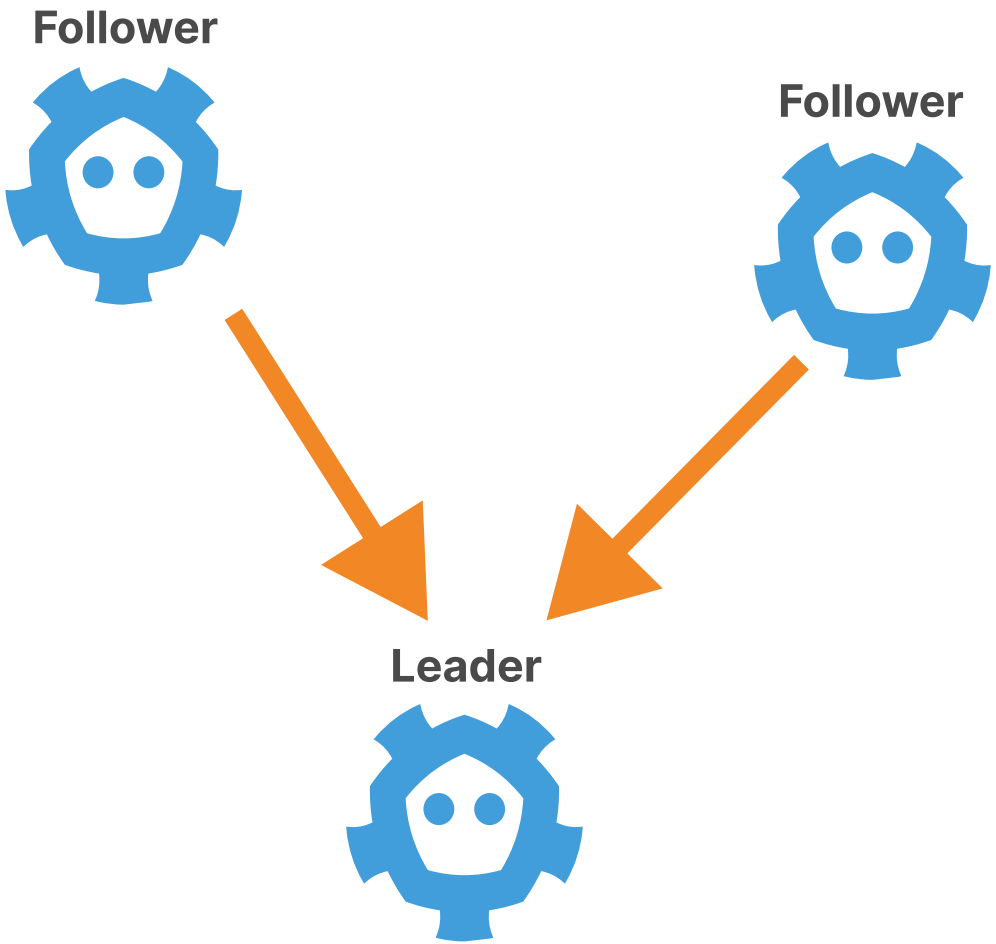


Writing values

RAFT


Follower
Follower
Leader
MY_KEY = 1


Follower
Follower
MY_KEY = 1
MY_KEY = 1
Leader


Follower
Follower
Leader
MY_KEY = 1

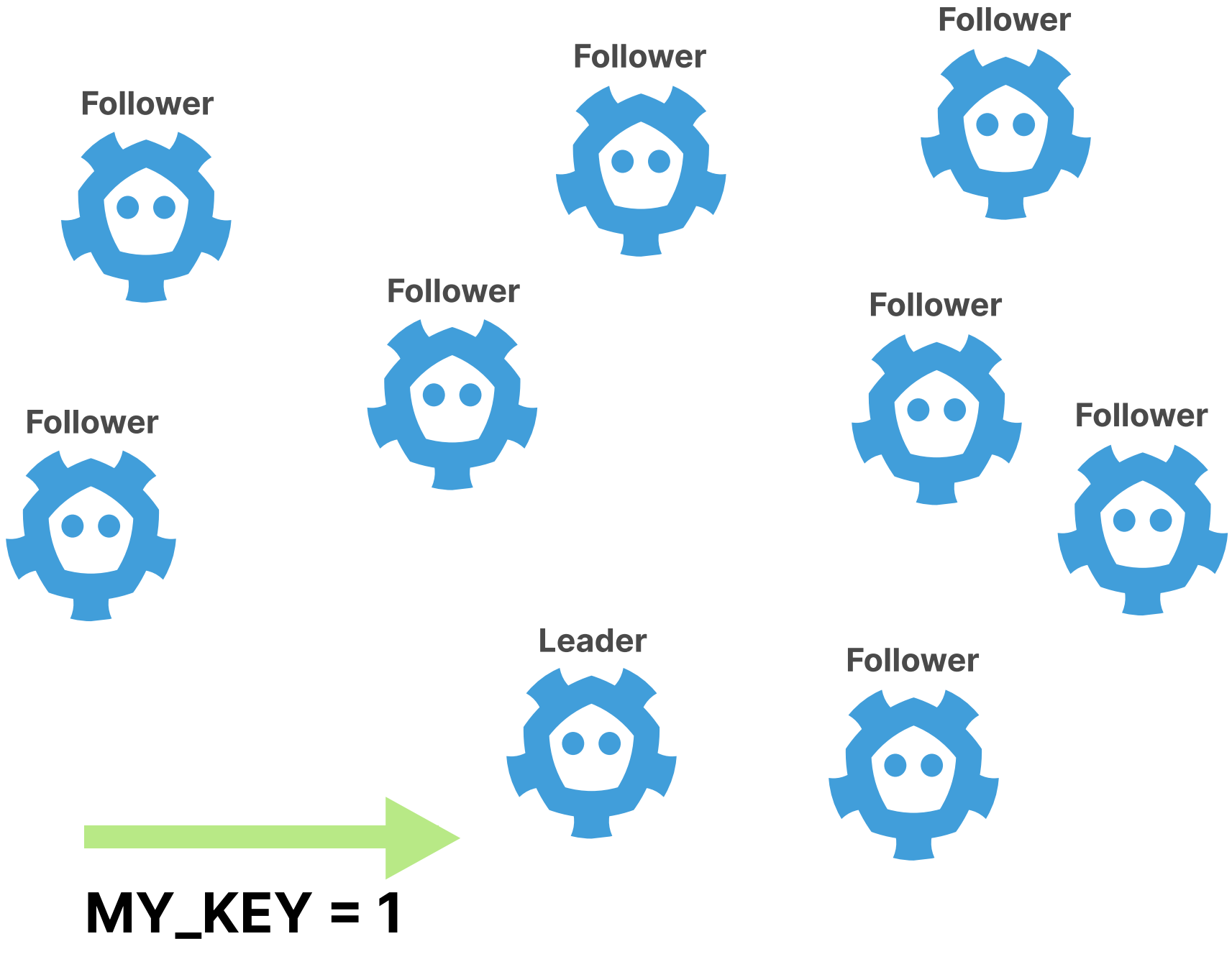
Availability

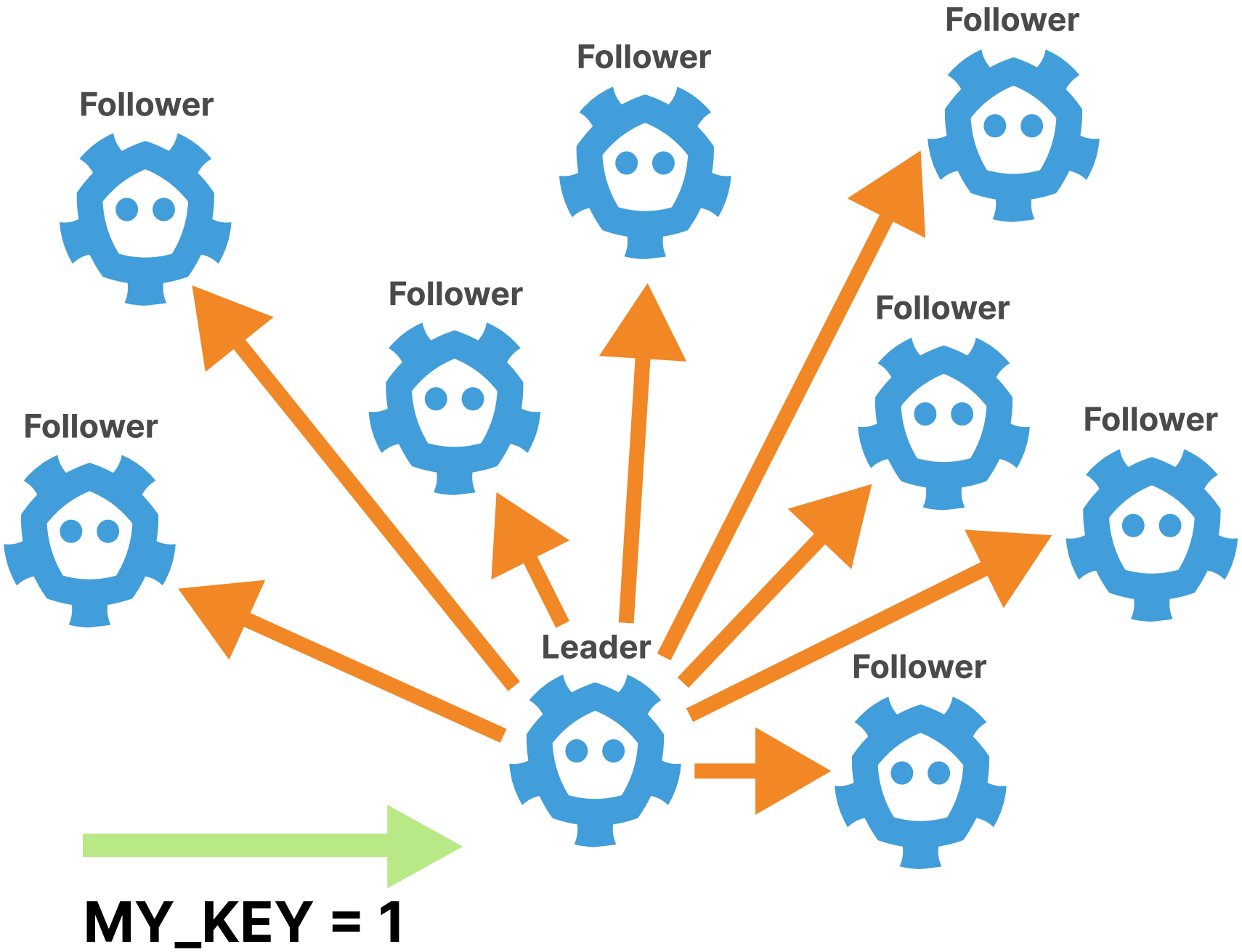
etcd

CLUSTER SIZE	MAJORITY	FAILURE TOLERANCE
1	1	0
3	2	1
4	3	1
5	3	**2**
6	4	2
7	4	**3**
8	5	3
9	5	**4**

Trade-offs

etcd


Follower
Follower
Follower
Follower
Follower
Follower
Follower
Leader
Follower
MY_KEY = 1


Follower
Follower
Follower
Follower
Follower
Follower
Follower
Leader
Follower
MY_KEY = 1

etcd trade-offs

Latency

Payload size

Bandwidth

etcd trade-offs

Latency

Payload size

Bandwidth

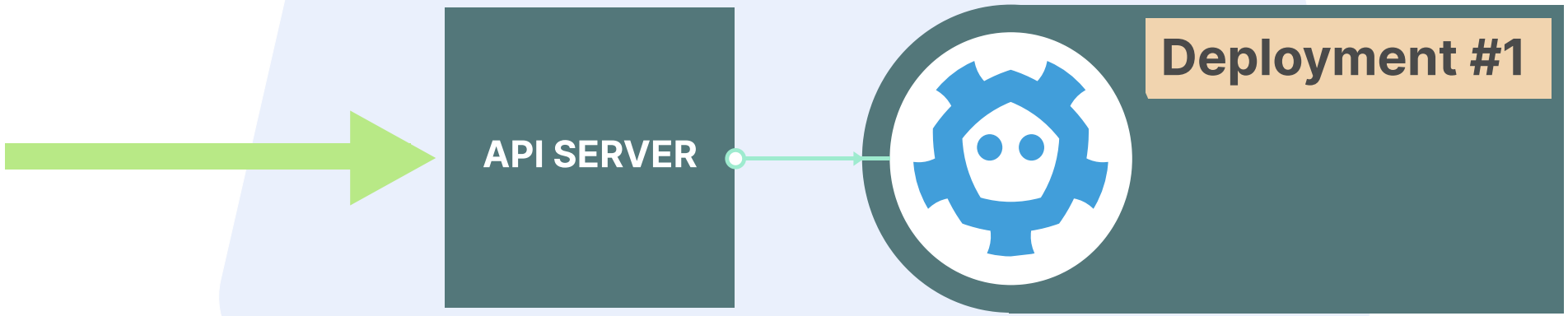
etcd trade-offs

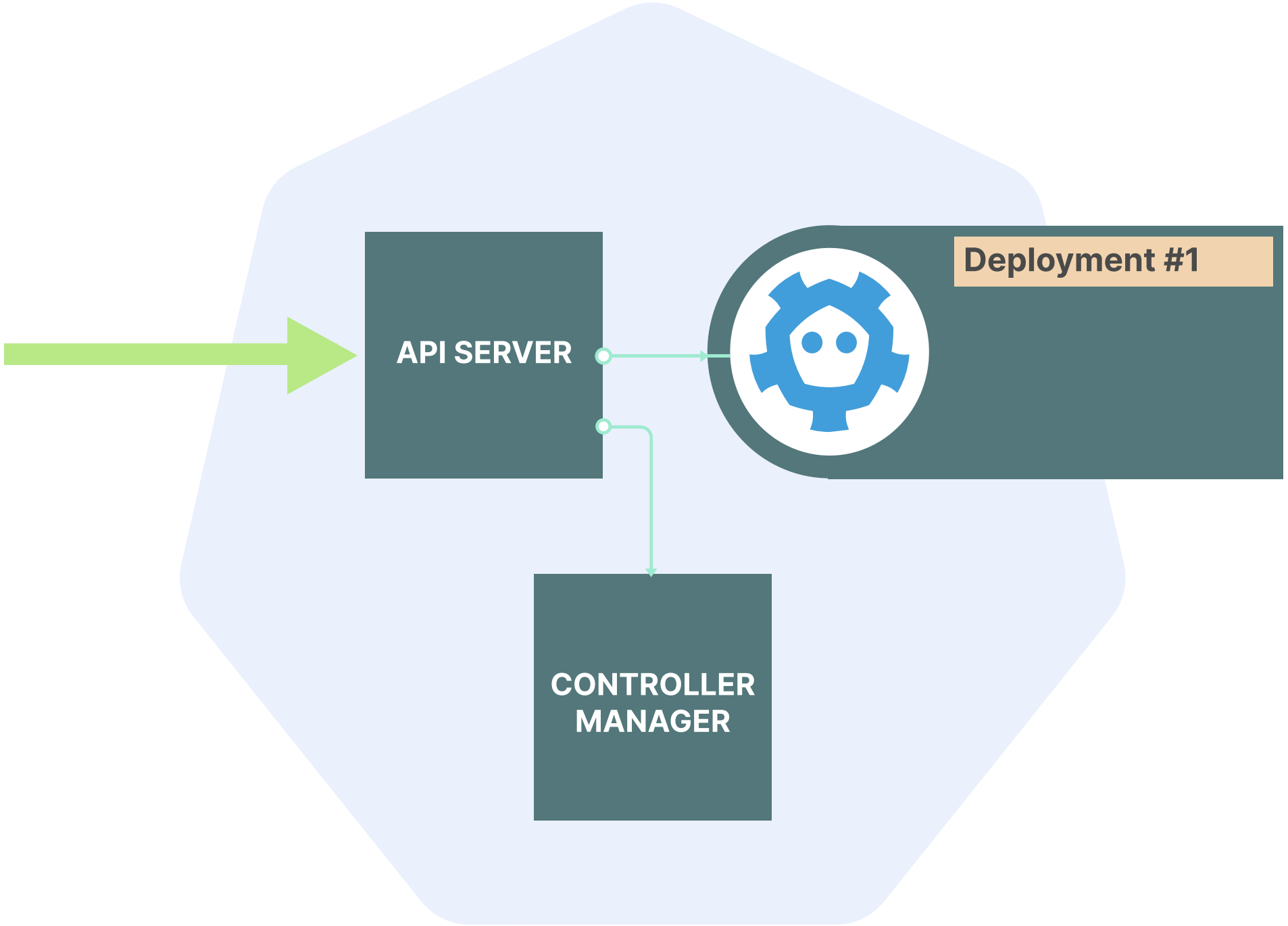
Latency

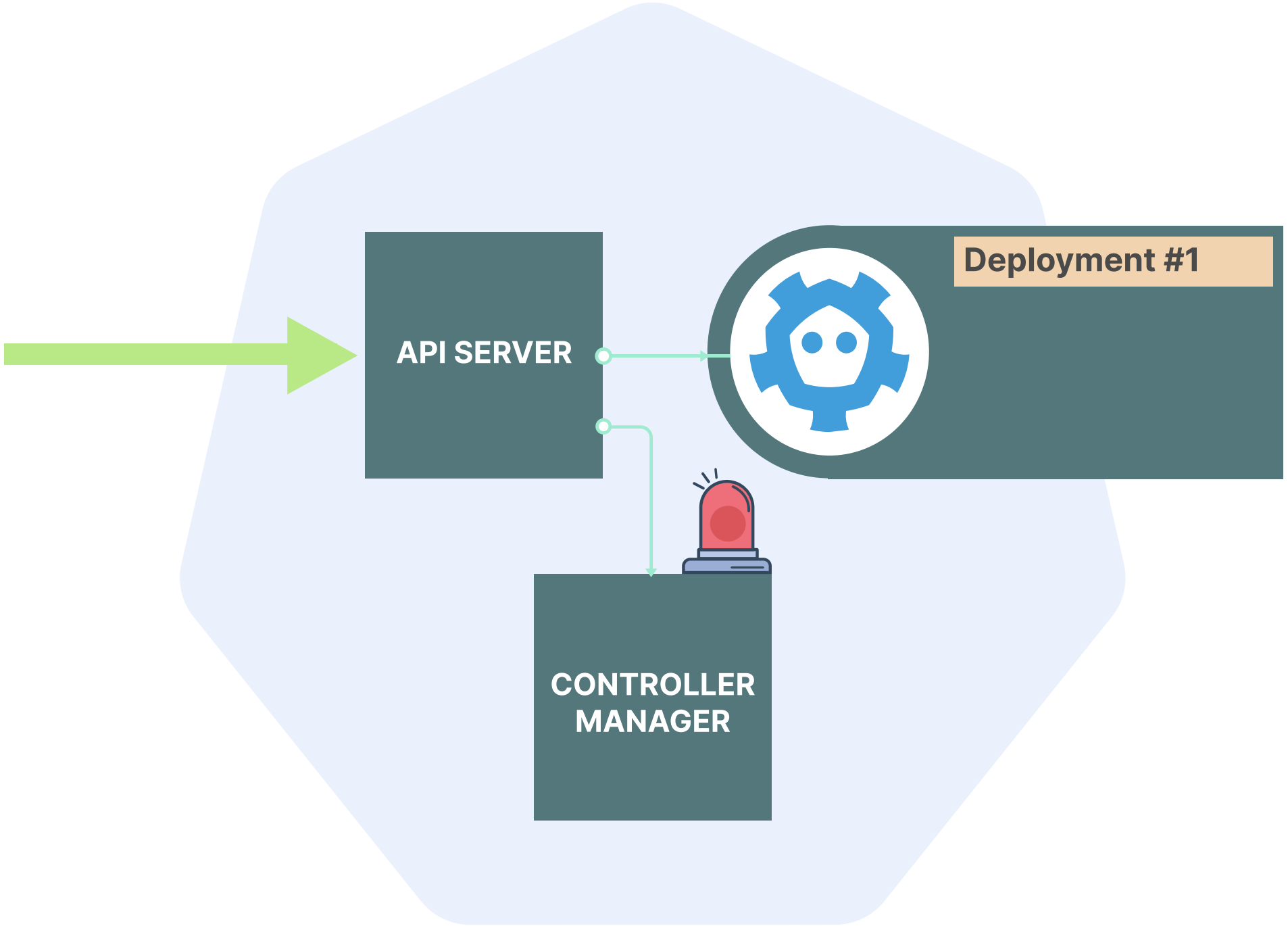
Payload size

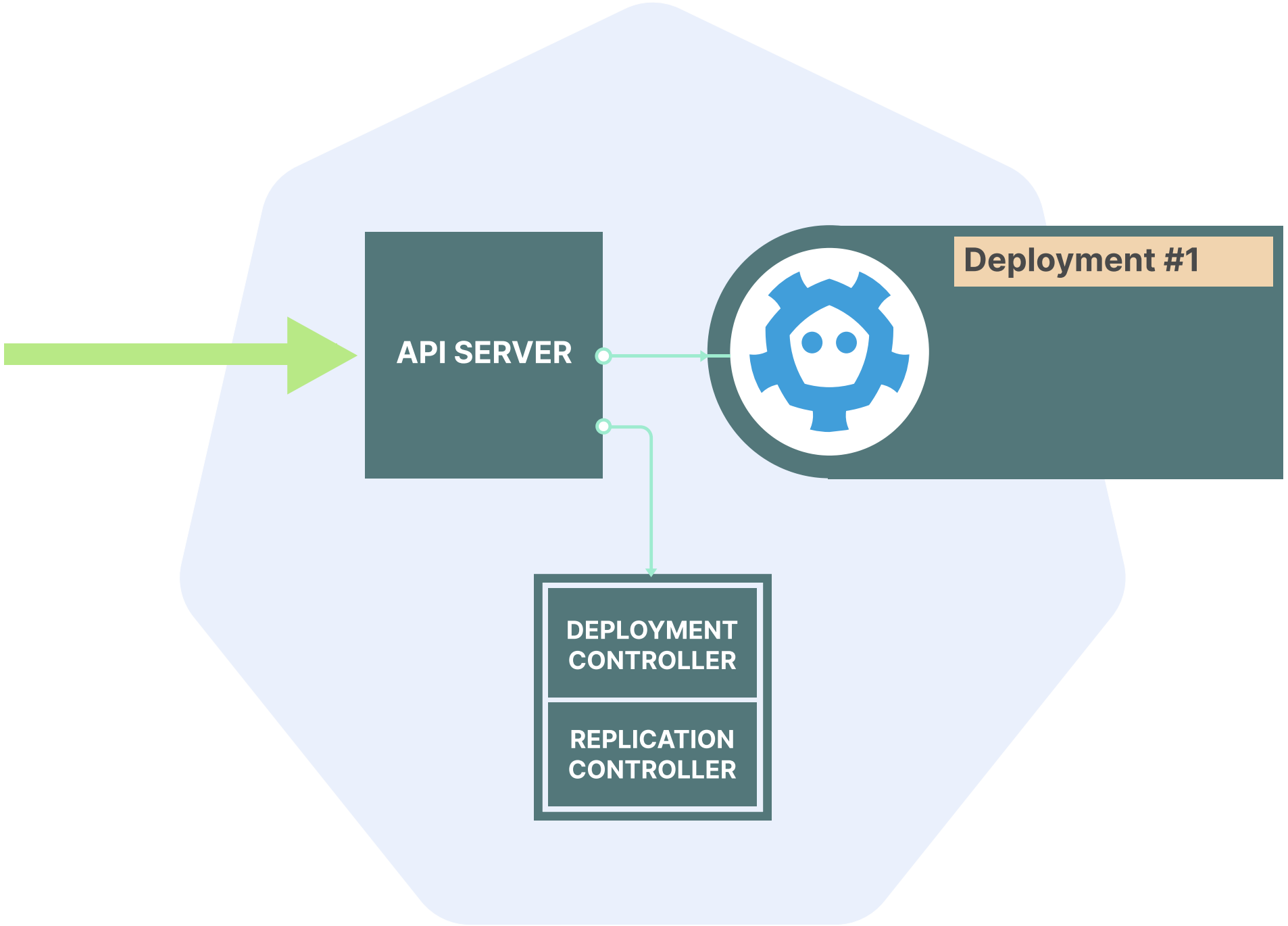
Bandwidth

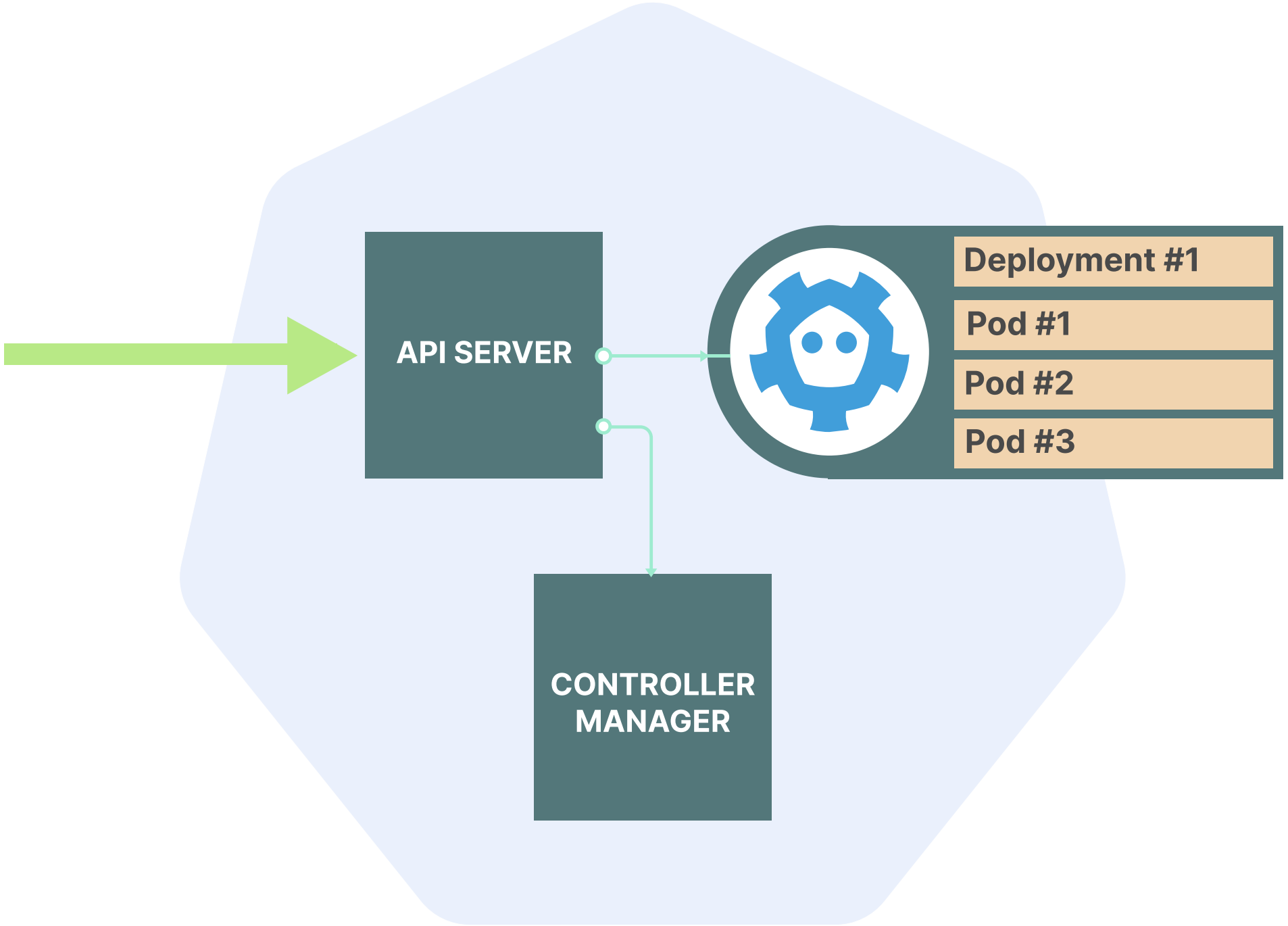
Creating resources

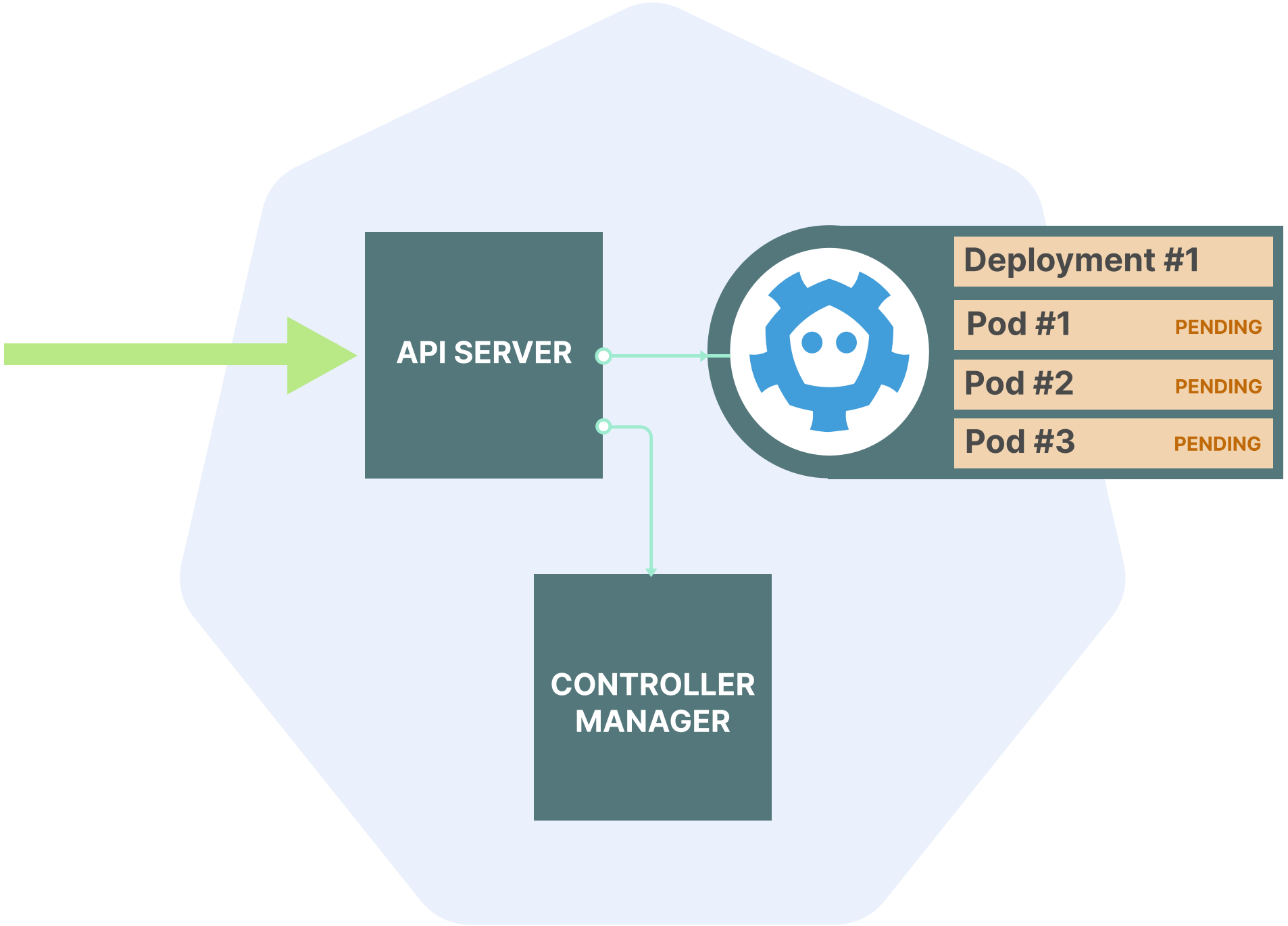

API SERVER
Deployment #1

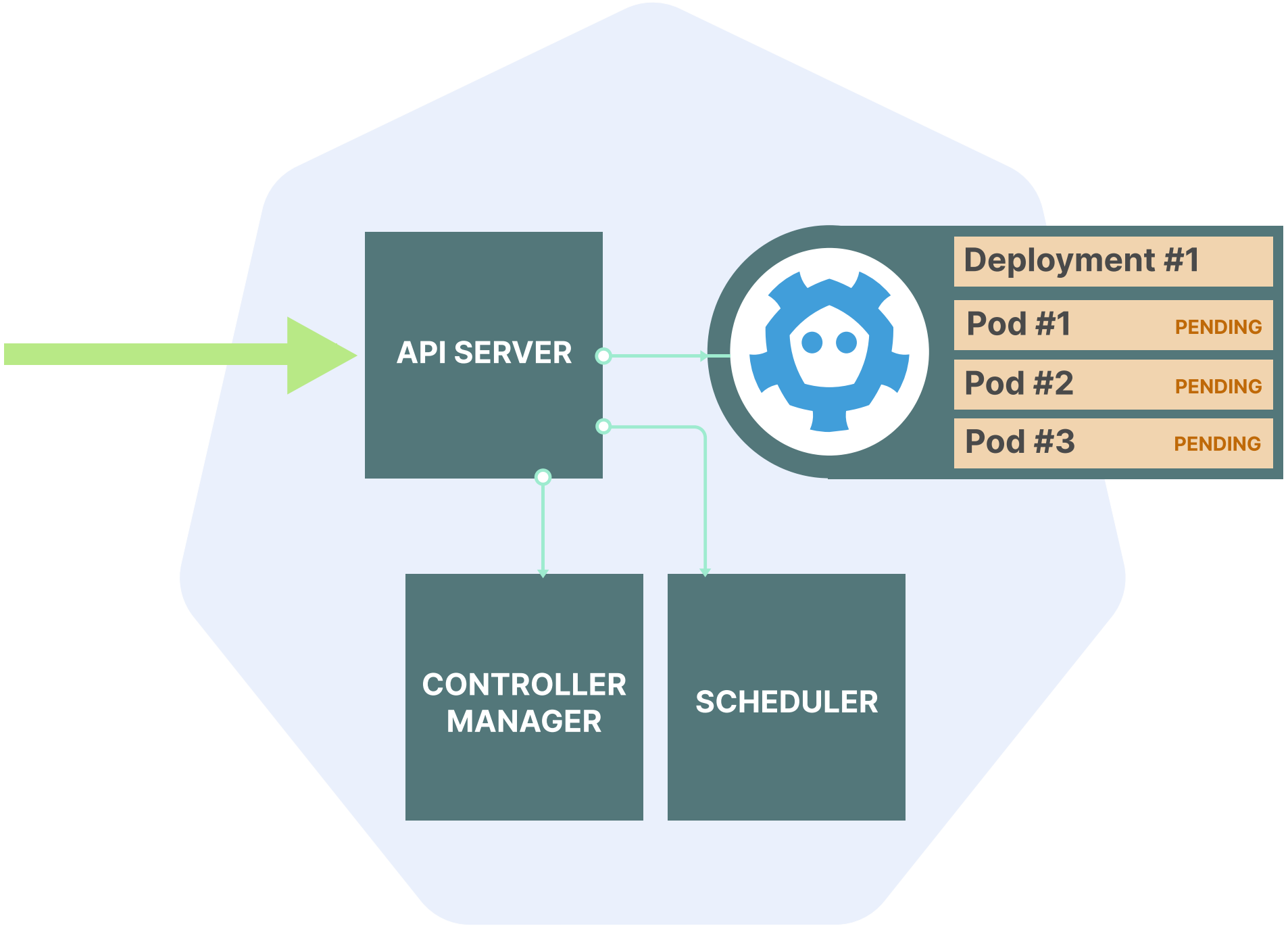

API SERVER
Deployment #1
CONTROLLER
MANAGER

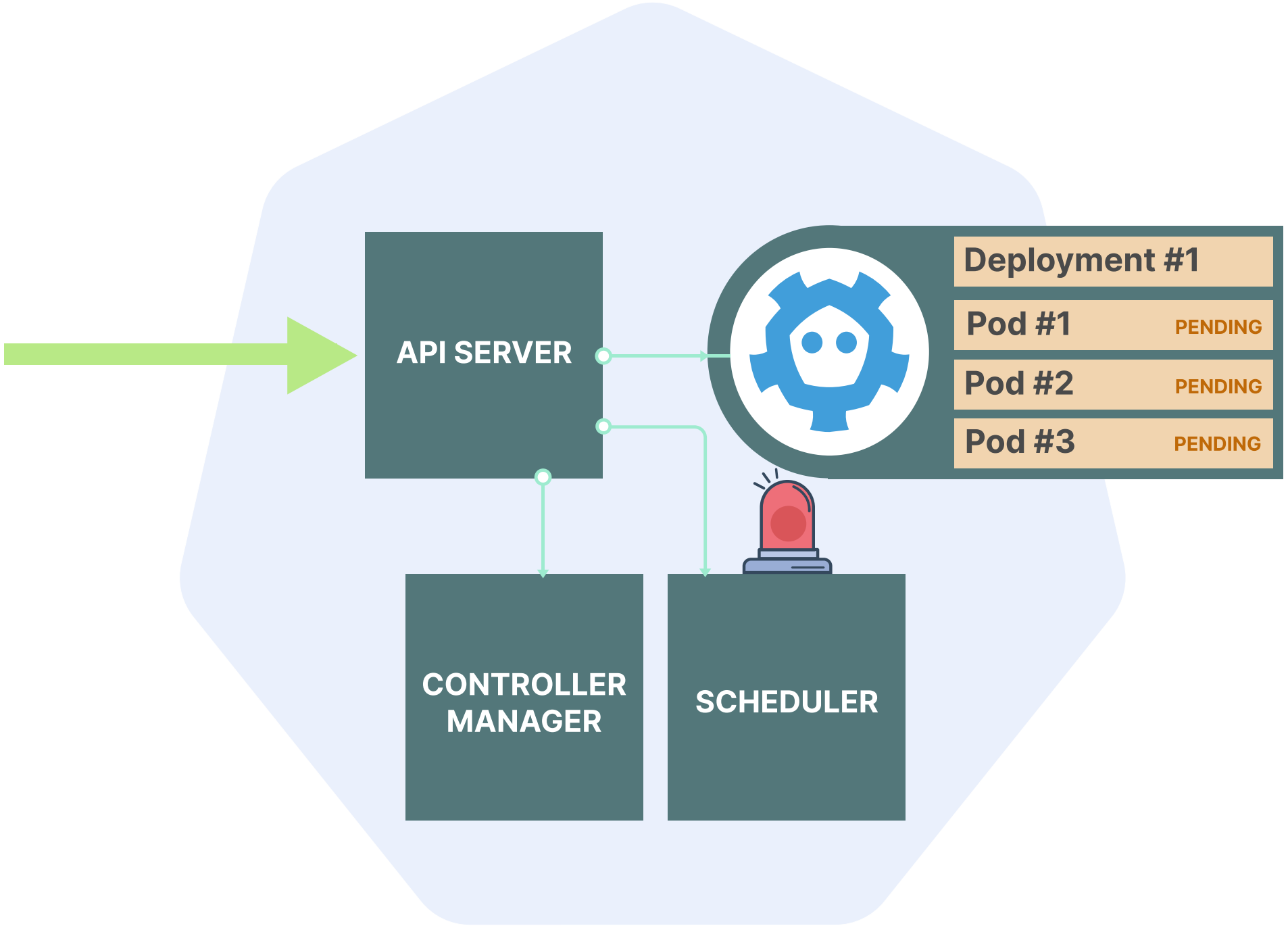

API SERVER
Deployment #1
CONTROLLER
MANAGER

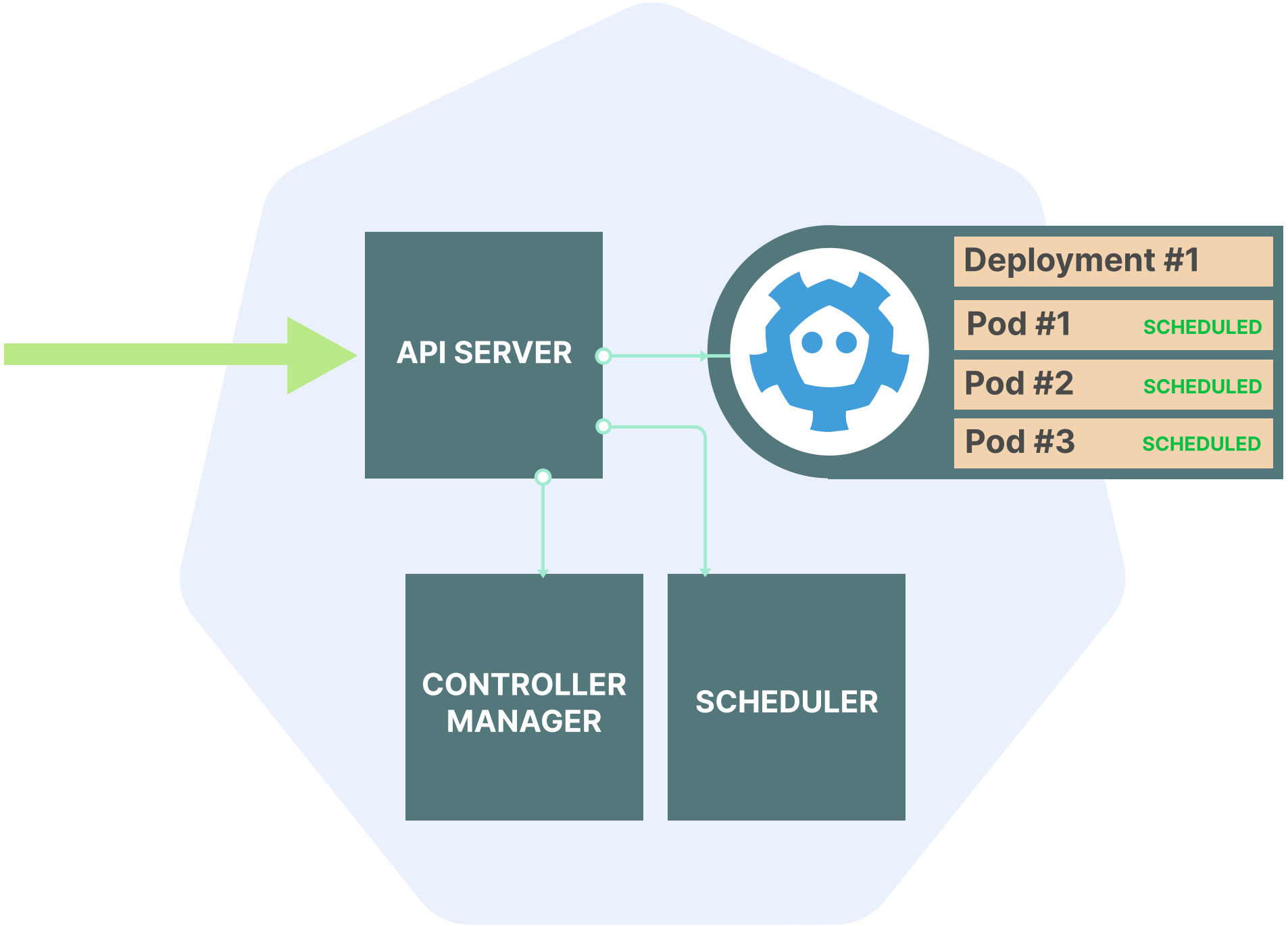

API SERVER
Deployment #1
DEPLOYMENT CONTROLLER
REPLICATION CONTROLLER


API SERVER
Deployment #1
Pod #1
Pod #2
Pod #3
CONTROLLER MANAGER


API SERVER
Deployment #1
Pod #1
PENDING
Pod #2
PENDING
Pod #3
PENDING
CONTROLLER MANAGER


API SERVER
Deployment #1
Pod #1
PENDING
Pod #2
PENDING
Pod #3
PENDING
CONTROLLER MANAGER
SCHEDULER


API SERVER
Deployment #1
Pod #1
PENDING
Pod #2
PENDING
Pod #3
PENDING
CONTROLLER MANAGER
SCHEDULER


API SERVER
Deployment #1
Pod #1
SCHEDULED
Pod #2
SCHEDULED
Pod #3
SCHEDULED
CONTROLLER MANAGER
SCHEDULER

Control plane

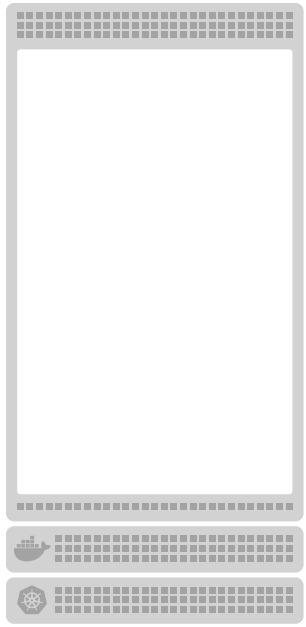
Not a single resource was created.

kubelet



Deploy 3 replicas, please

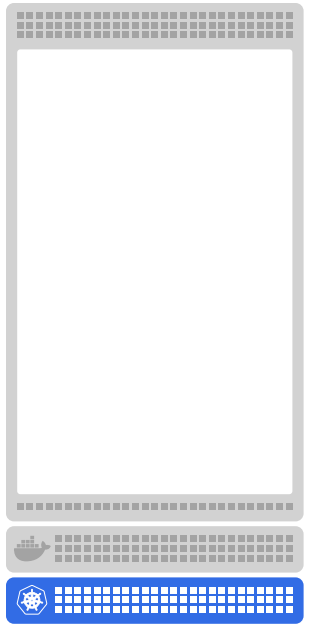






Deploy 3 replicas, please

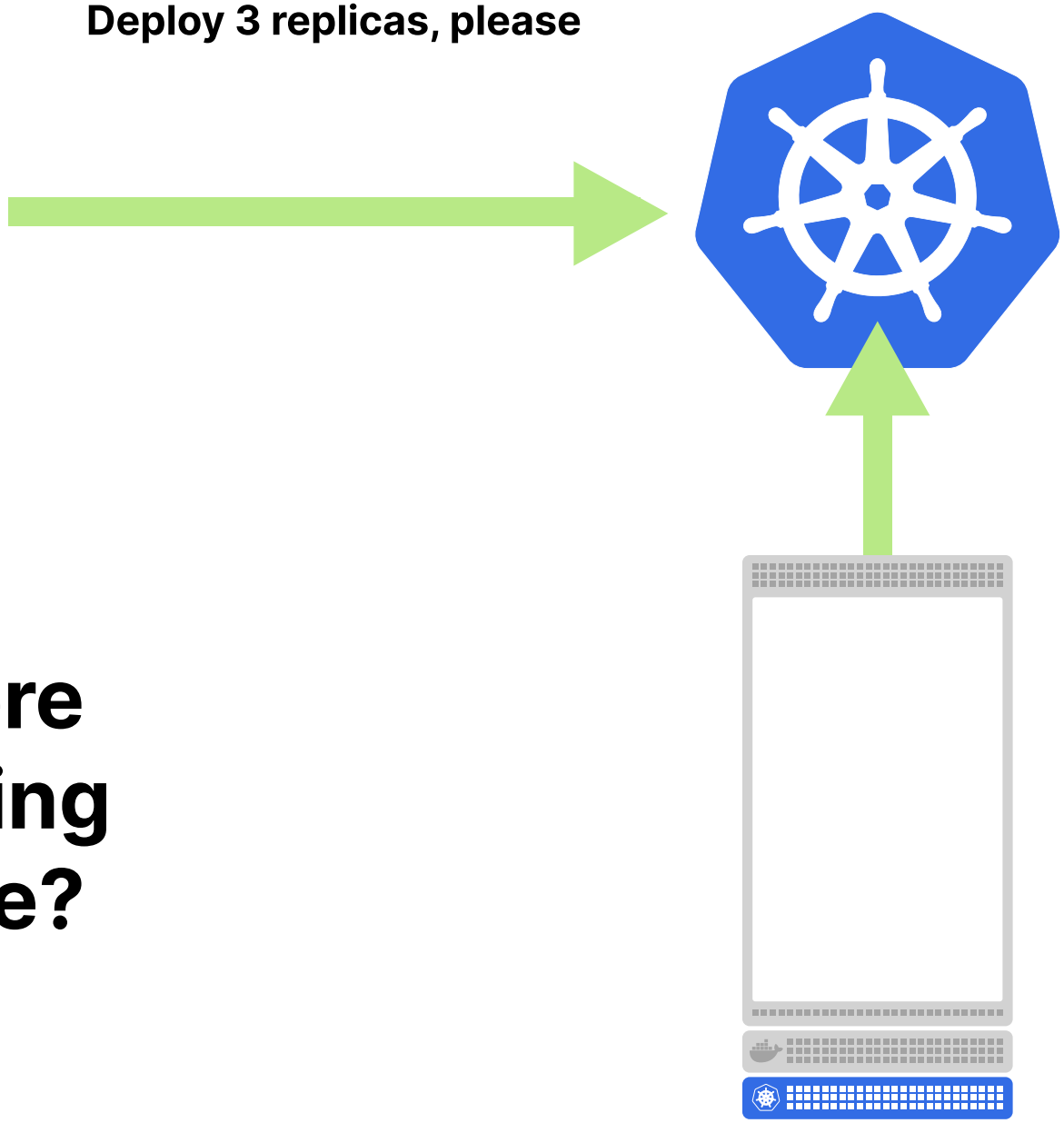




kubelet



Deploy 3 replicas, please

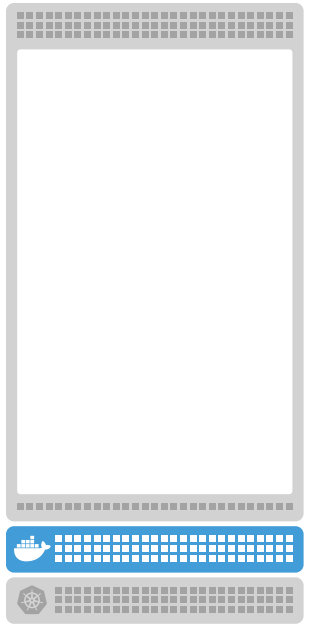


Is there anything for me?



Deploy 3 replicas, please





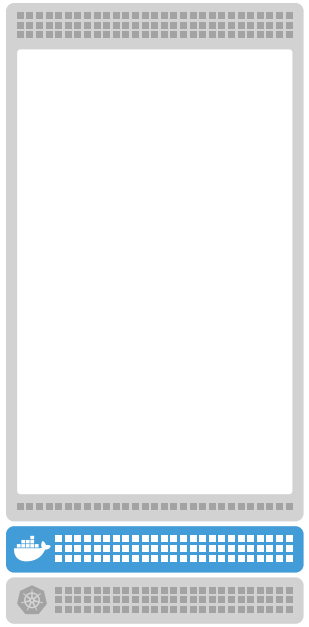
Docker daemon



Deploy 3 replicas, please



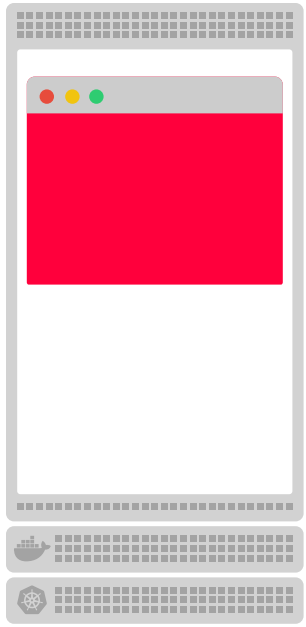
docker run -d mycontainer





Deploy 3 replicas, please

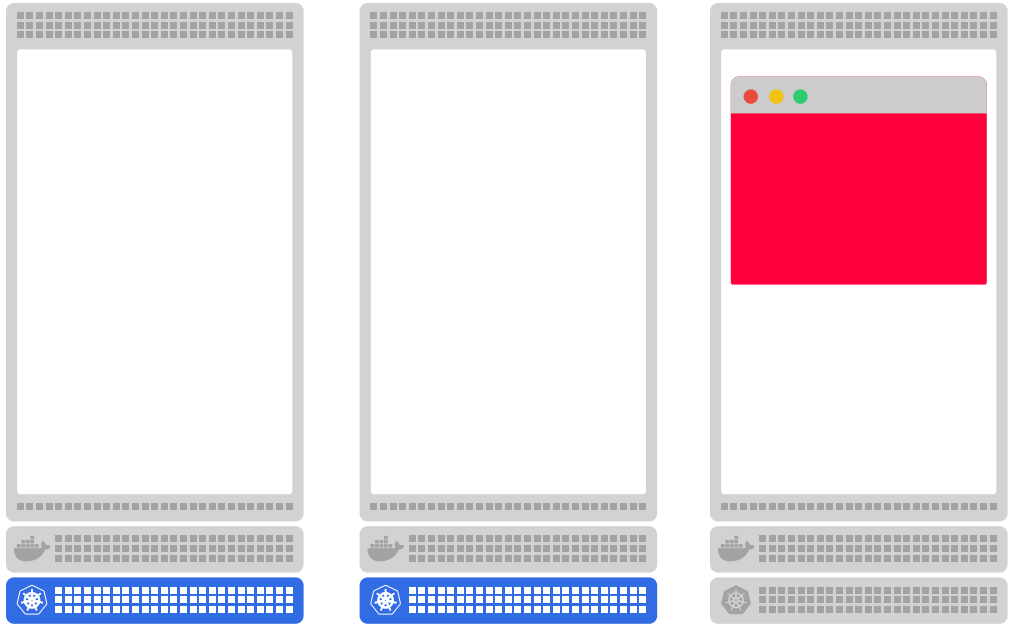






Deploy 3 replicas, please

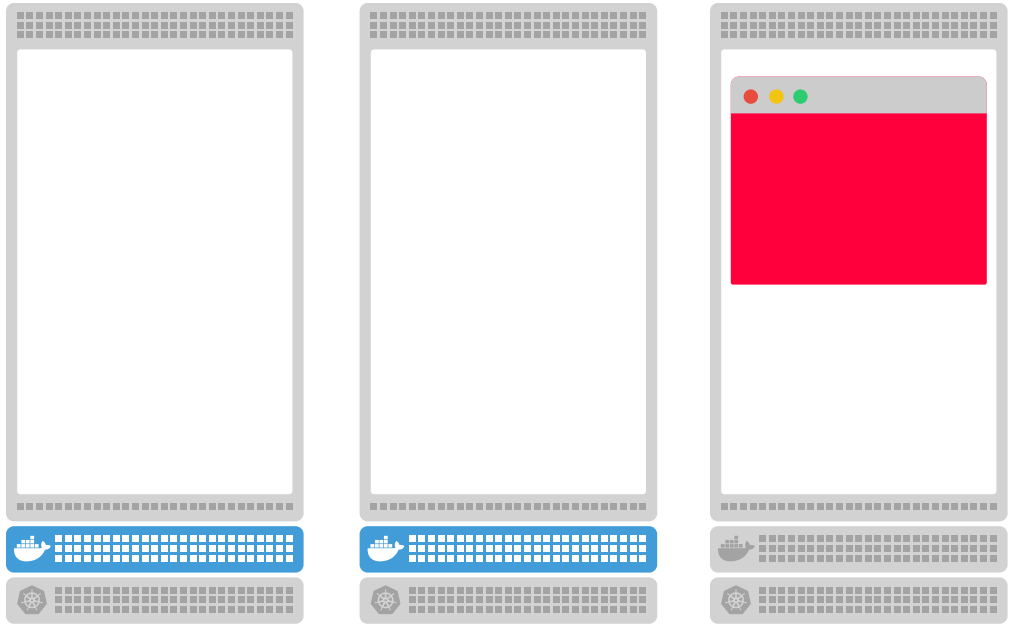






Deploy 3 replicas, please

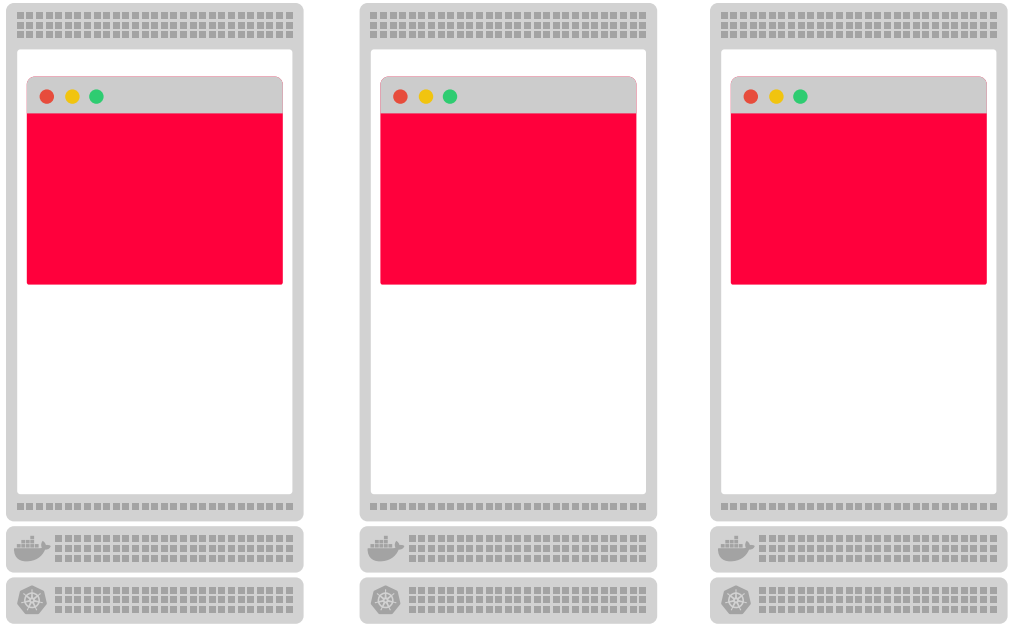




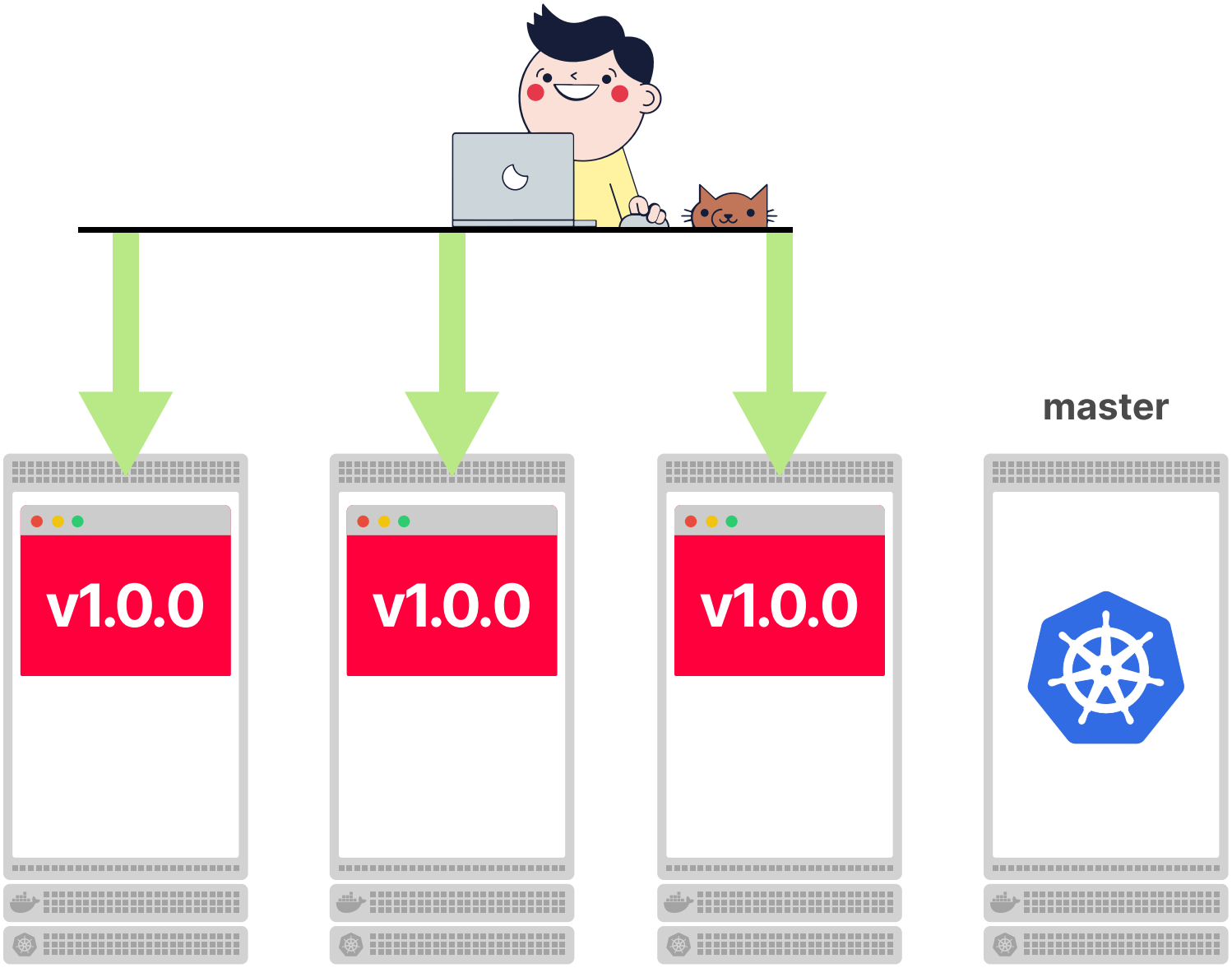


Deploy 3 replicas, please





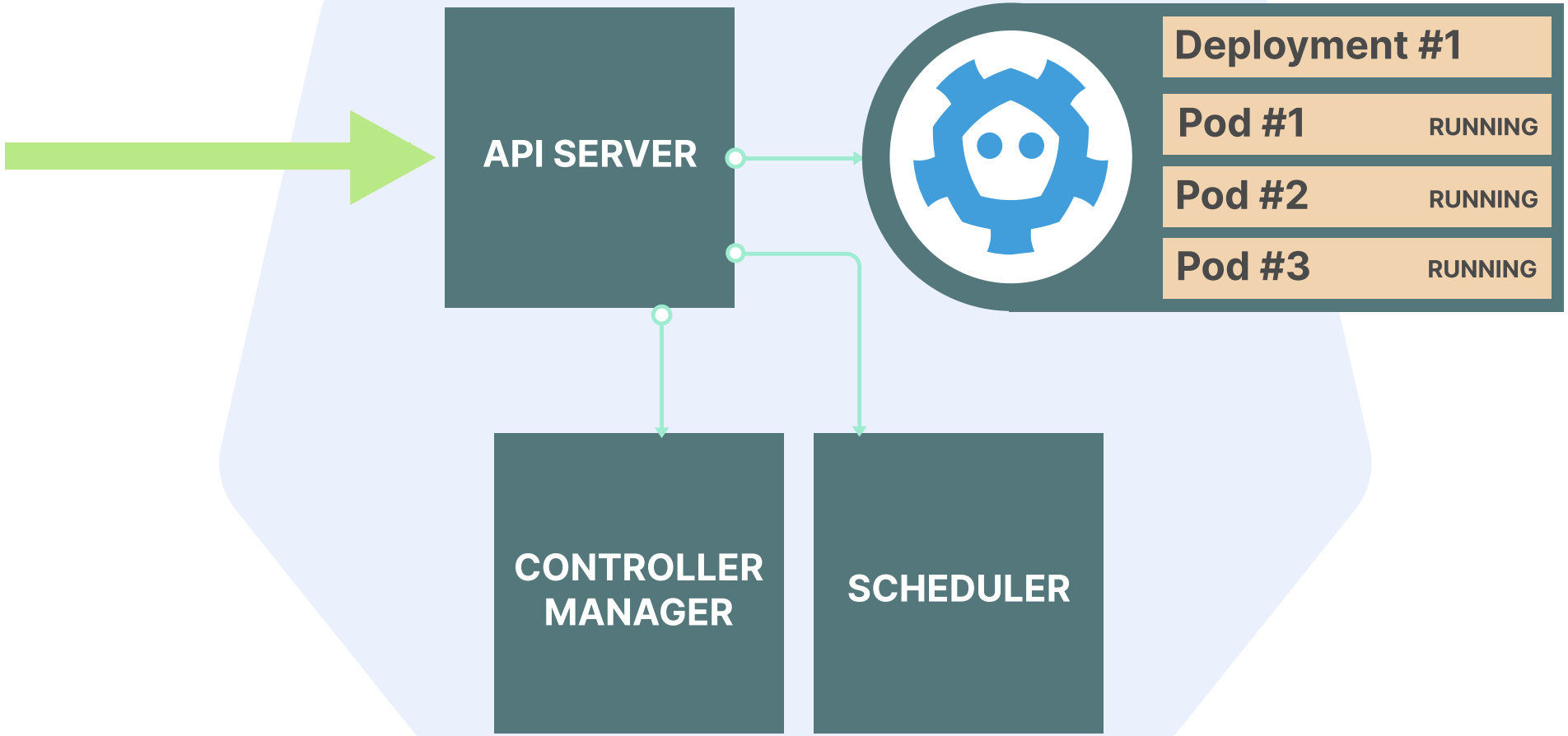
Rolling updates revisited

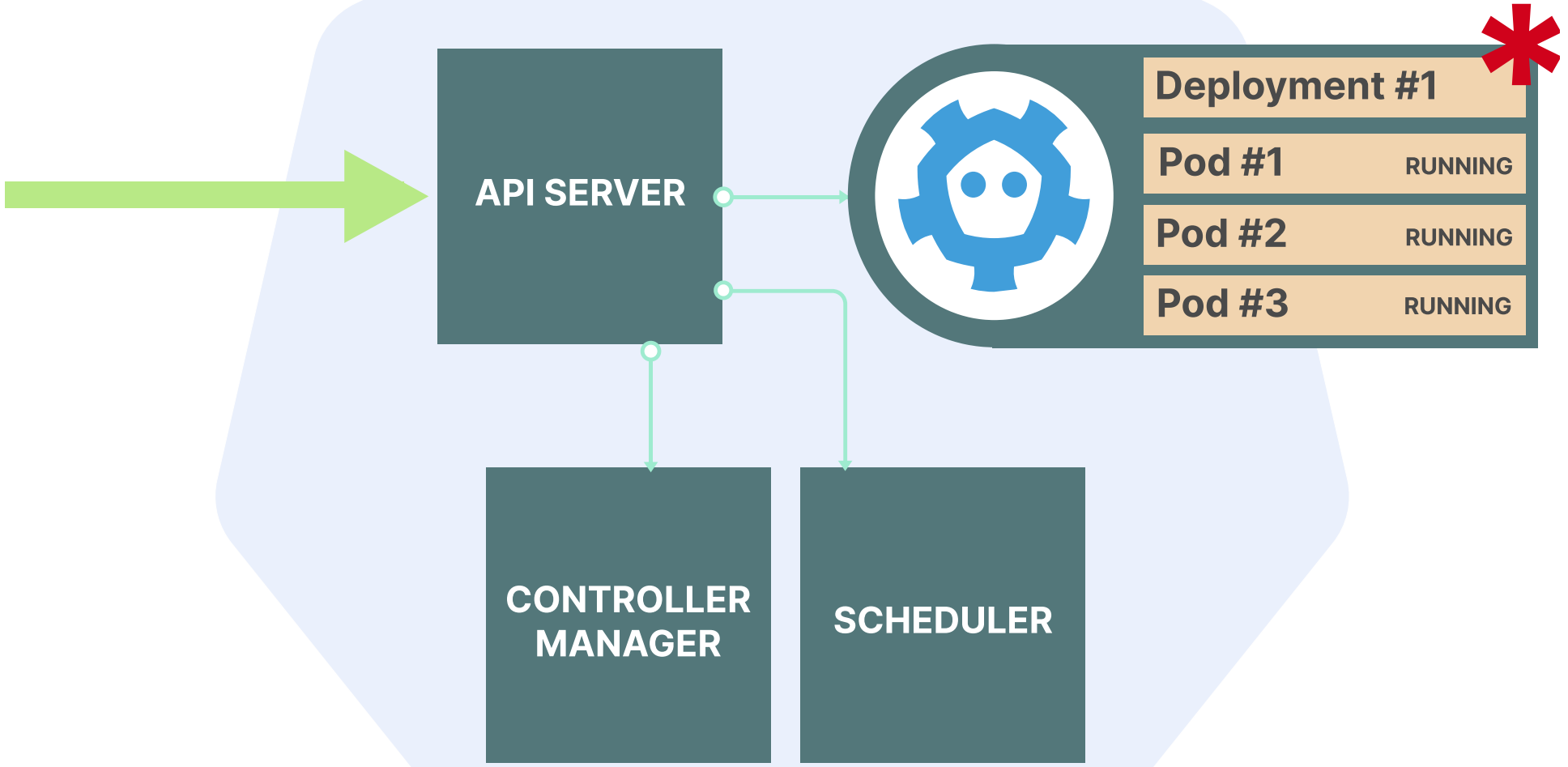

v1.0.0
v1.0.0
v1.0.0
master

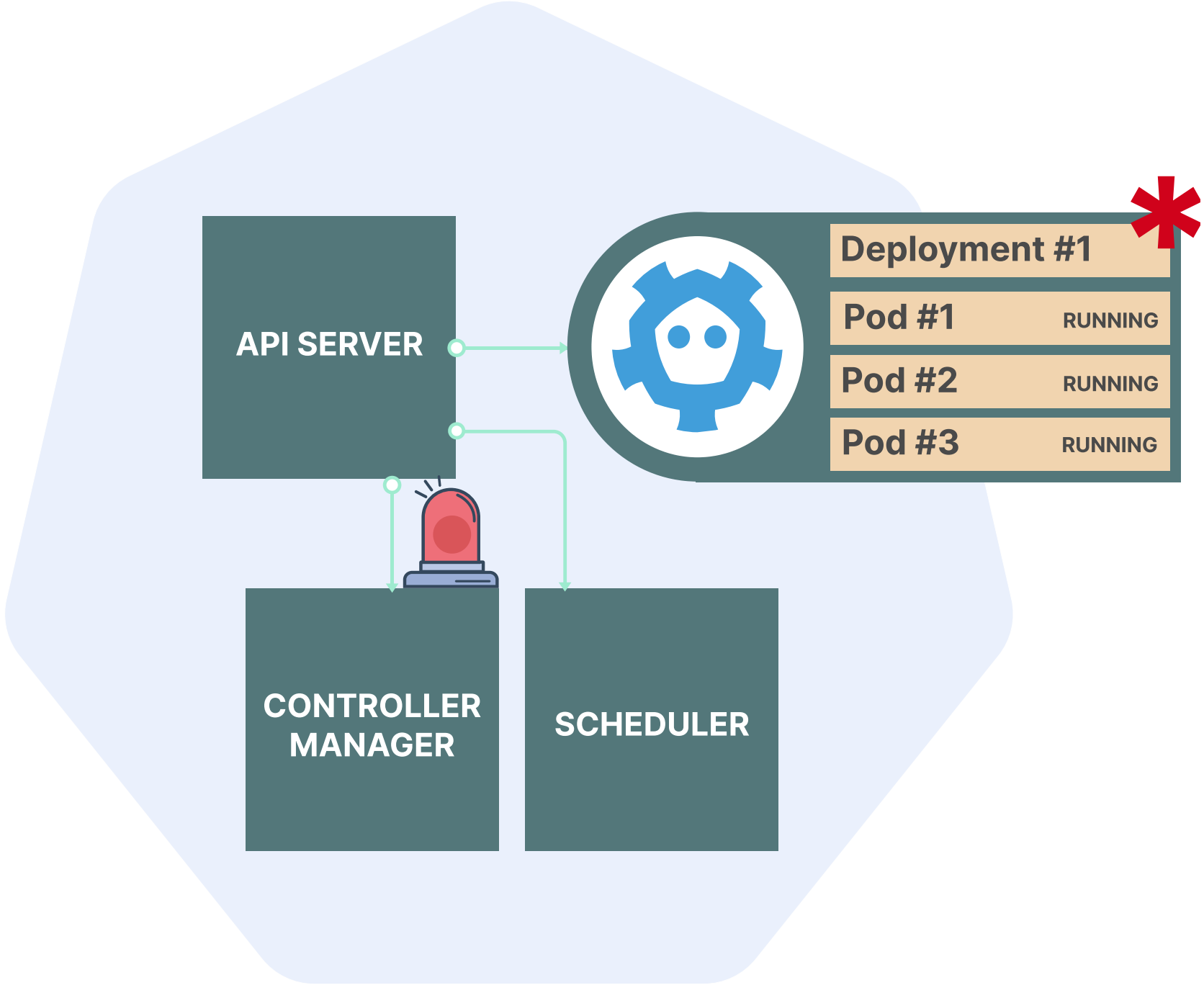
Update to image:v2.0.0

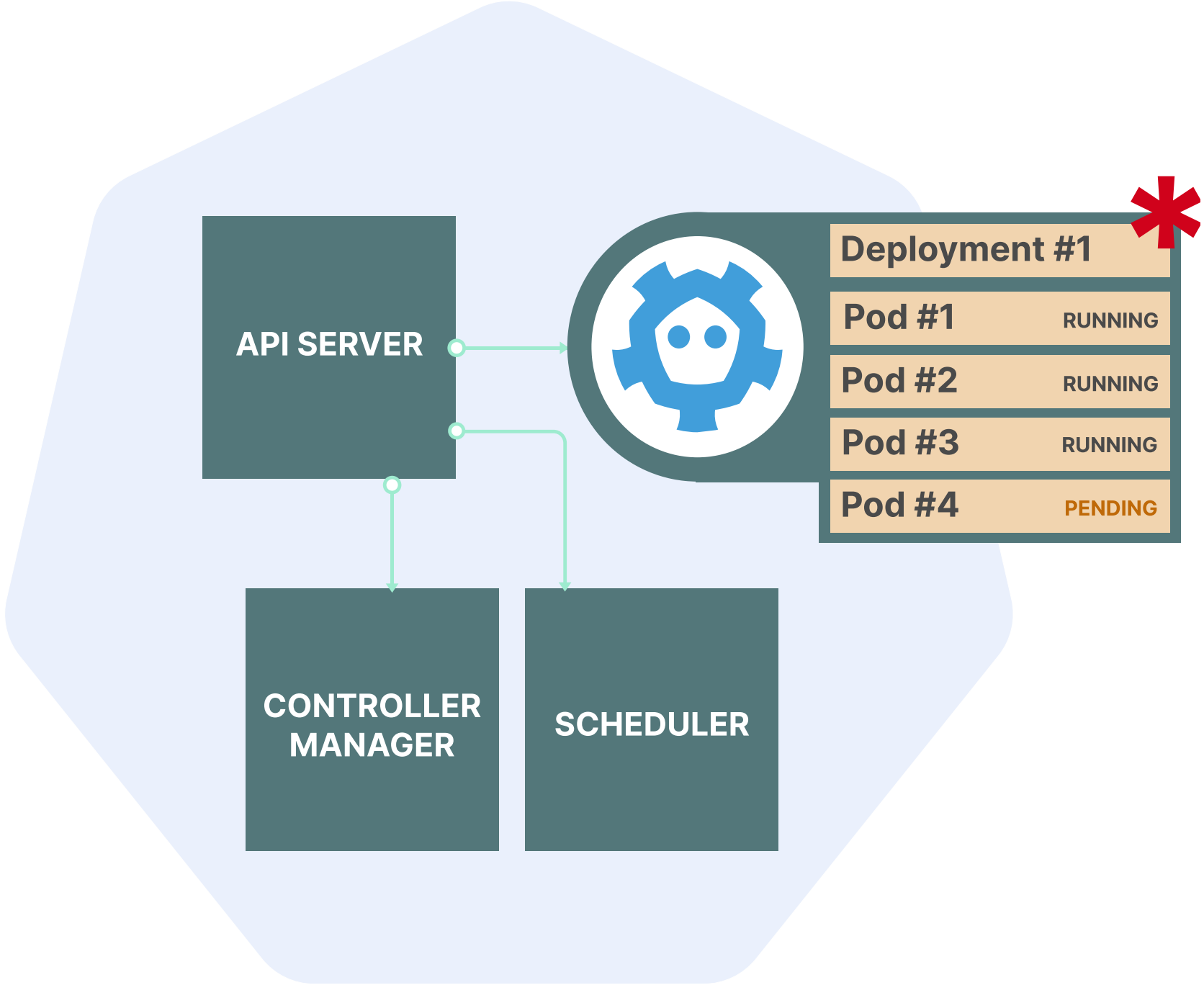


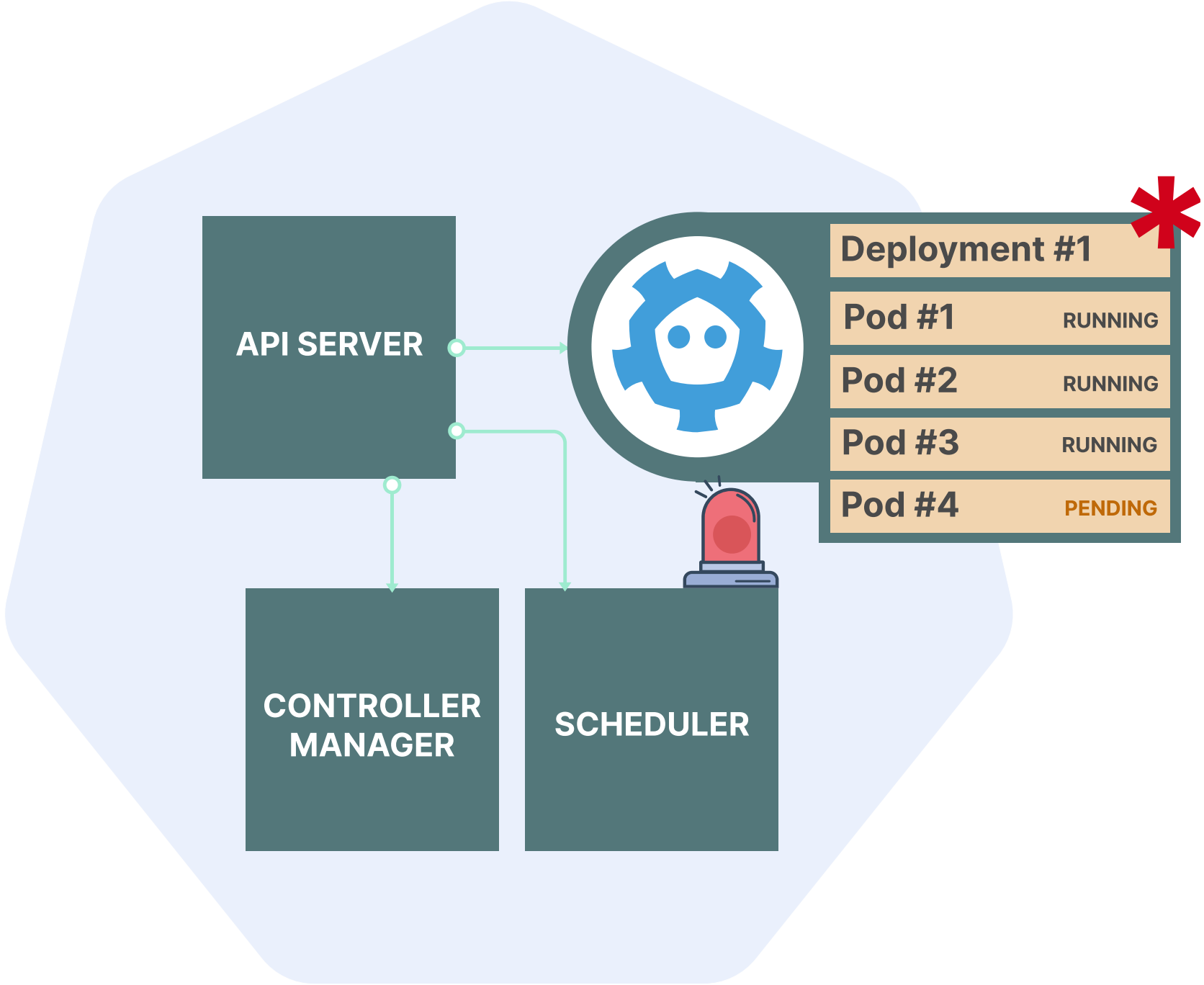


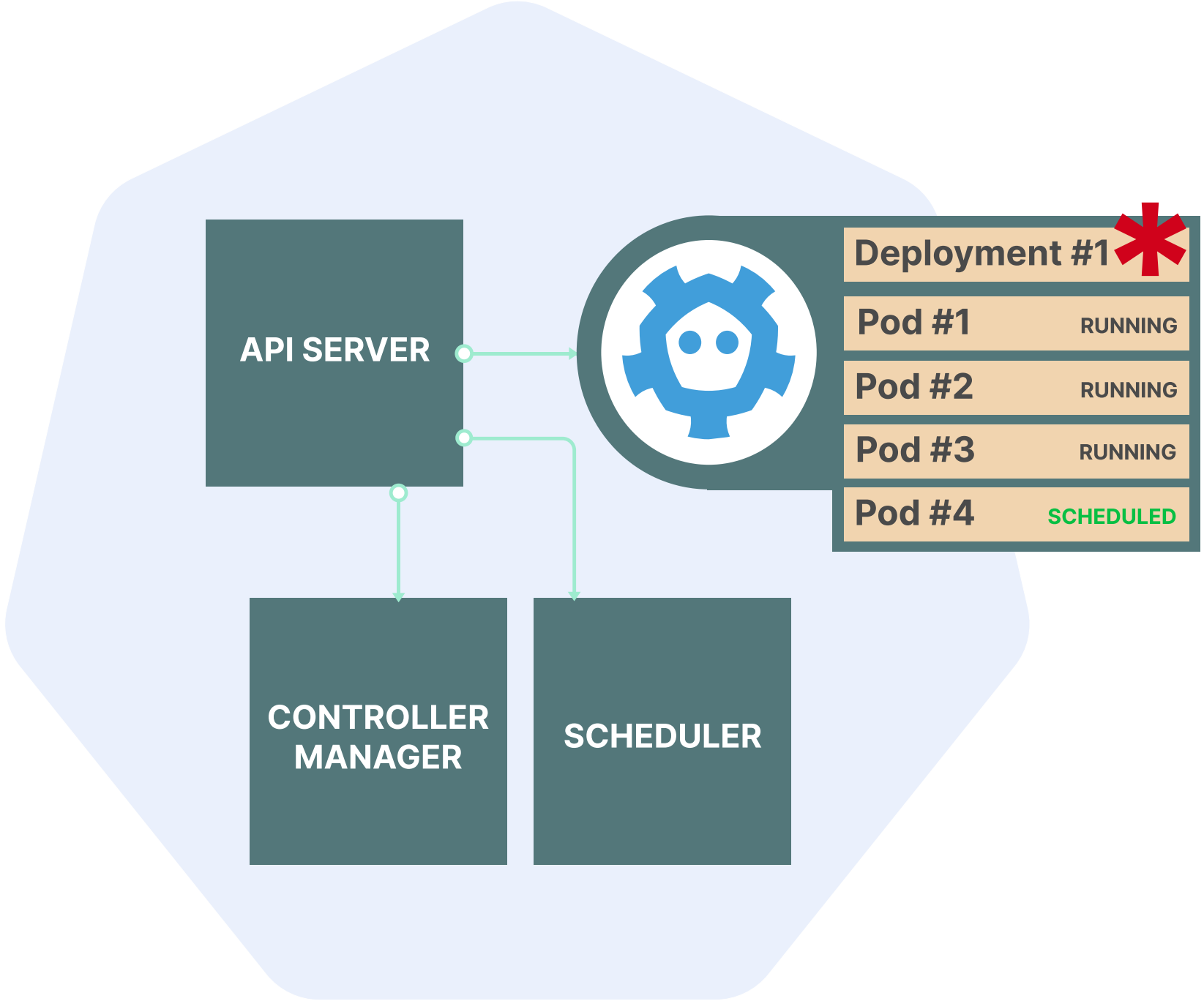

API SERVER
Deployment #1
Pod #1
RUNNING
Pod #2
RUNNING
Pod #3
RUNNING
CONTROLLER MANAGER
SCHEDULER

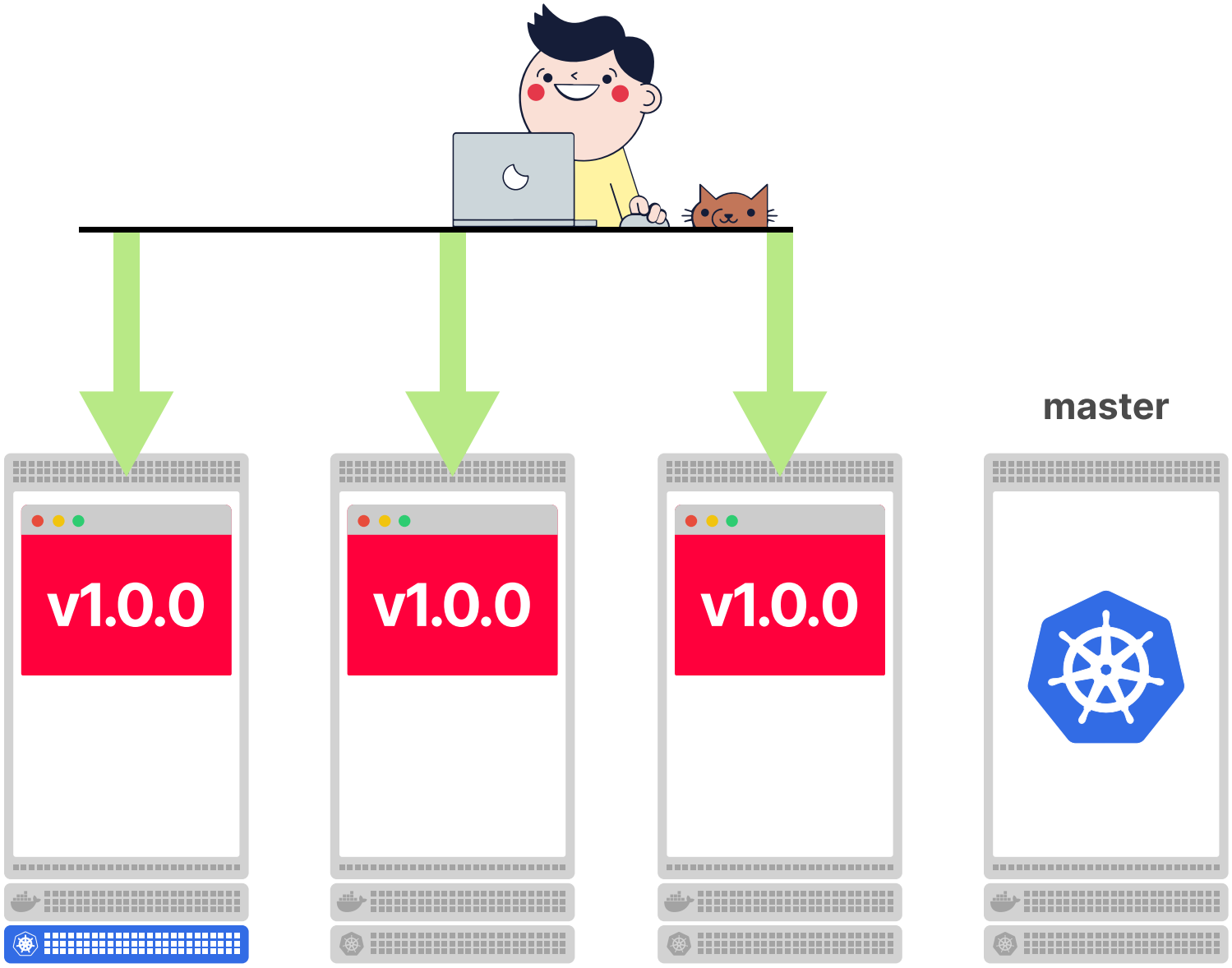

API SERVER
Deployment #1
Pod #1
RUNNING
Pod #2
RUNNING
Pod #3
RUNNING
CONTROLLER MANAGER
SCHEDULER

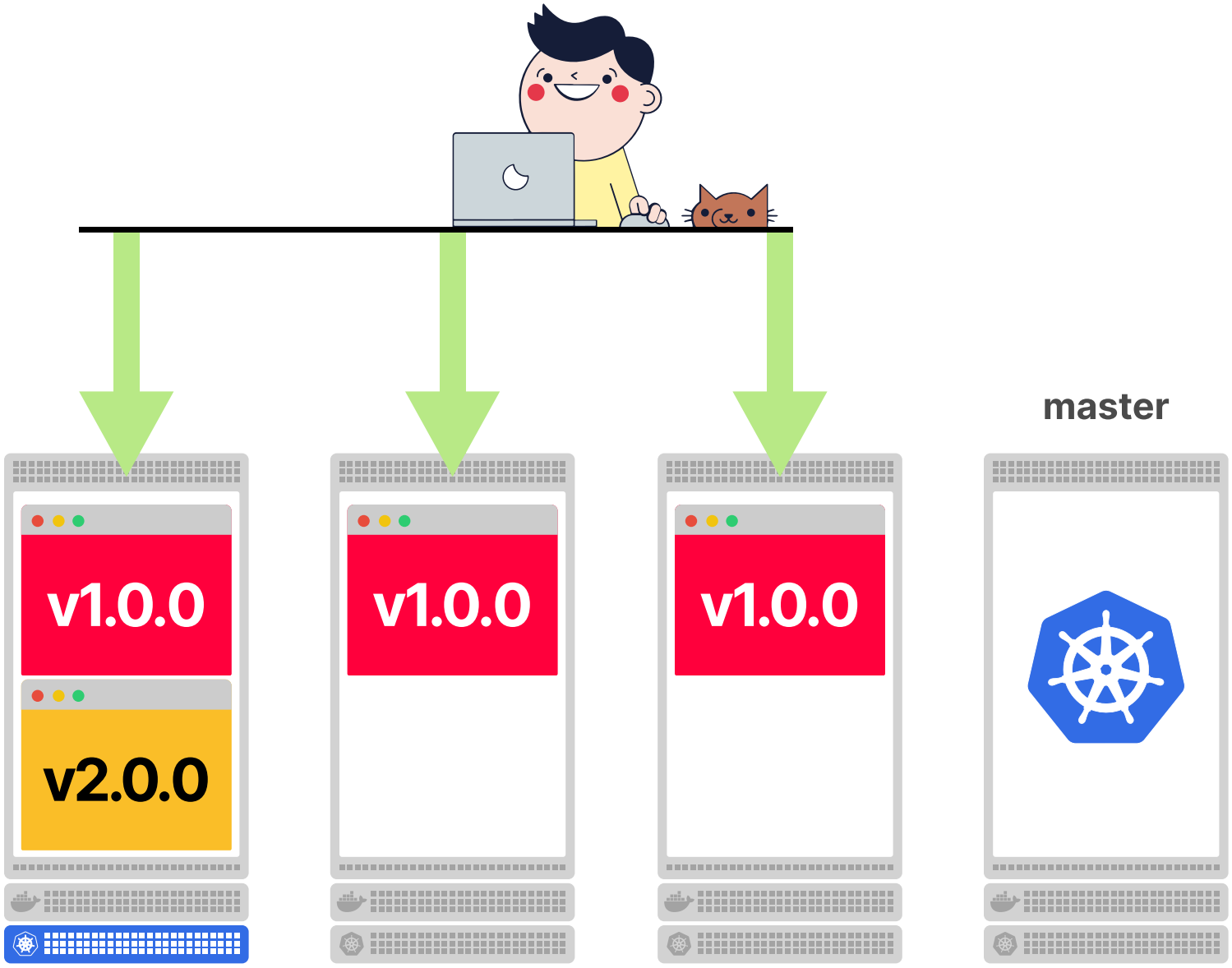

API SERVER
Deployment #1
Pod #1
RUNNING
Pod #2
RUNNING
Pod #3
RUNNING
CONTROLLER MANAGER
SCHEDULER

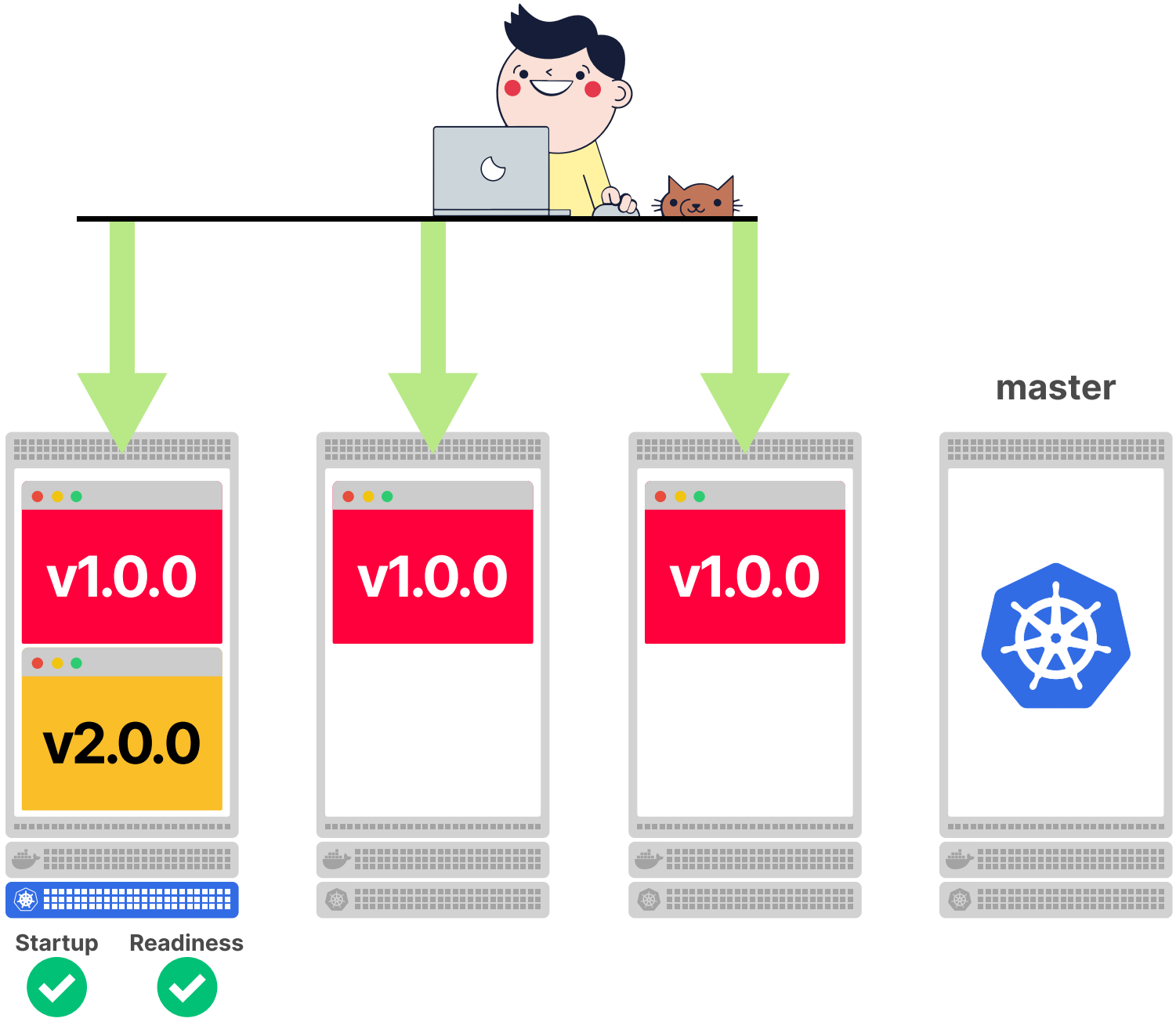

API SERVER
CONTROLLER MANAGER
SCHEDULER
Deployment #1
Pod #1 RUNNING
Pod #2 RUNNING
Pod #3 RUNNING
Pod #4 PENDING

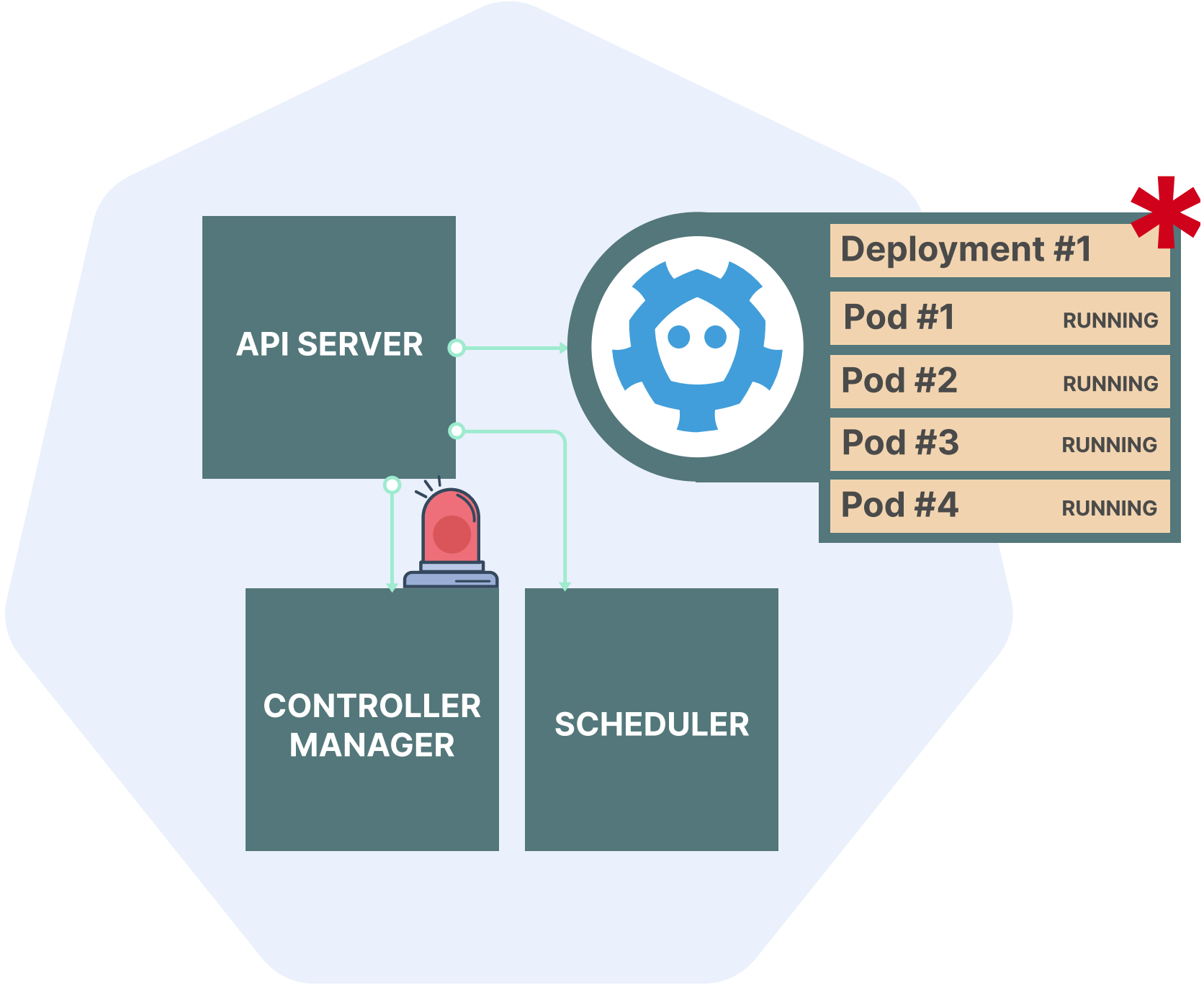

API SERVER
CONTROLLER MANAGER
SCHEDULER
Deployment #1
Pod #1 RUNNING
Pod #2 RUNNING
Pod #3 RUNNING
Pod #4 PENDING

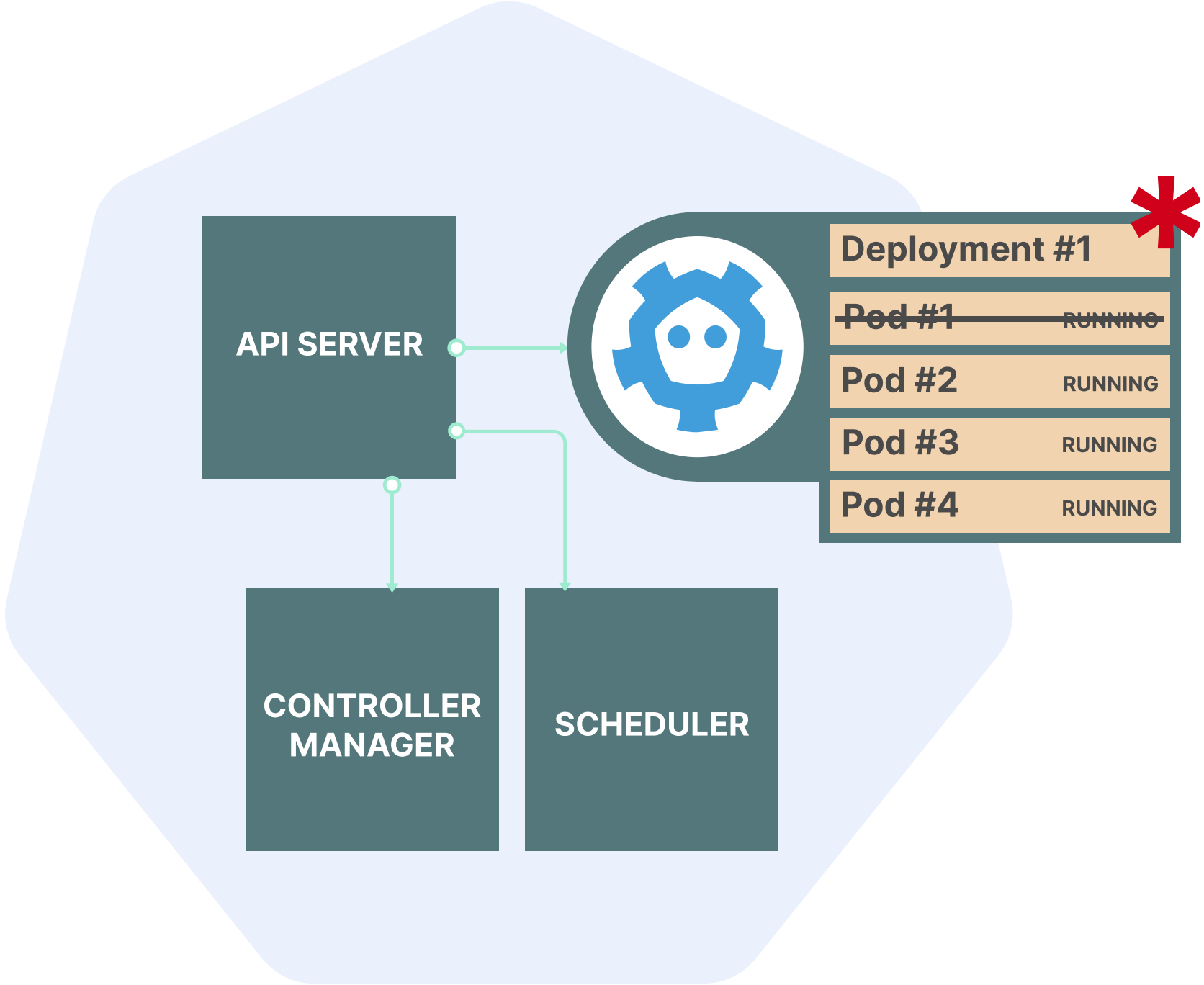

API SERVER
CONTROLLER MANAGER
SCHEDULER
Deployment #1
Pod #1 RUNNING
Pod #2 RUNNING
Pod #3 RUNNING
Pod #4 SCHEDULED

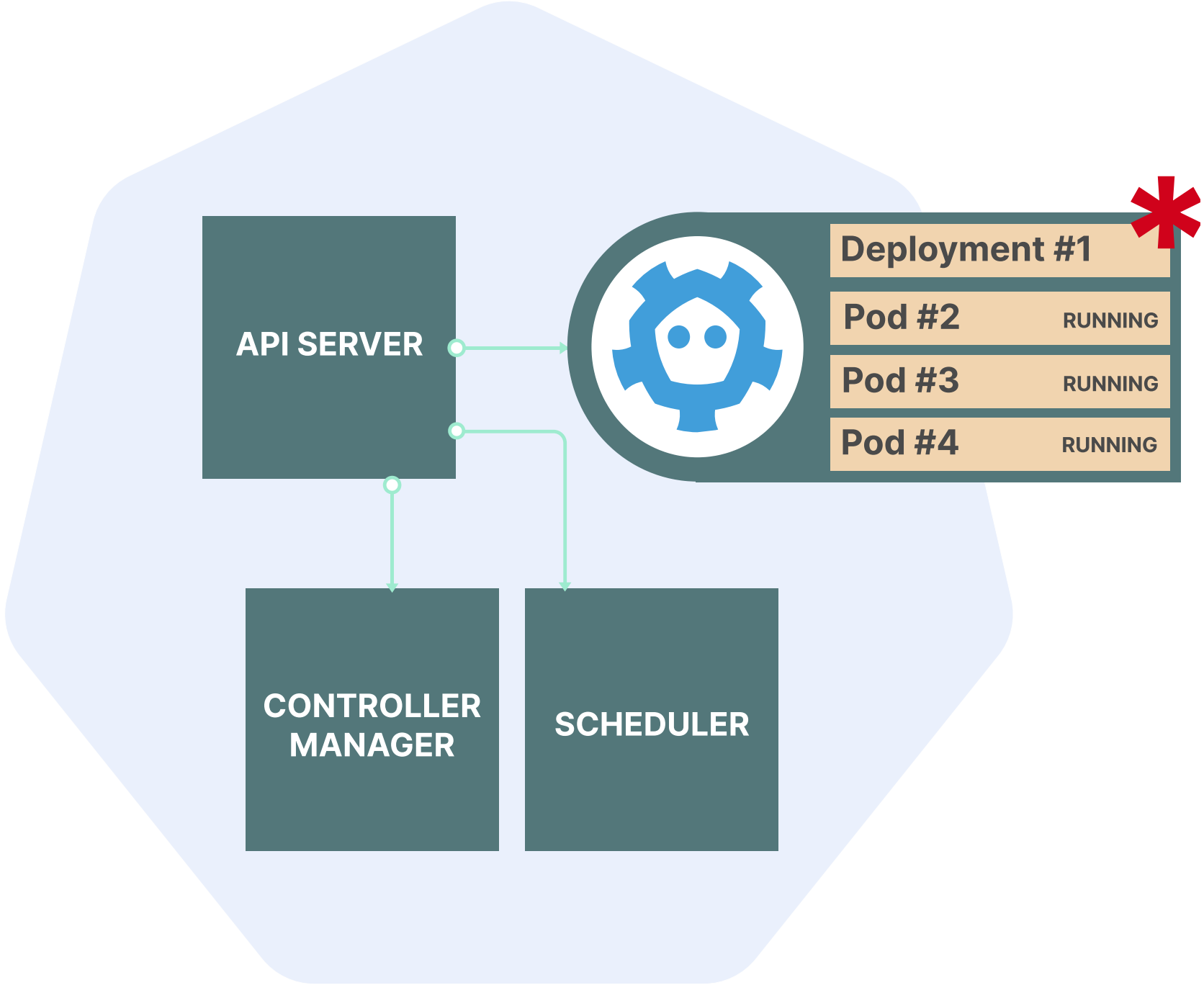

v1.0.0
v1.0.0
v1.0.0
master

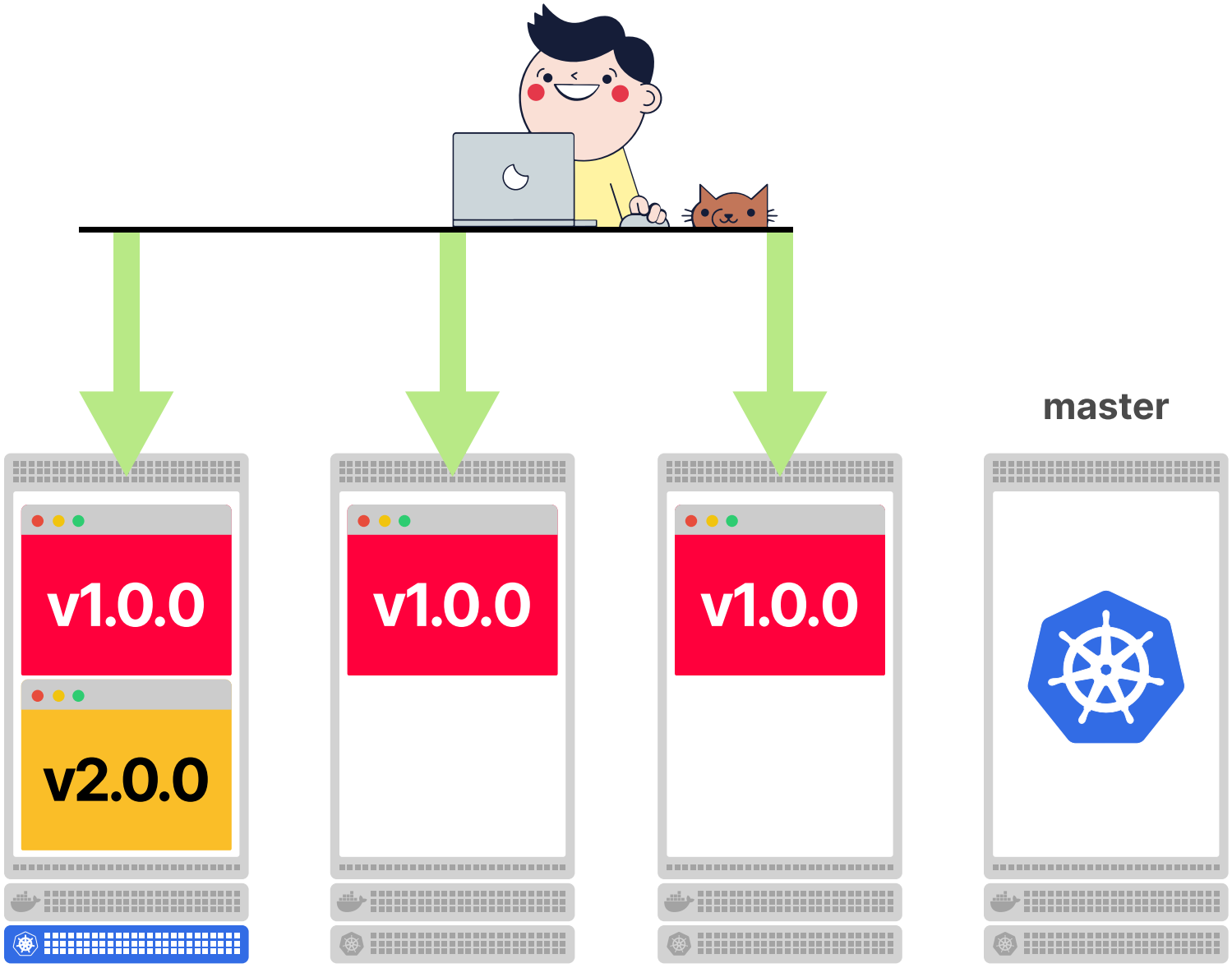

v1.0.0
v2.0.0
v1.0.0
v1.0.0
master

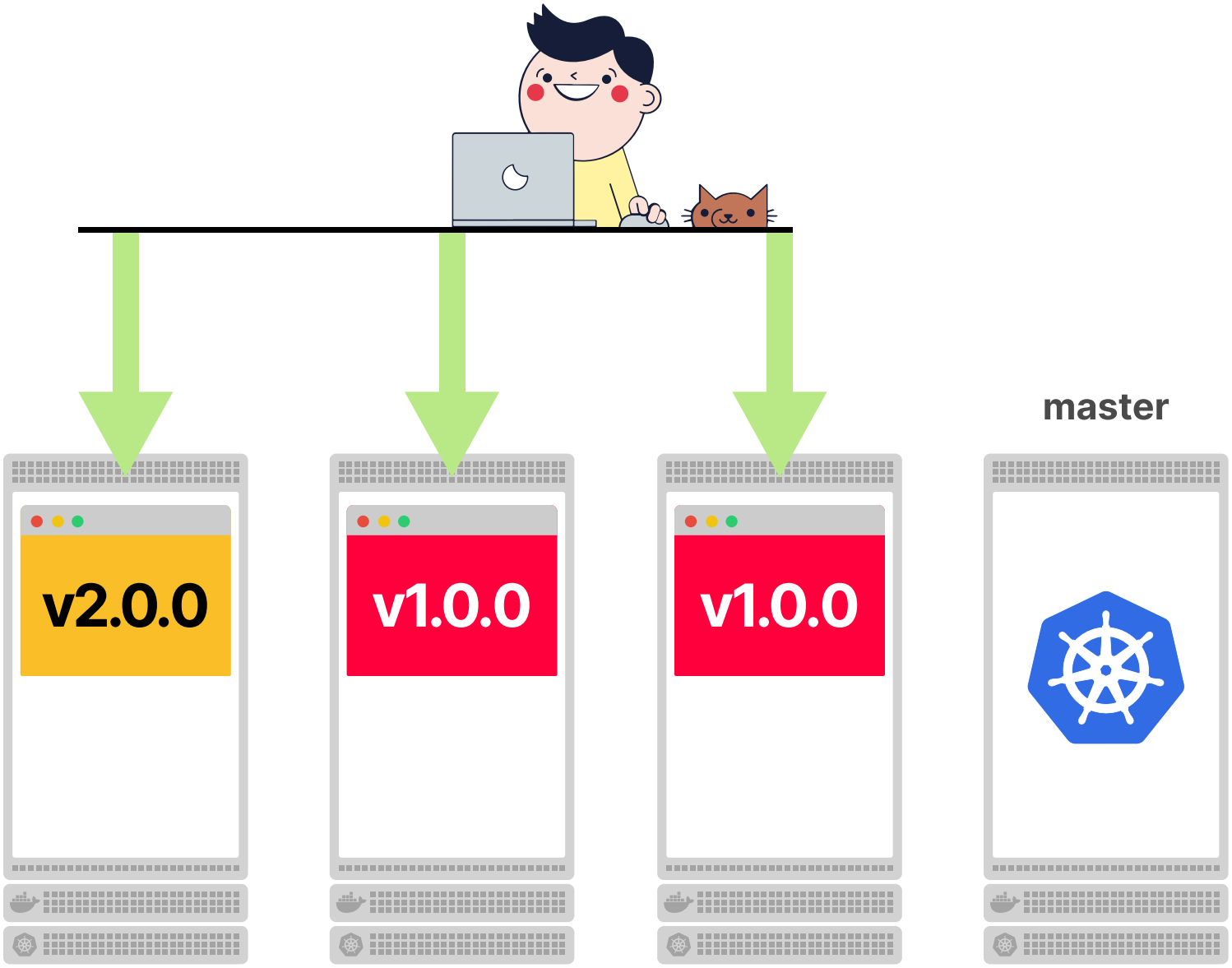

master
v1.0.0
v2.0.0
v1.0.0
v1.0.0
Startup
Readiness

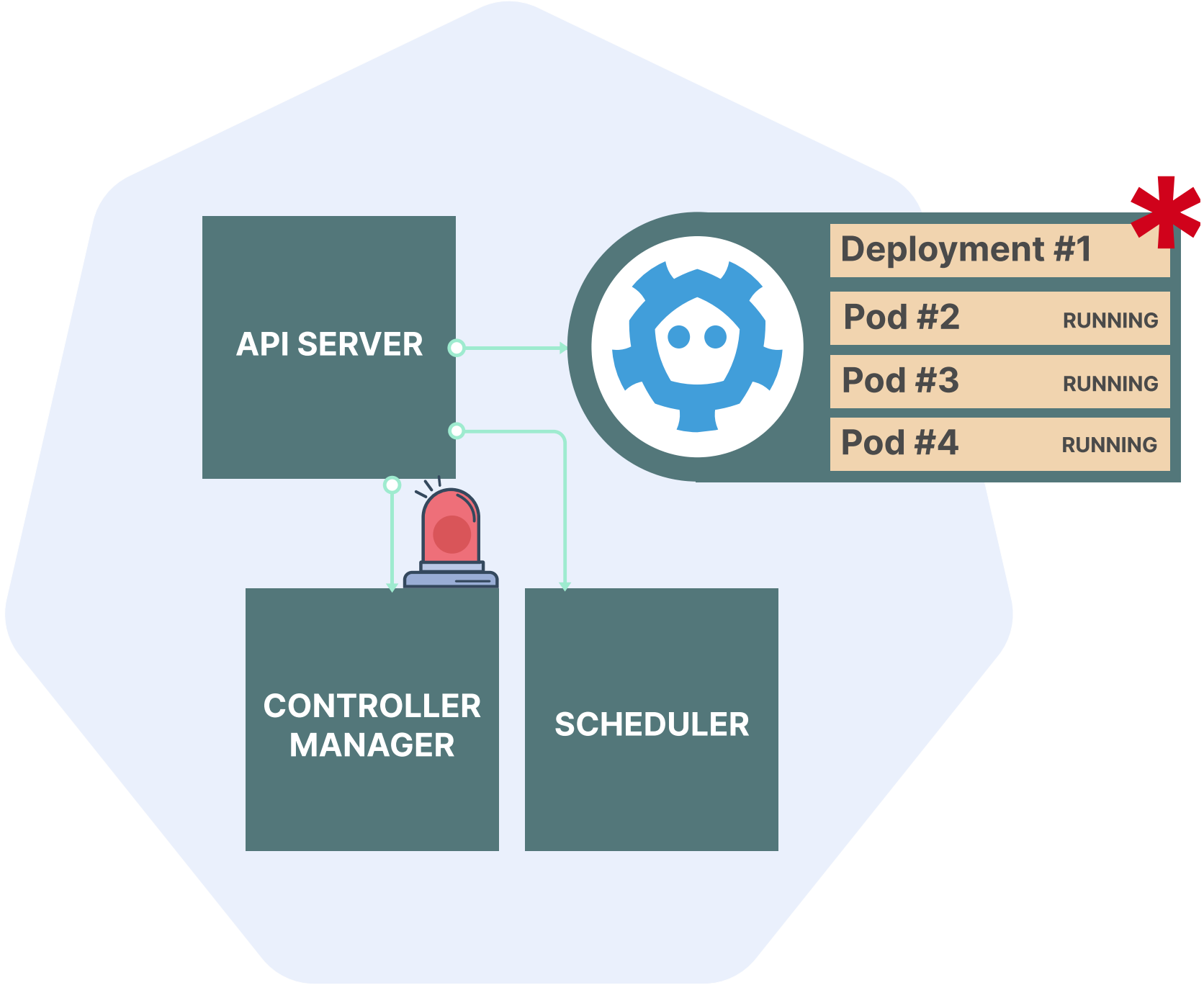

API SERVER
CONTROLLER MANAGER
SCHEDULER
Deployment #1
Pod #1 RUNNING
Pod #2 RUNNING
Pod #3 RUNNING
Pod #4 RUNNING

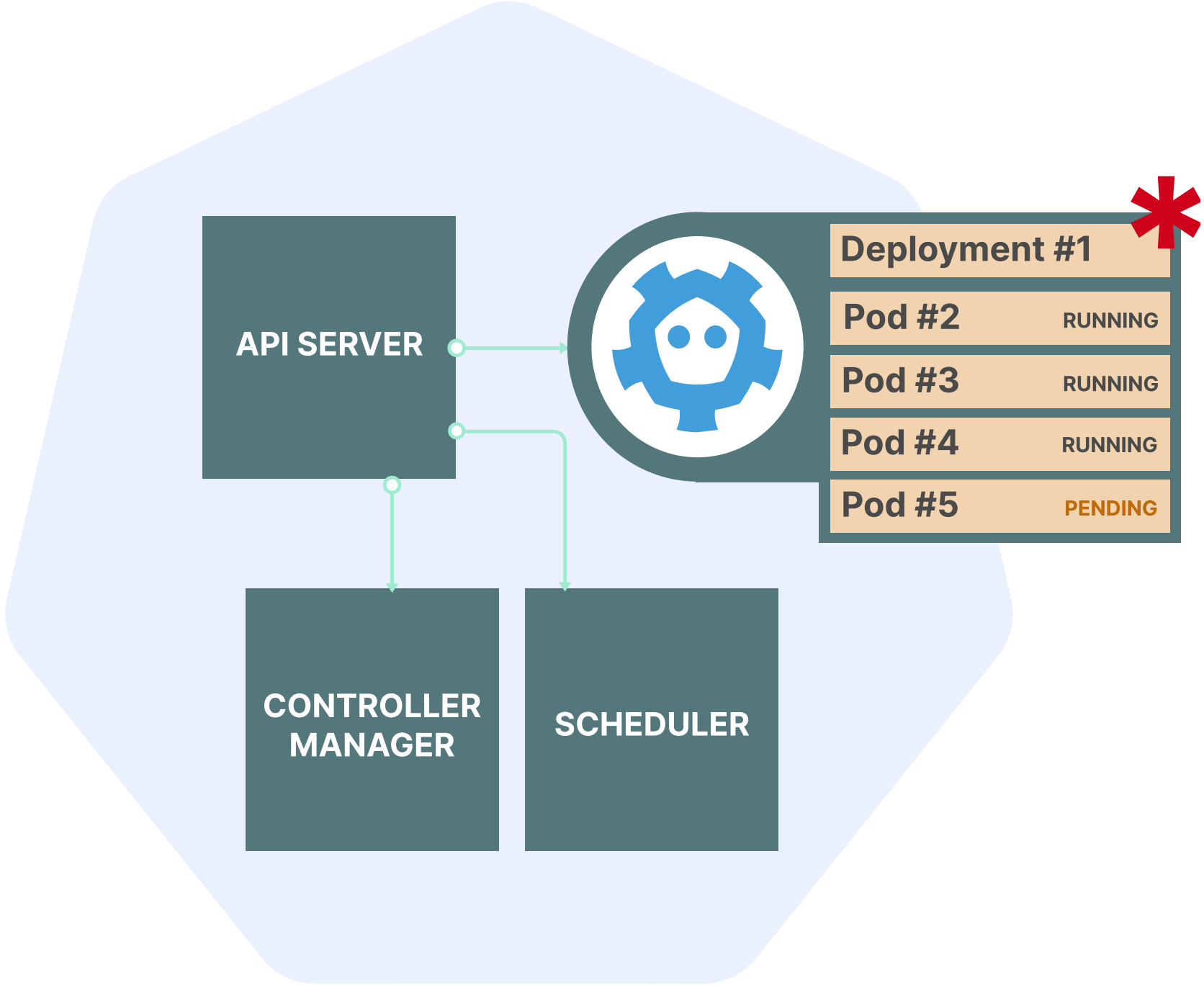

API SERVER
CONTROLLER MANAGER
SCHEDULER
Deployment #1
Pod #1 RUNNING
Pod #2 RUNNING
Pod #3 RUNNING
Pod #4 RUNNING

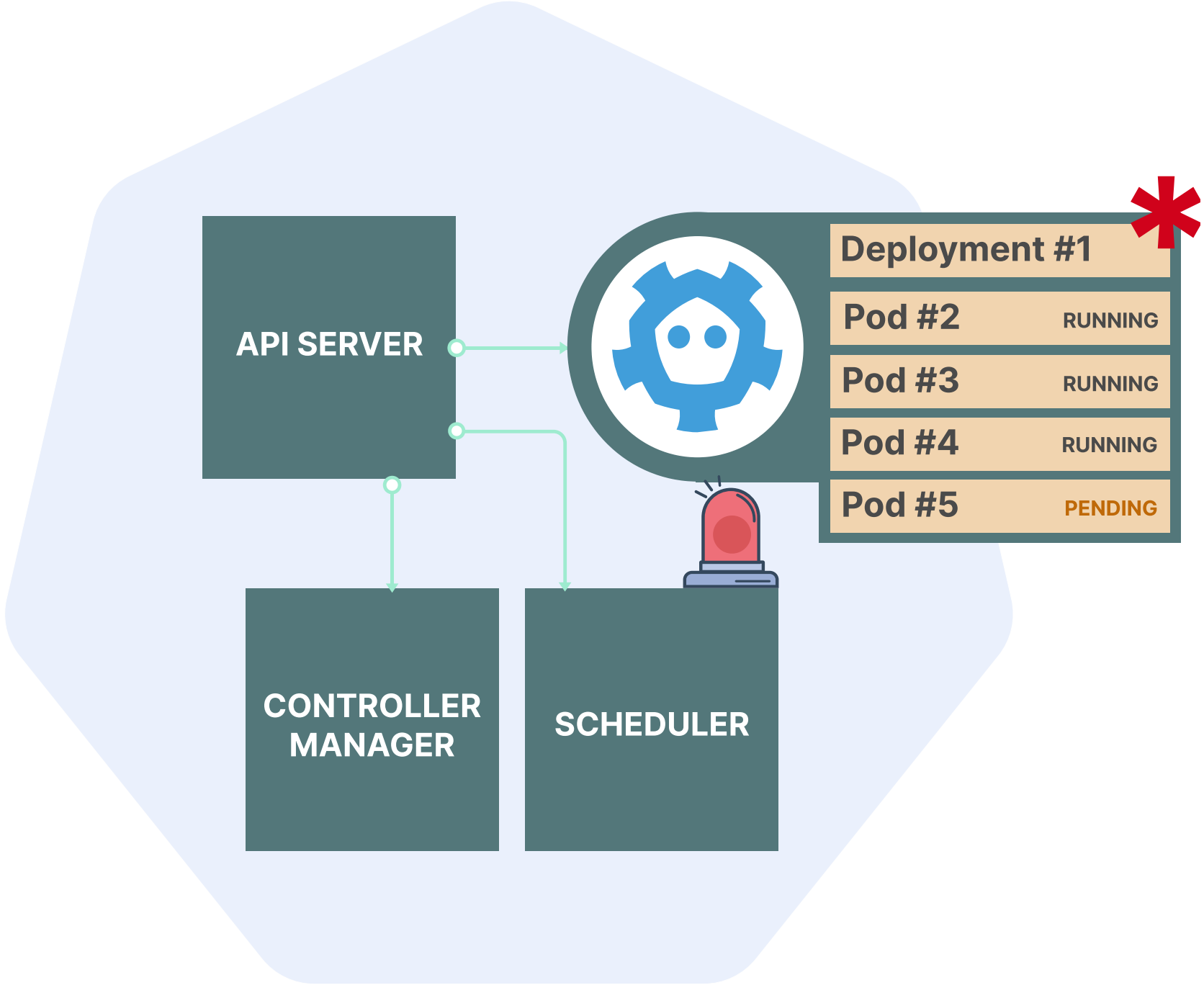

API SERVER
Deployment #1
Pod #2
RUNNING
Pod #3
RUNNING
Pod #4
RUNNING
CONTROLLER MANAGER
SCHEDULER

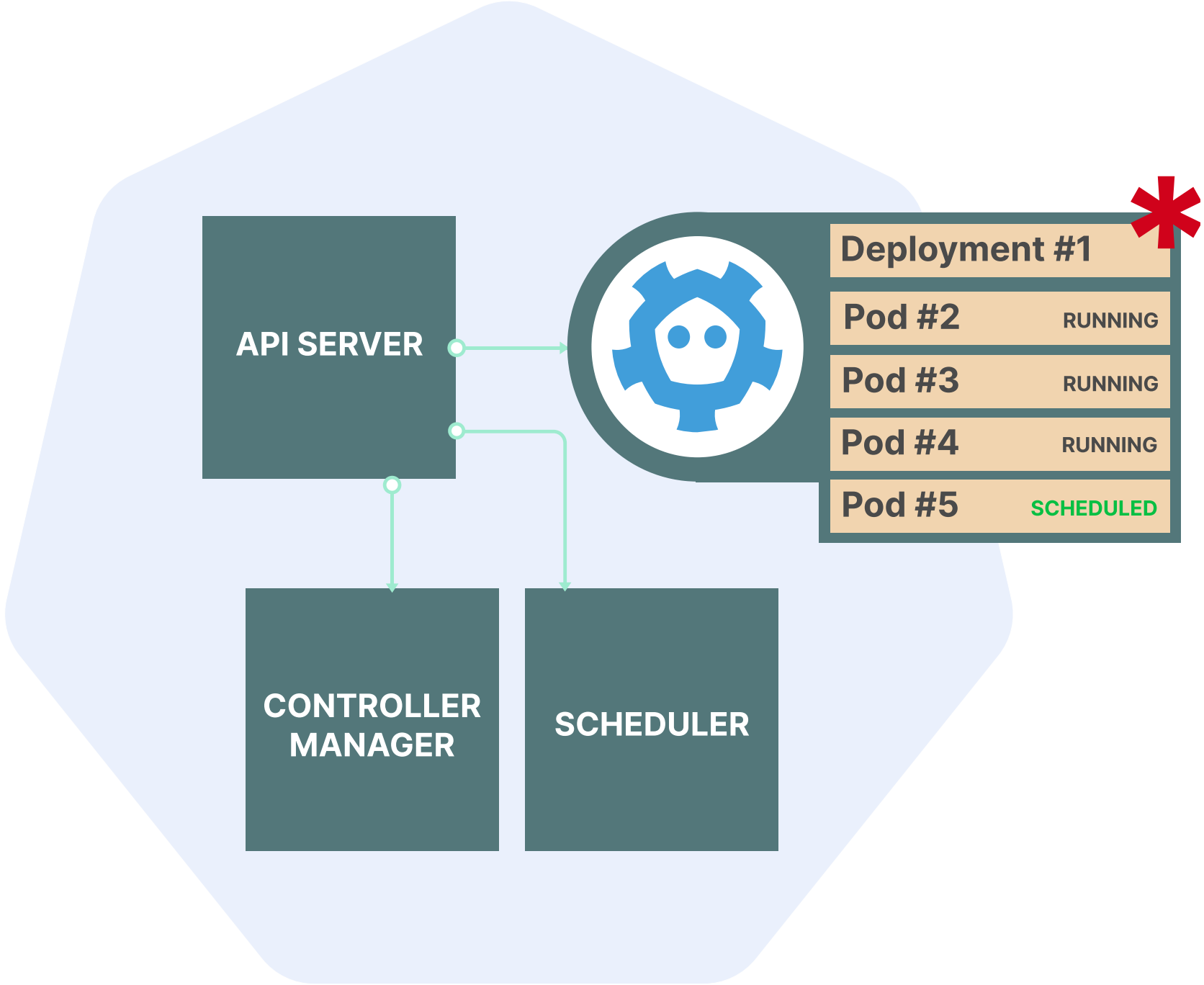

v1.0.0
v2.0.0
v1.0.0
v1.0.0
master

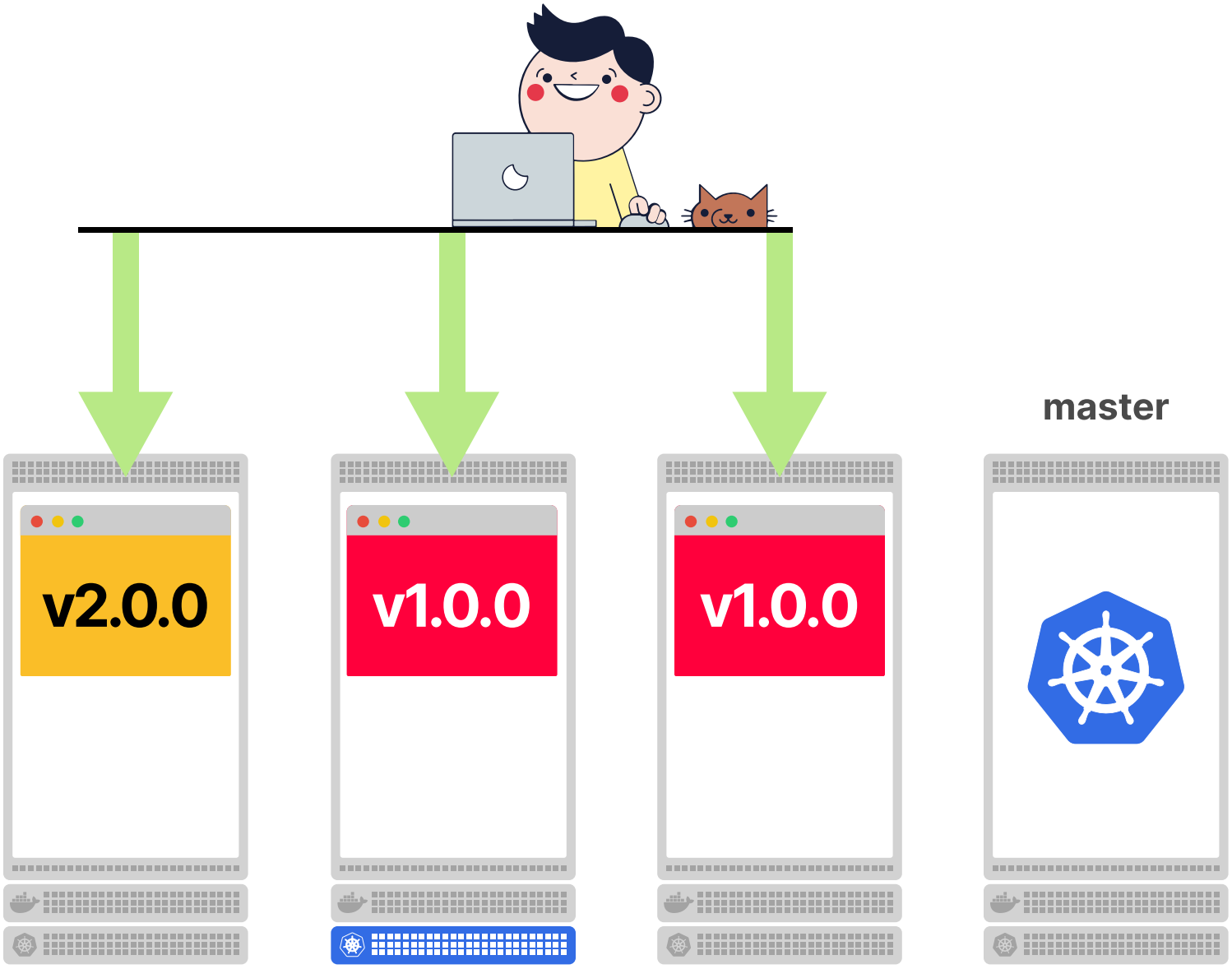

v2.0.0
v1.0.0
v1.0.0
master

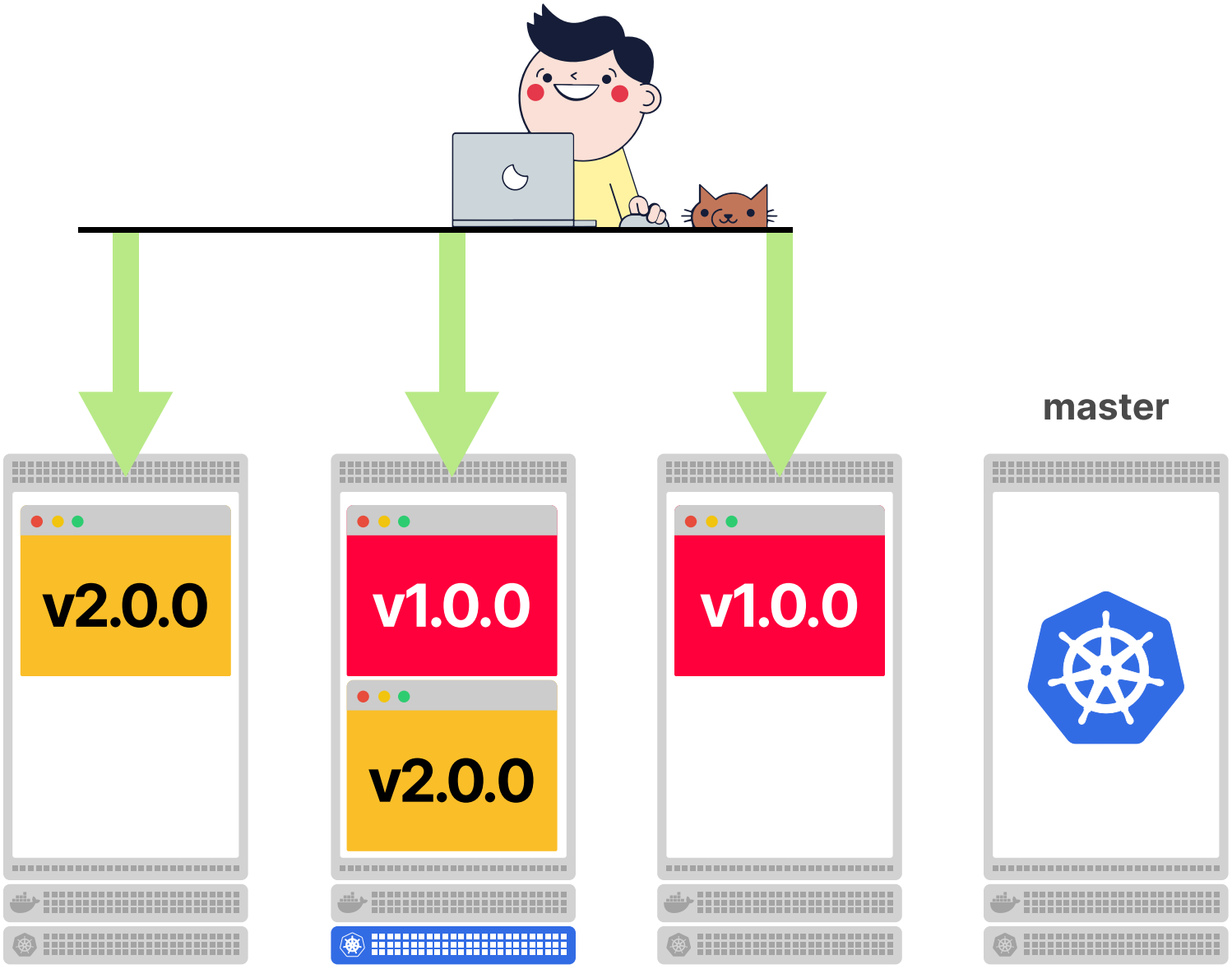

API SERVER
Deployment #1
Pod #2
RUNNING
Pod #3
RUNNING
Pod #4
RUNNING
CONTROLLER MANAGER
SCHEDULER

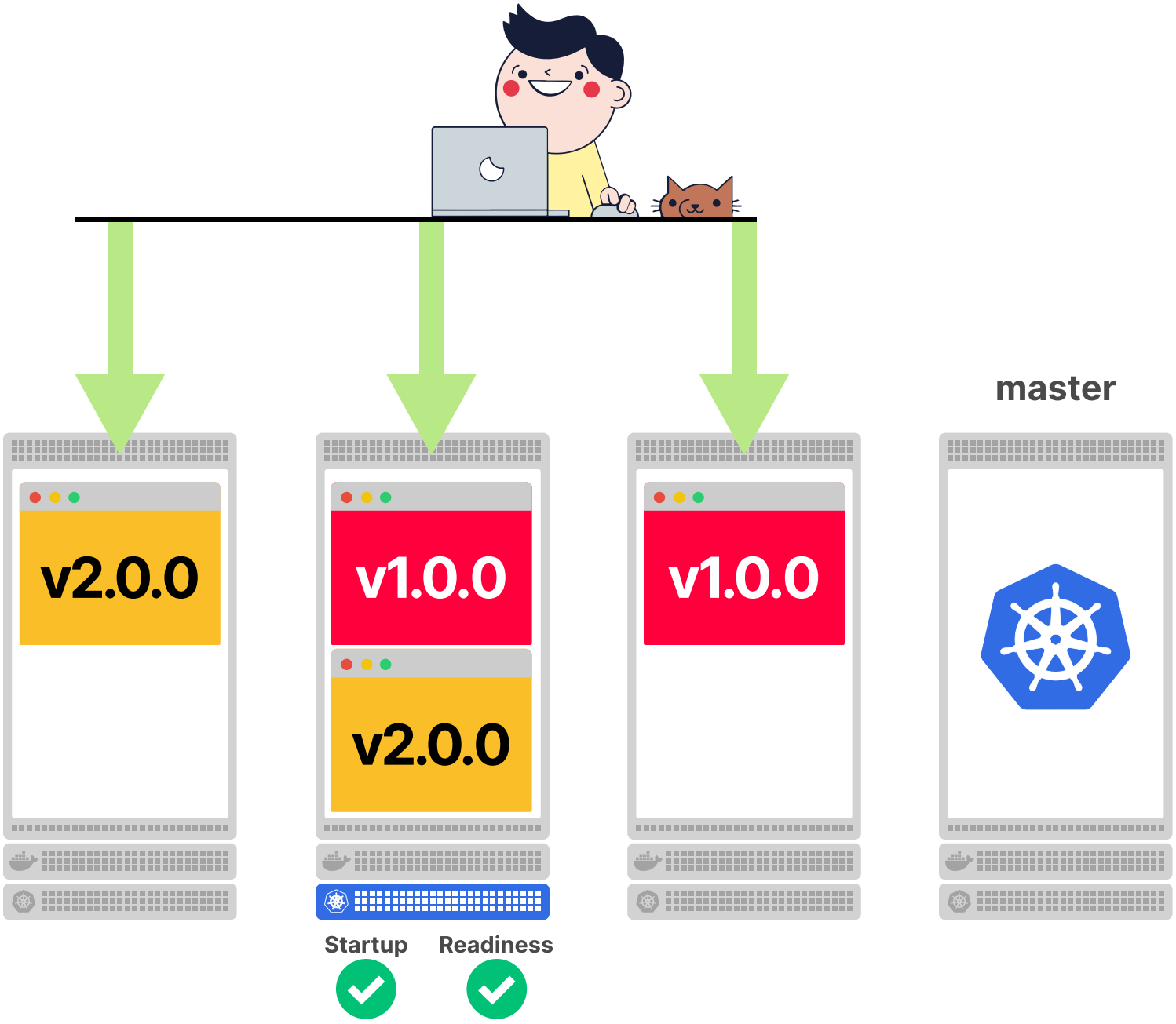

API SERVER
CONTROLLER MANAGER
SCHEDULER
Deployment #1
Pod #2 RUNNING
Pod #3 RUNNING
Pod #4 RUNNING
Pod #5 PENDING

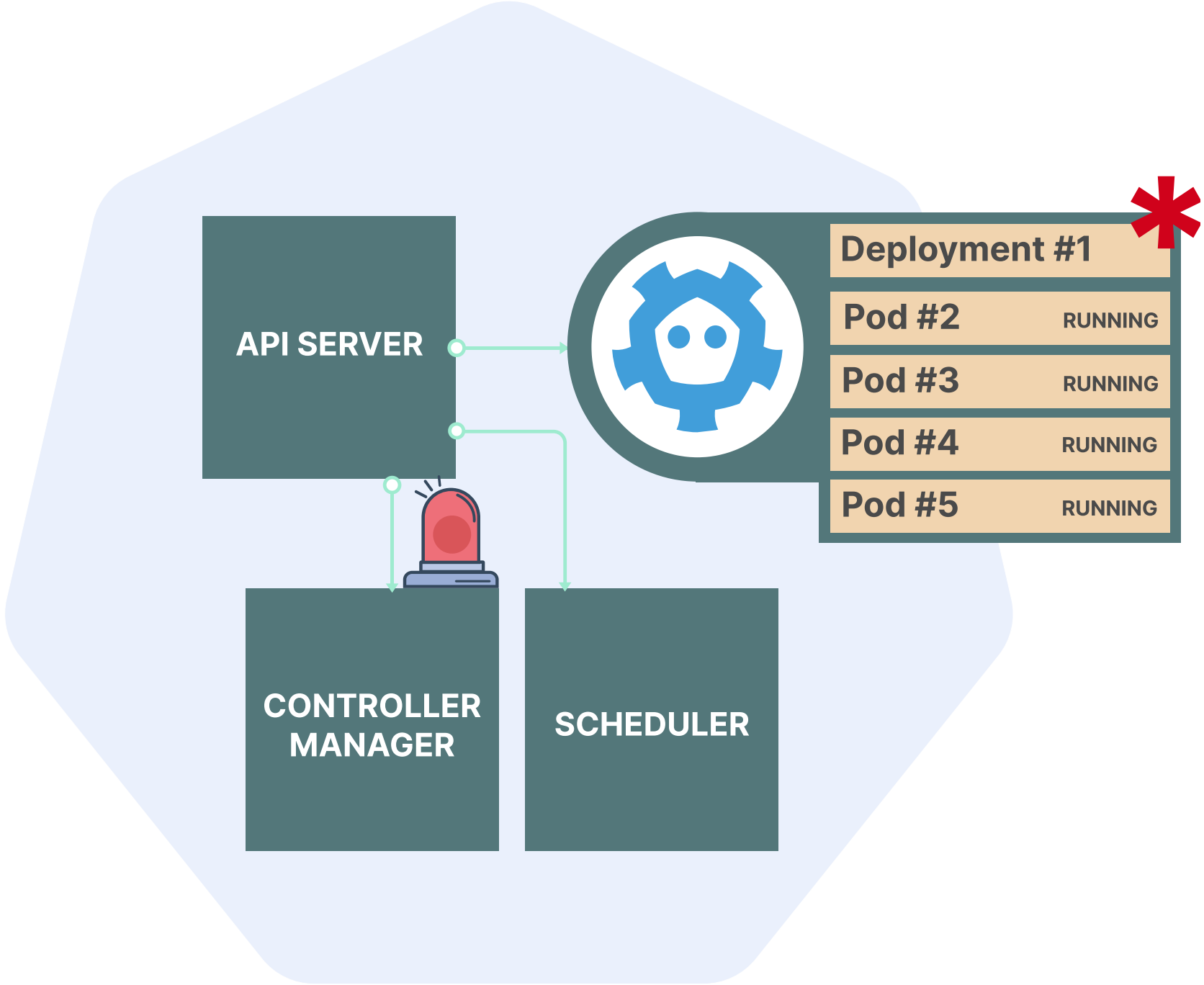

API SERVER
CONTROLLER MANAGER
SCHEDULER
Deployment #1
Pod #2 RUNNING
Pod #3 RUNNING
Pod #4 RUNNING
Pod #5 PENDING

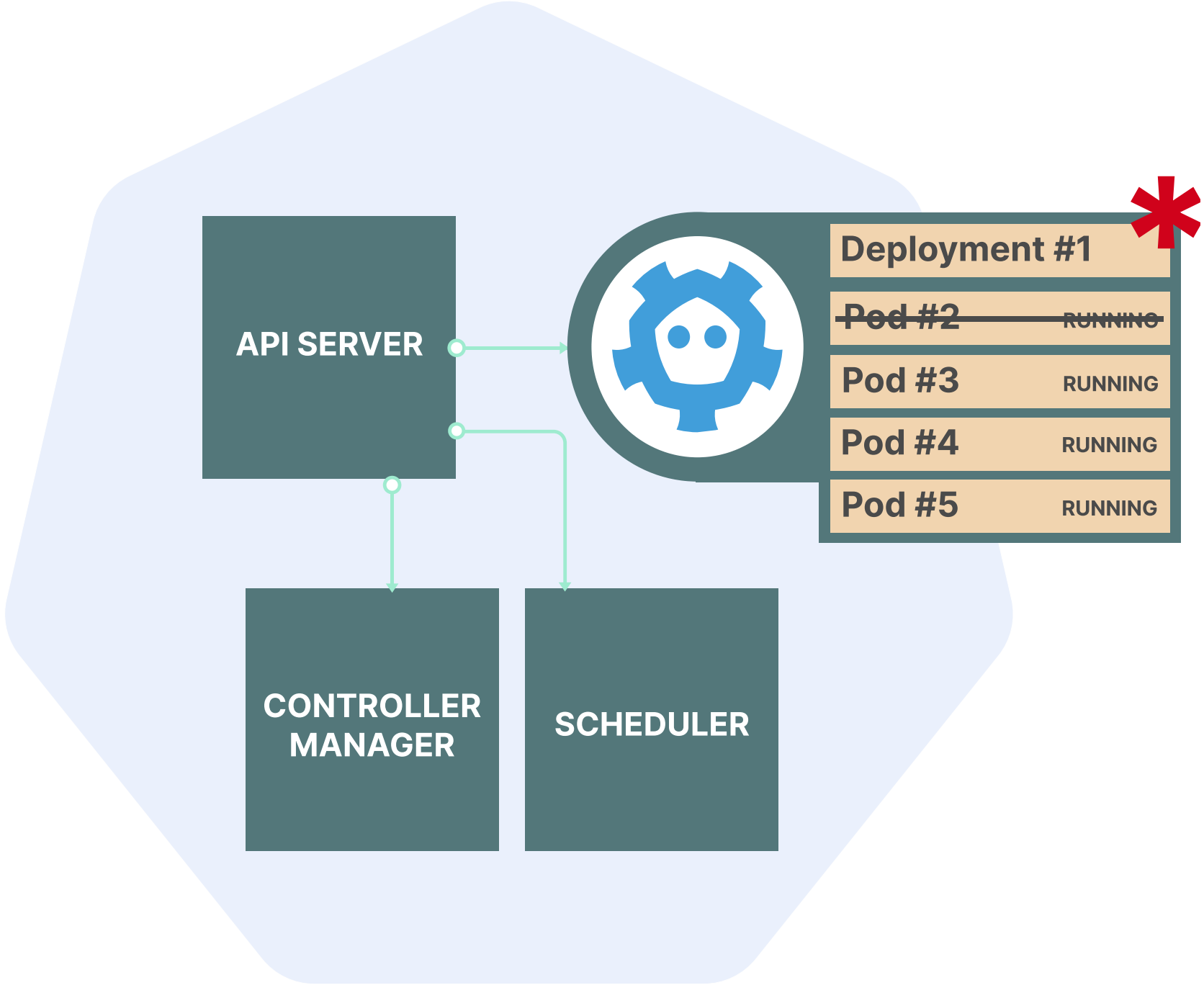

API SERVER
CONTROLLER MANAGER
SCHEDULER
Deployment #1
Pod #2 RUNNING
Pod #3 RUNNING
Pod #4 RUNNING
Pod #5 SCHEDULED

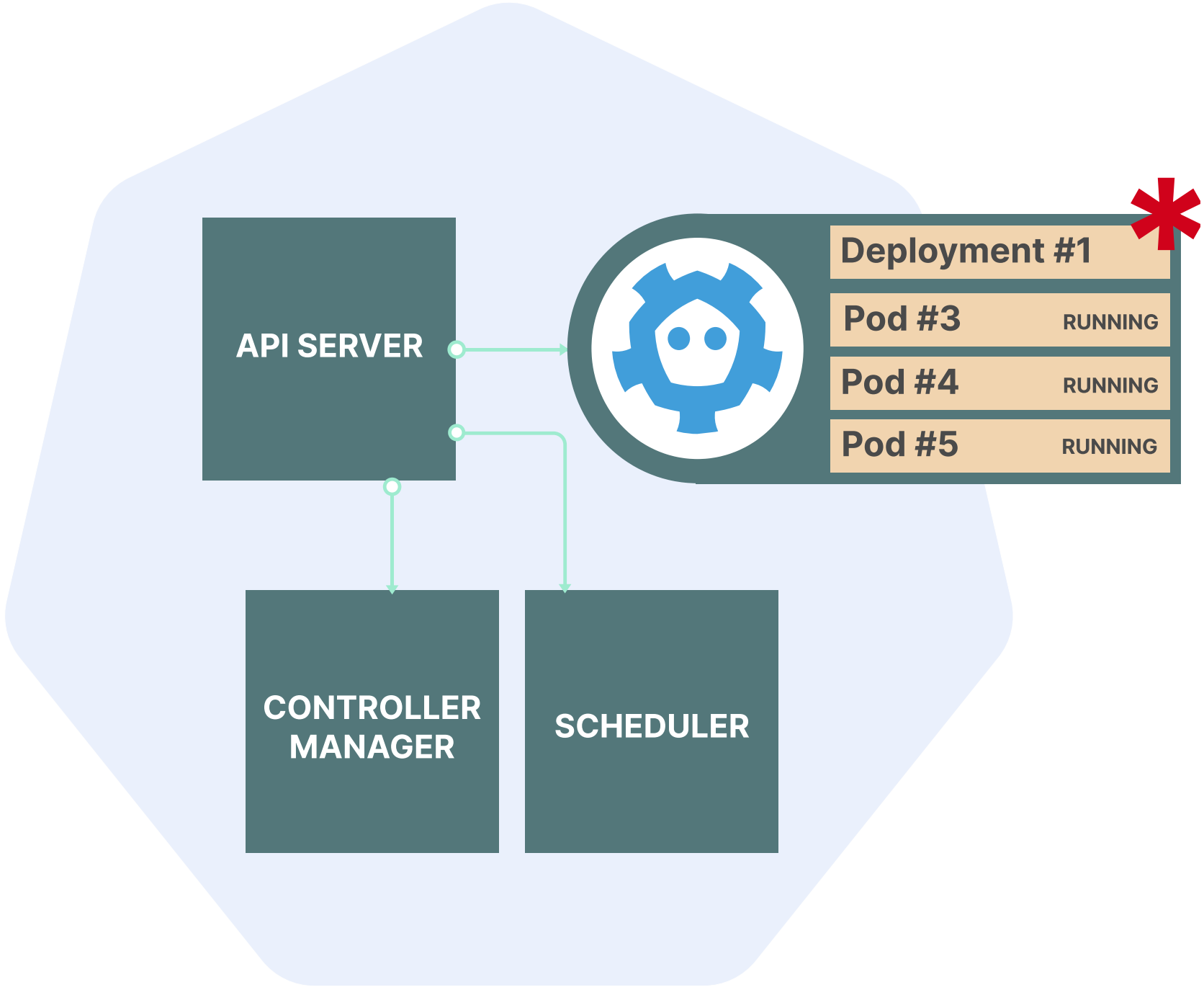

v2.0.0
v1.0.0
v1.0.0
master

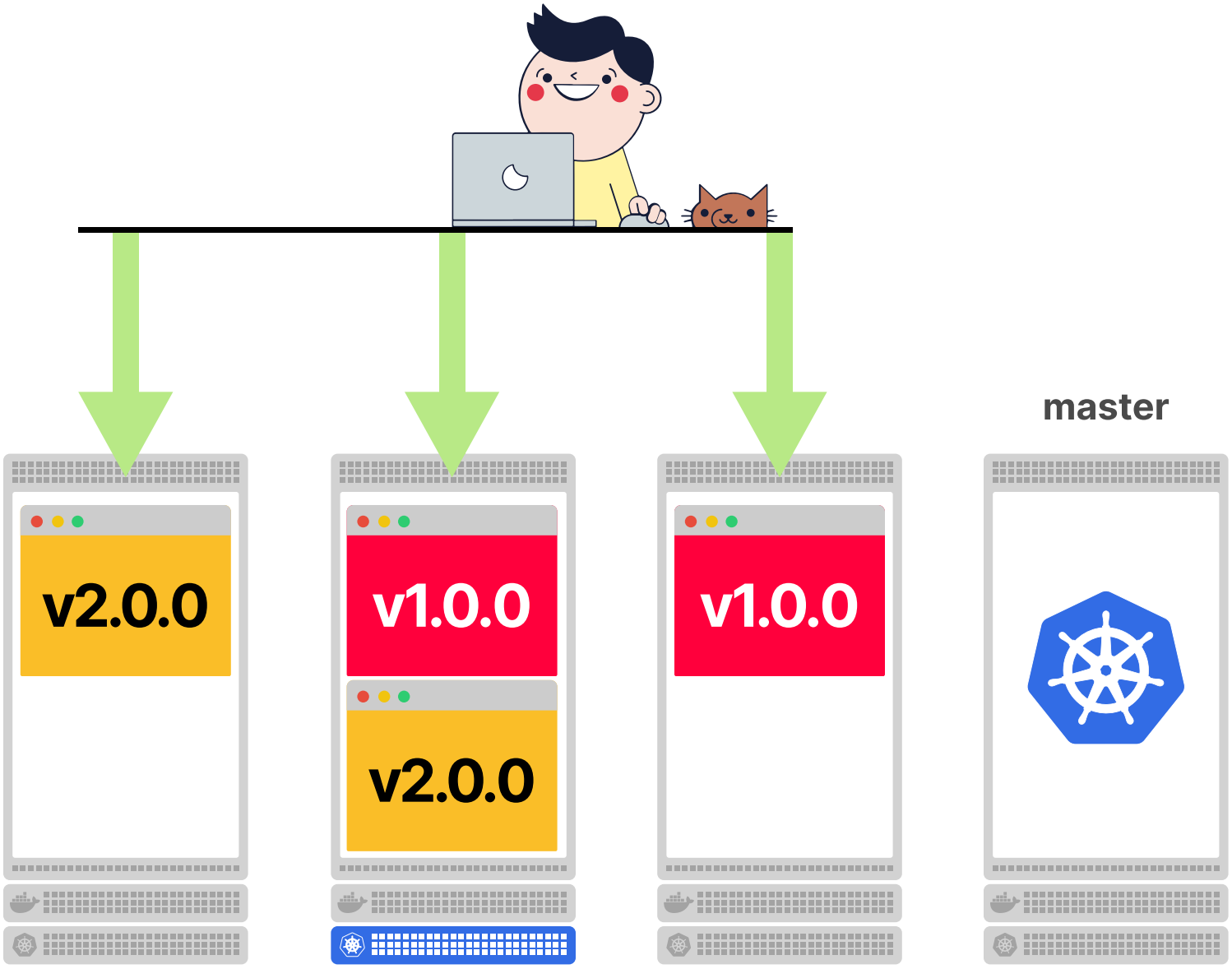

v2.0.0
v1.0.0
v2.0.0
v1.0.0
master

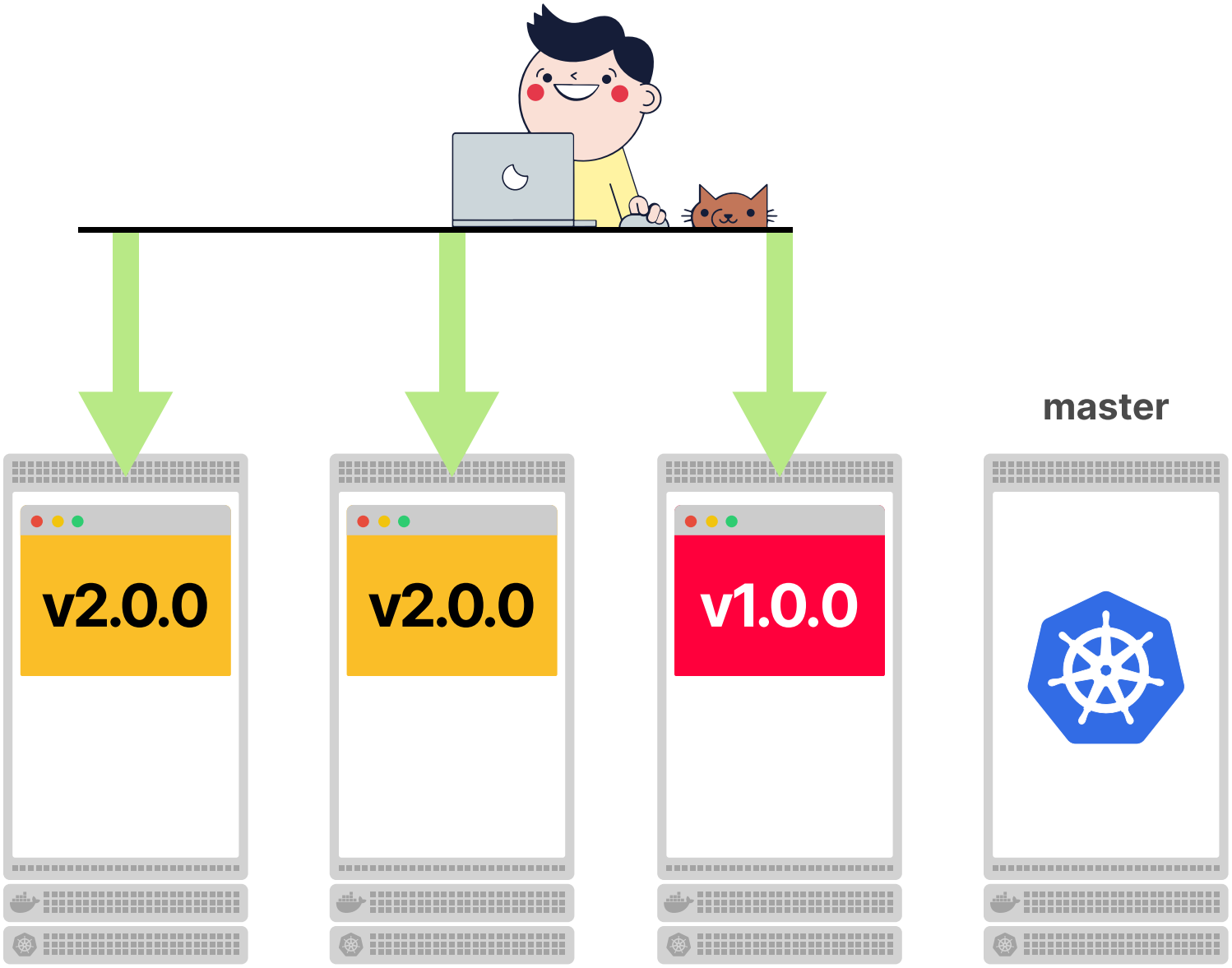

v2.0.0
v1.0.0
v2.0.0
v1.0.0
master
Startup
Readiness

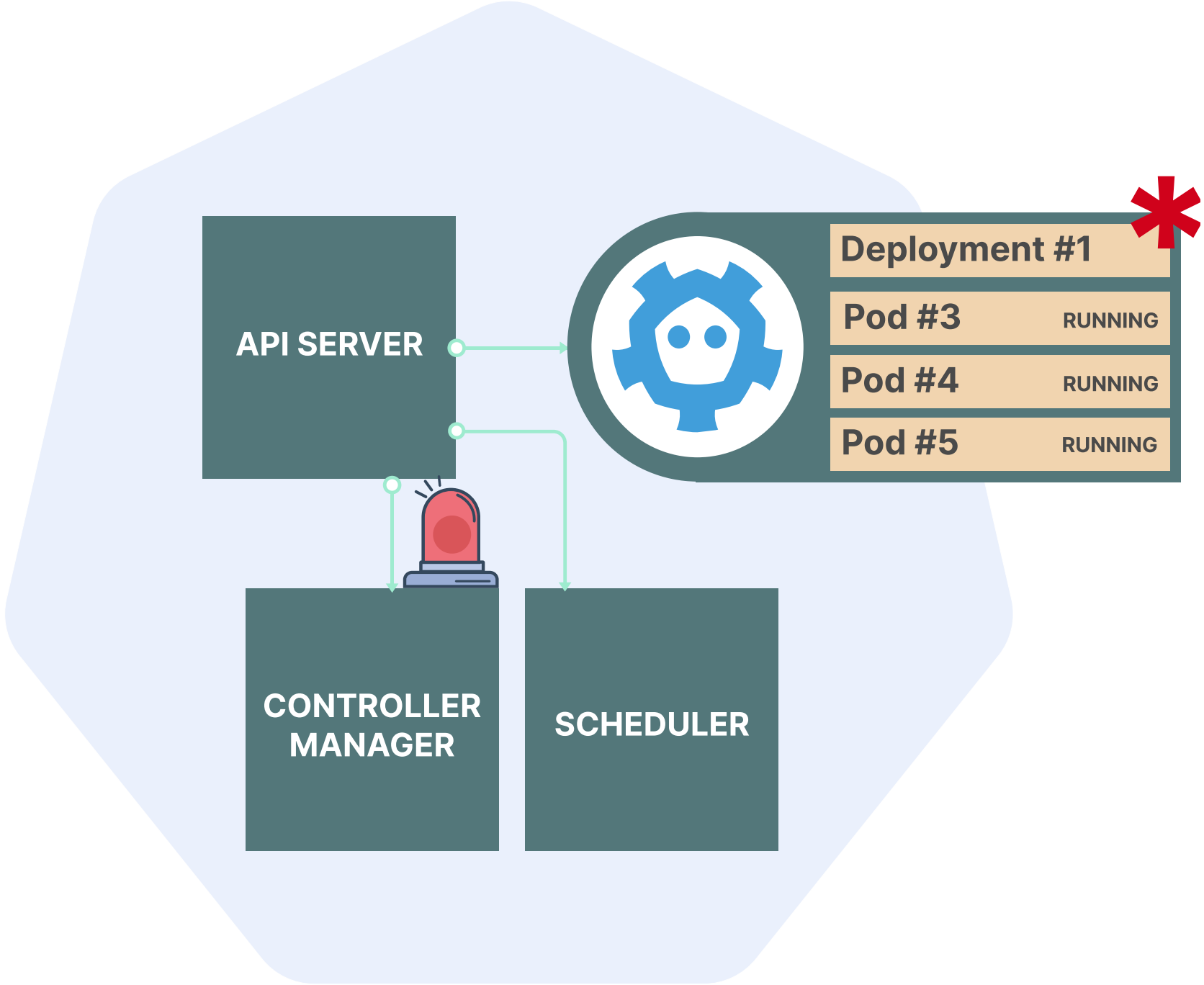

API SERVER
CONTROLLER MANAGER
SCHEDULER
Deployment #1
Pod #2 RUNNING
Pod #3 RUNNING
Pod #4 RUNNING
Pod #5 RUNNING

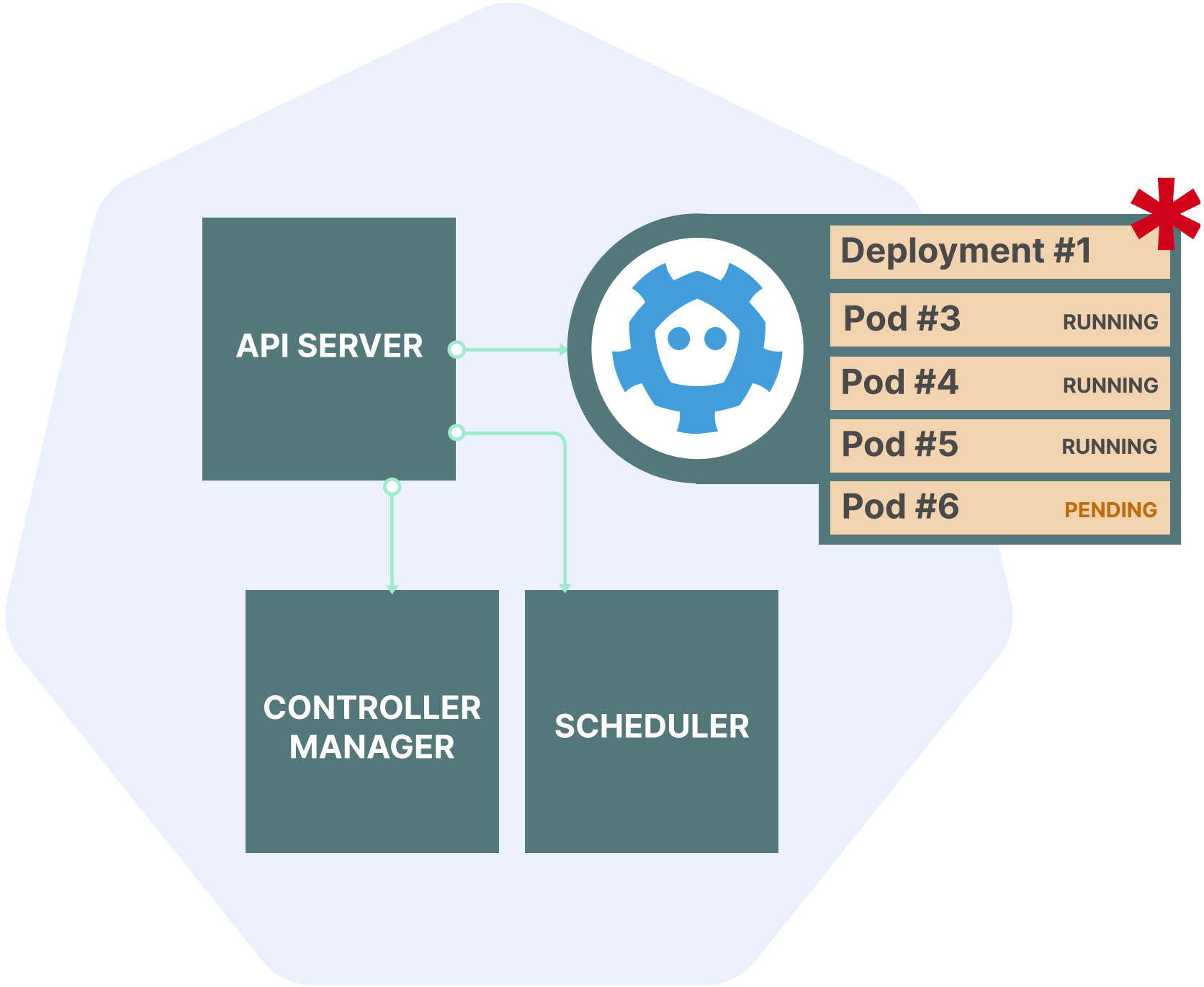

API SERVER
CONTROLLER MANAGER
SCHEDULER
Deployment #1
Pod #2 RUNNING
Pod #3 RUNNING
Pod #4 RUNNING
Pod #5 RUNNING

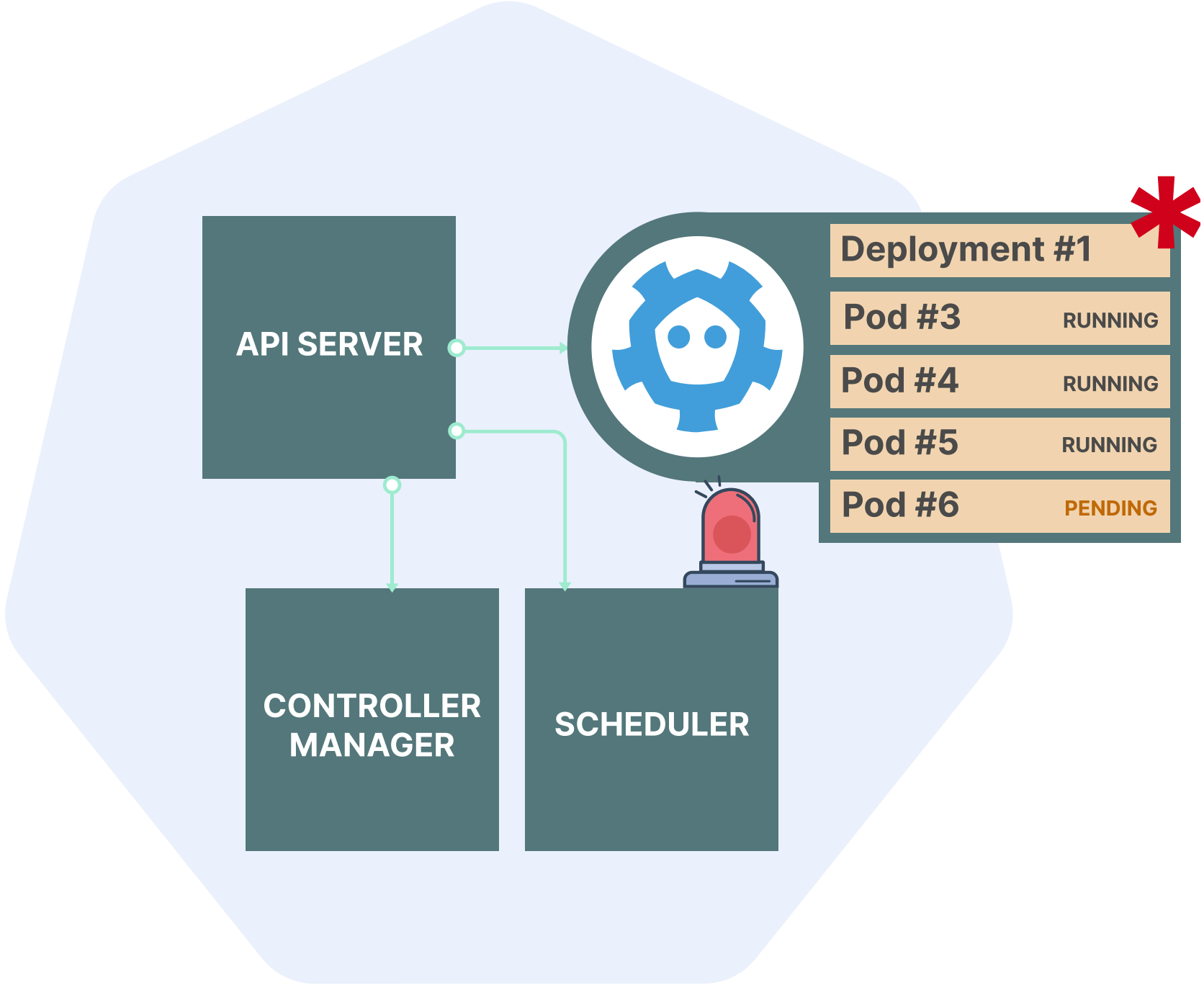

API SERVER
Deployment #1
Pod #3
RUNNING
Pod #4
RUNNING
Pod #5
RUNNING
CONTROLLER MANAGER
SCHEDULER

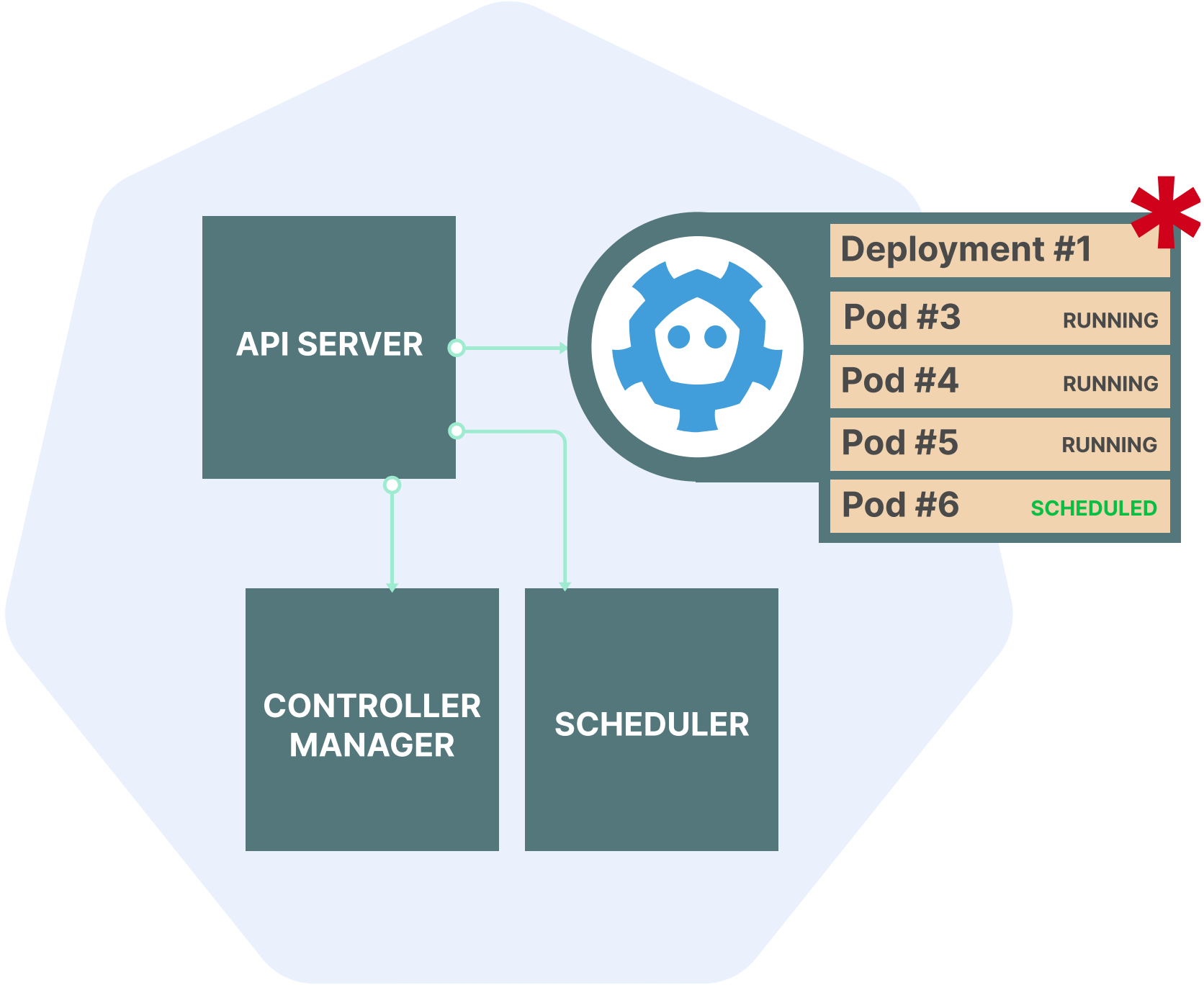

v2.0.0
v1.0.0
v2.0.0
v1.0.0
master

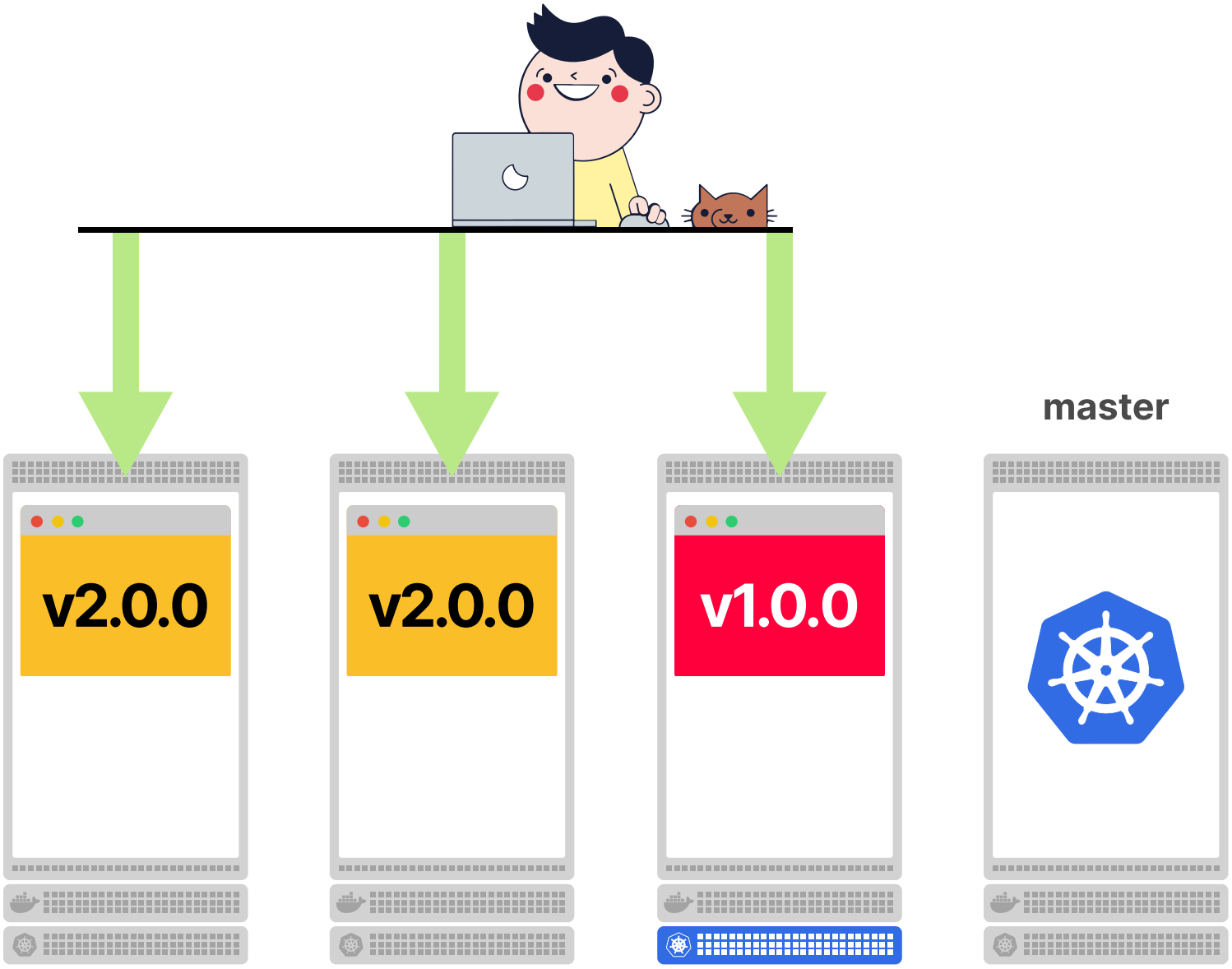

v2.0.0
v2.0.0
v1.0.0
master

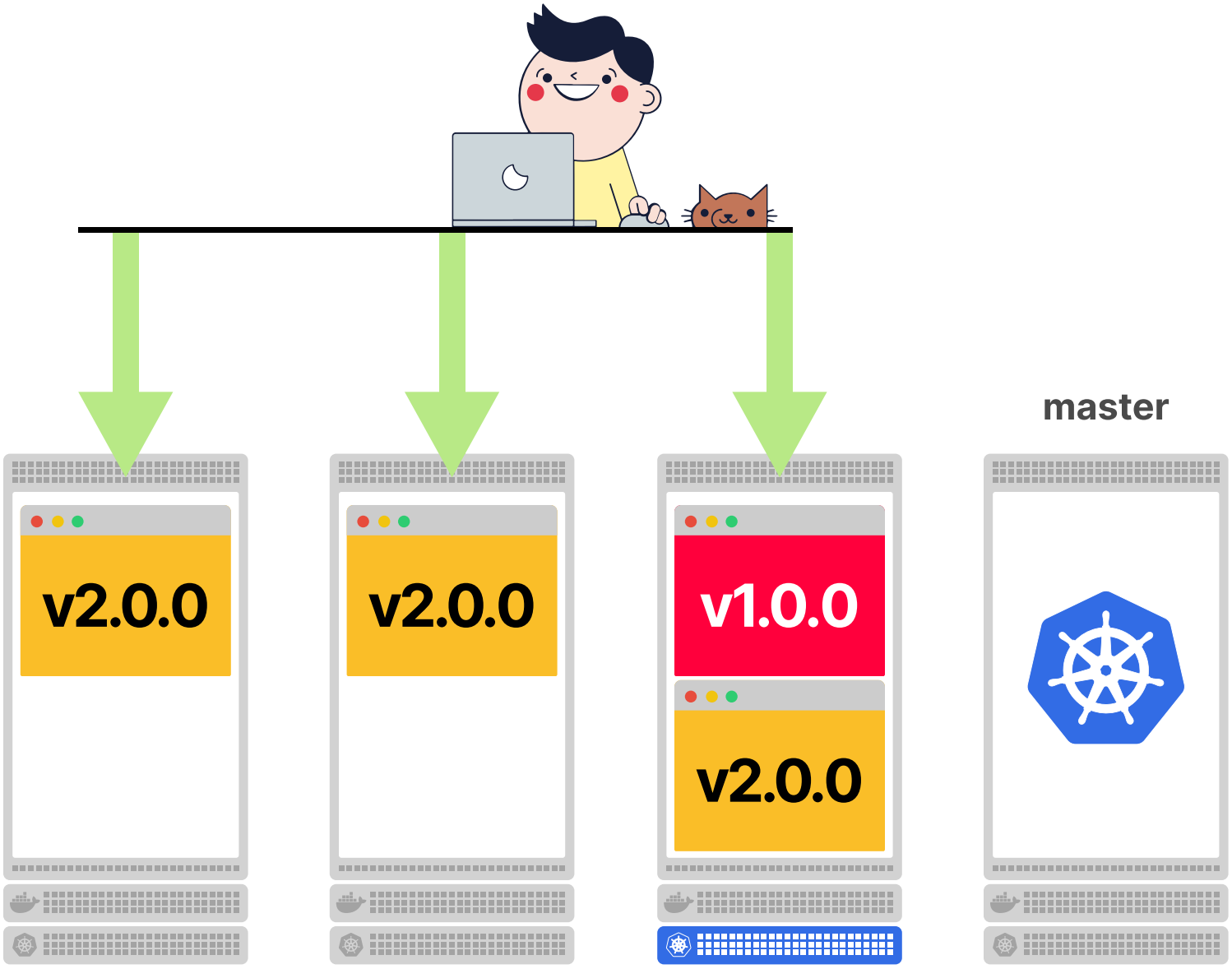

API SERVER
Deployment #1
Pod #3
RUNNING
Pod #4
RUNNING
Pod #5
RUNNING
CONTROLLER MANAGER
SCHEDULER

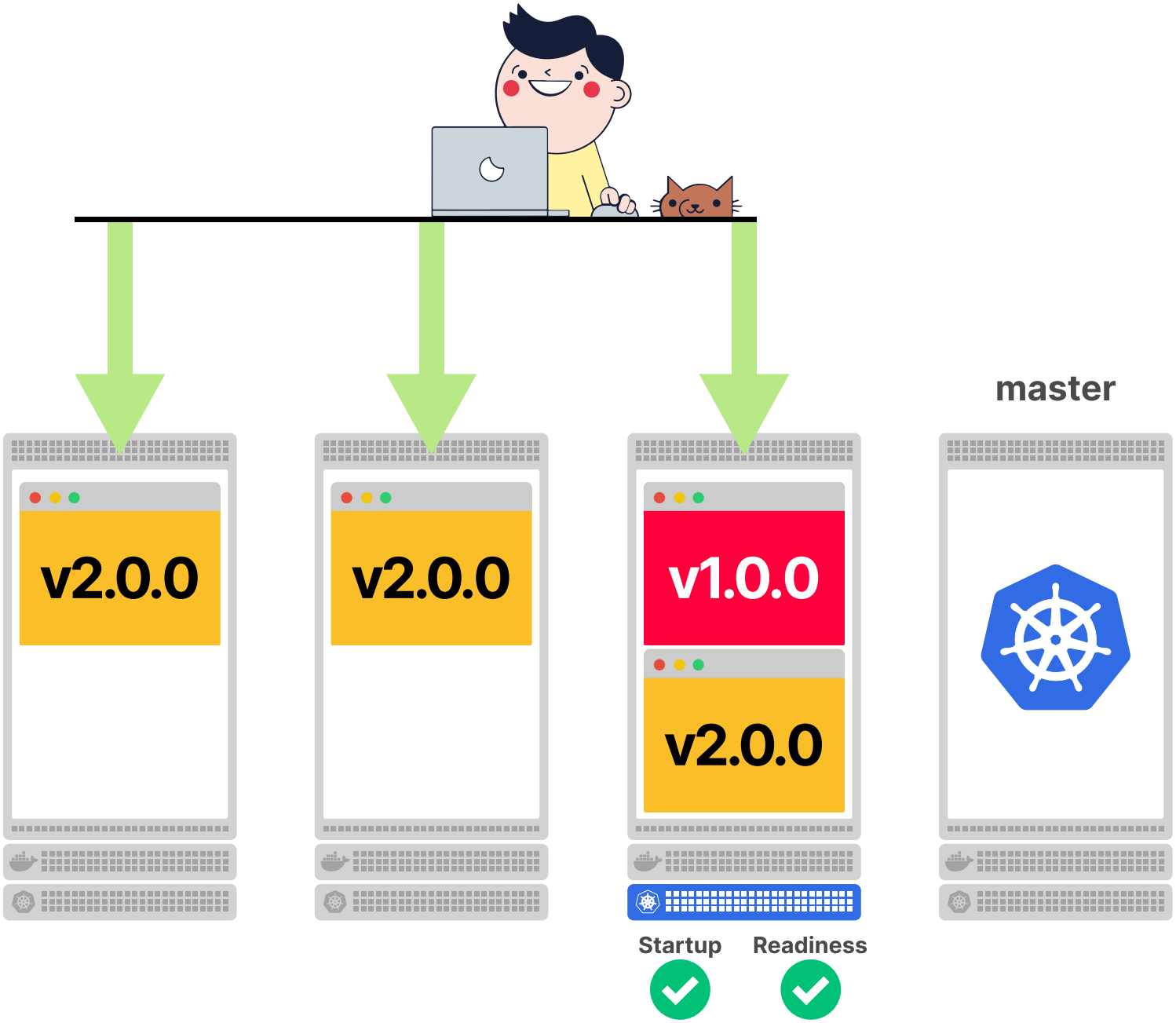

API SERVER
CONTROLLER MANAGER
SCHEDULER
Deployment #1
Pod #3 RUNNING
Pod #4 RUNNING
Pod #5 RUNNING
Pod #6 PENDING

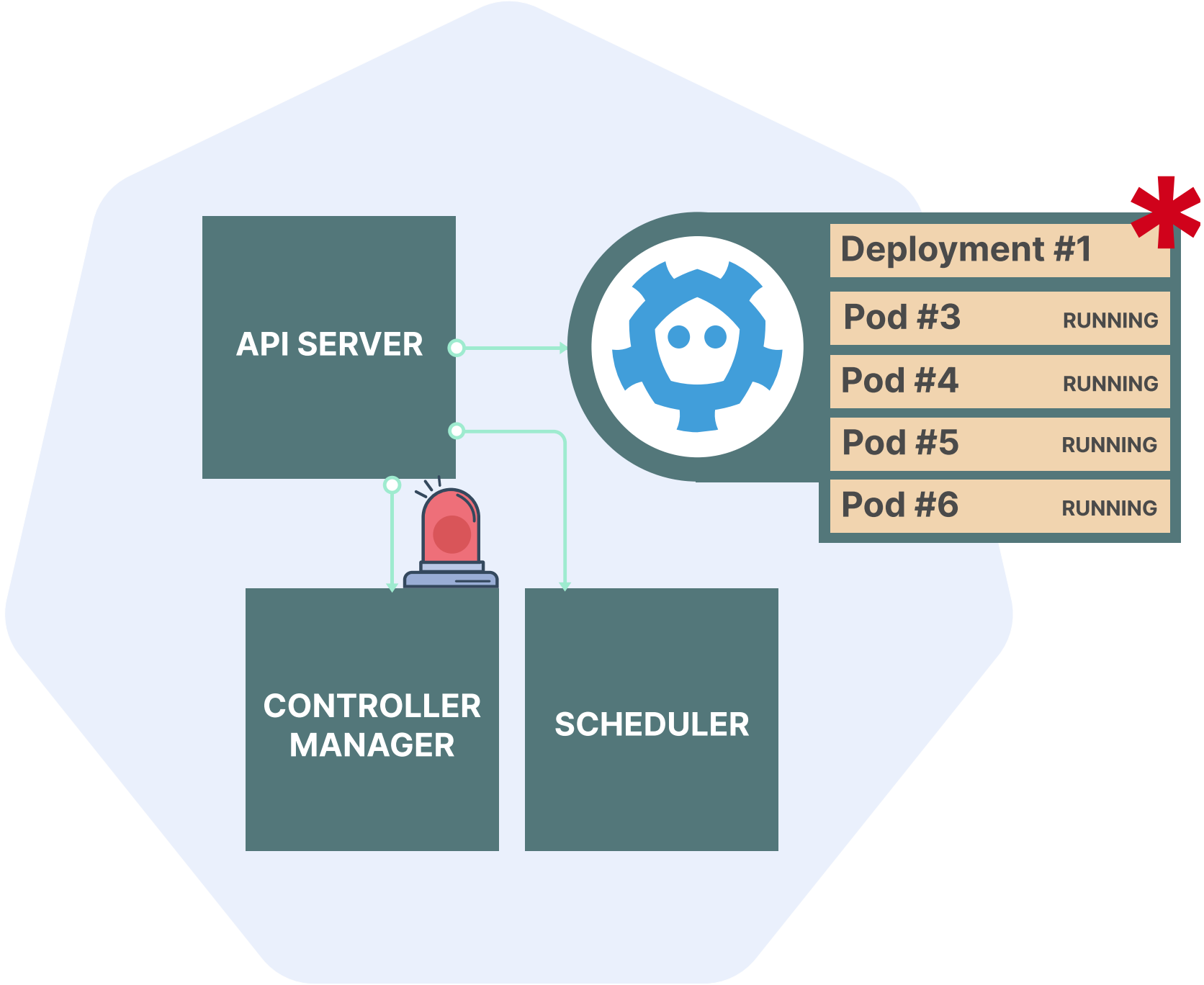

API SERVER
CONTROLLER MANAGER
SCHEDULER
Deployment #1
Pod #3 RUNNING
Pod #4 RUNNING
Pod #5 RUNNING
Pod #6 PENDING

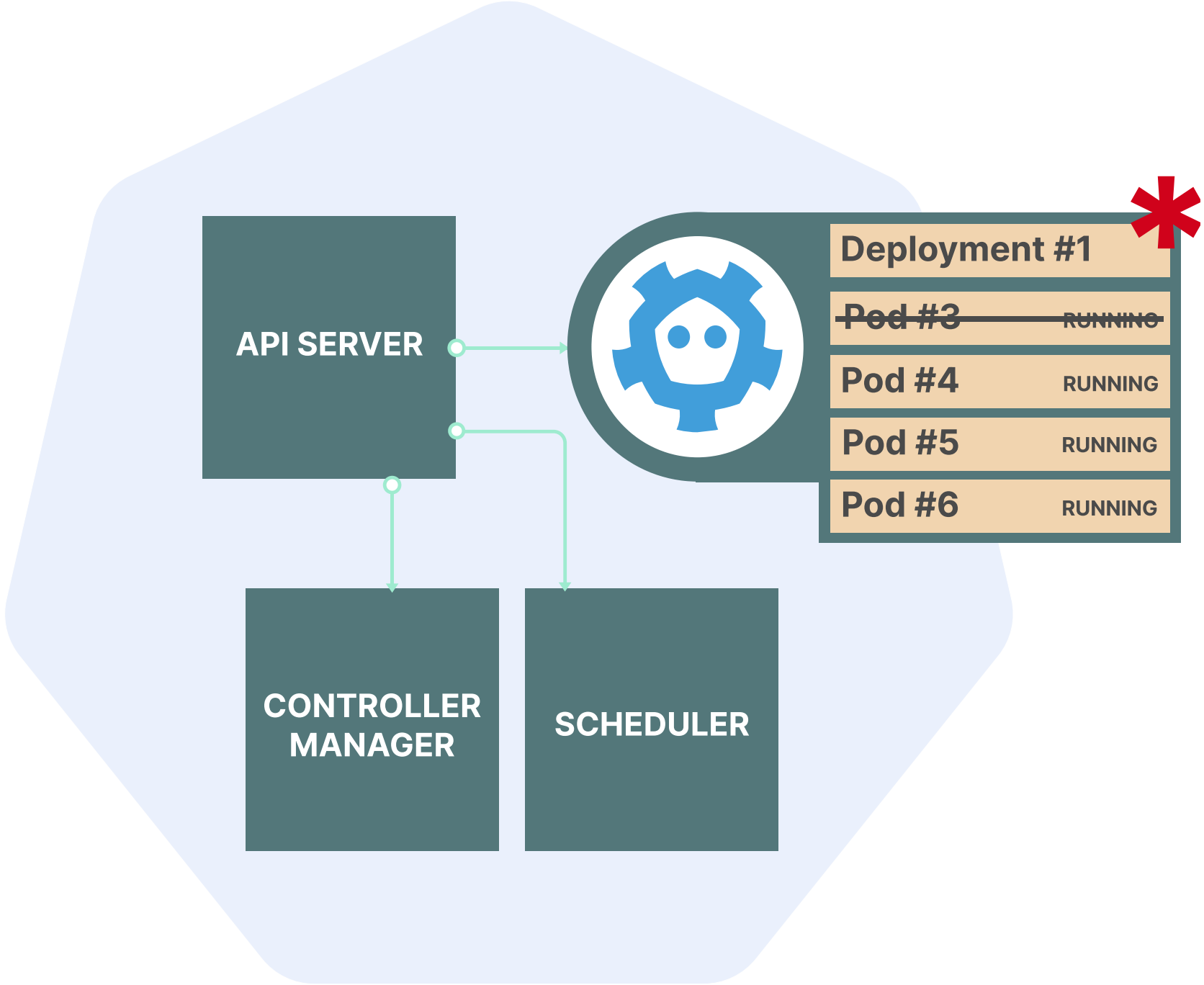

API SERVER
CONTROLLER MANAGER
SCHEDULER
Deployment #1
Pod #3
RUNNING
Pod #4
RUNNING
Pod #5
RUNNING
Pod #6
SCHEDULED

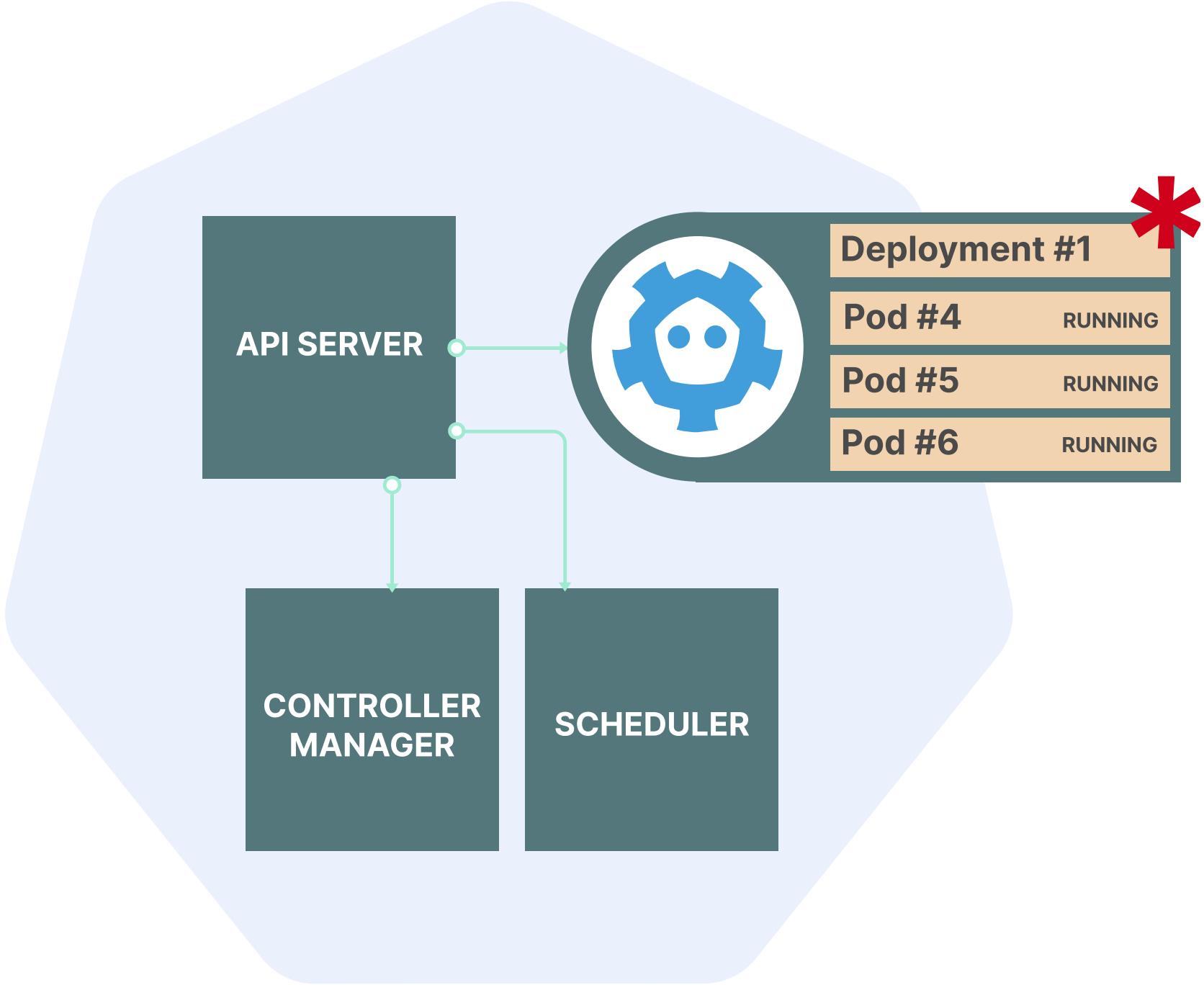

v2.0.0
v2.0.0
v1.0.0
master

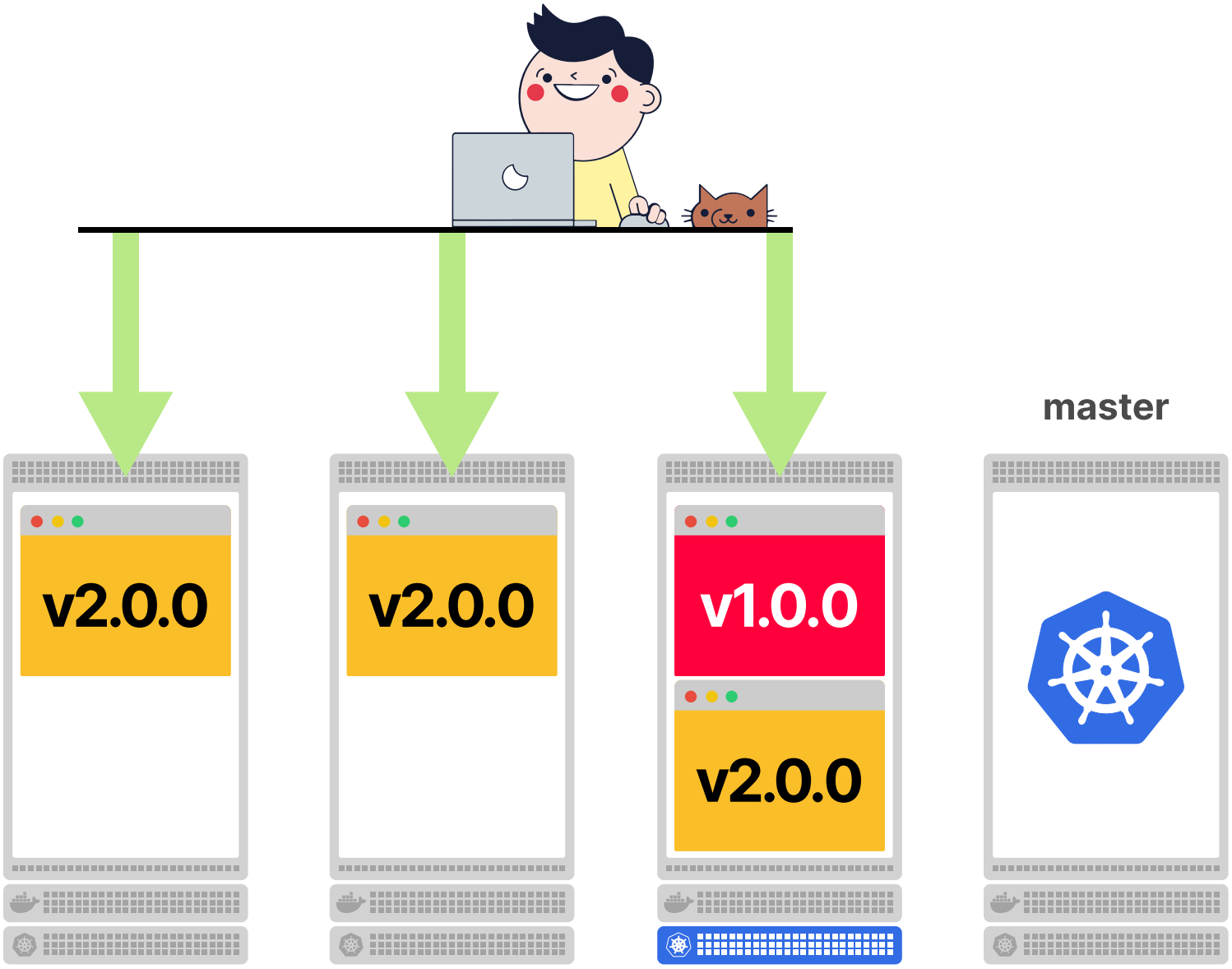

v2.0.0
v2.0.0
v1.0.0
v2.0.0
master

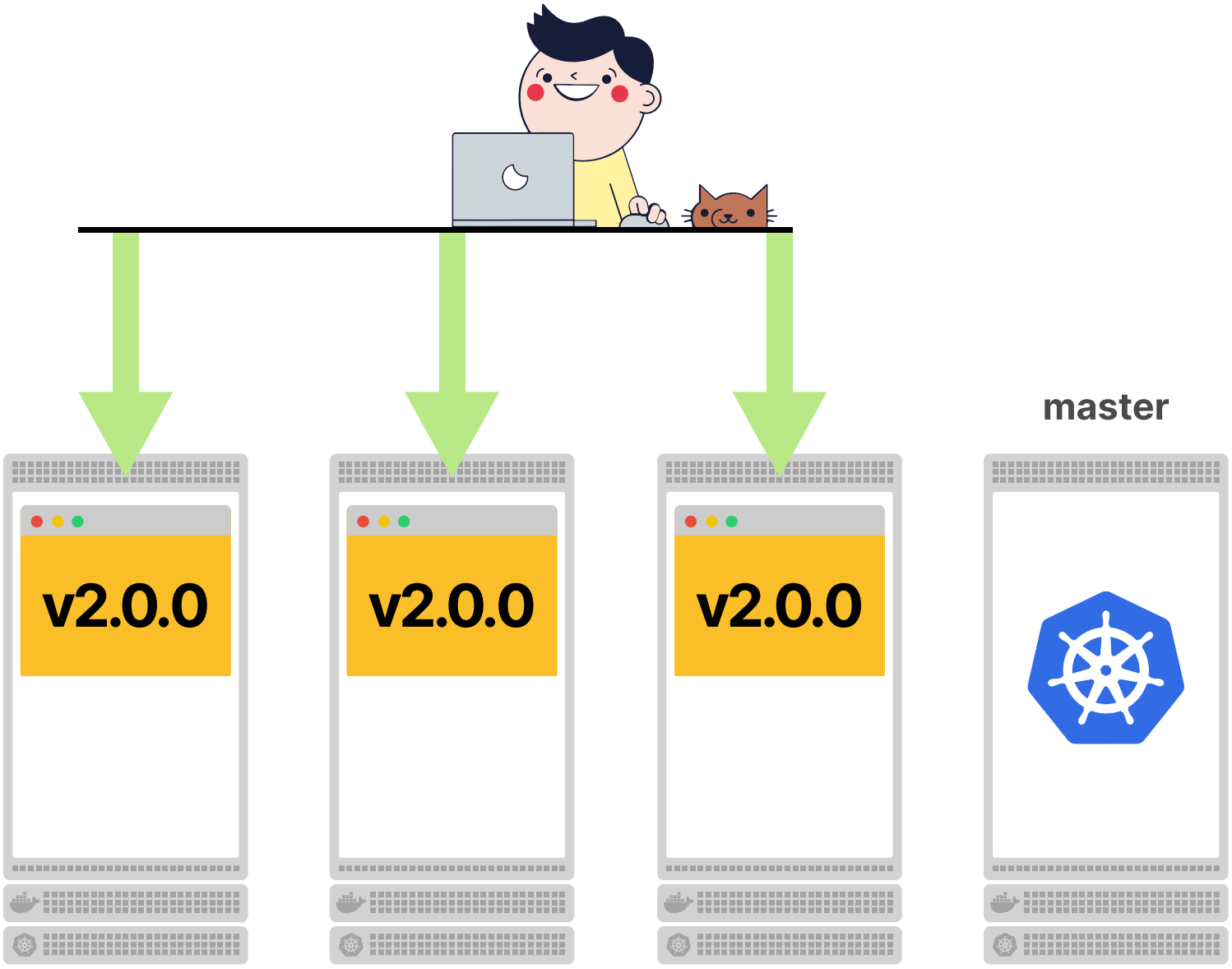

v2.0.0
v2.0.0
v1.0.0
v2.0.0
master
Startup
Readiness


API SERVER
CONTROLLER MANAGER
SCHEDULER
Deployment #1
Pod #3 RUNNING
Pod #4 RUNNING
Pod #5 RUNNING
Pod #6 RUNNING

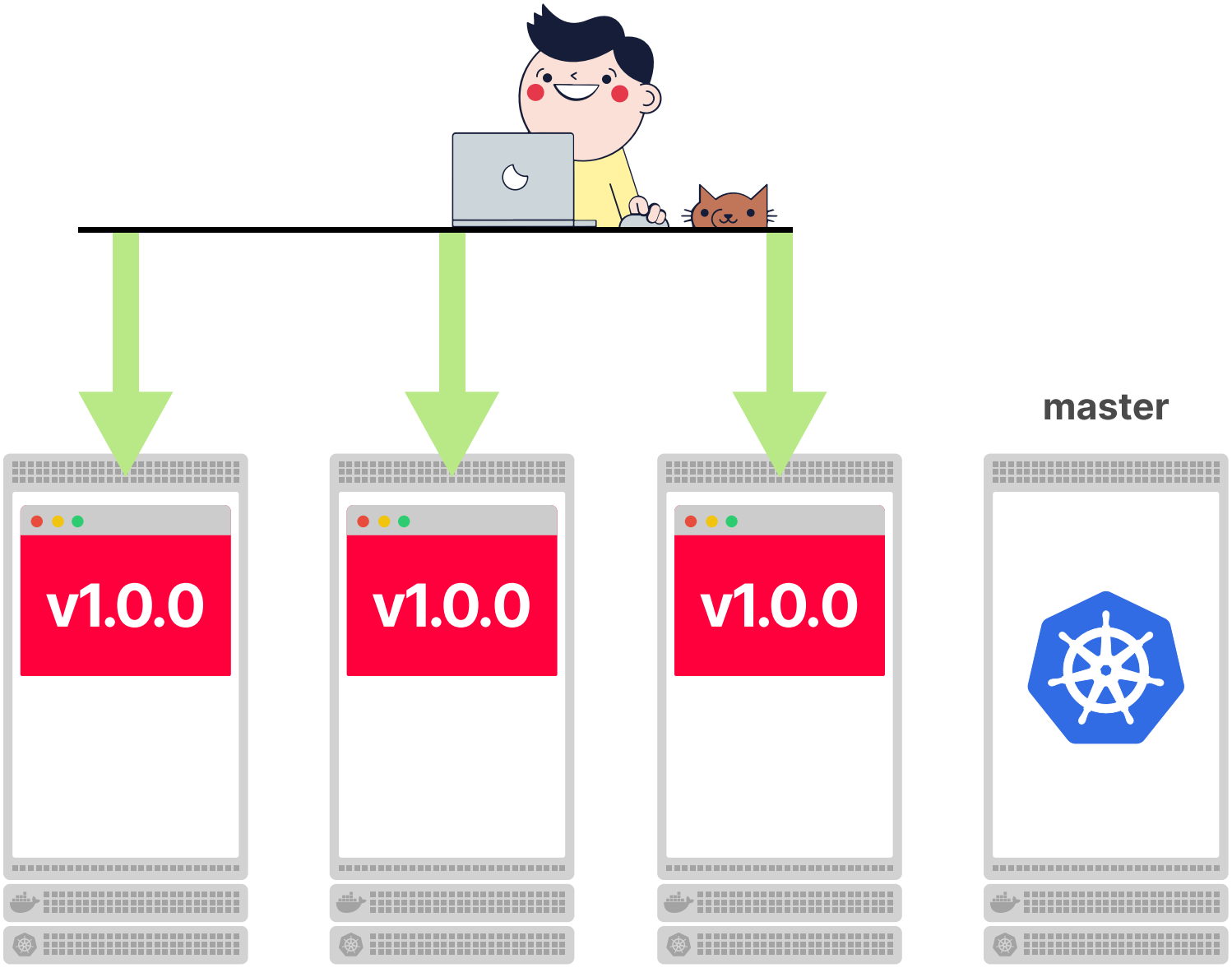

API SERVER
CONTROLLER MANAGER
SCHEDULER
Deployment #1
Pod #3 RUNNING
Pod #4 RUNNING
Pod #5 RUNNING
Pod #6 RUNNING


API SERVER
Deployment #1
Pod #4
RUNNING
Pod #5
RUNNING
Pod #6
RUNNING
CONTROLLER MANAGER
SCHEDULER


v2.0.0
v2.0.0
v1.0.0
v2.0.0
master


v2.0.0
v2.0.0
v2.0.0
master

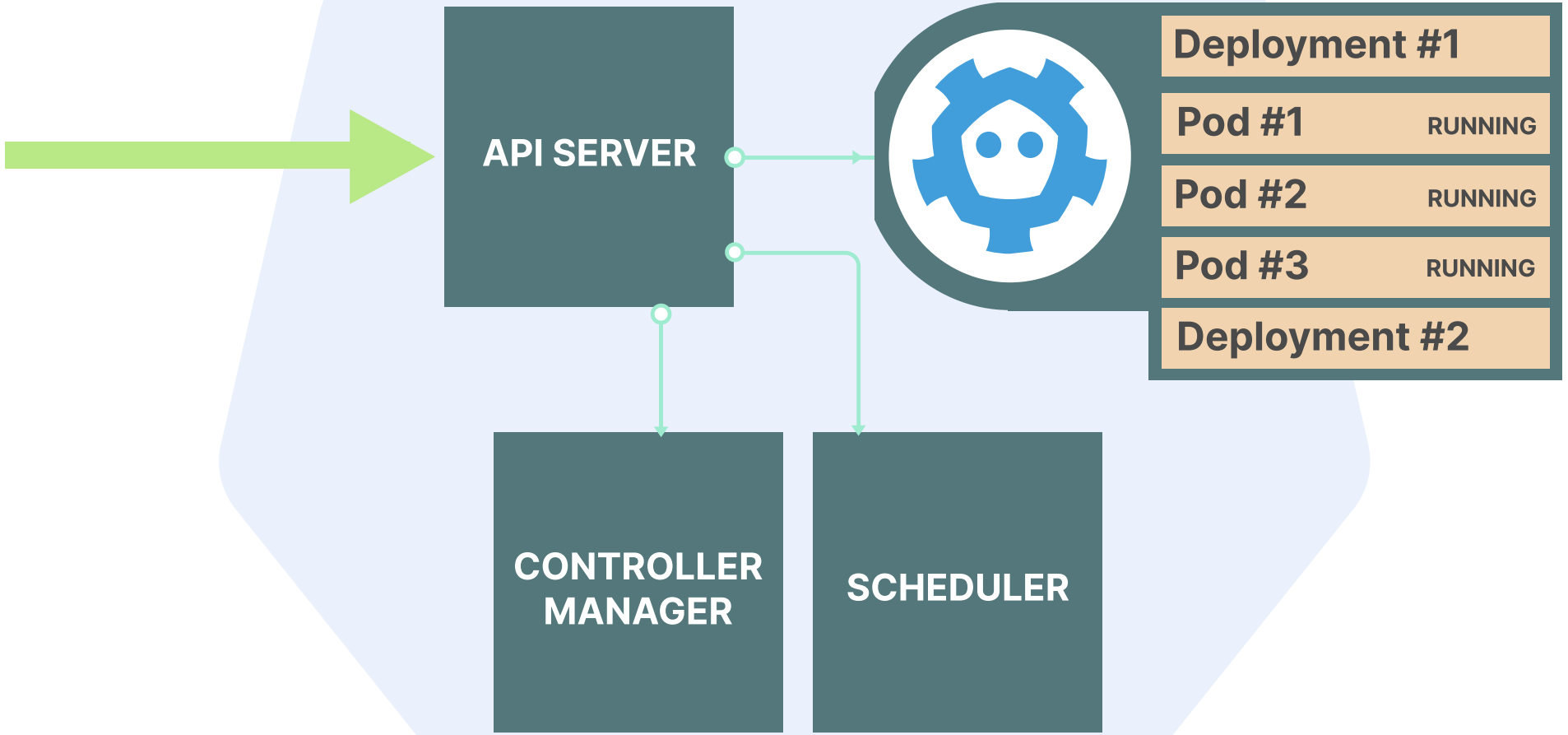
Blue-green deployment revisited

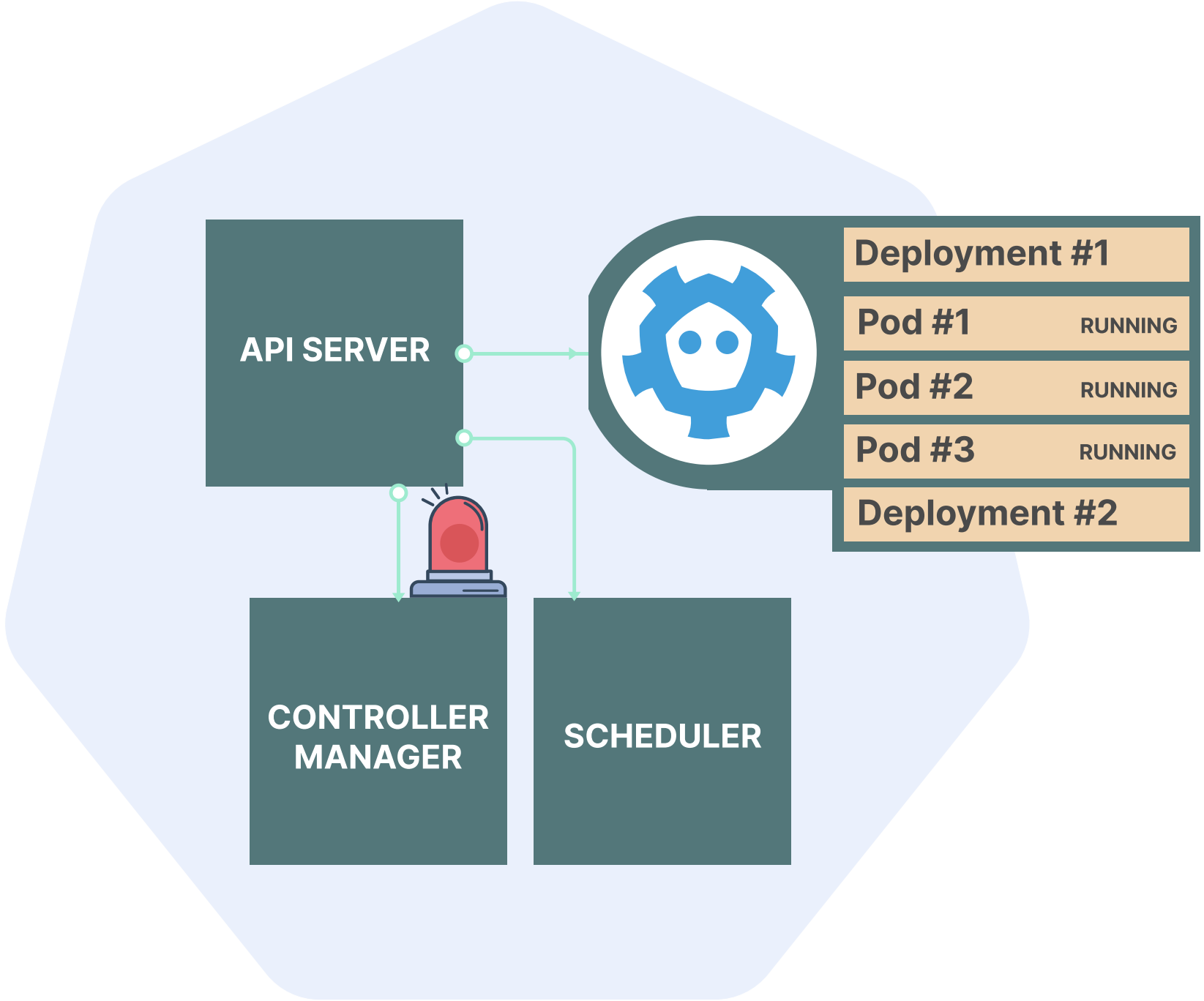

v1.0.0
v1.0.0
v1.0.0
master

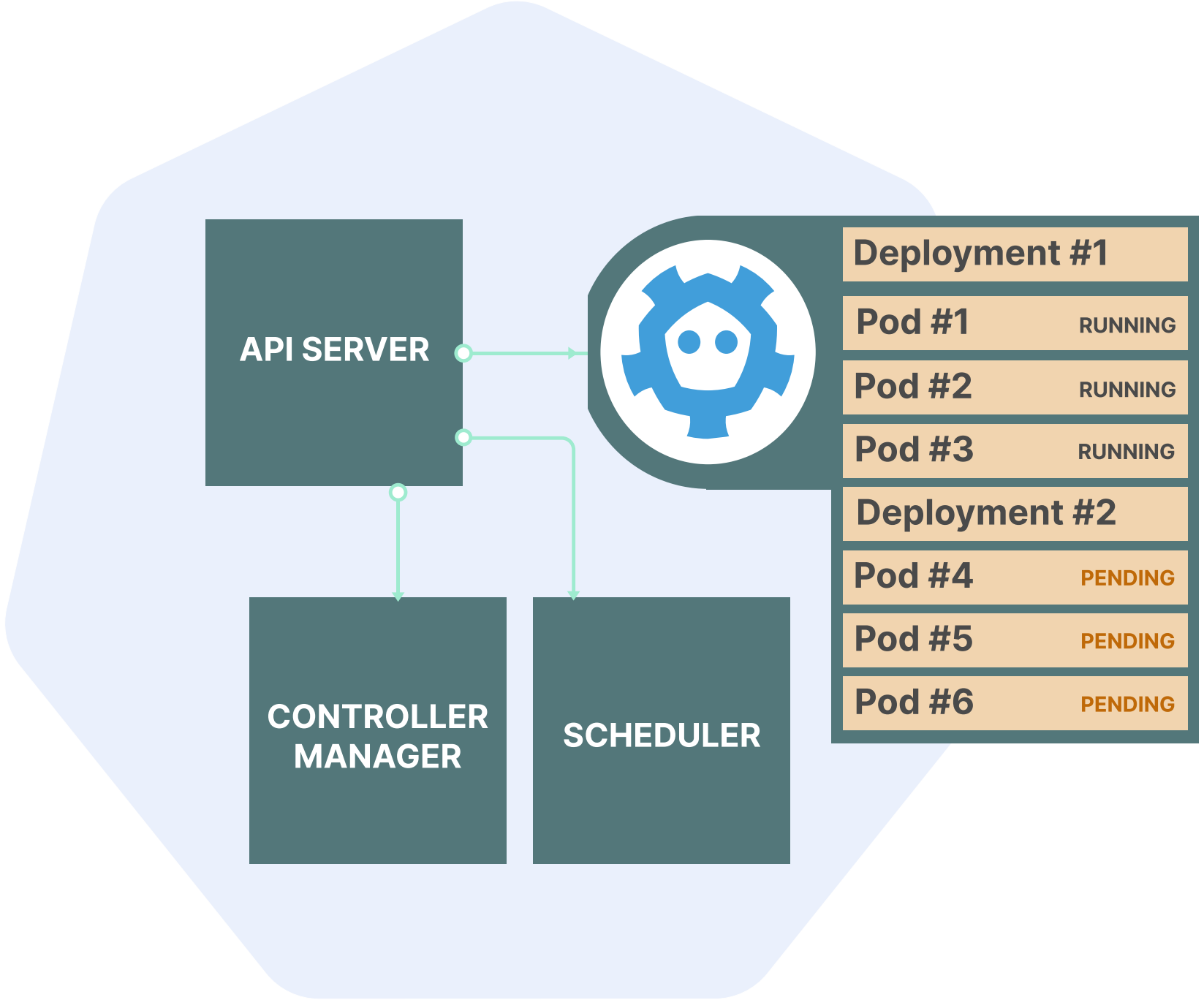
Create Deployment for v2.0.0

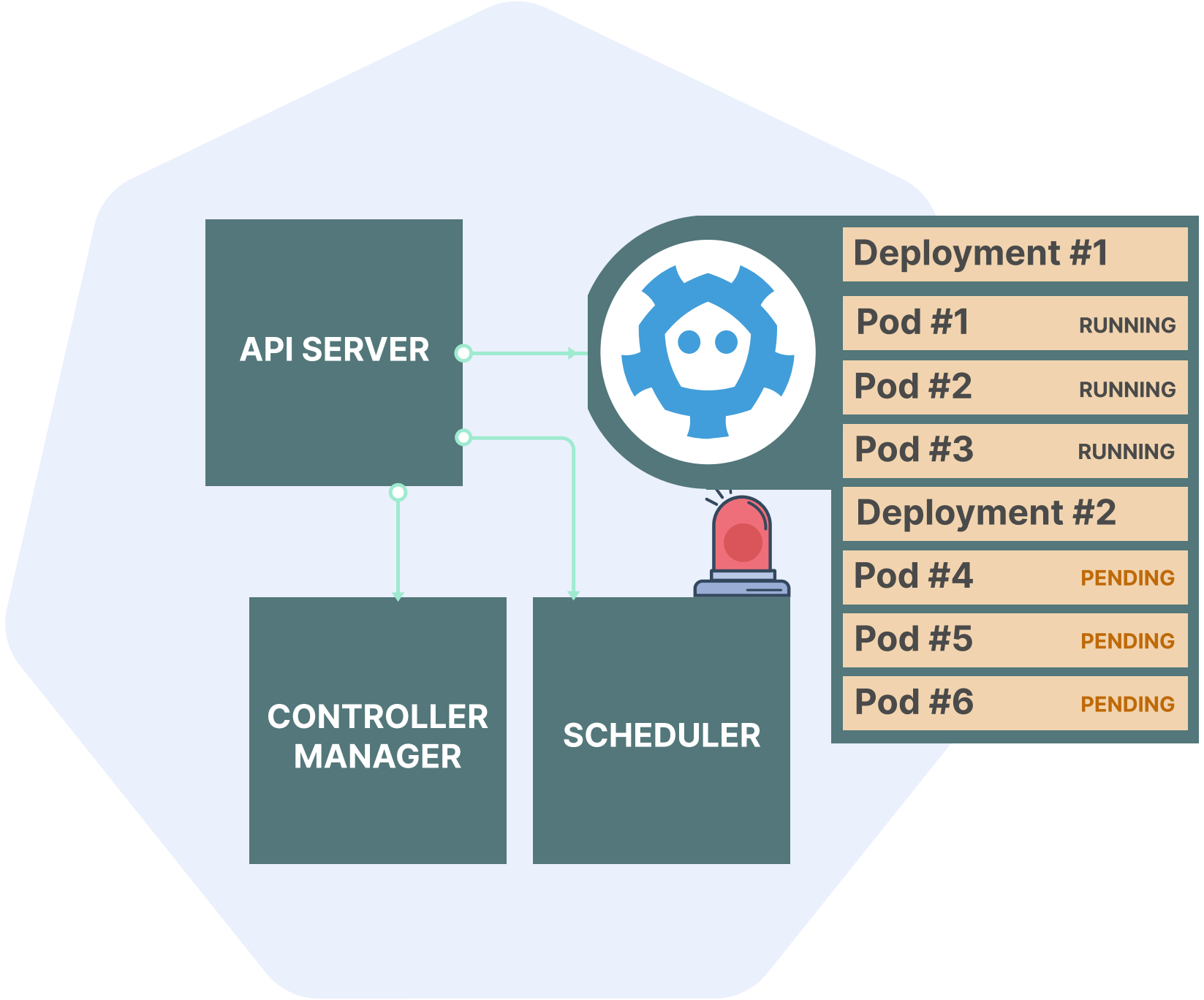


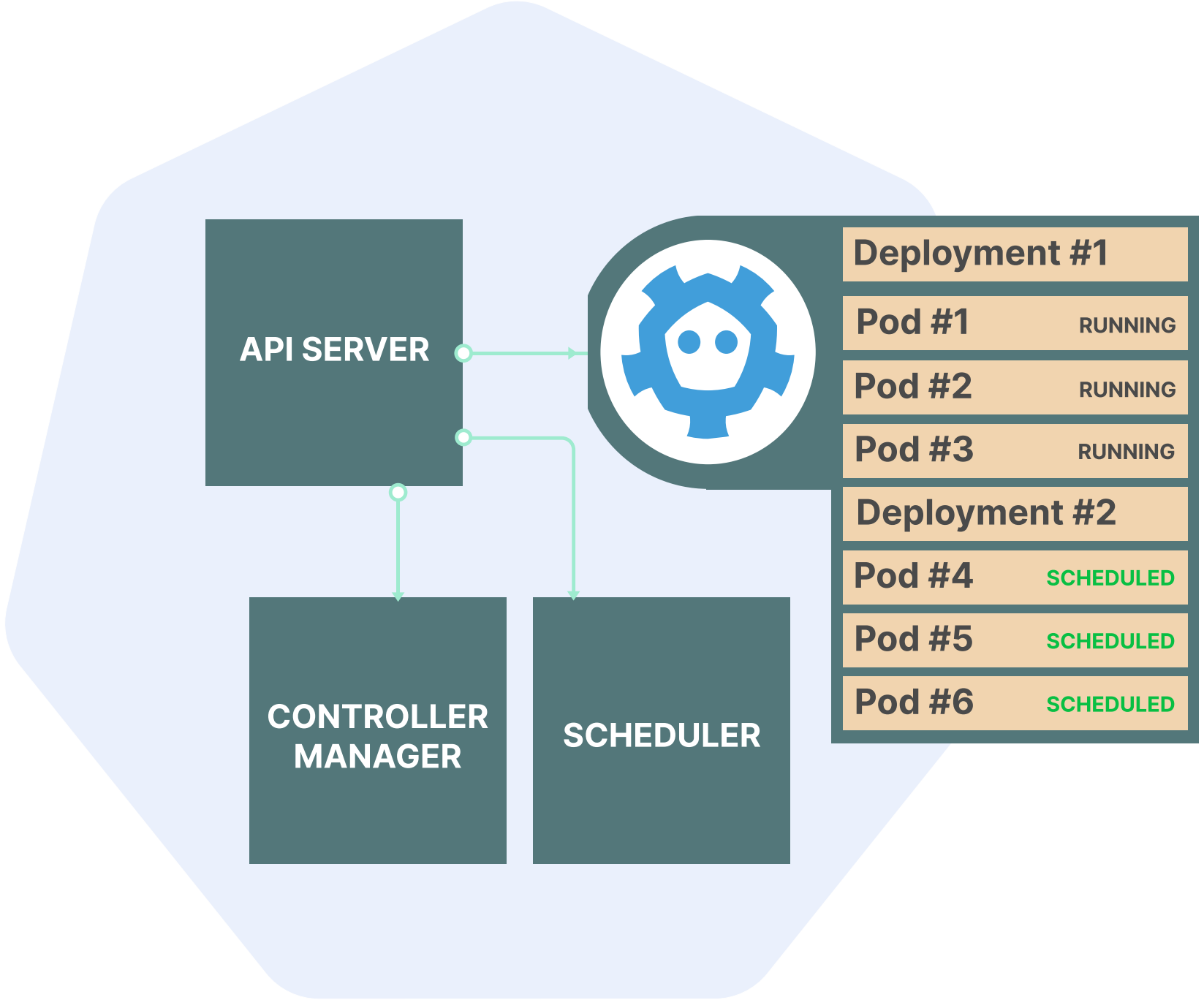


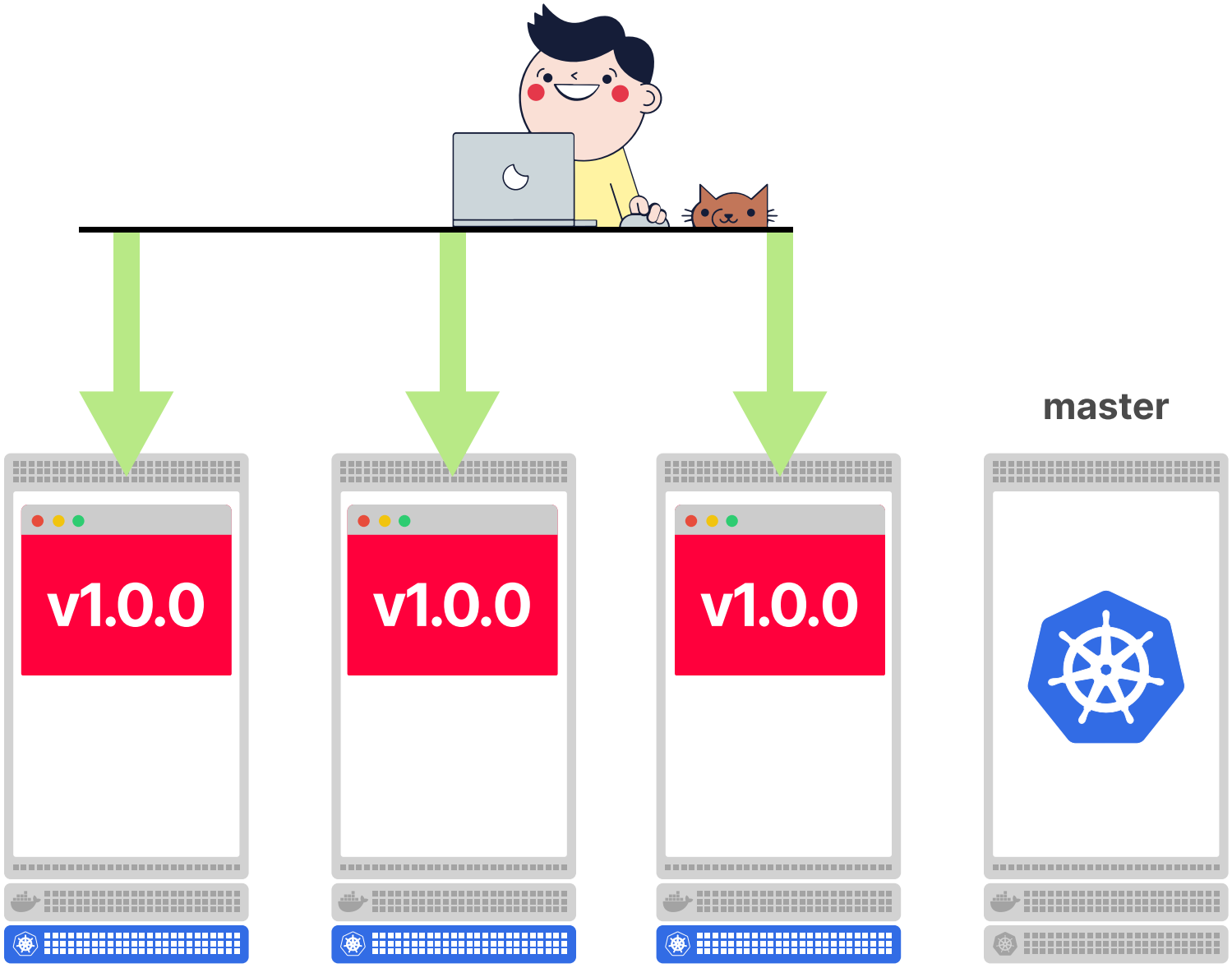

API SERVER
Deployment #1
Pod #1
RUNNING
Pod #2
RUNNING
Pod #3
RUNNING
Deployment #2
CONTROLLER MANAGER
SCHEDULER

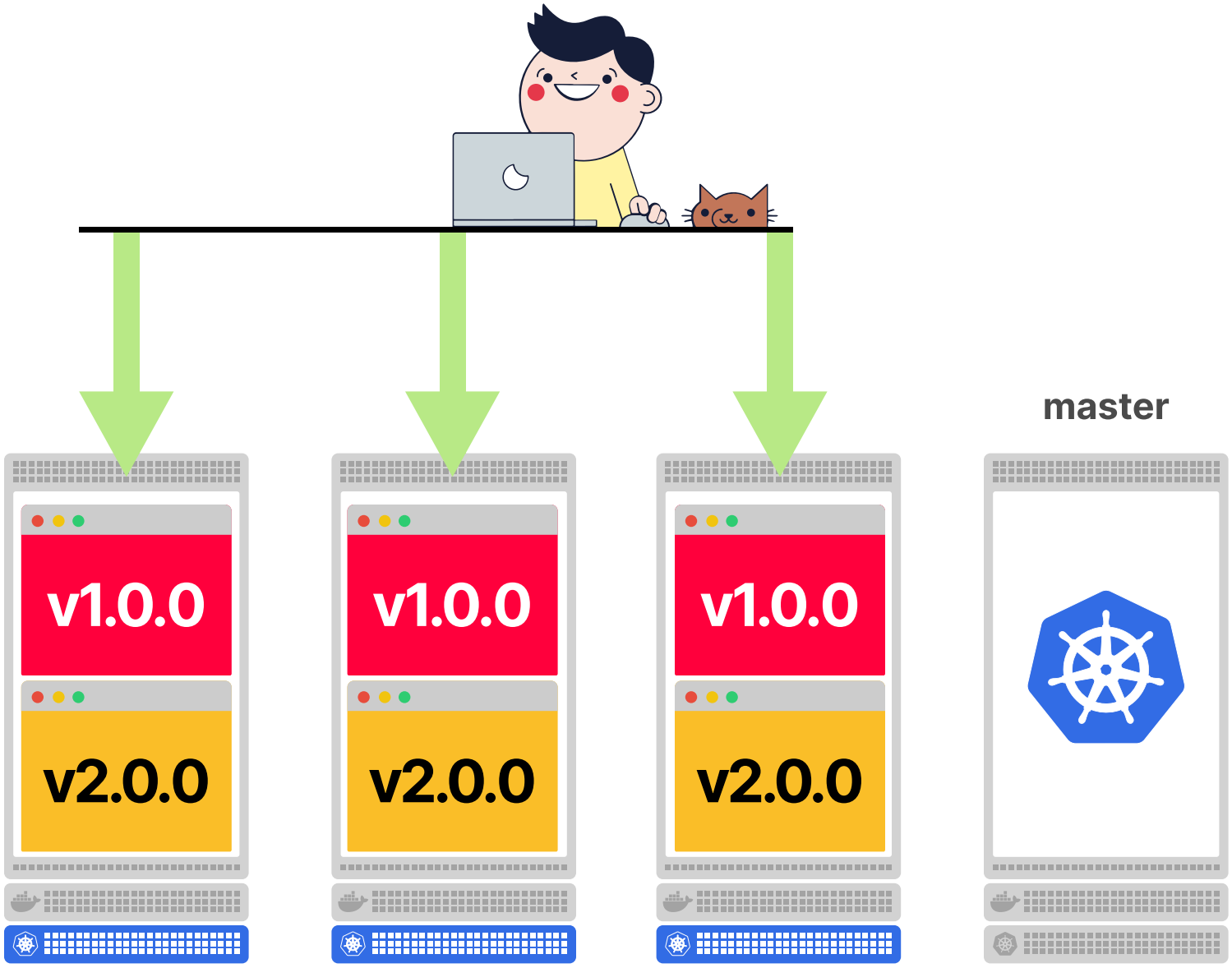

API SERVER
CONTROLLER MANAGER
SCHEDULER
Deployment #1
Pod #1
RUNNING
Pod #2
RUNNING
Pod #3
RUNNING
Deployment #2

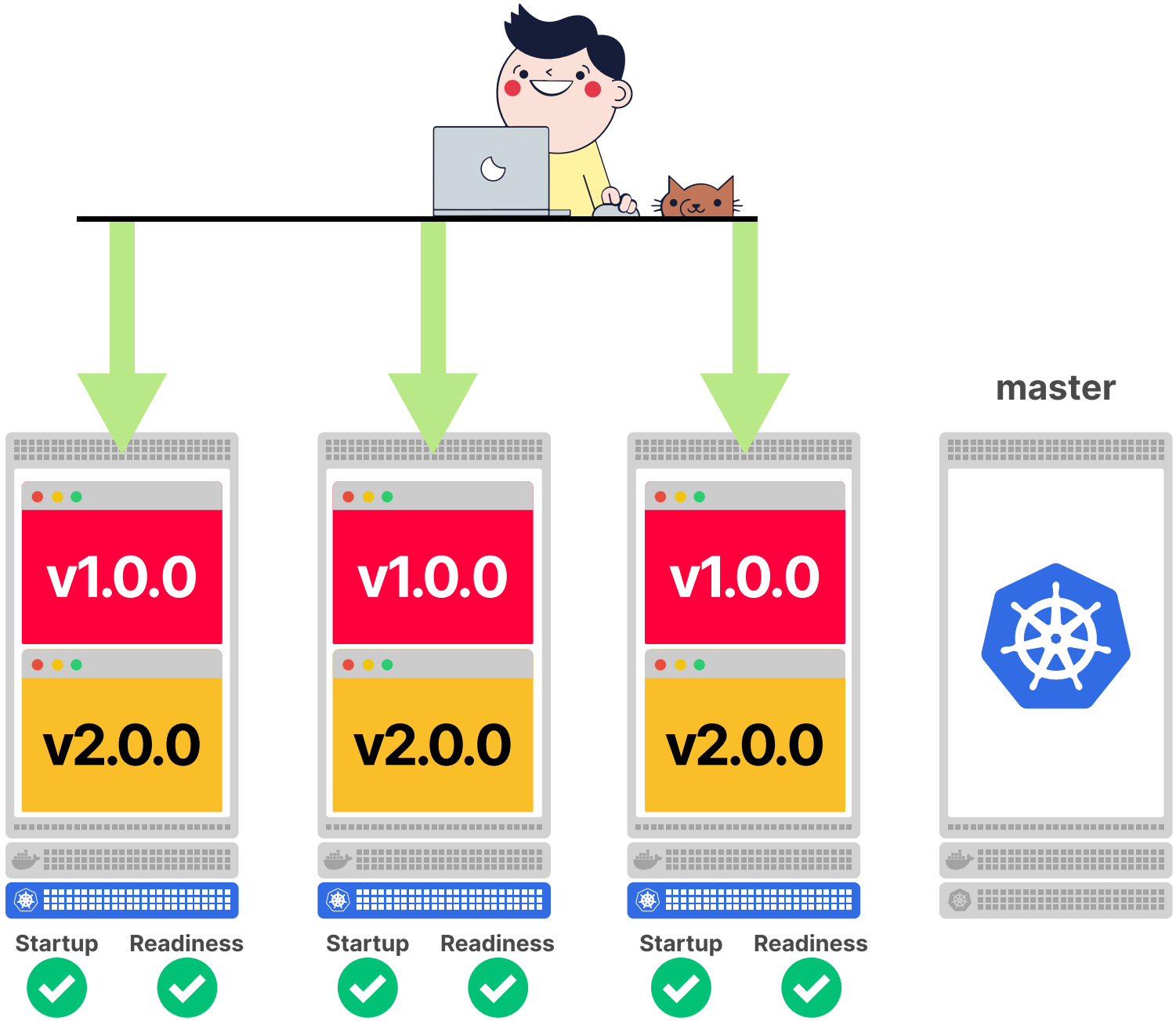

API SERVER
CONTROLLER MANAGER
SCHEDULER
Deployment #1
Pod #1 RUNNING
Pod #2 RUNNING
Pod #3 RUNNING
Deployment #2
Pod #4 PENDING
Pod #5 PENDING
Pod #6 PENDING

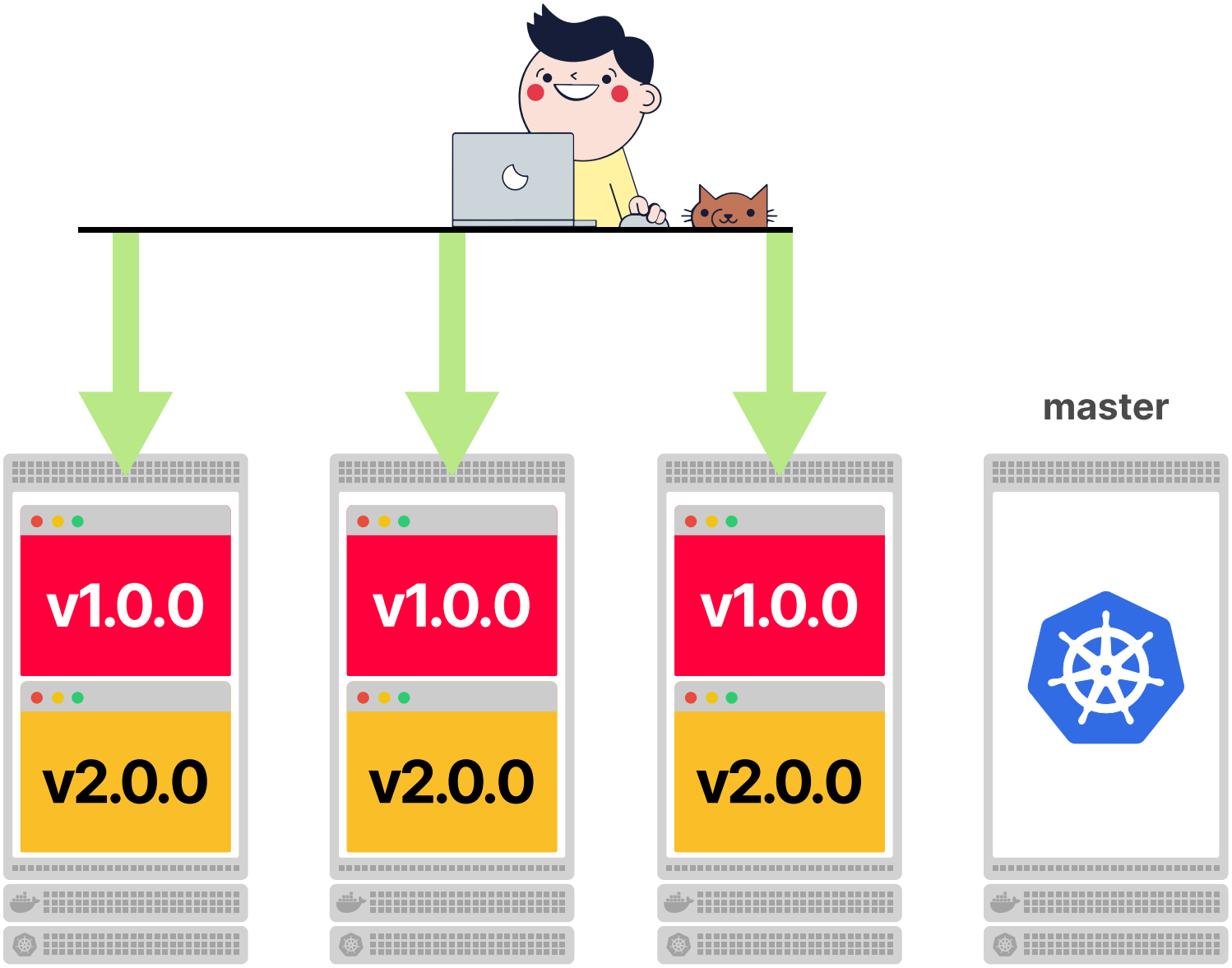

API SERVER
CONTROLLER MANAGER
SCHEDULER
Deployment #1
Pod #1 RUNNING
Pod #2 RUNNING
Pod #3 RUNNING
Deployment #2
Pod #4 PENDING
Pod #5 PENDING
Pod #6 PENDING


API SERVER
CONTROLLER MANAGER
SCHEDULER
Deployment #1
Pod #1 RUNNING
Pod #2 RUNNING
Pod #3 RUNNING
Deployment #2
Pod #4 SCHEDULED
Pod #5 SCHEDULED
Pod #6 SCHEDULED


v1.0.0
v1.0.0
v1.0.0
master


v1.0.0
v2.0.0
v1.0.0
v2.0.0
v1.0.0
v2.0.0
master

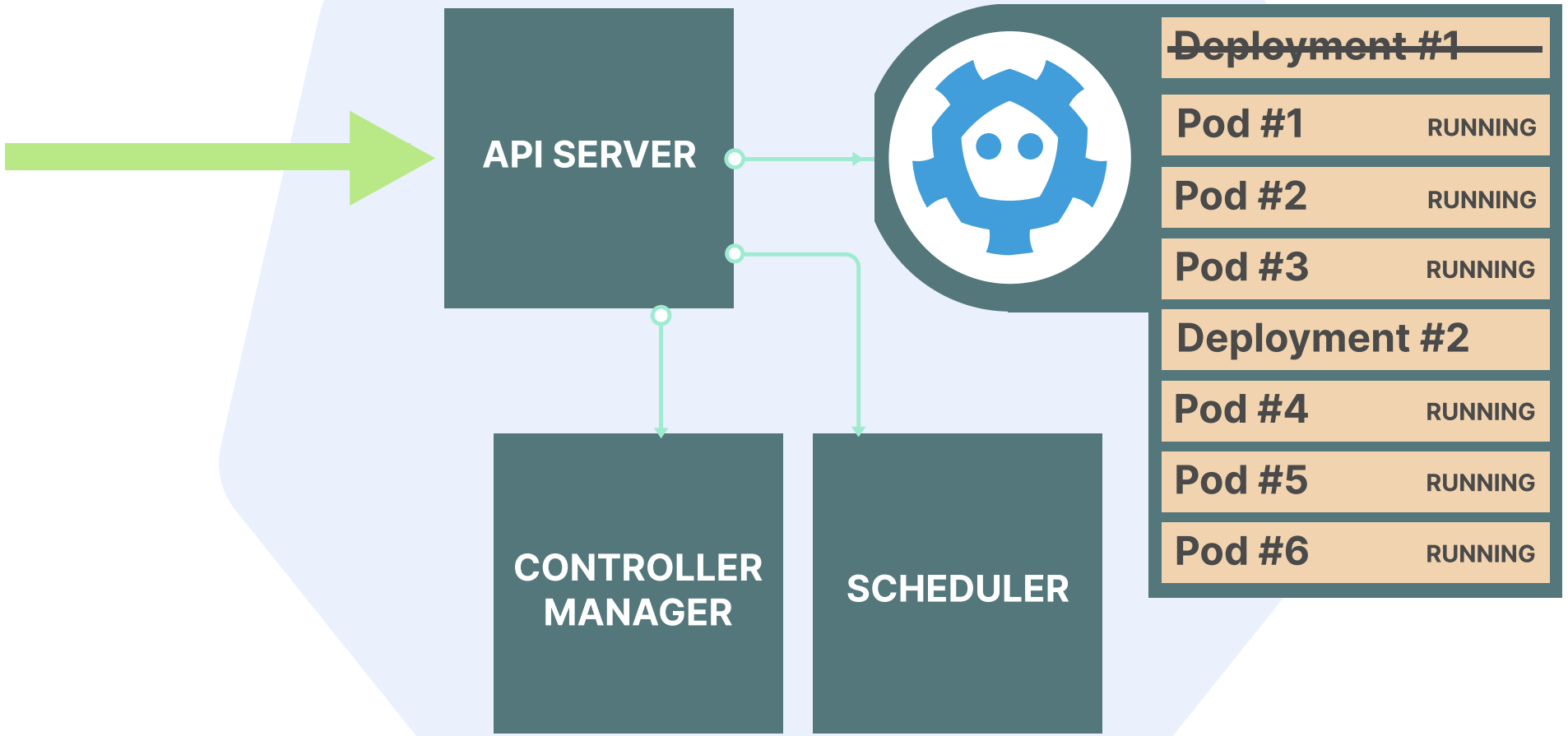

master
v1.0.0
v2.0.0
v1.0.0
v2.0.0
v1.0.0
v2.0.0
Startup
Readiness
Startup
Readiness
Startup
Readiness

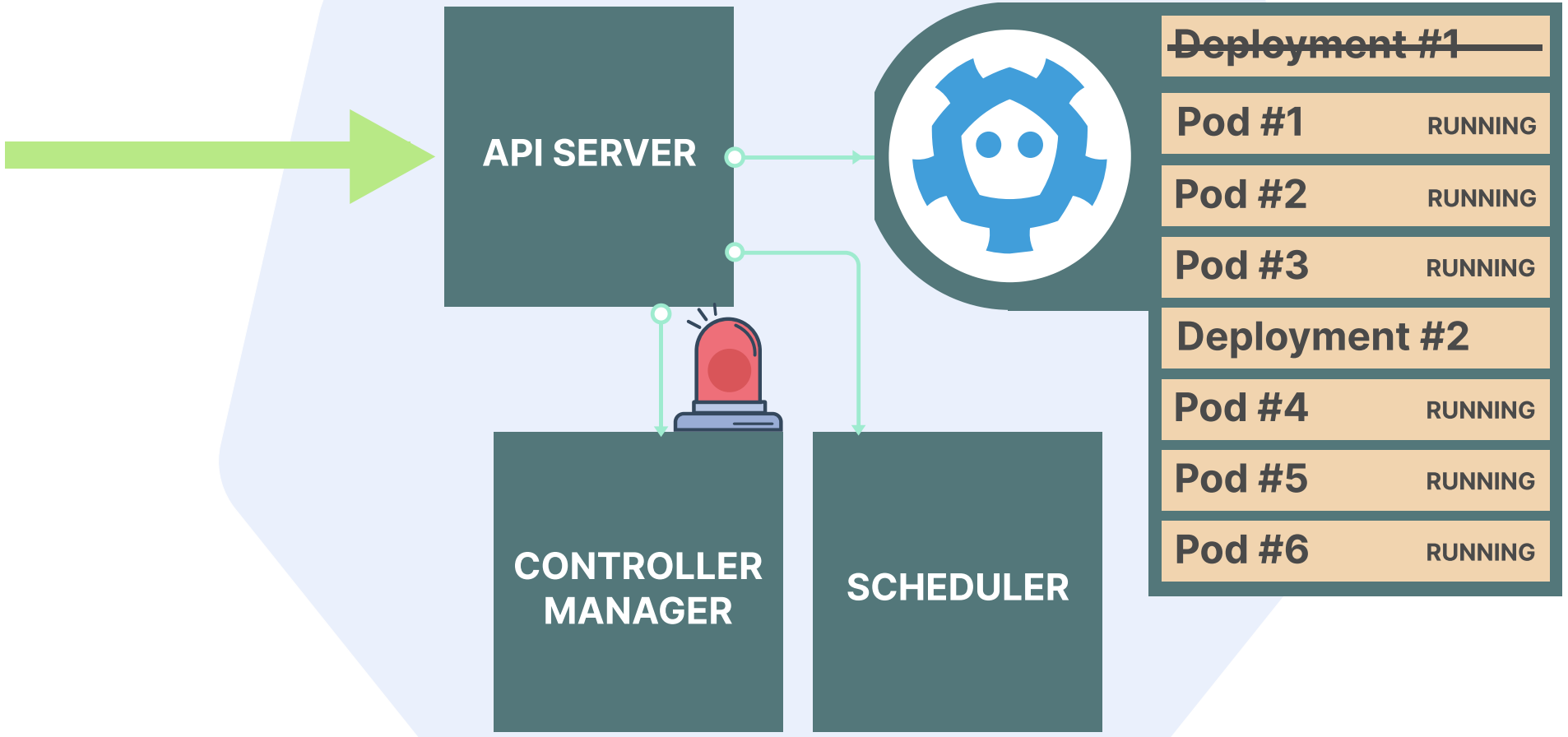

v1.0.0
v2.0.0
v1.0.0
v2.0.0
v1.0.0
v2.0.0
master

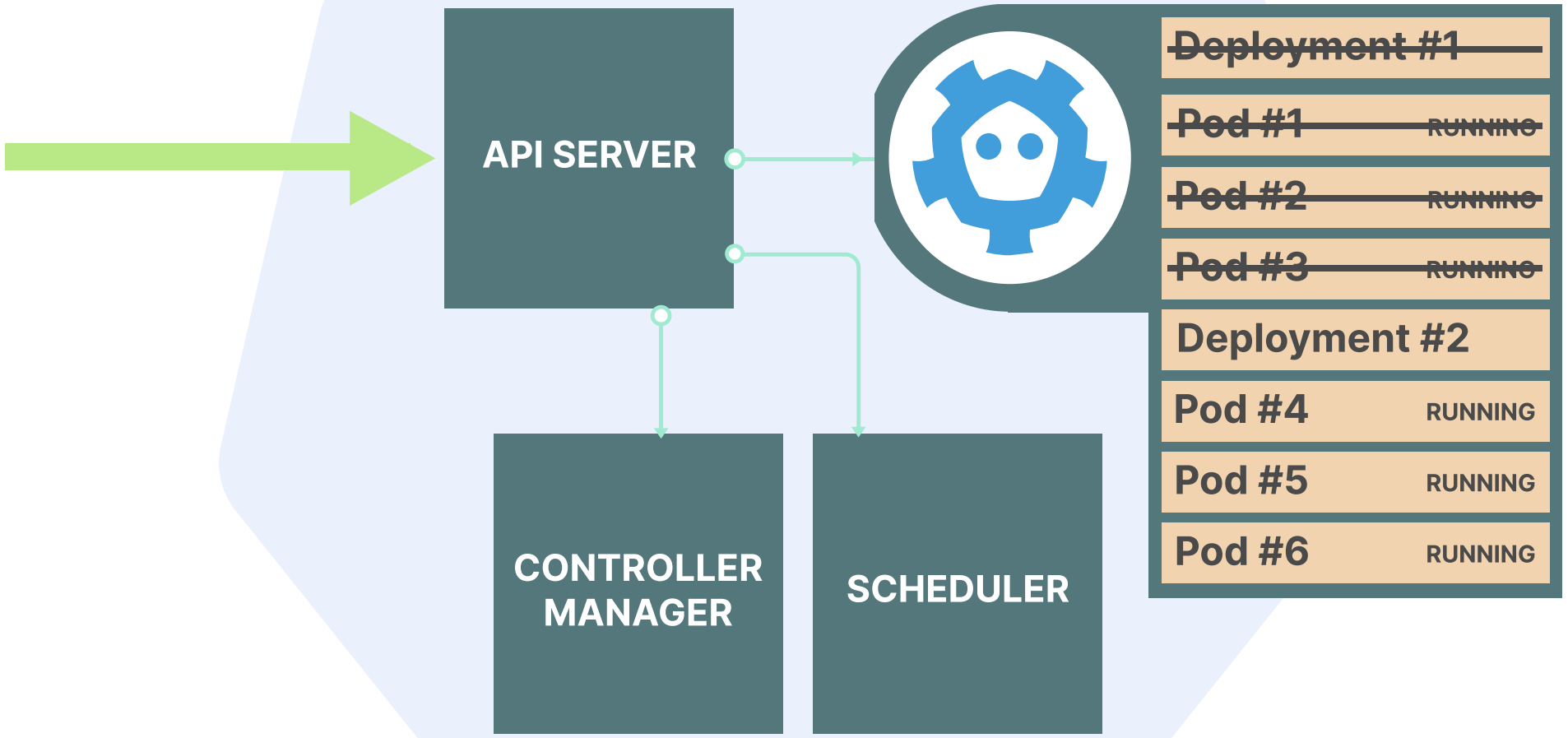
Delete Deployment for v1.0.0

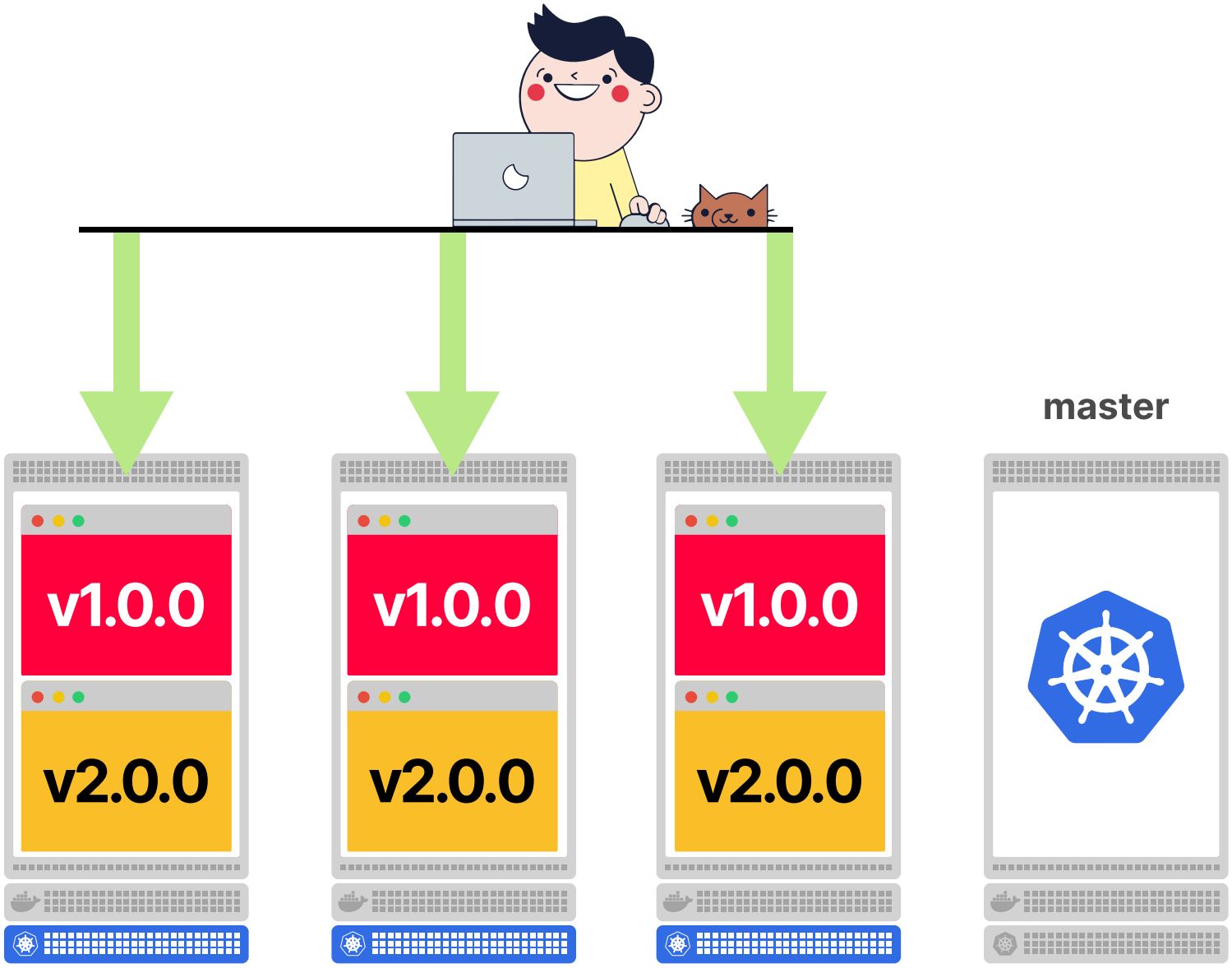


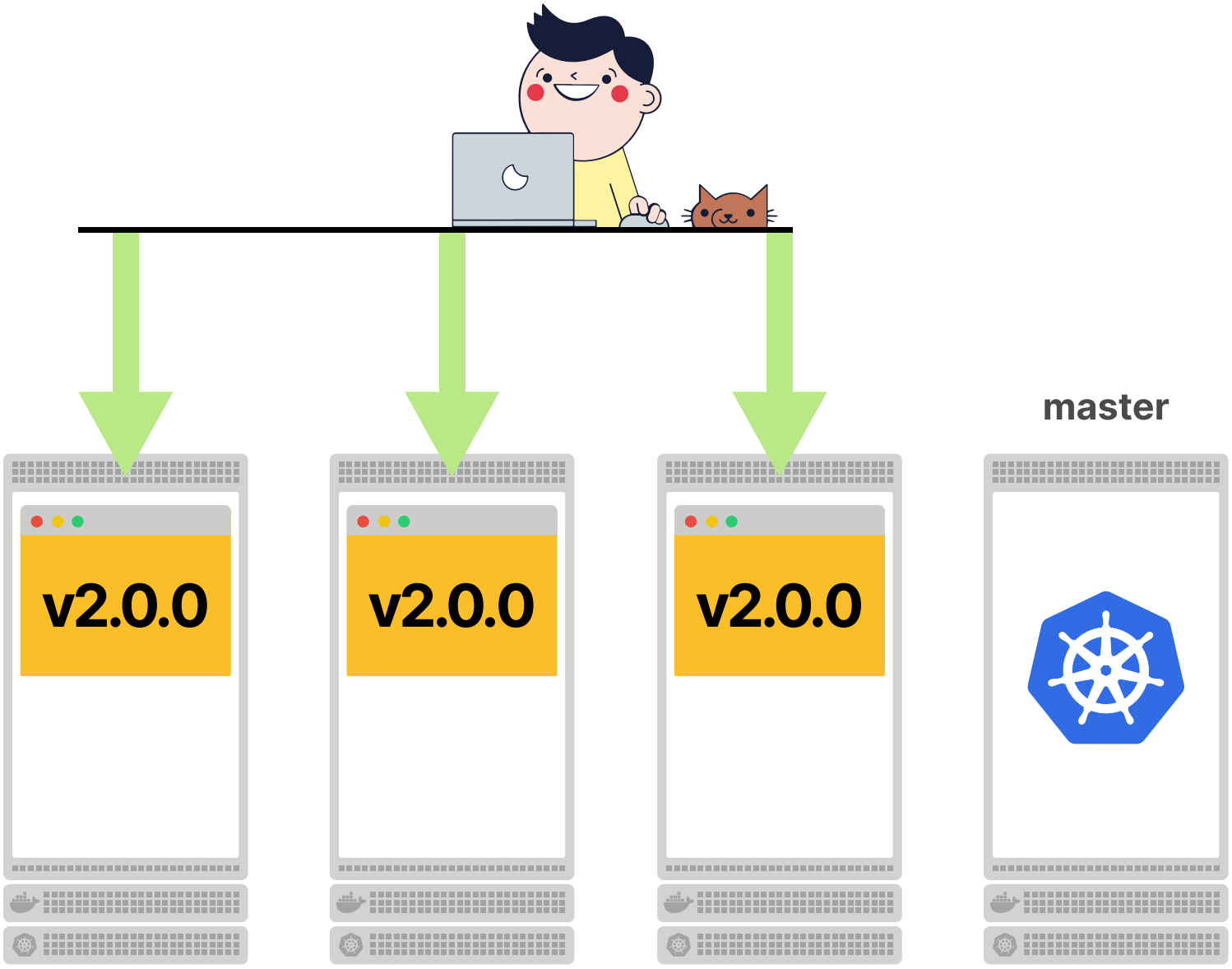



API SERVER
CONTROLLER MANAGER
SCHEDULER
Deployment #1
Pod #1 RUNNING
Pod #2 RUNNING
Pod #3 RUNNING
Deployment #2
Pod #4 RUNNING
Pod #5 RUNNING
Pod #6 RUNNING


API SERVER
CONTROLLER MANAGER
SCHEDULER
Deployment #1
Pod #1 RUNNING
Pod #2 RUNNING
Pod #3 RUNNING
Deployment #2
Pod #4 RUNNING
Pod #5 RUNNING
Pod #6 RUNNING


API SERVER
CONTROLLER MANAGER
SCHEDULER
Deployment #1
Pod #1 RUNNING
Pod #2 RUNNING
Pod #3 RUNNING
Deployment #2
Pod #4 RUNNING
Pod #5 RUNNING
Pod #6 RUNNING


v1.0.0
v2.0.0
v1.0.0
v2.0.0
v1.0.0
v2.0.0
master


v2.0.0
v2.0.0
v2.0.0
master

Recap

Control plane

Recap — control plane

1. API server
2. etcd
3. Controller manager
4. Scheduler
5. Kubelet

Recap — control plane

1. API server

2. etcd

3. Controller manager

4. Scheduler

5. Kubelet

Recap — control plane

1. API server

2. etcd

3. Controller manager

4. Scheduler

5. Kubelet

Recap — control plane

1. API server

2. etcd

3. Controller manager

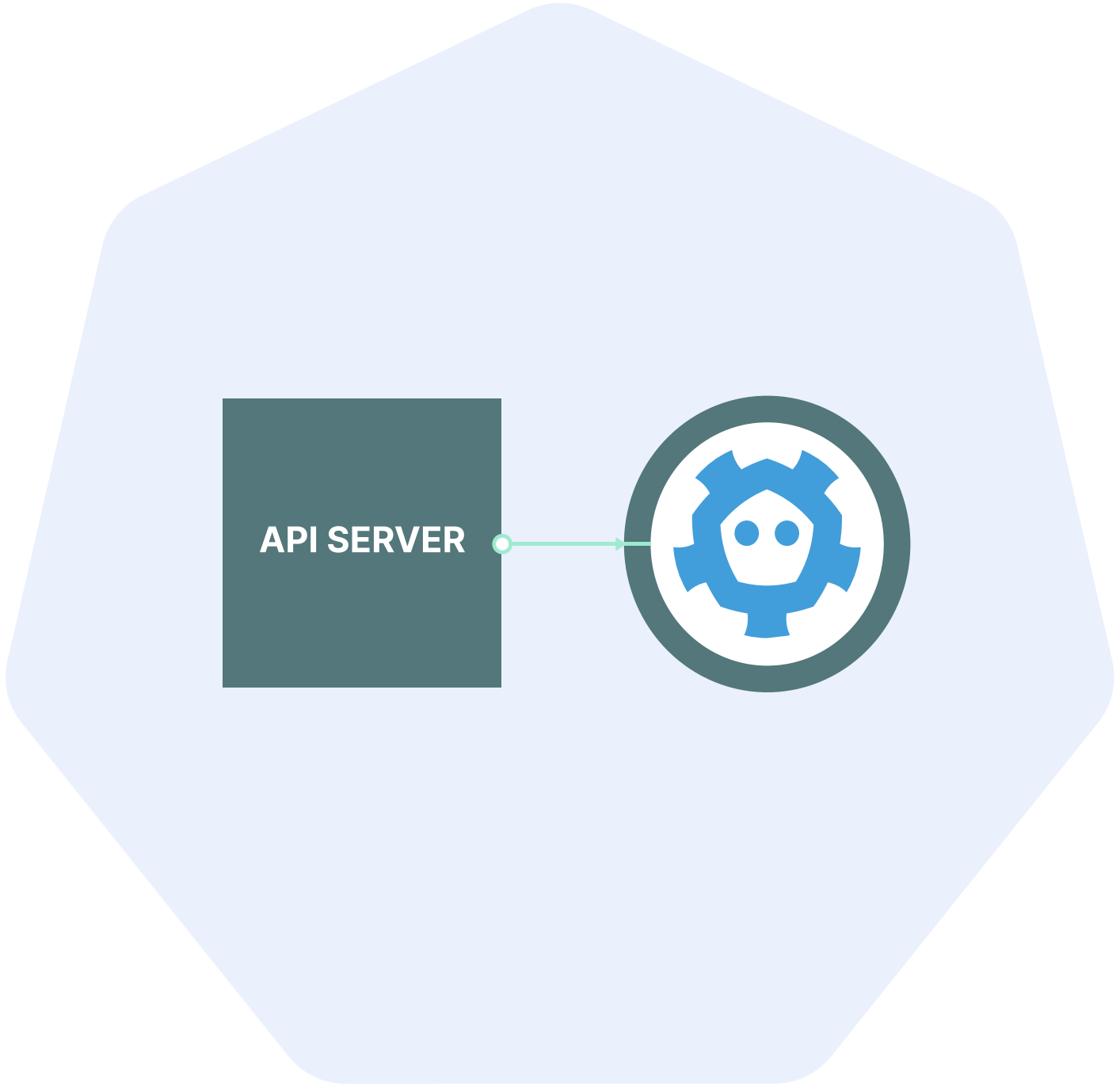
4. Scheduler

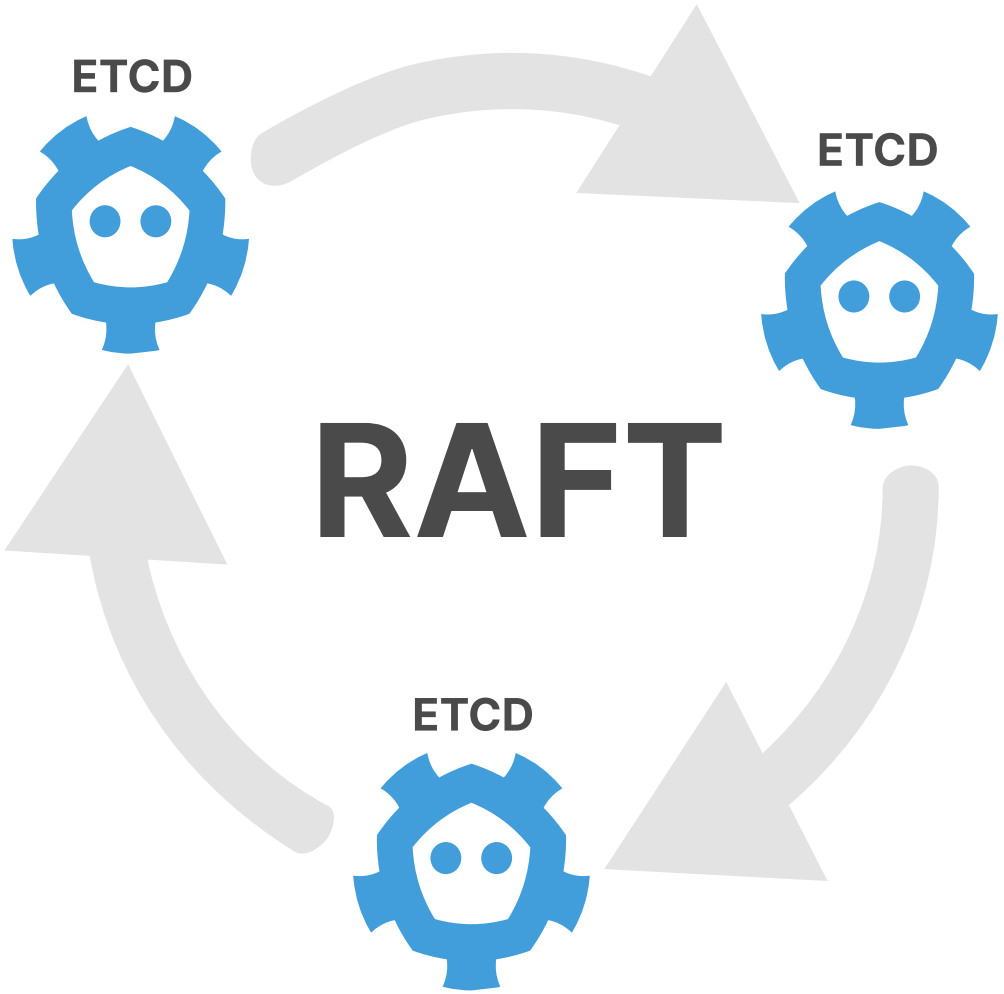
5. Kubelet

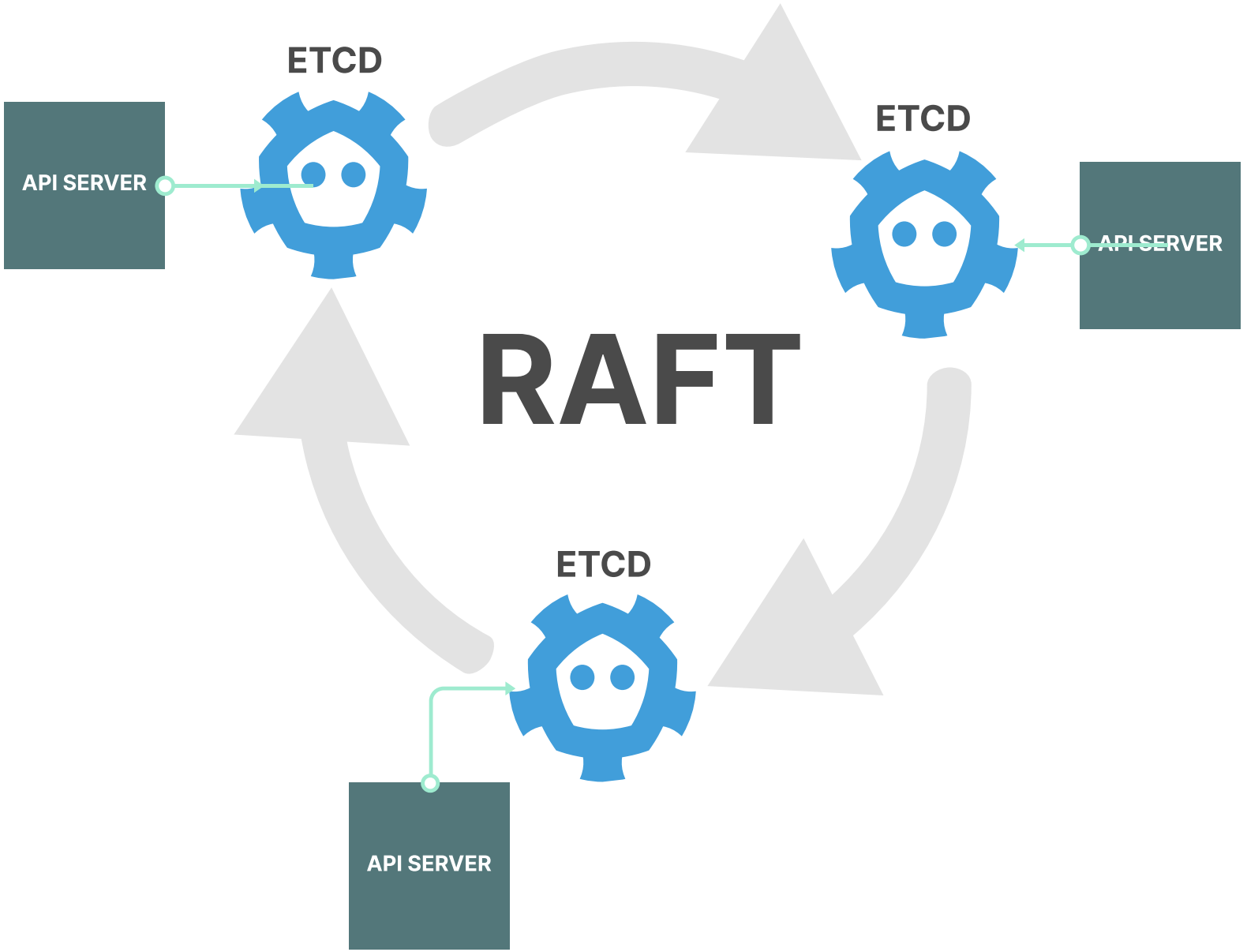
Recap — control plane

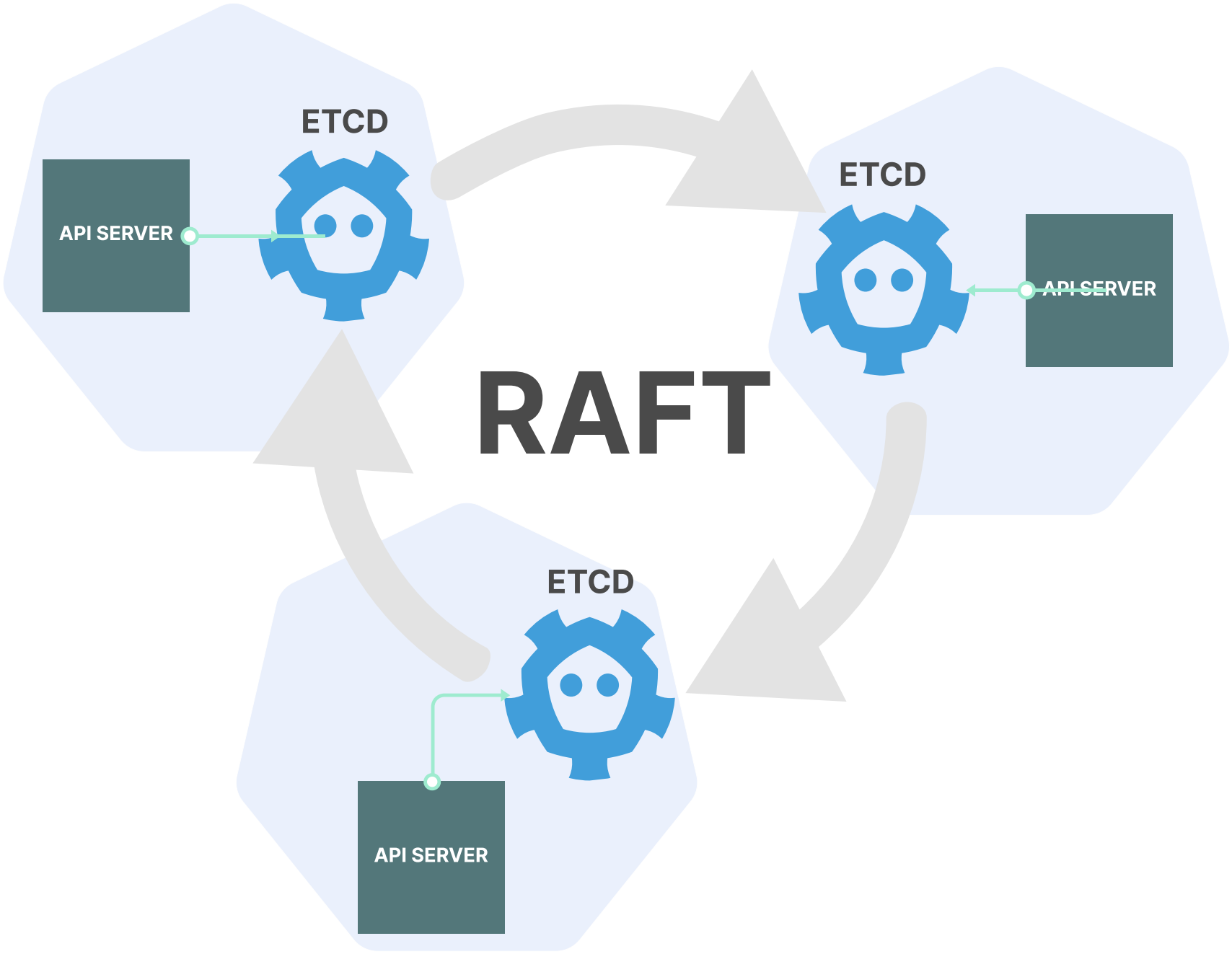
1. API server
2. etcd
3. Controller manager
4. Scheduler
5. **Kubelet**

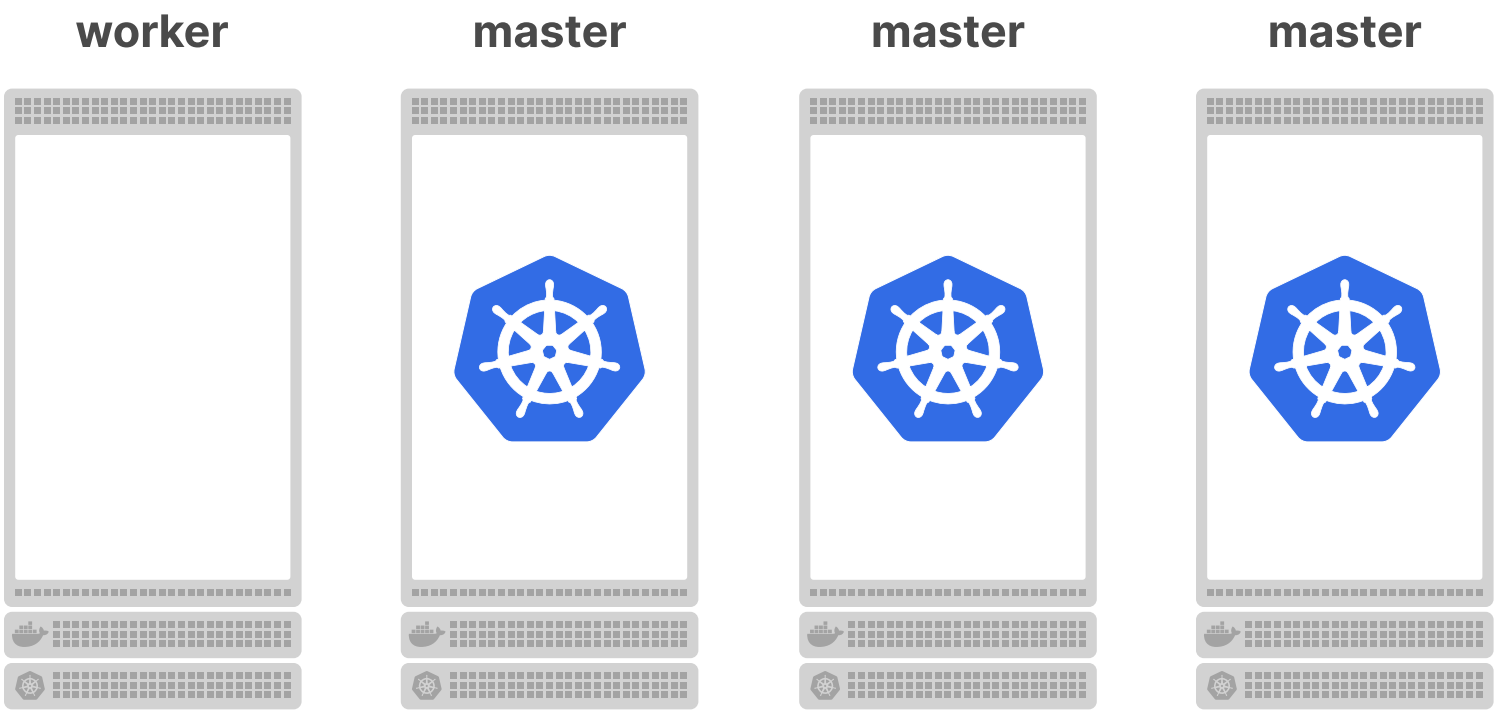
Control plane high availability

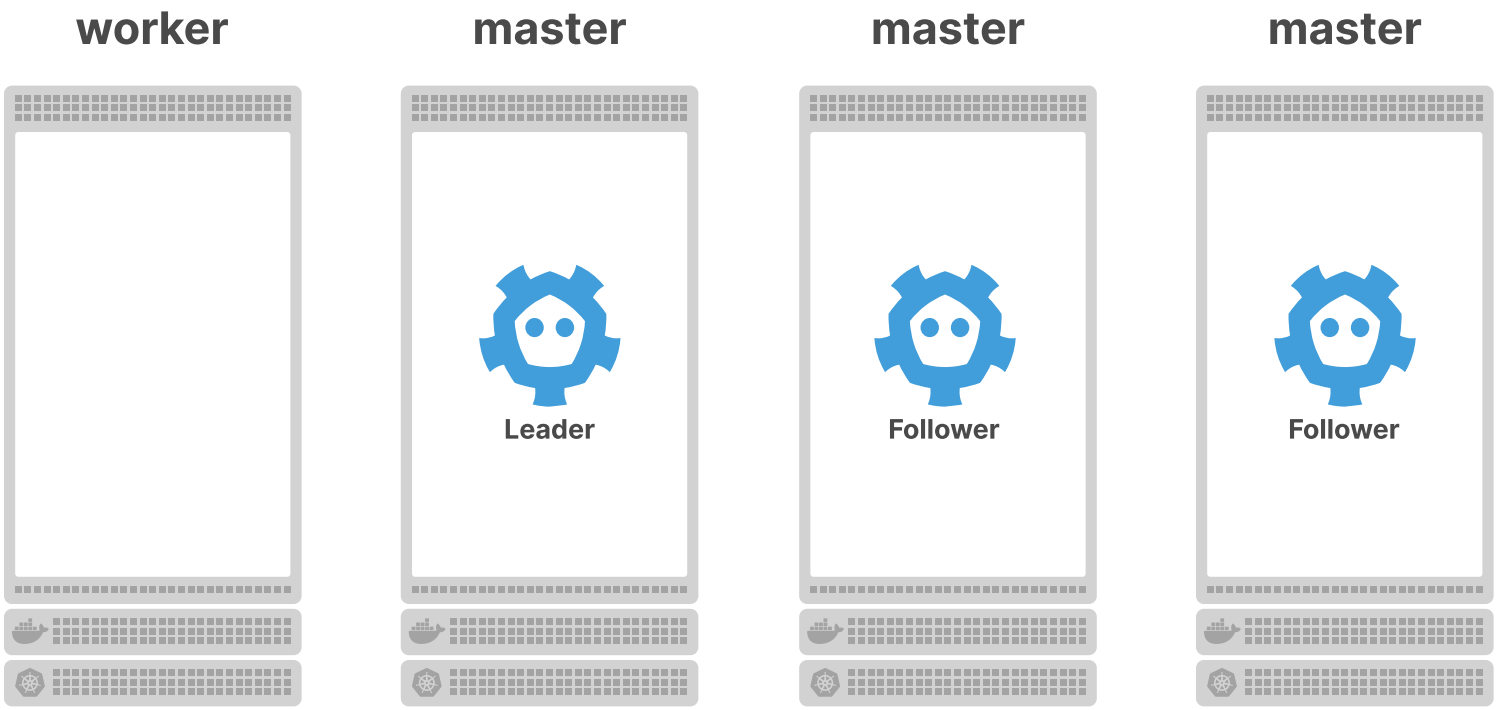

API SERVER

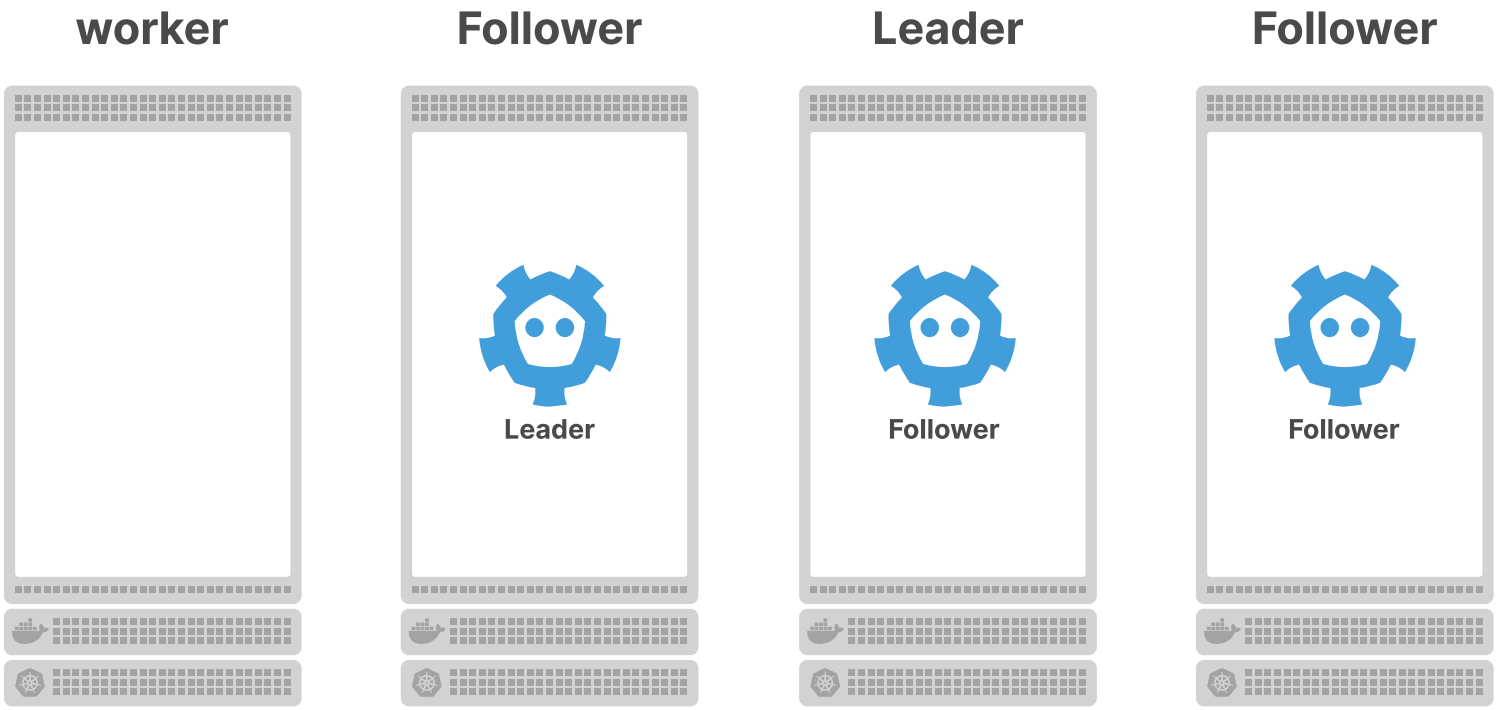

ETCD
ETCD
RAFT
ETCD

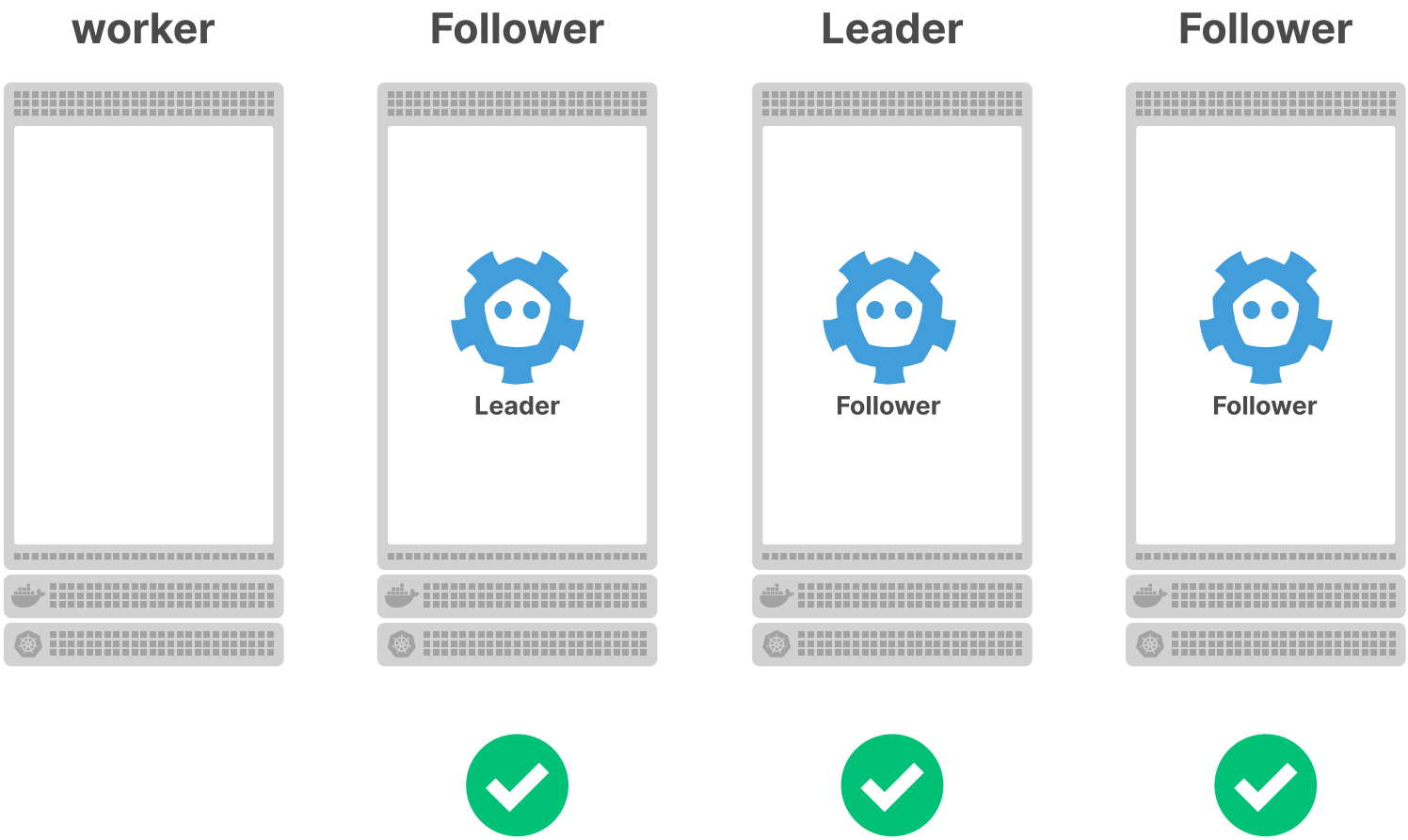

ETCD
API SERVER
ETCD
API SERVER
RAFT
ETCD
API SERVER

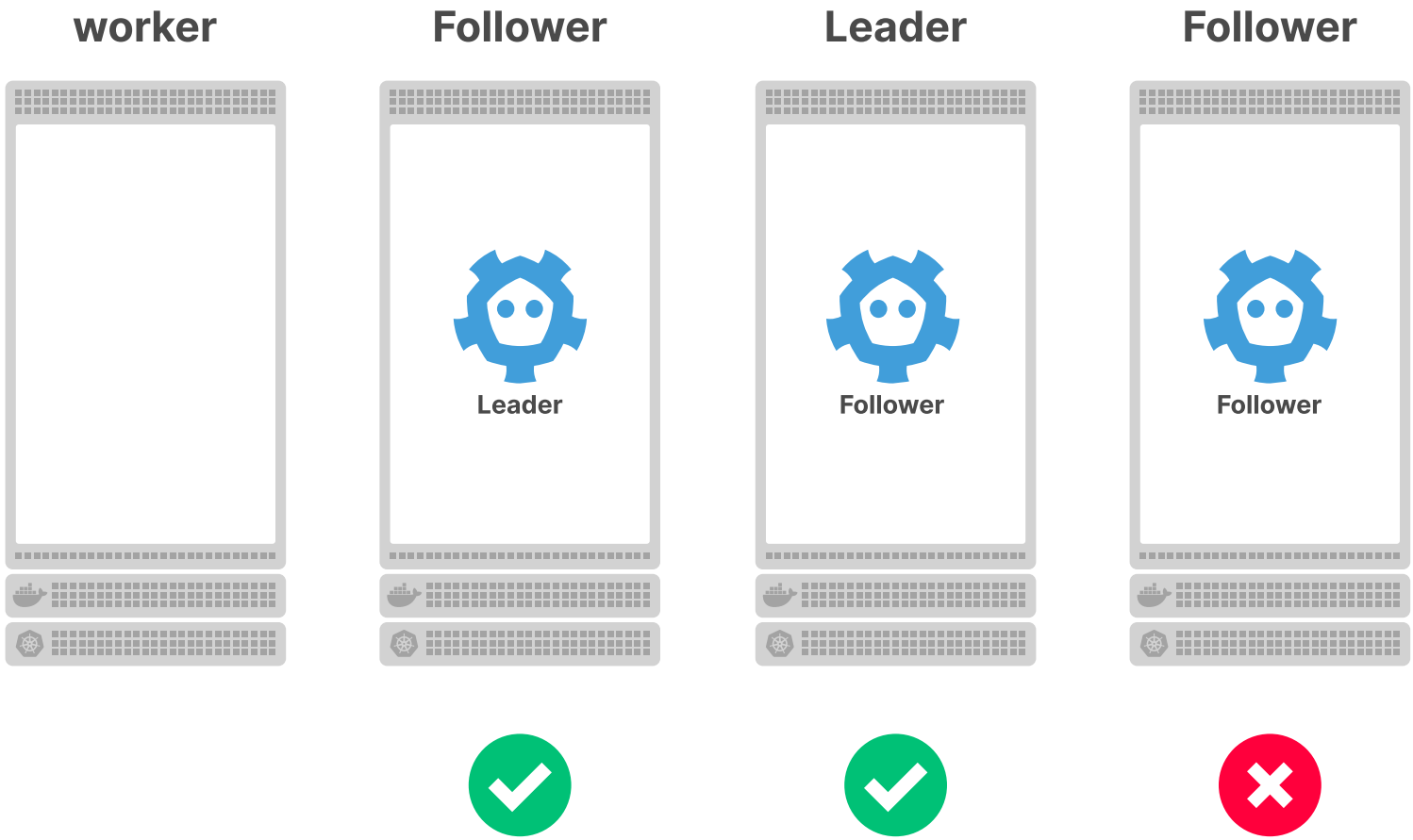

ETCD
API SERVER
ETCD
API SERVER
RAFT
ETCD
API SERVER


worker
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master

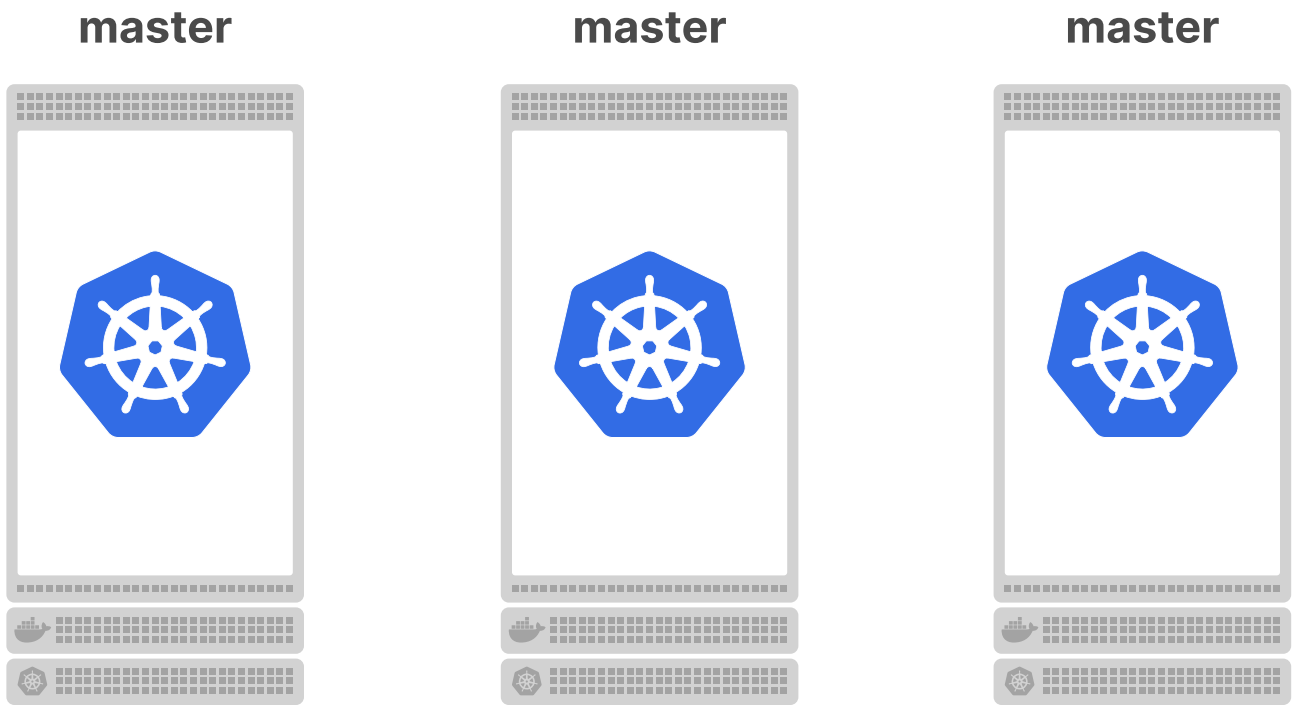

worker
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master
master
Leader
Follower
Follower

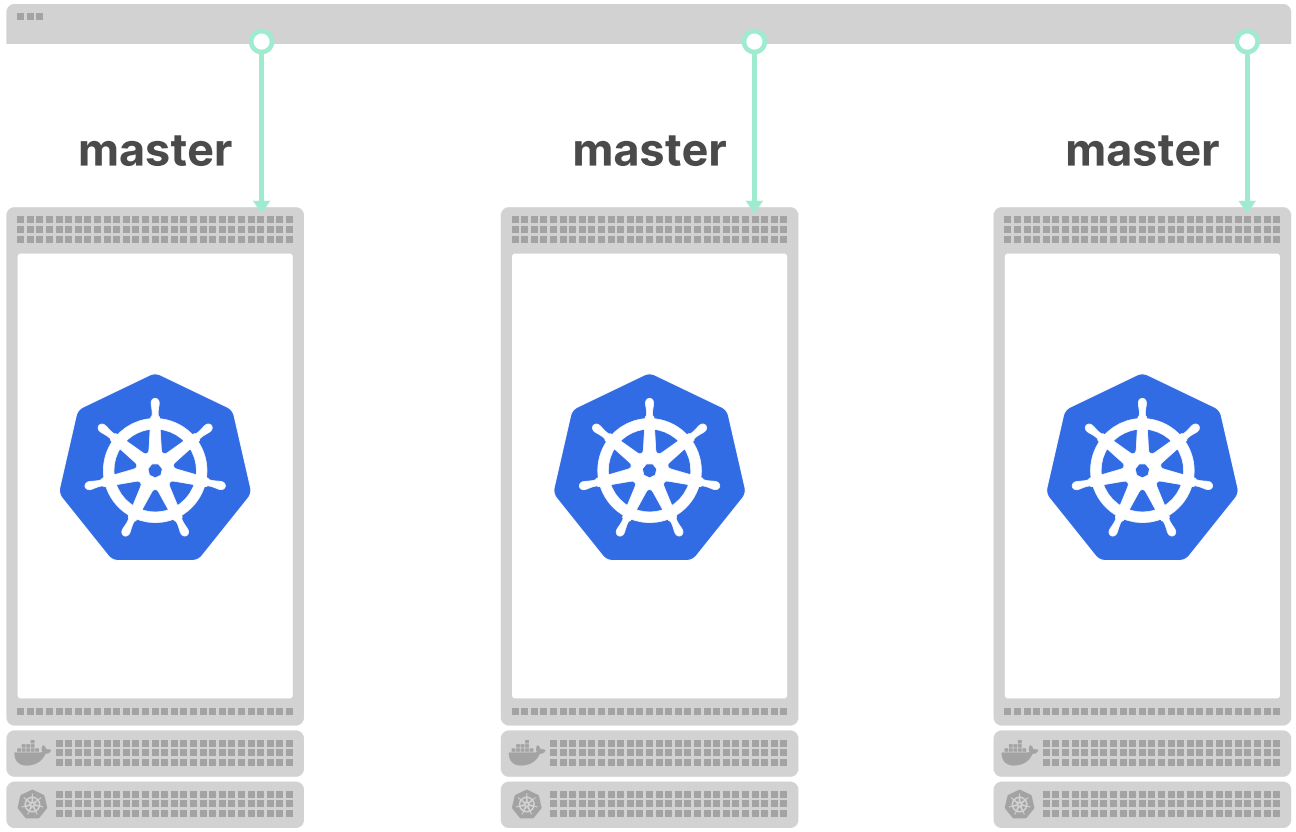

worker
Follower
Leader
Leader
Follower
Follower
Follower

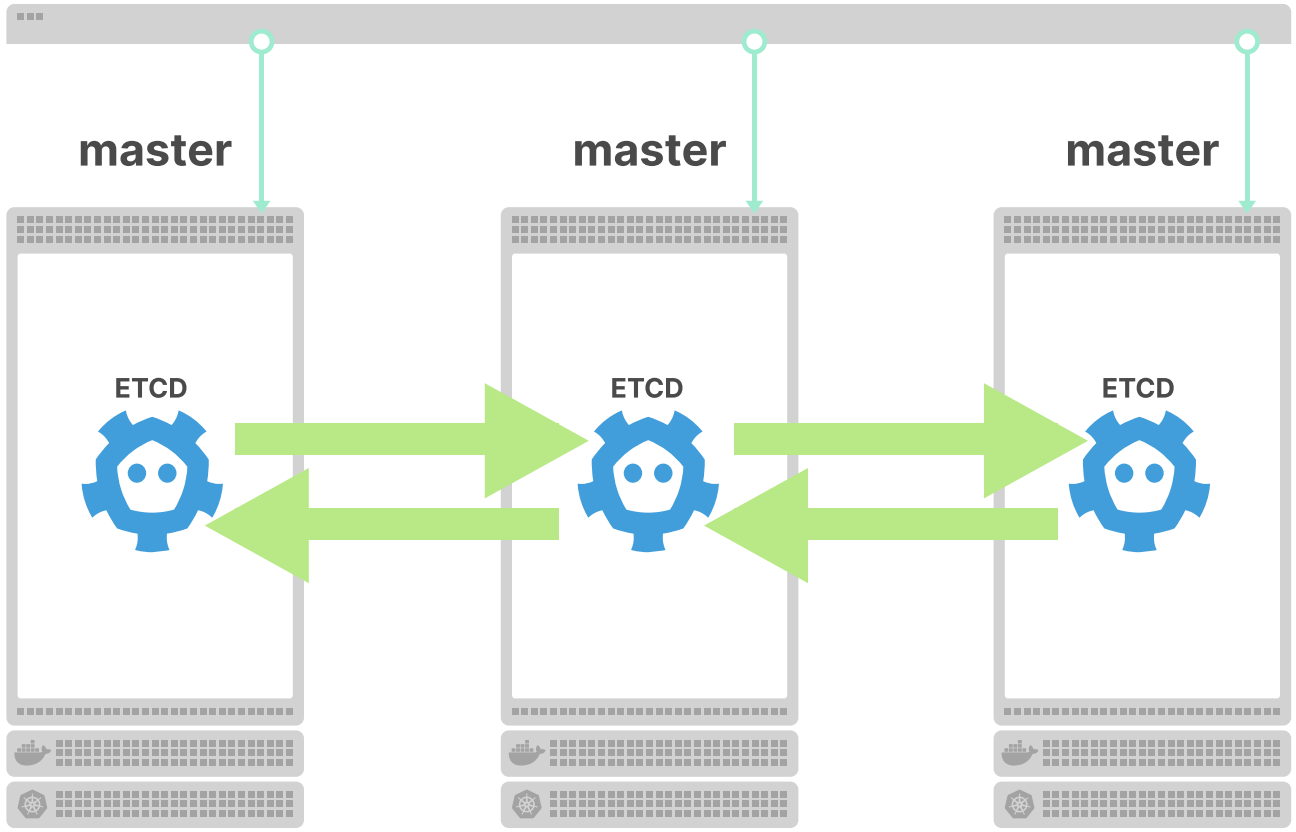

worker
Follower
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Leader
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Follower


worker
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Amazon EKS

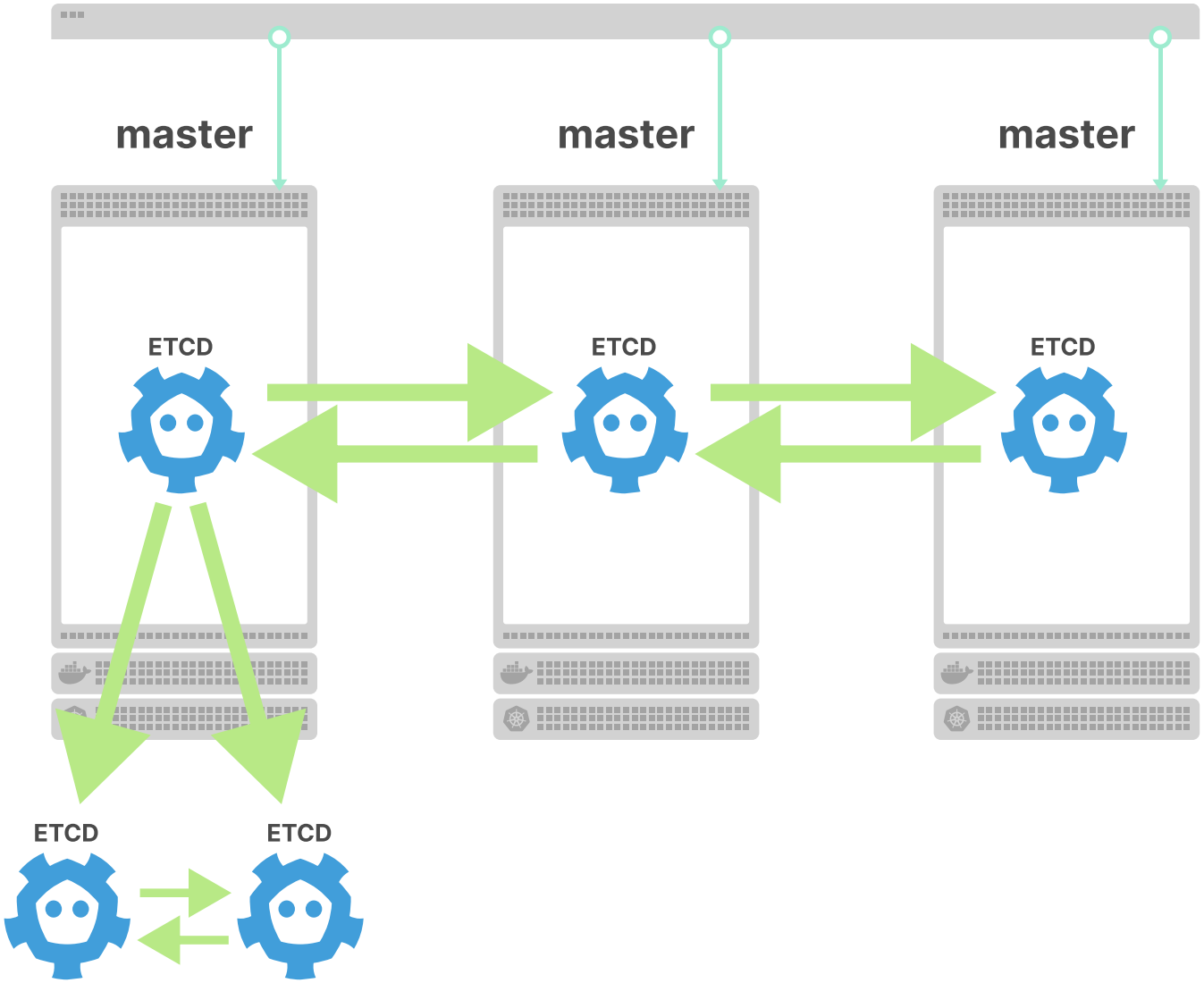

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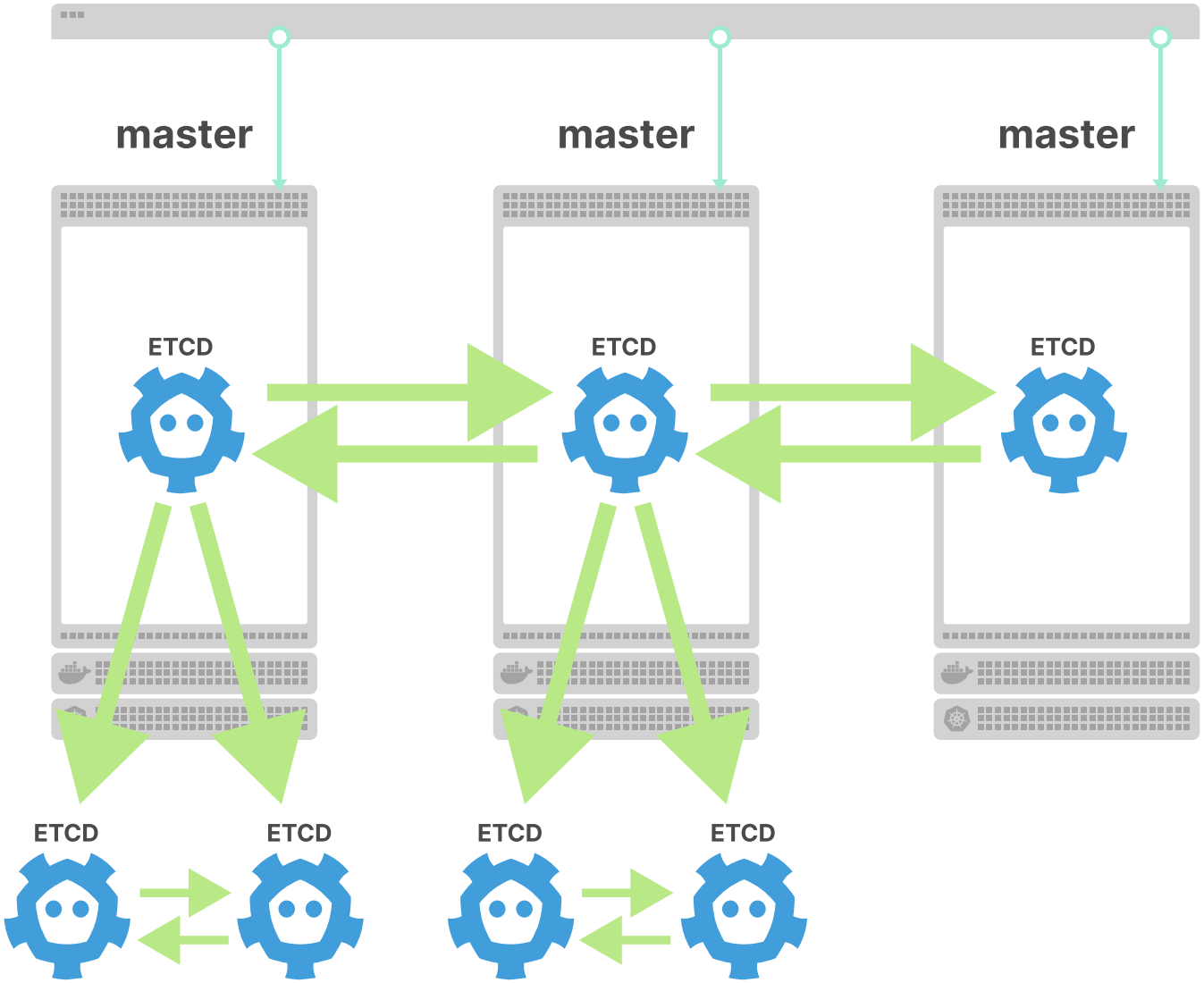

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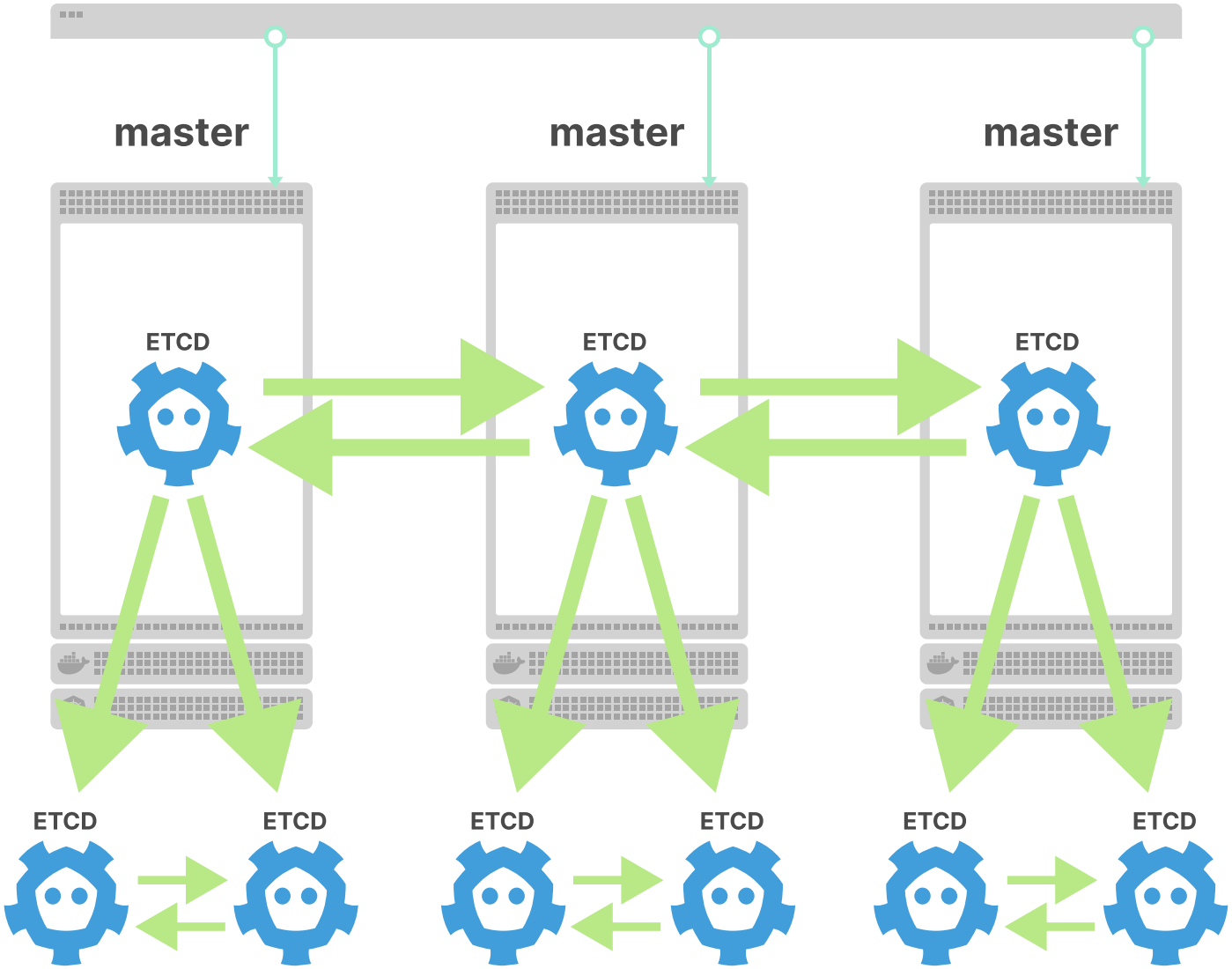

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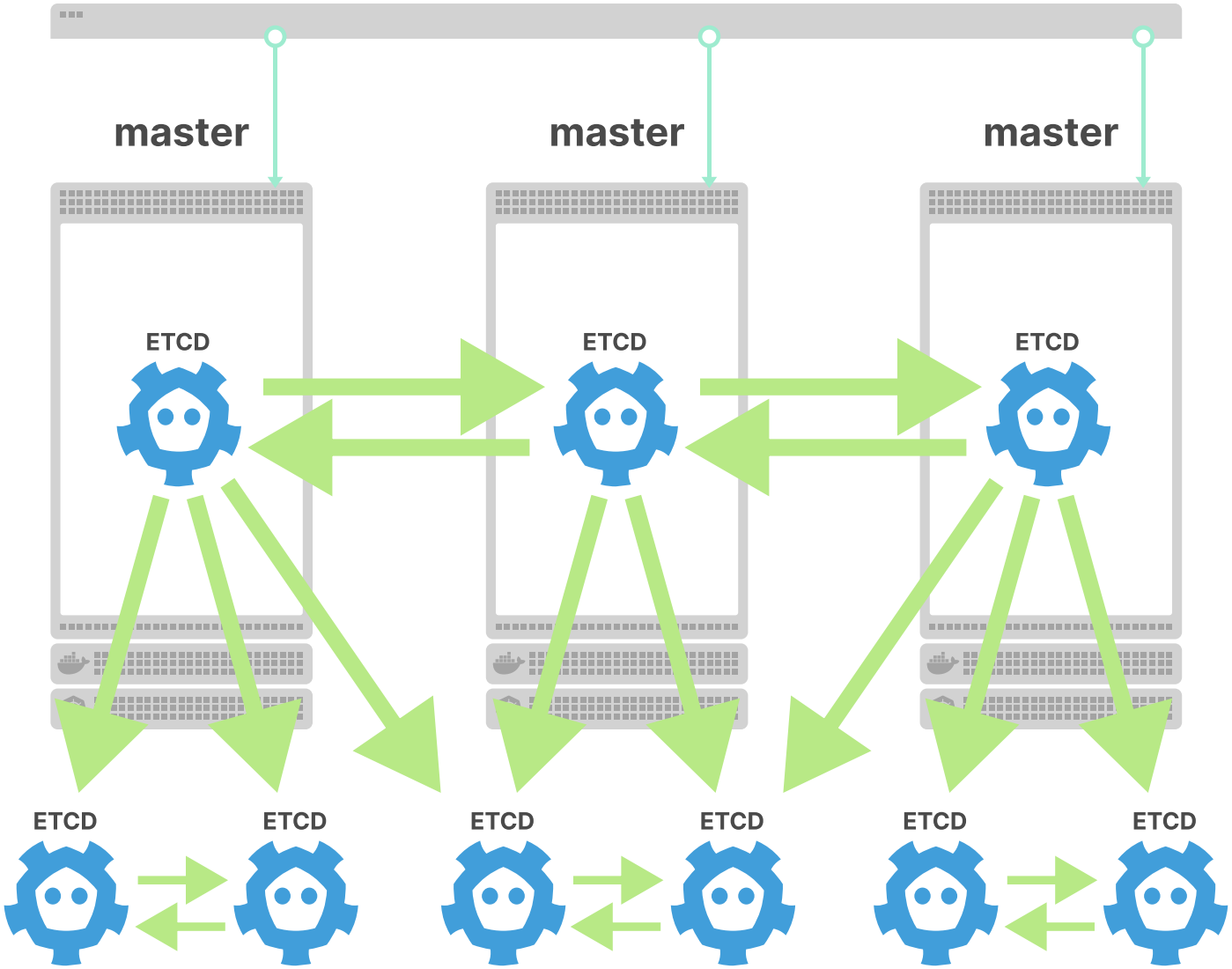


monzo


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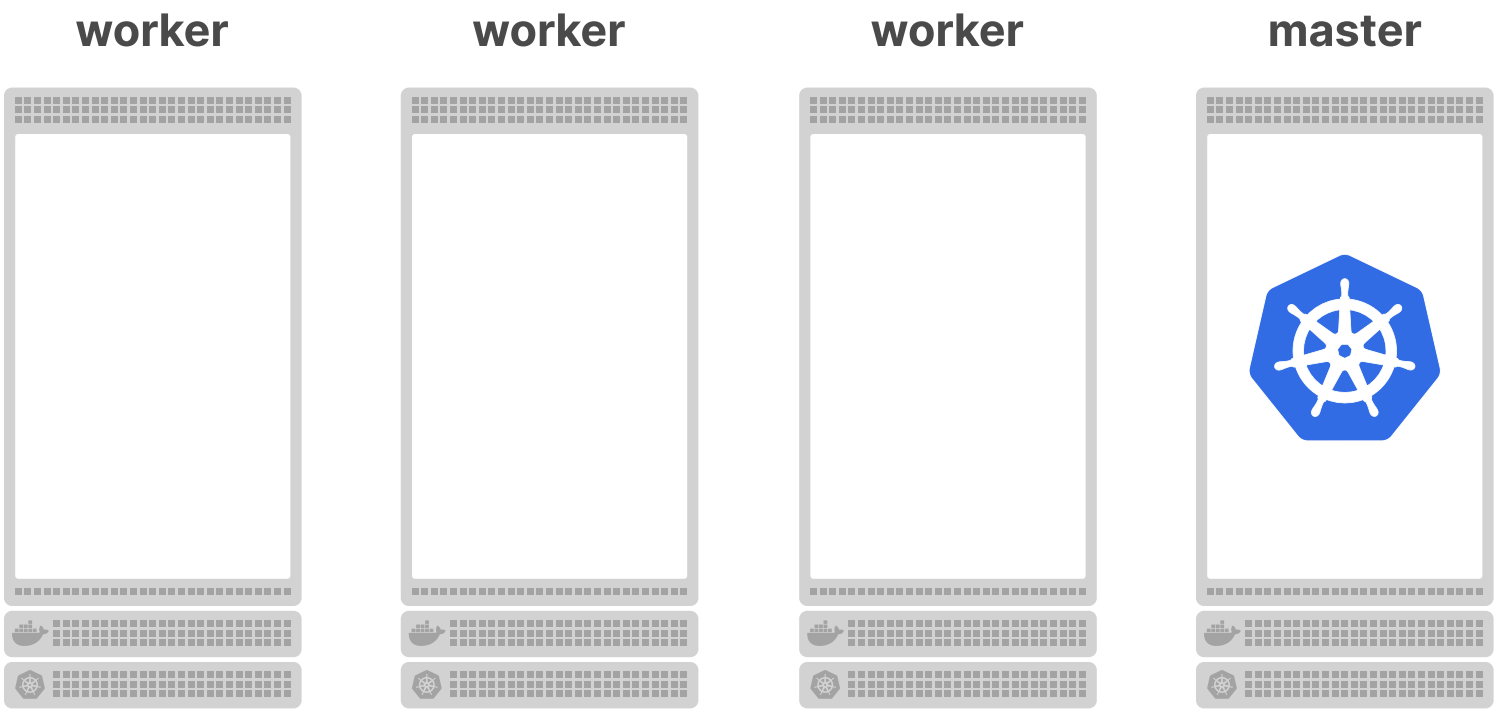

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CLUSTER SIZE	MAJORITY	FAILURE TOLERANCE
1	1	0
3	2	1
4	3	1
5	3	**2**
6	4	2
7	4	**3**
8	5	3
9	5	**4**

Breaking the cluster

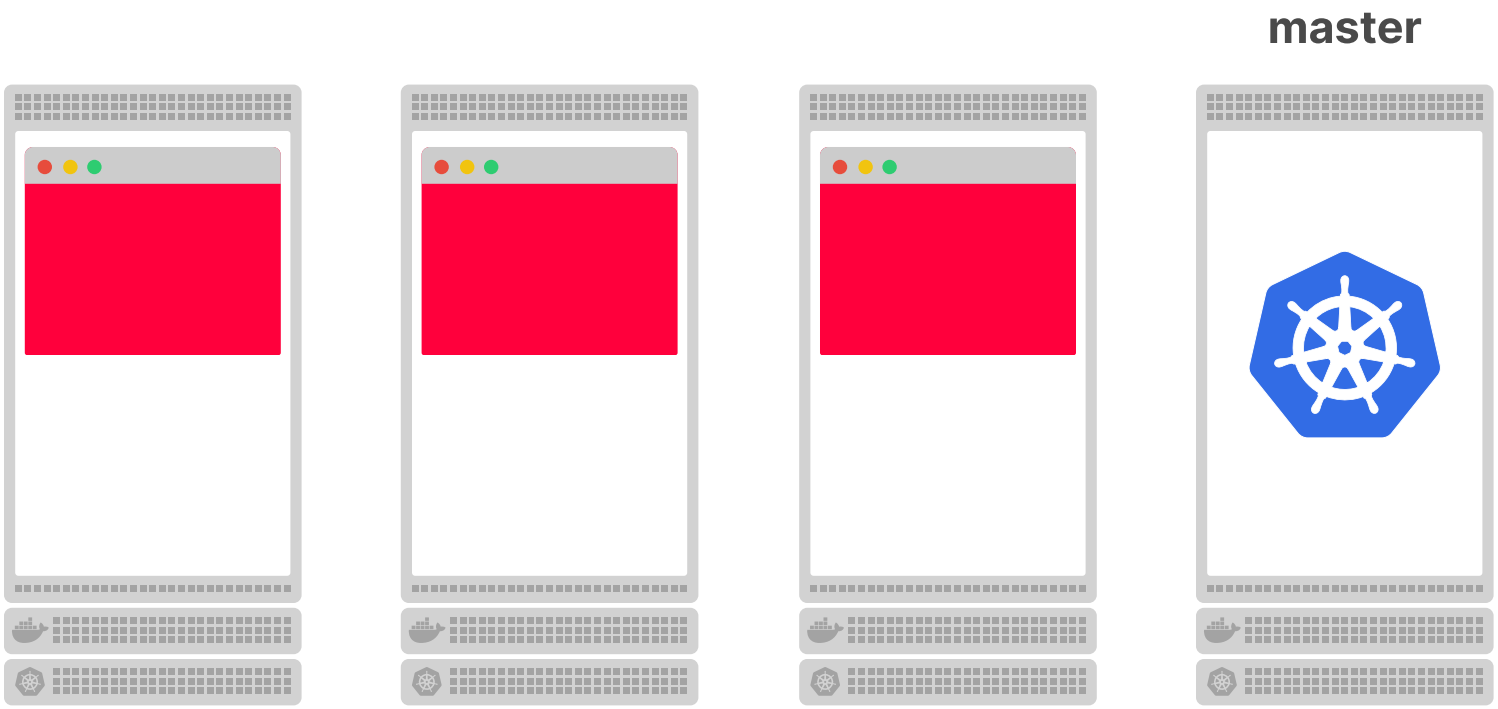

worker
worker
worker
master

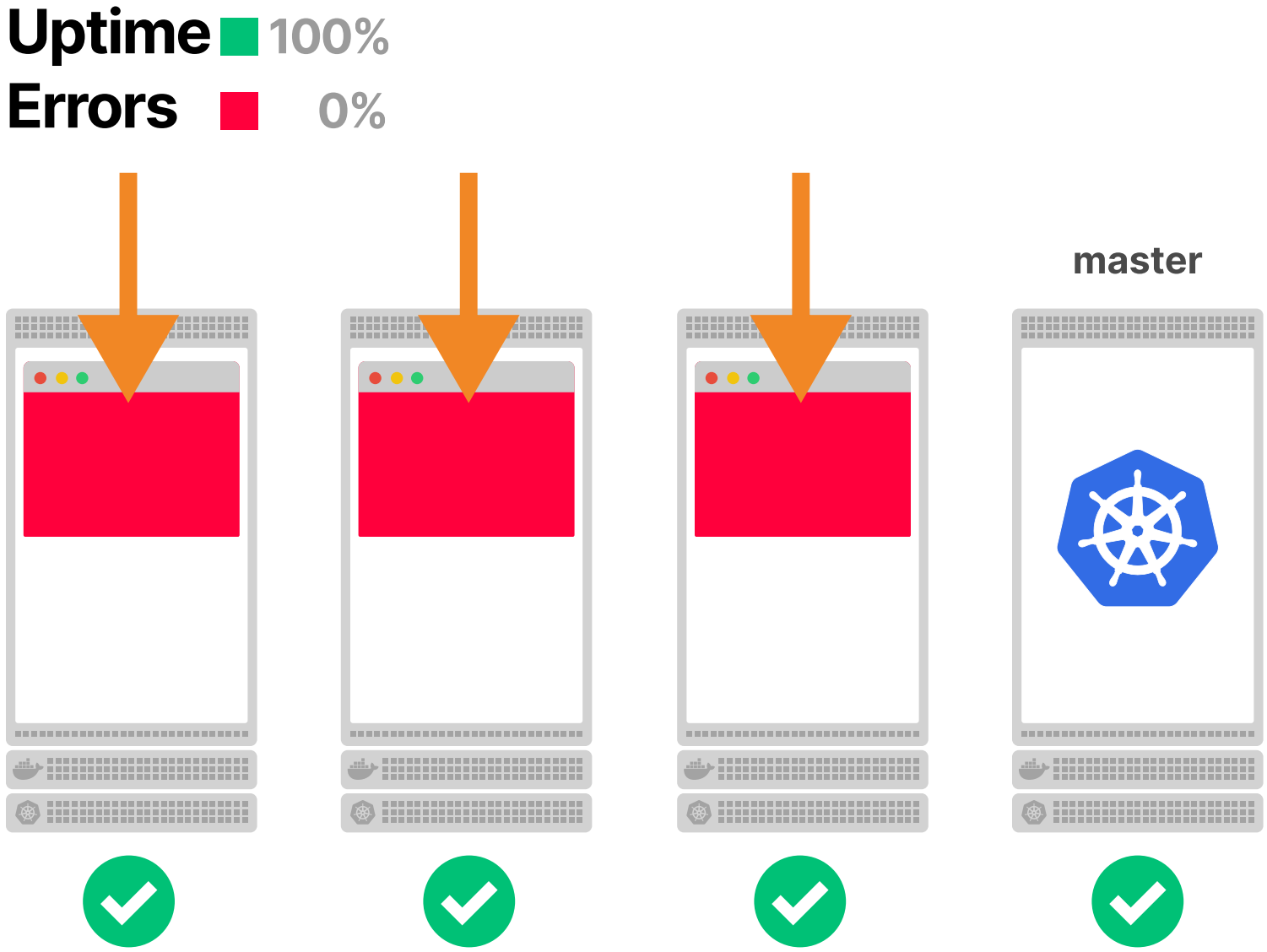
Deploy 3 replicas

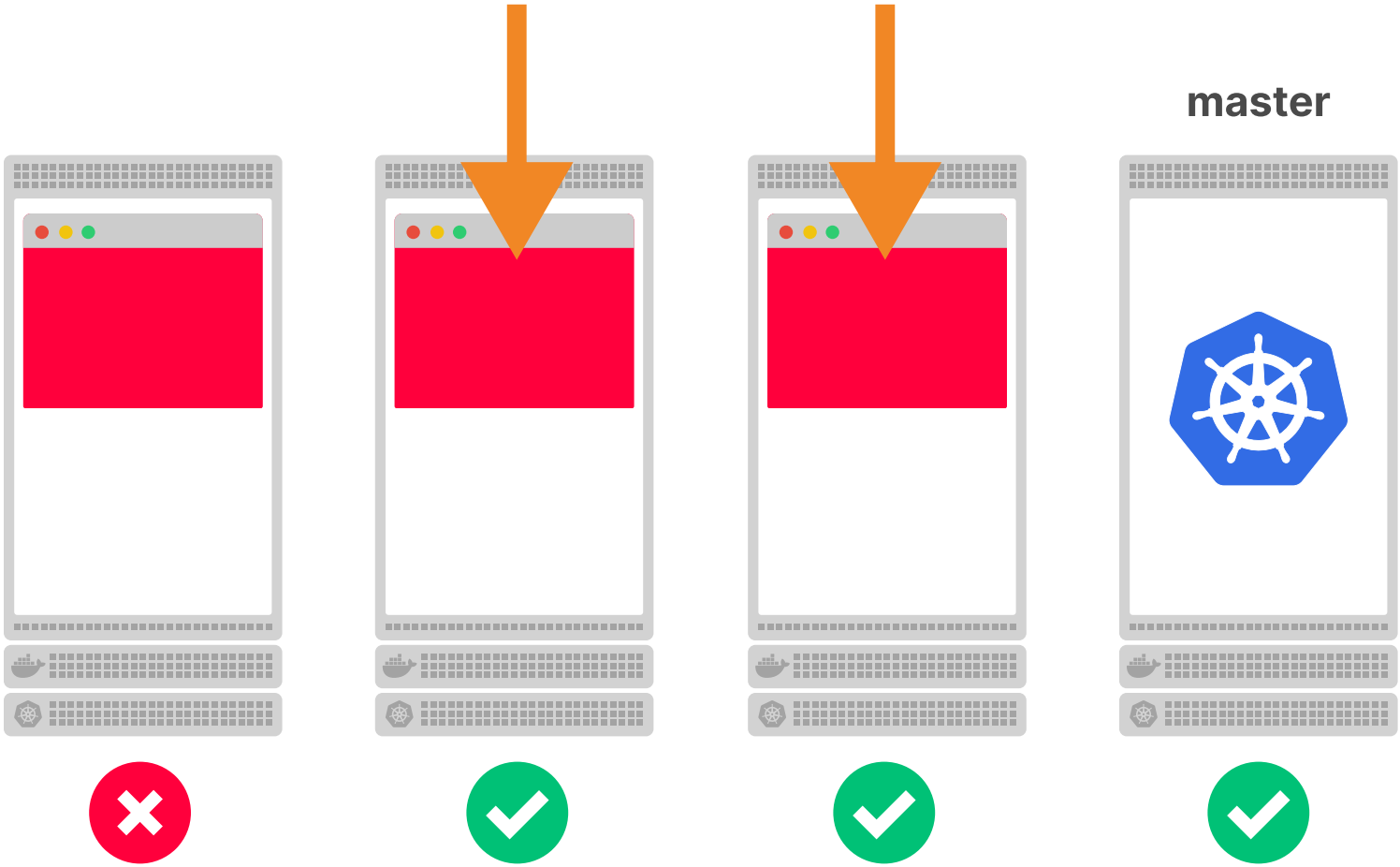
Breaking the cluster

Uptime 100%

Errors 0%

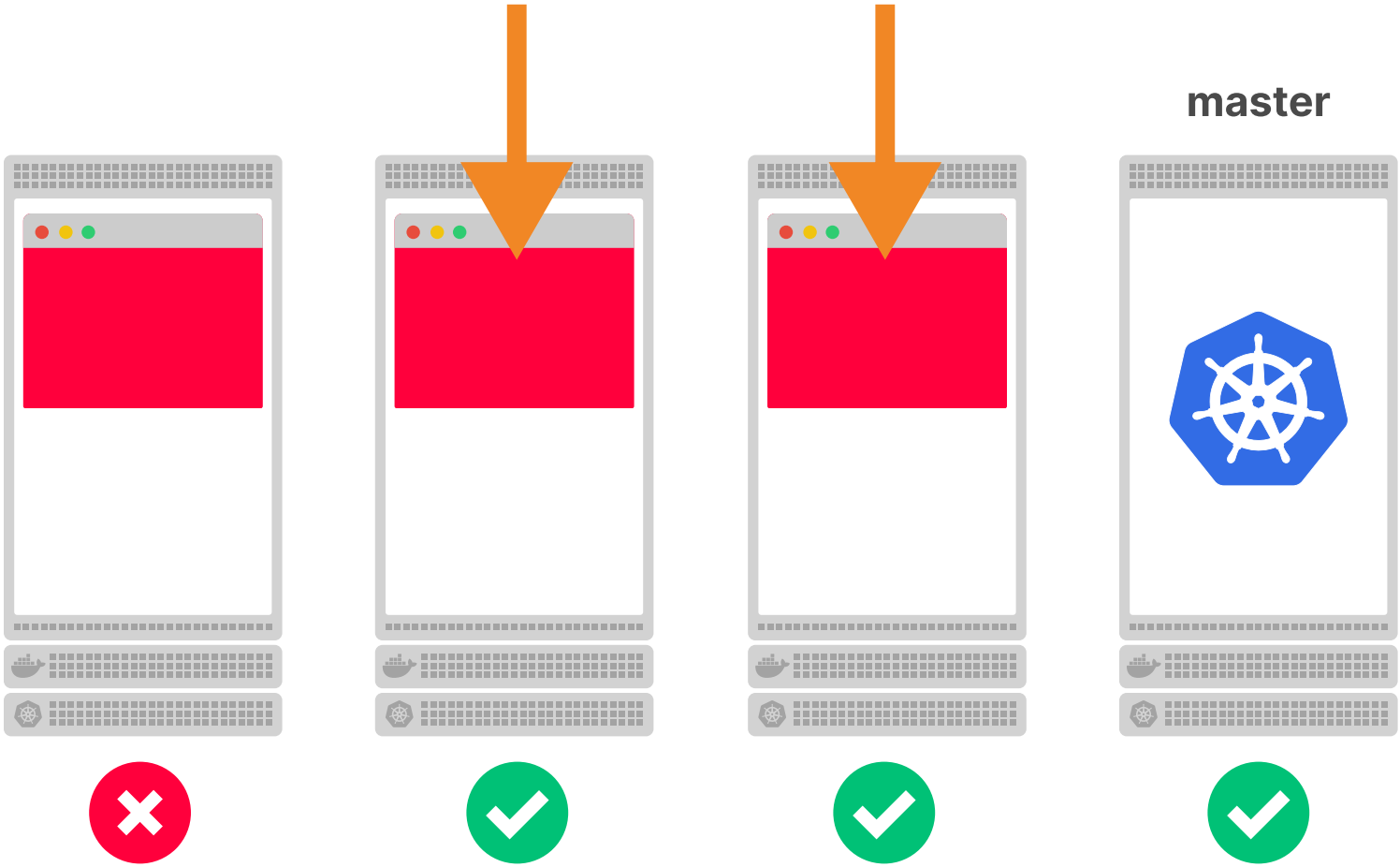


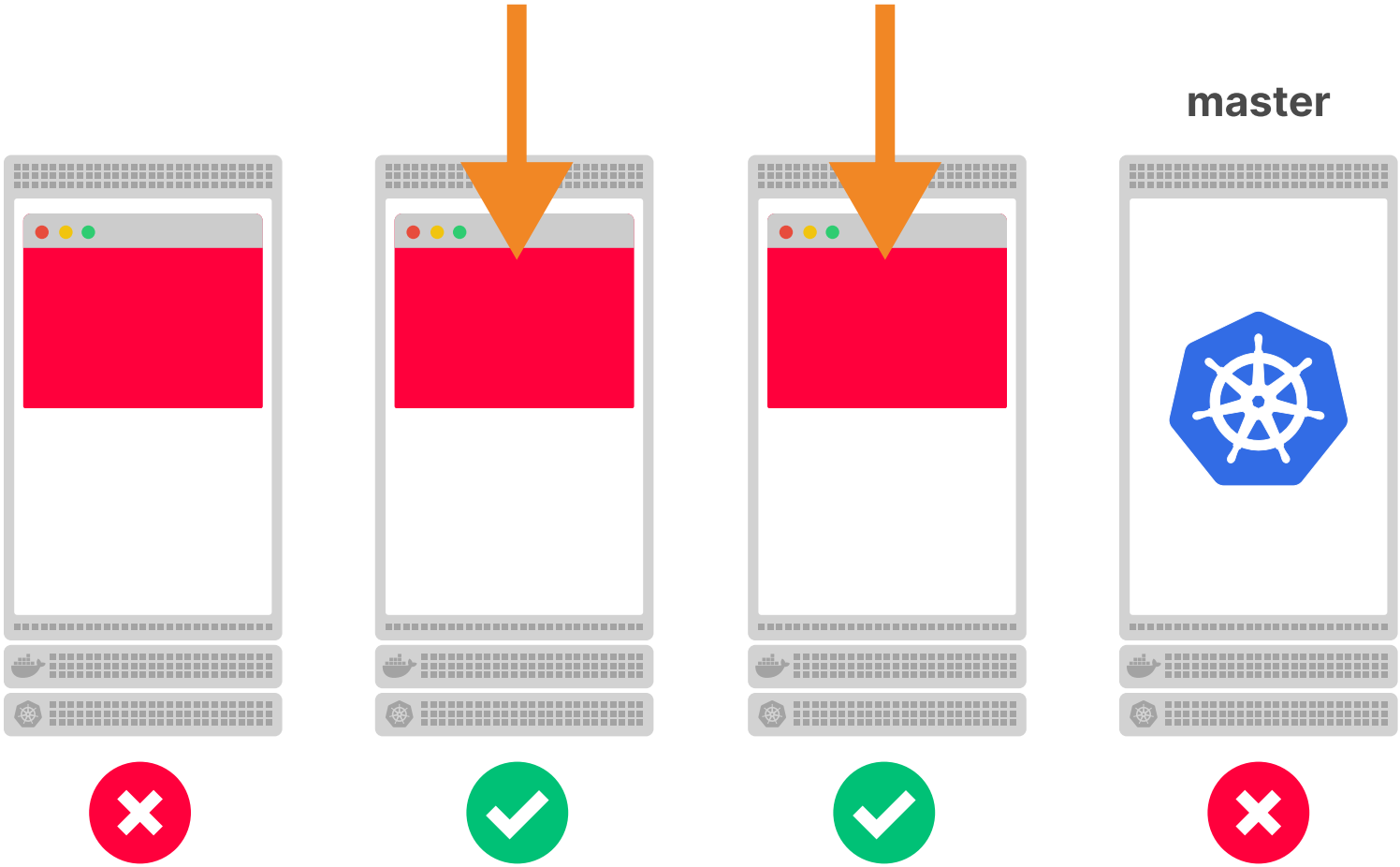

Uptime
100%
Errors
0%
master


master

Uptime 100%

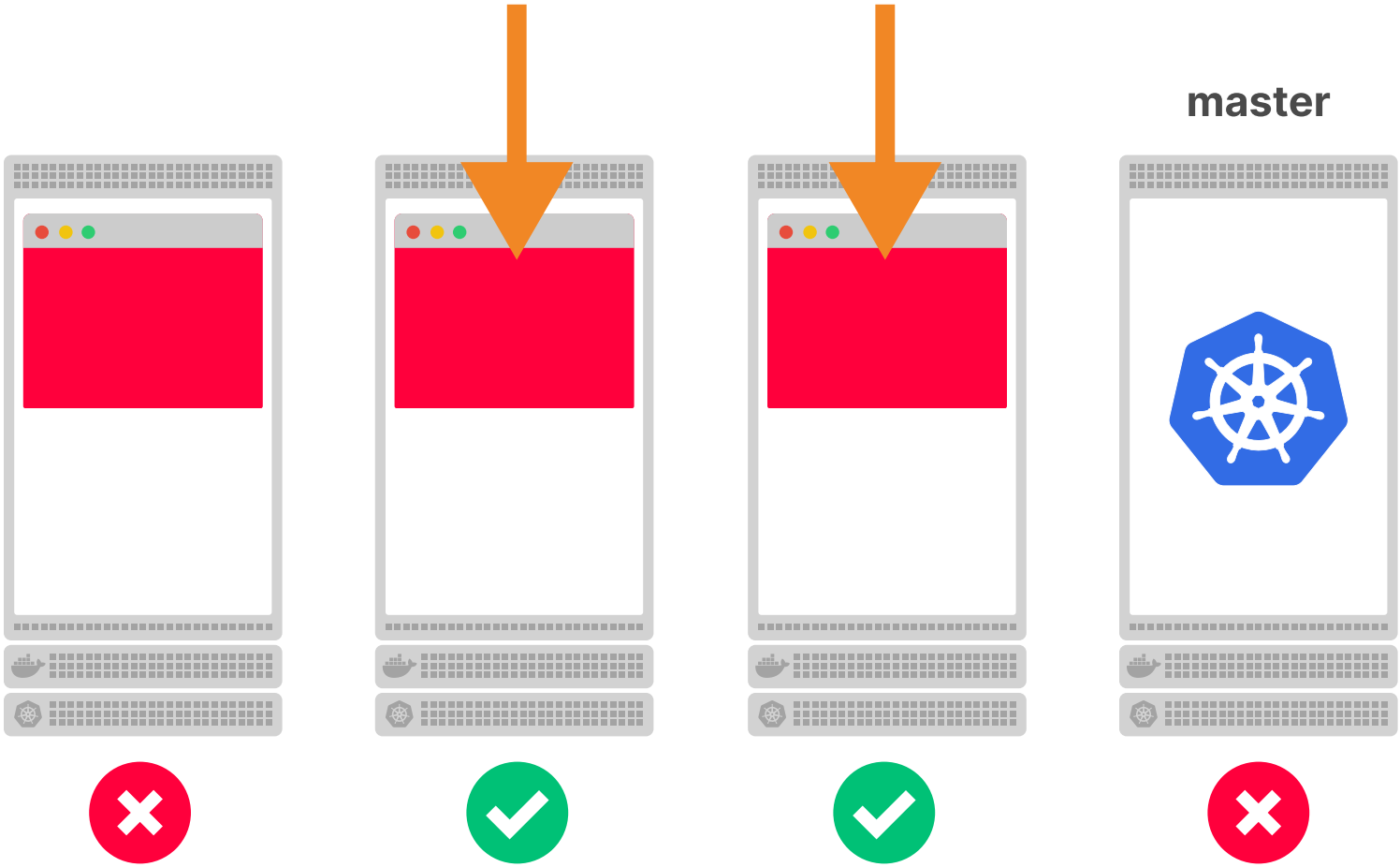
Errors 0%




master

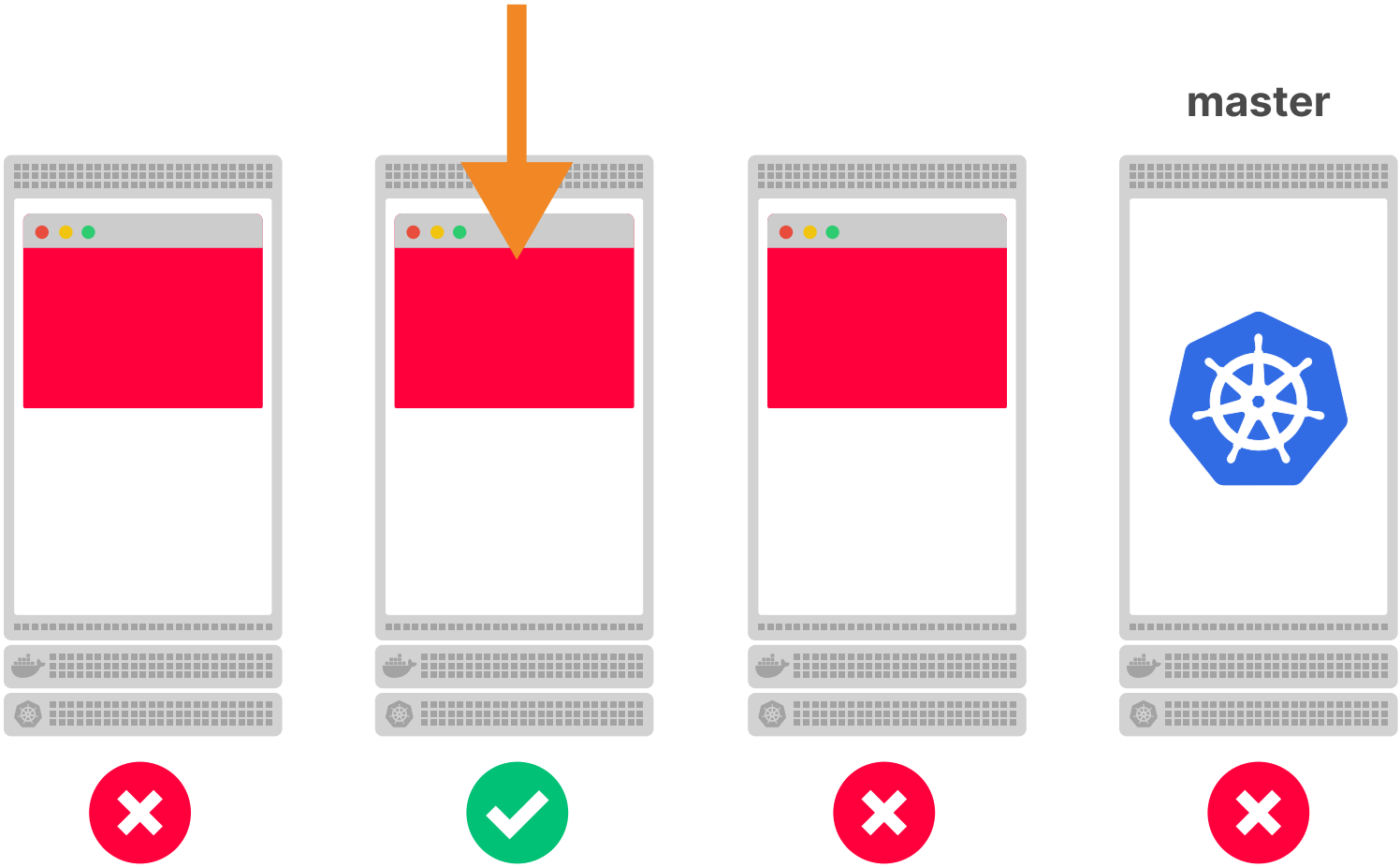
Uptime 100%

Errors 0%



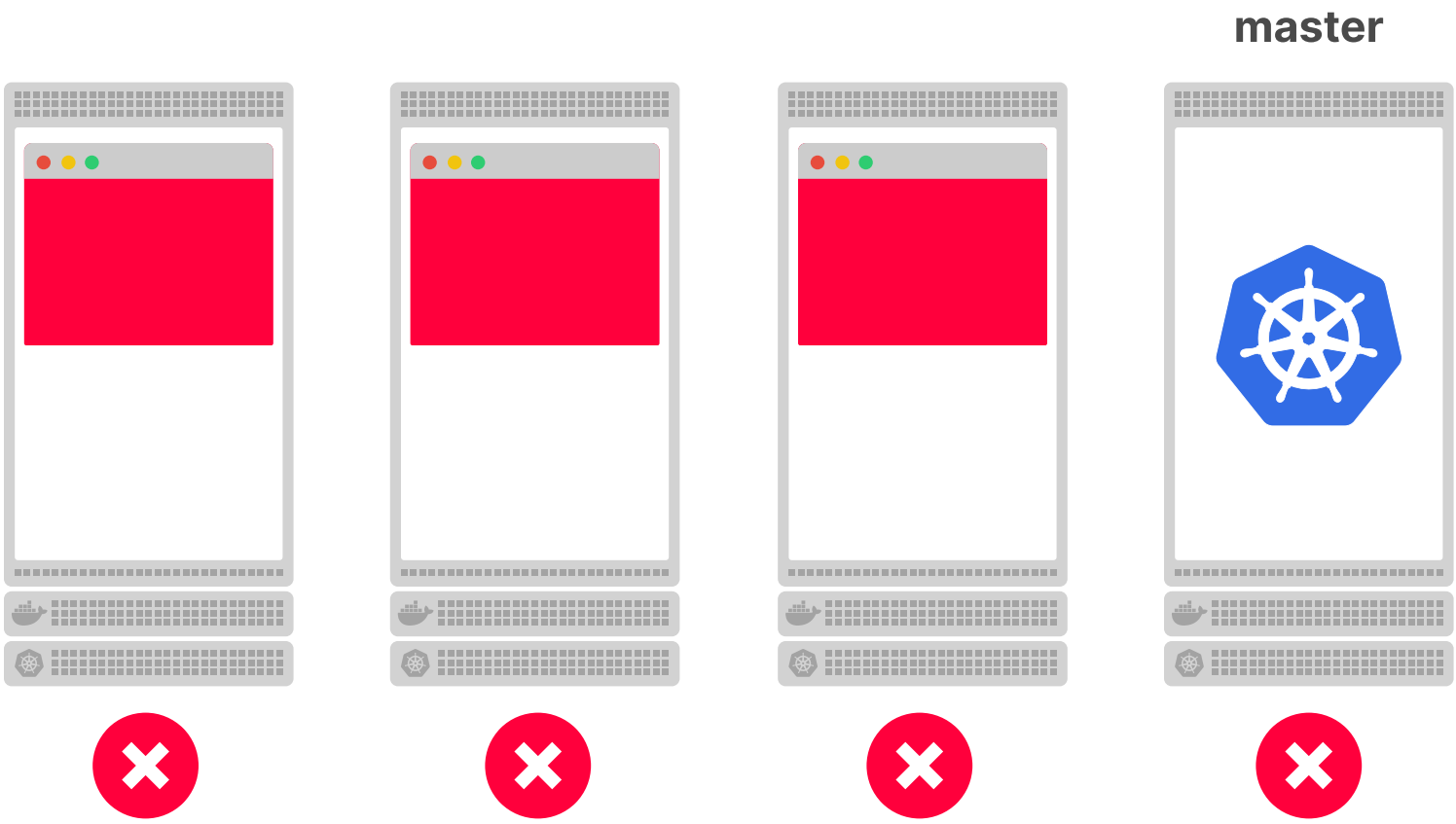
Uptime 100%

Errors 50%



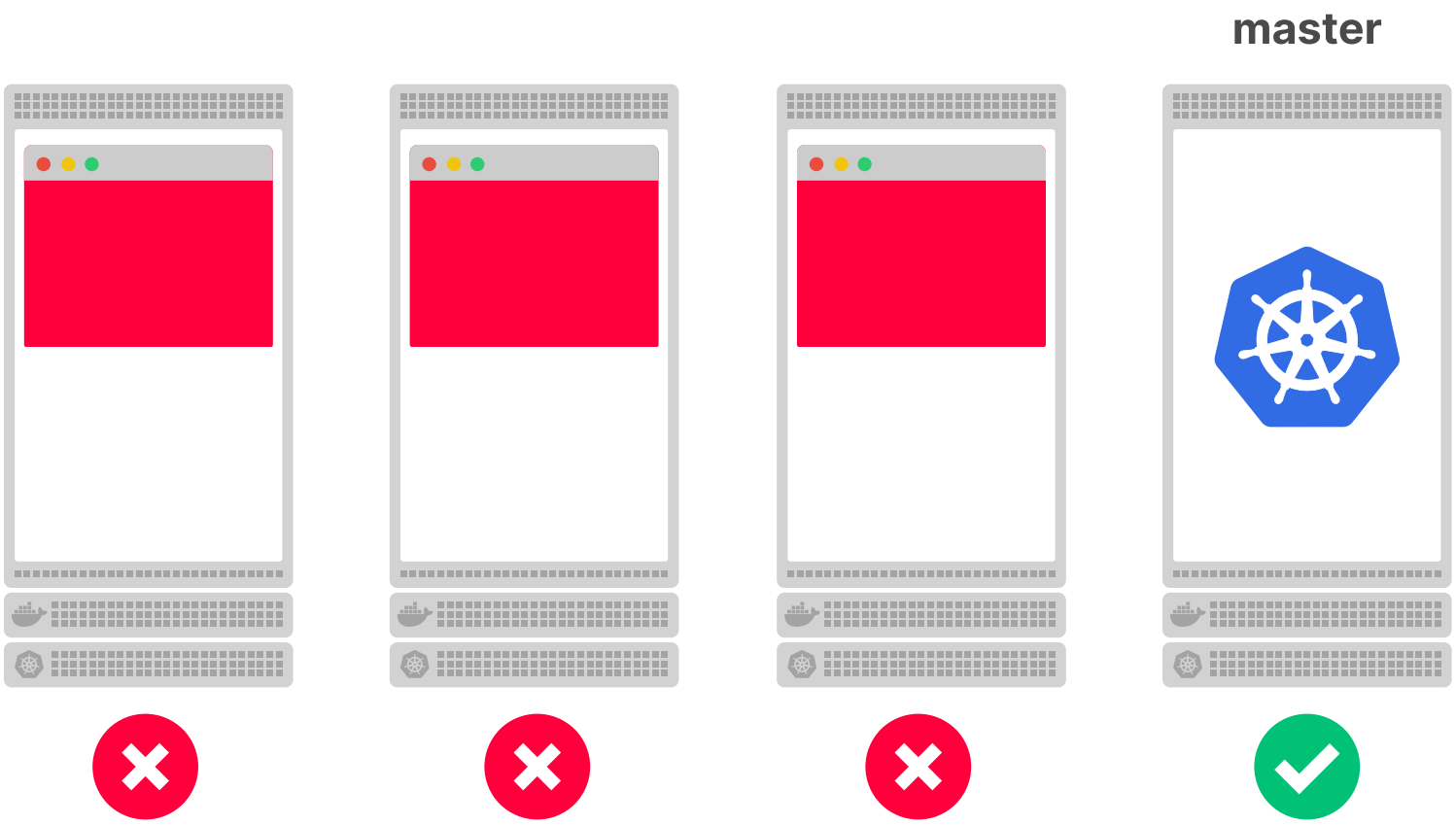
Uptime 0%

Errors 100%



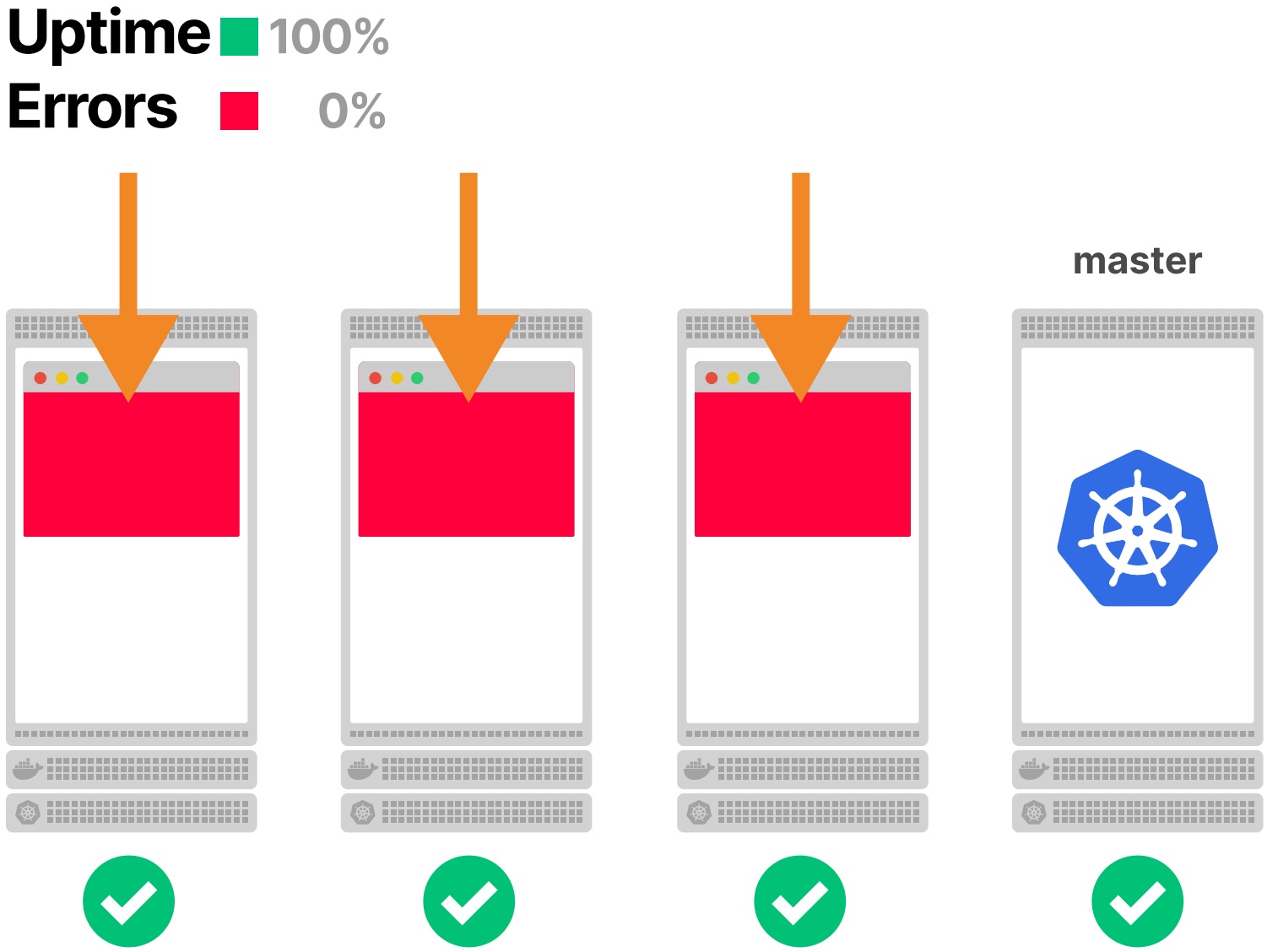
Uptime 0%

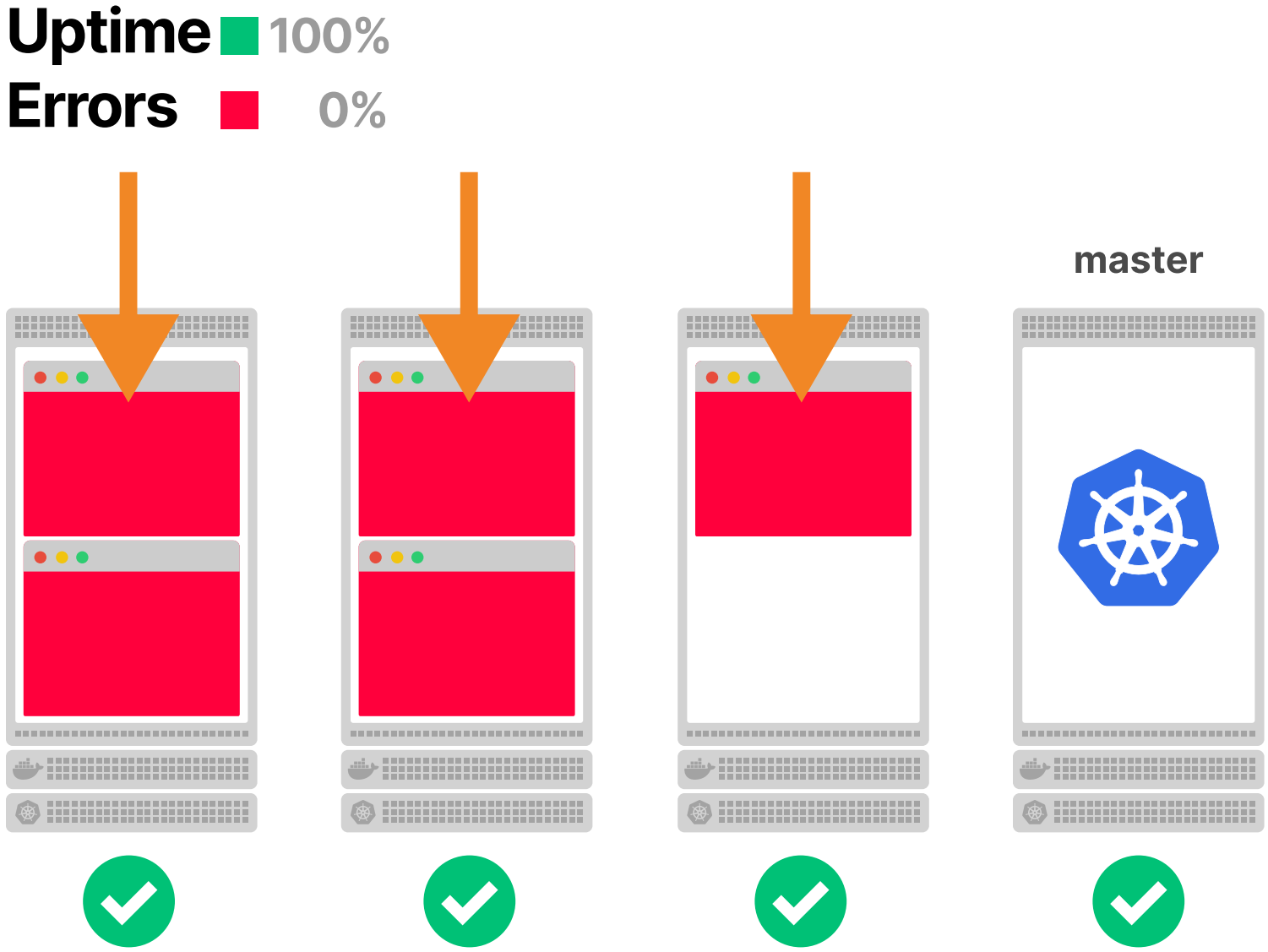
Errors 100%



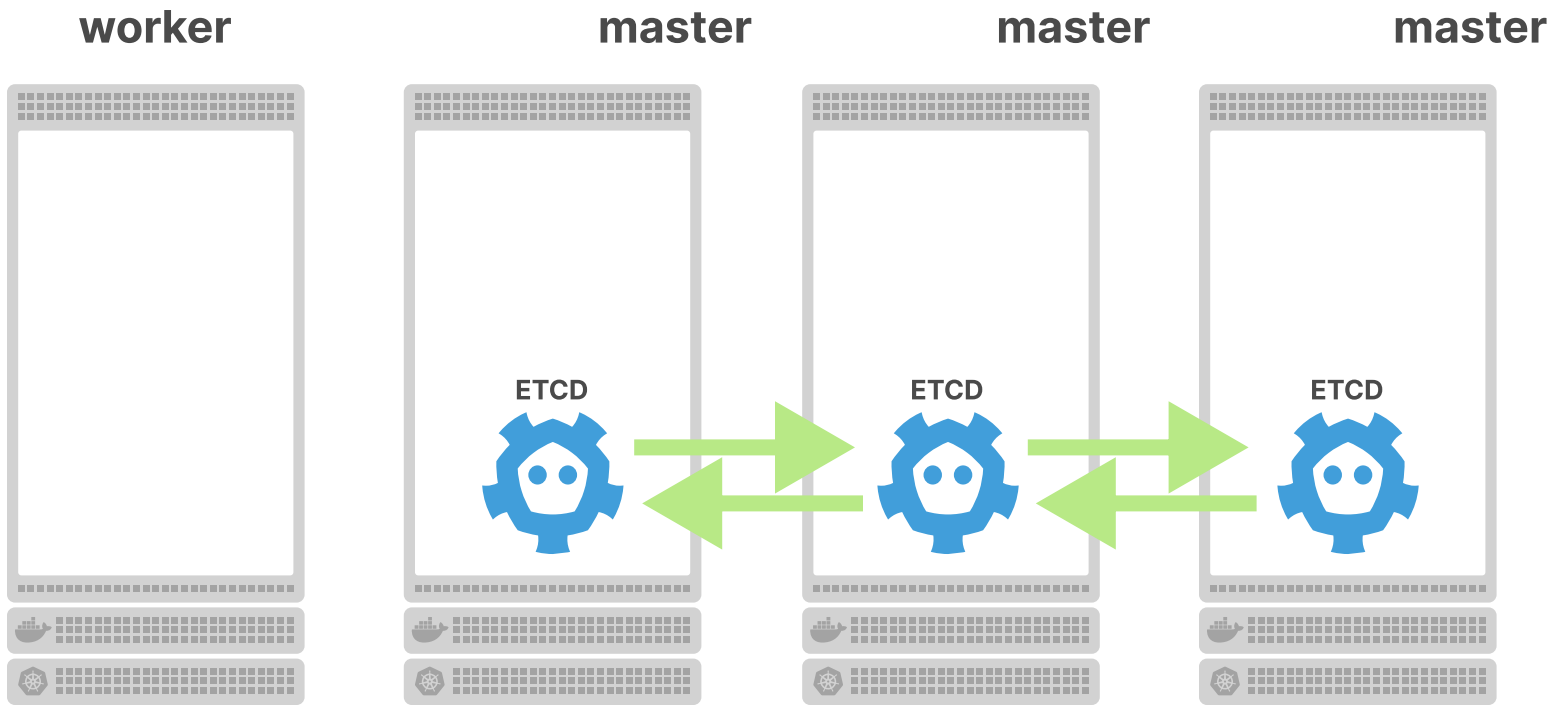
Scale to 5 replicas

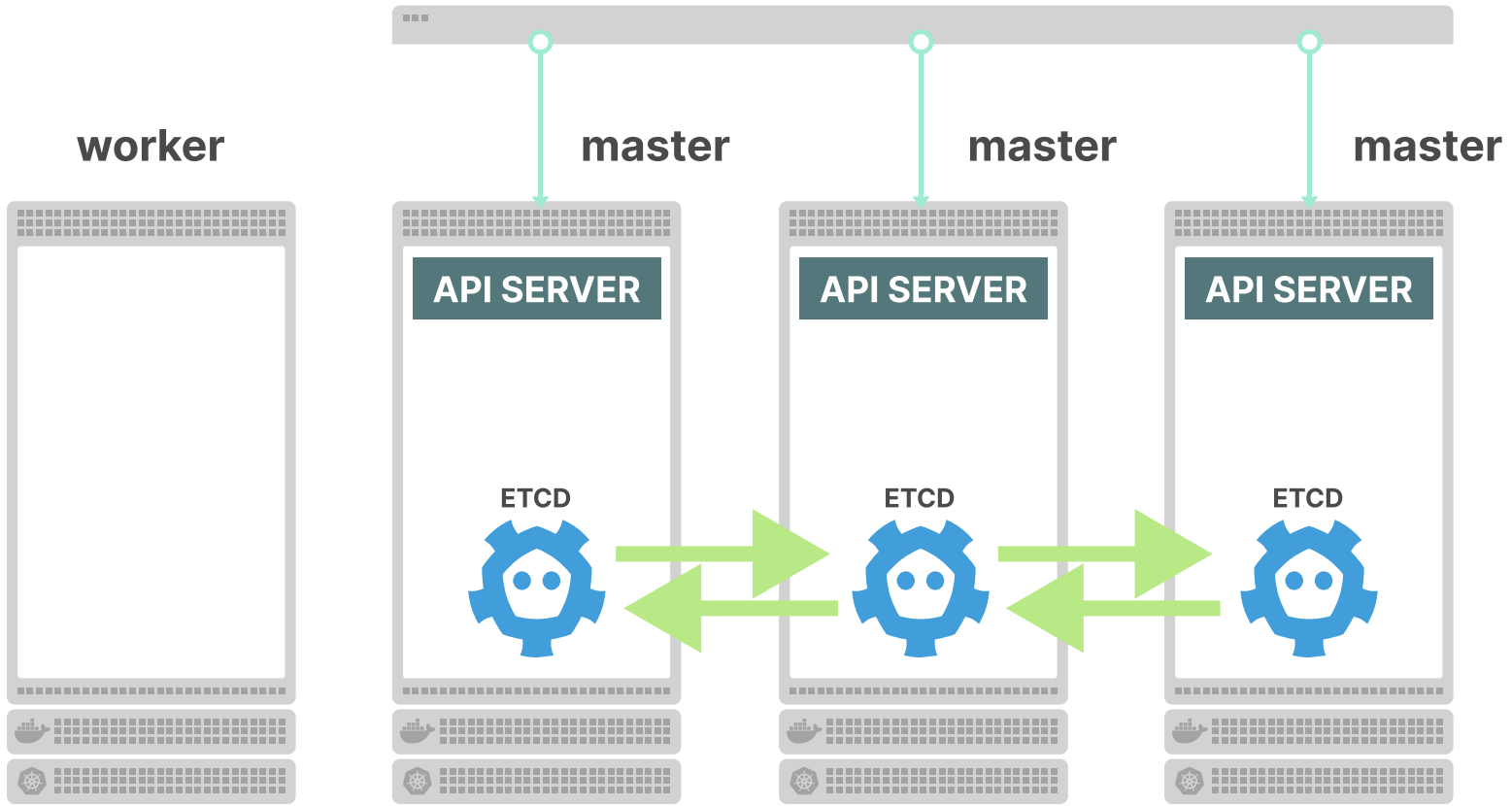
Breaking the cluster

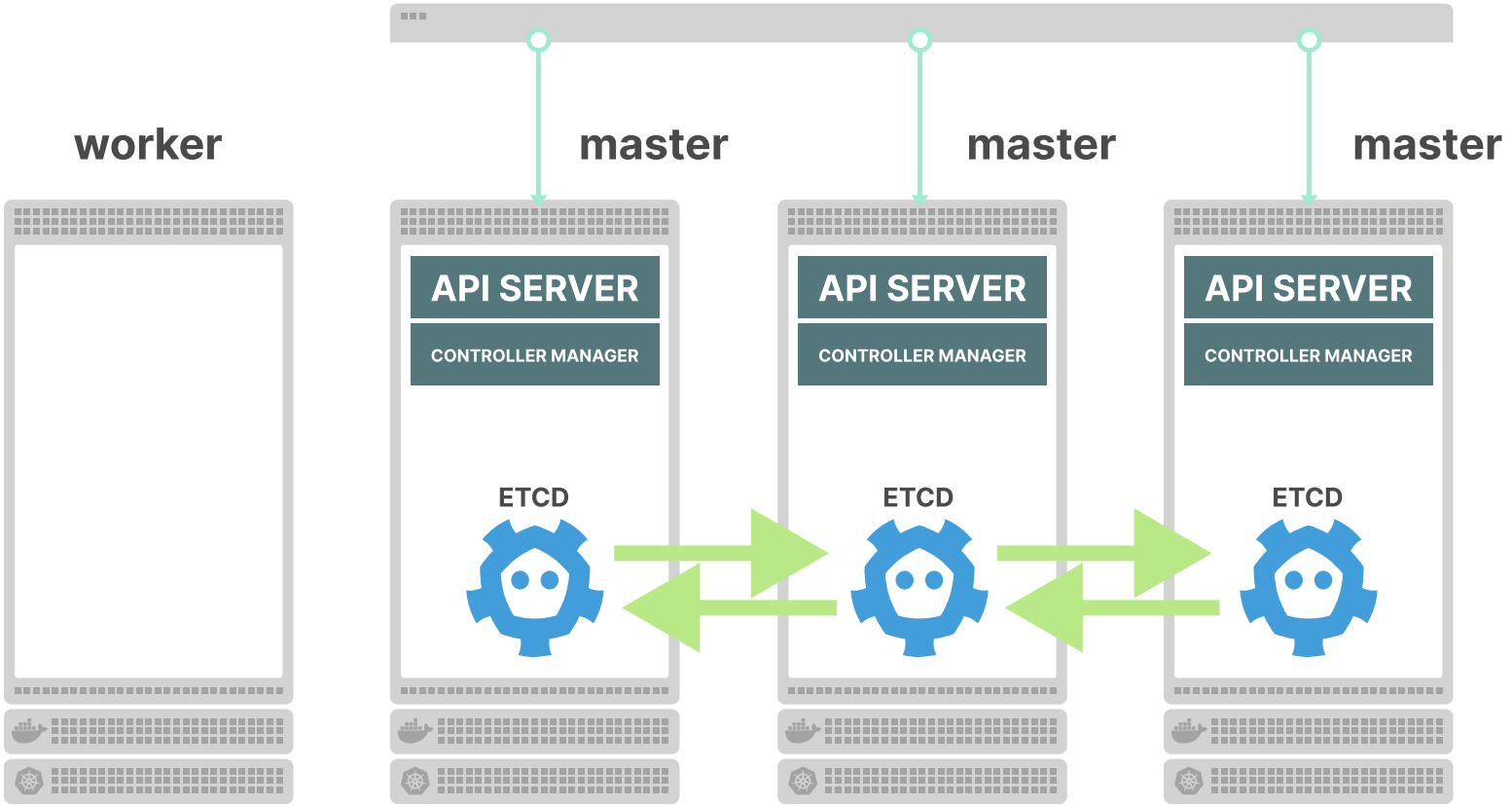

Uptime
100%
Errors
0%
master

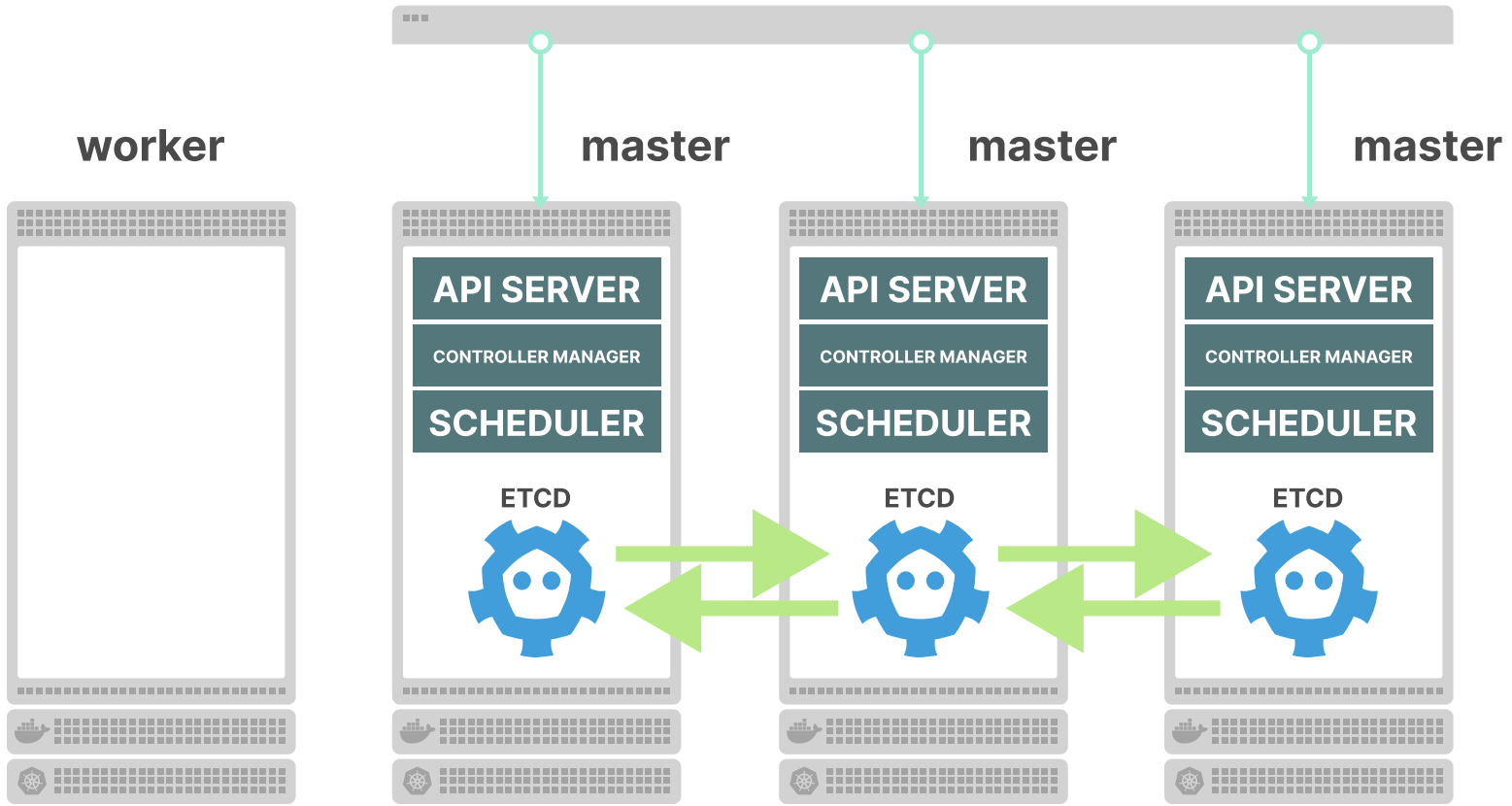

Uptime 100%
Errors 0%
master

Breaking the control plane

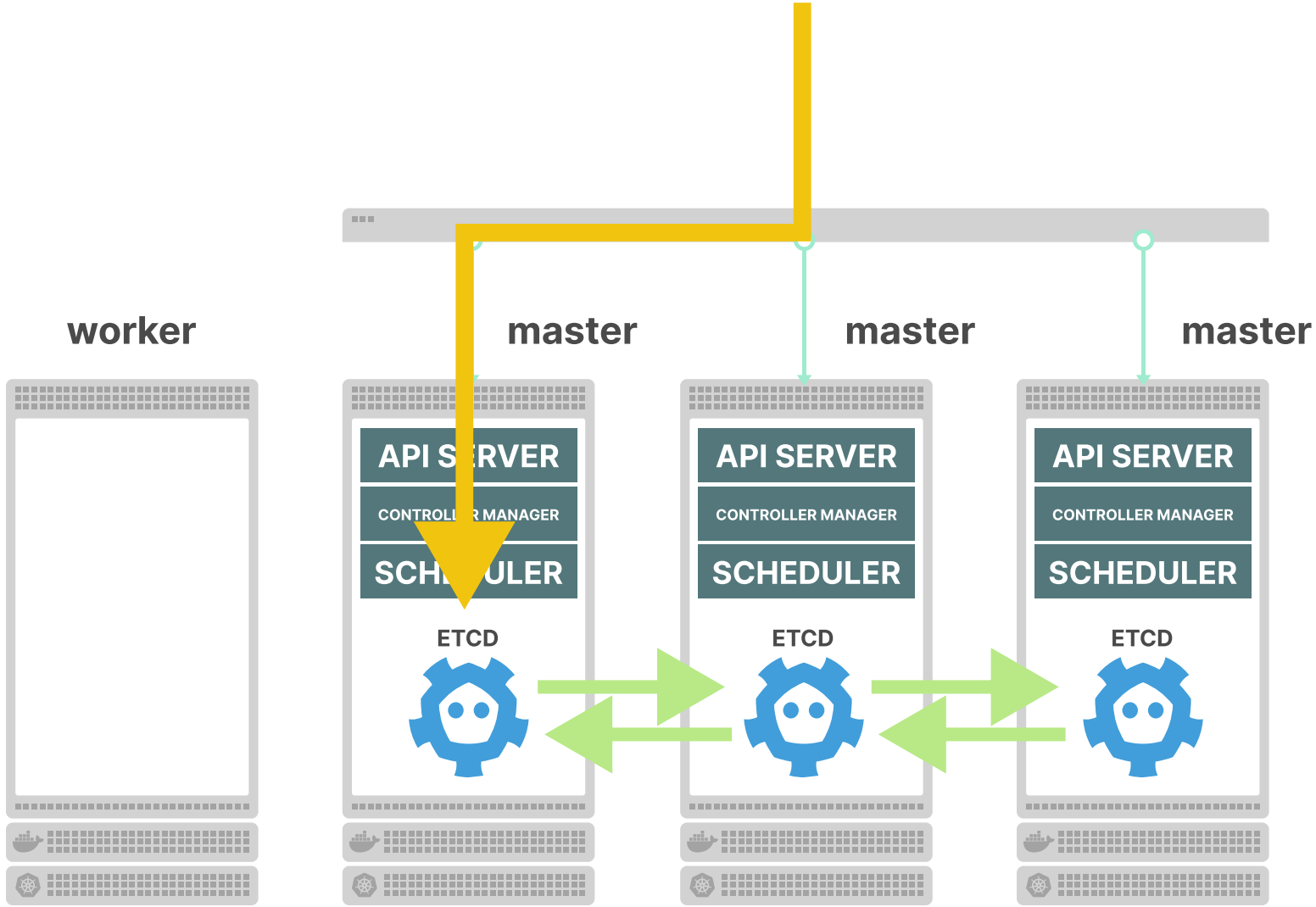

worker
master
master
master
ETCD
ETCD
ETCD


worker
master
master
master
API SERVER
API SERVER
API SERVER
ETCD
ETCD
ETCD

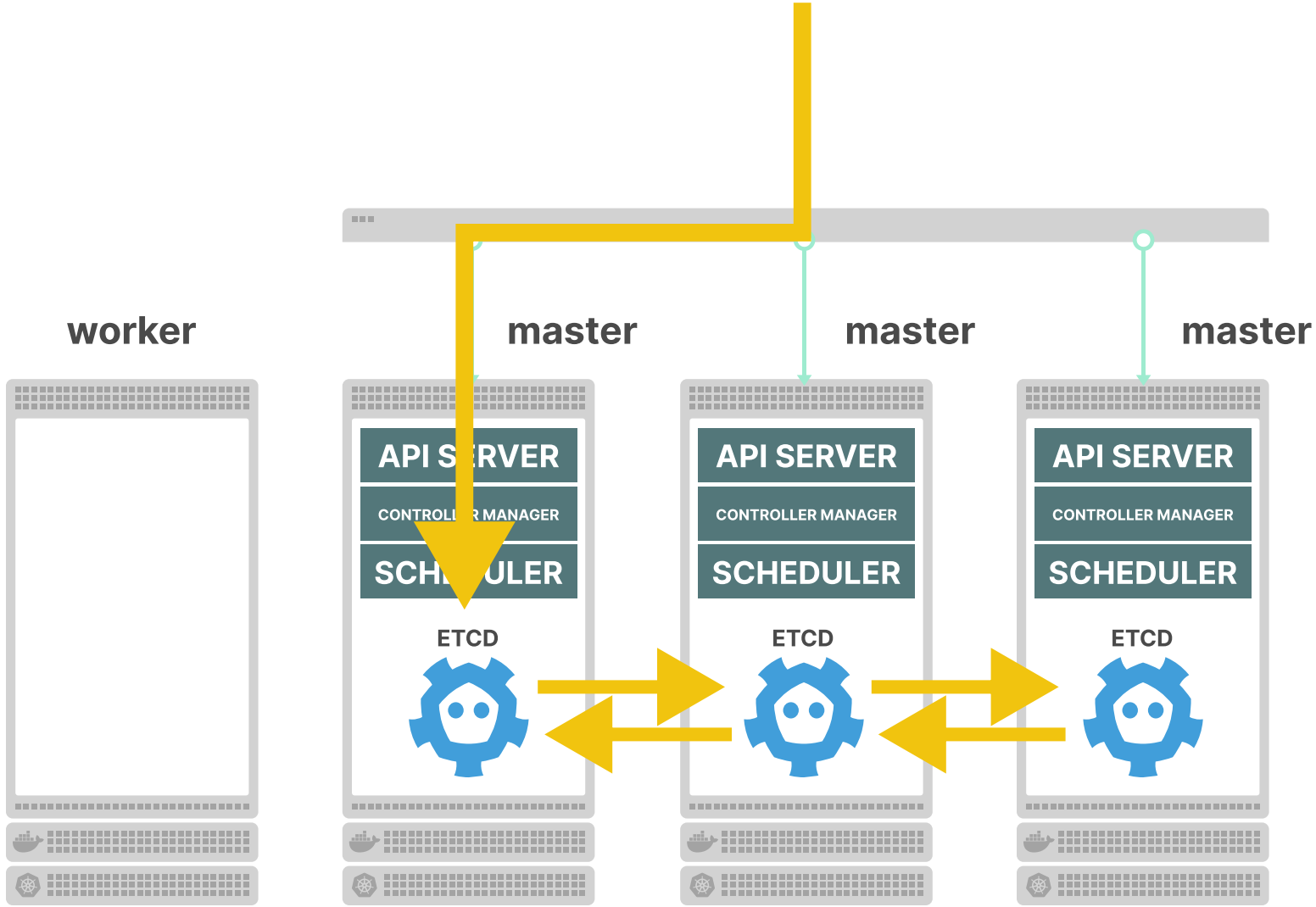

worker
master
master
master
API SERVER
CONTROLLER MANAGER
API SERVER
CONTROLLER MANAGER
API SERVER
CONTROLLER MANAGER
ETCD
ETCD
ETCD


worker
master
master
master
API SERVER
CONTROLLER MANAGER
SCHEDULER
ETCD
API SERVER
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API SERVER
CONTROLLER MANAGER
SCHEDULER
ETCD

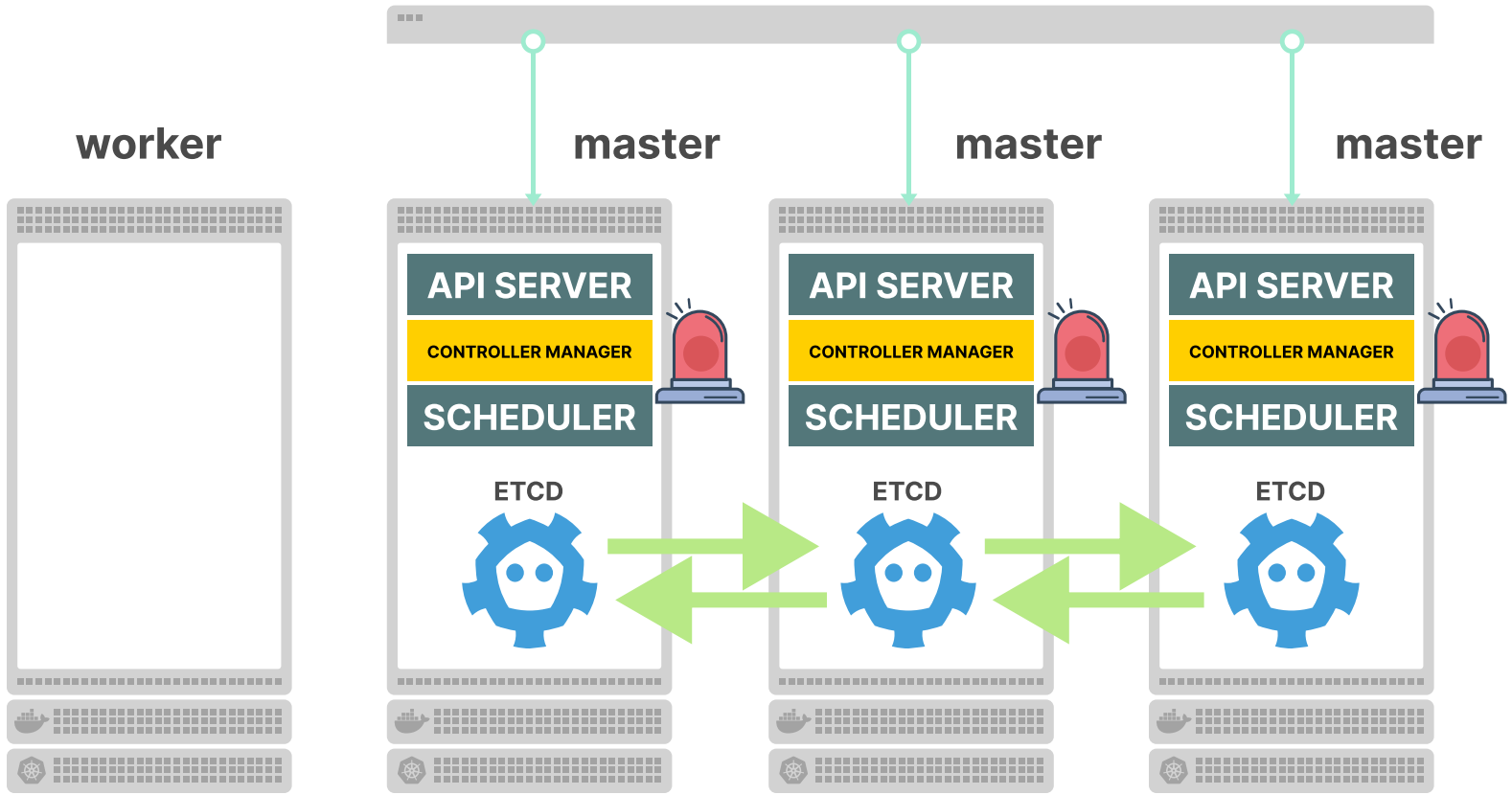
kubectl apply -f deployment.yaml



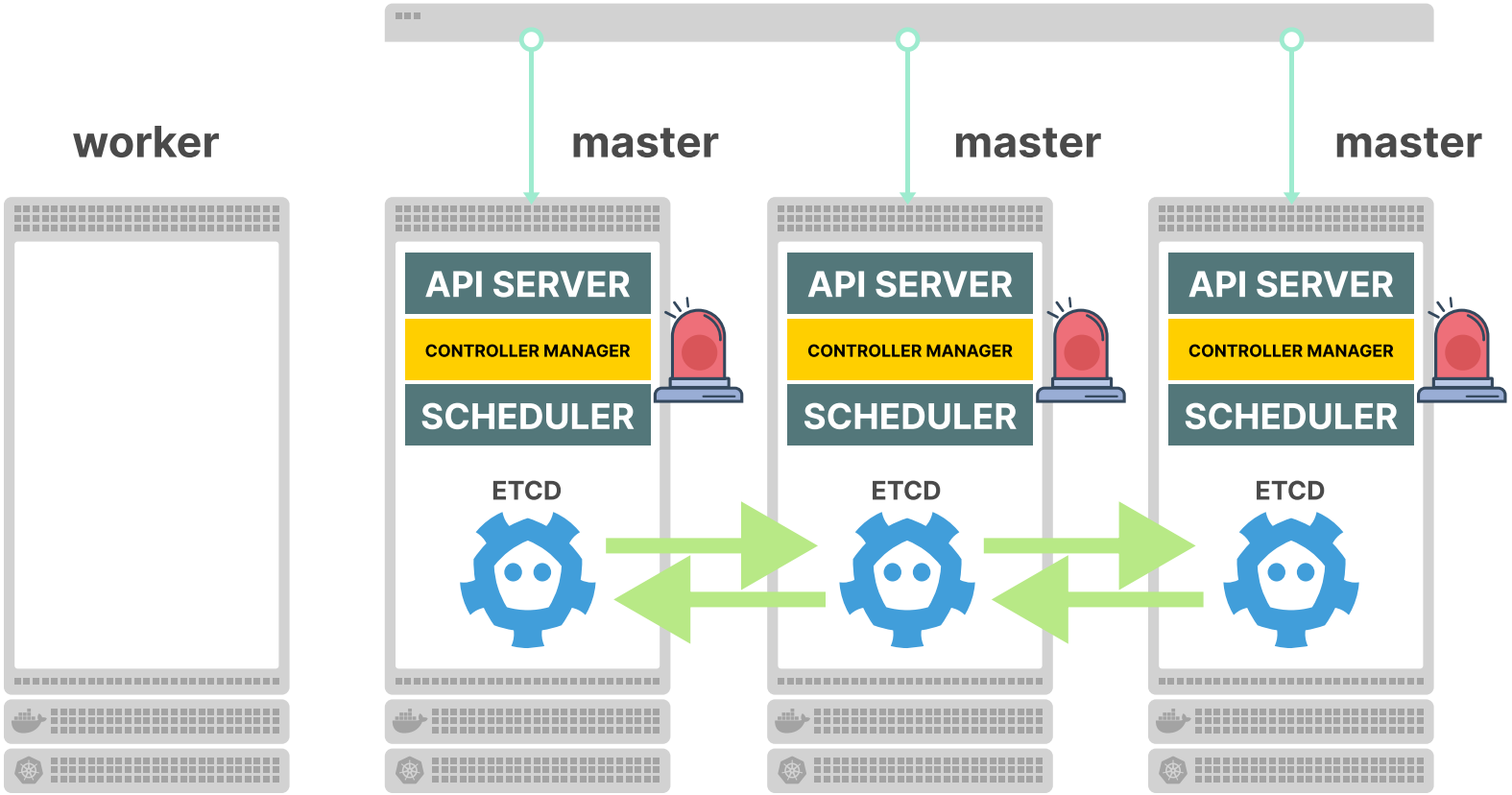
kubectl apply -f deployment.yaml



kubectl apply -f deployment.yaml

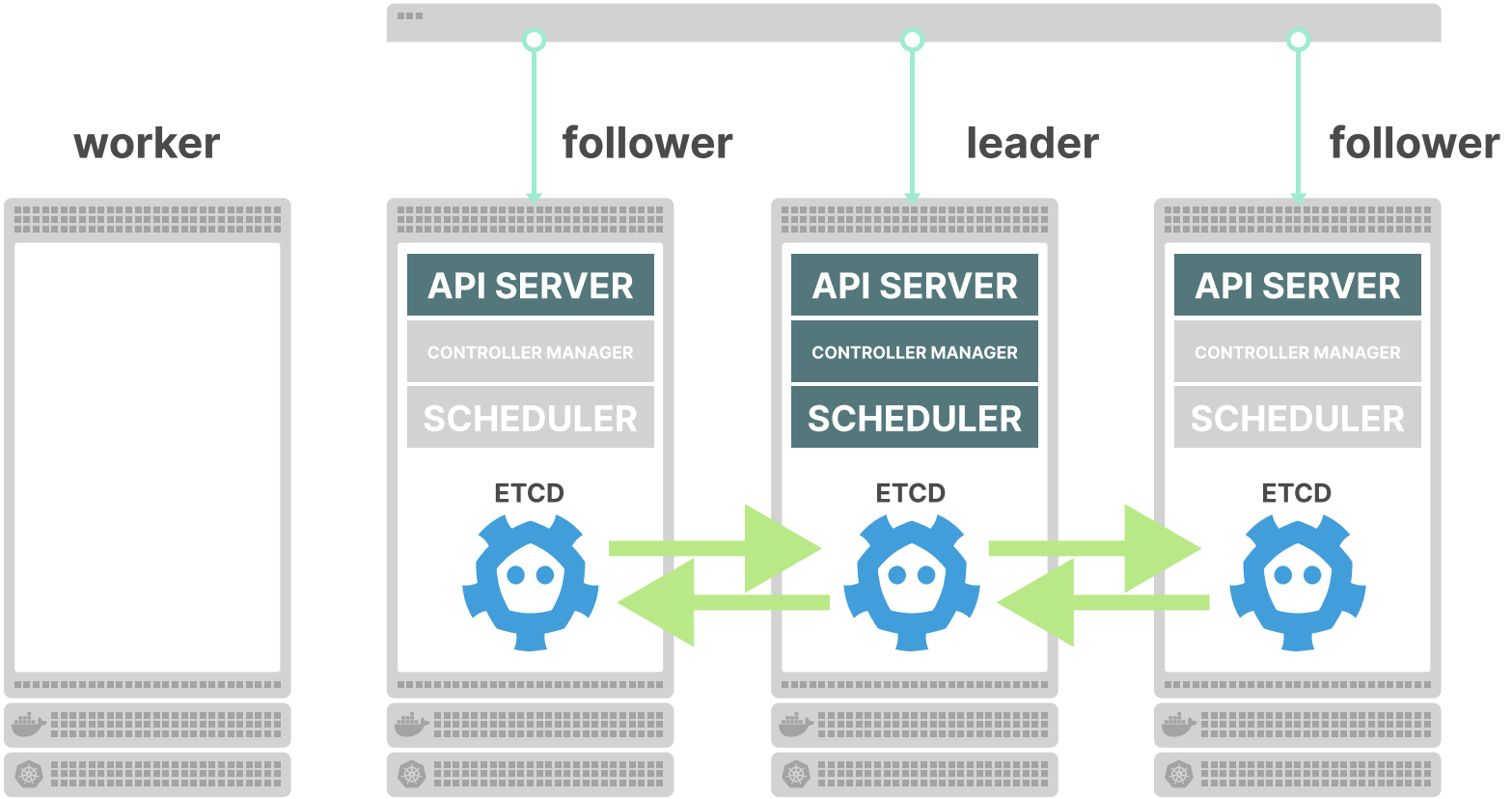


kubectl apply -f deployment.yaml

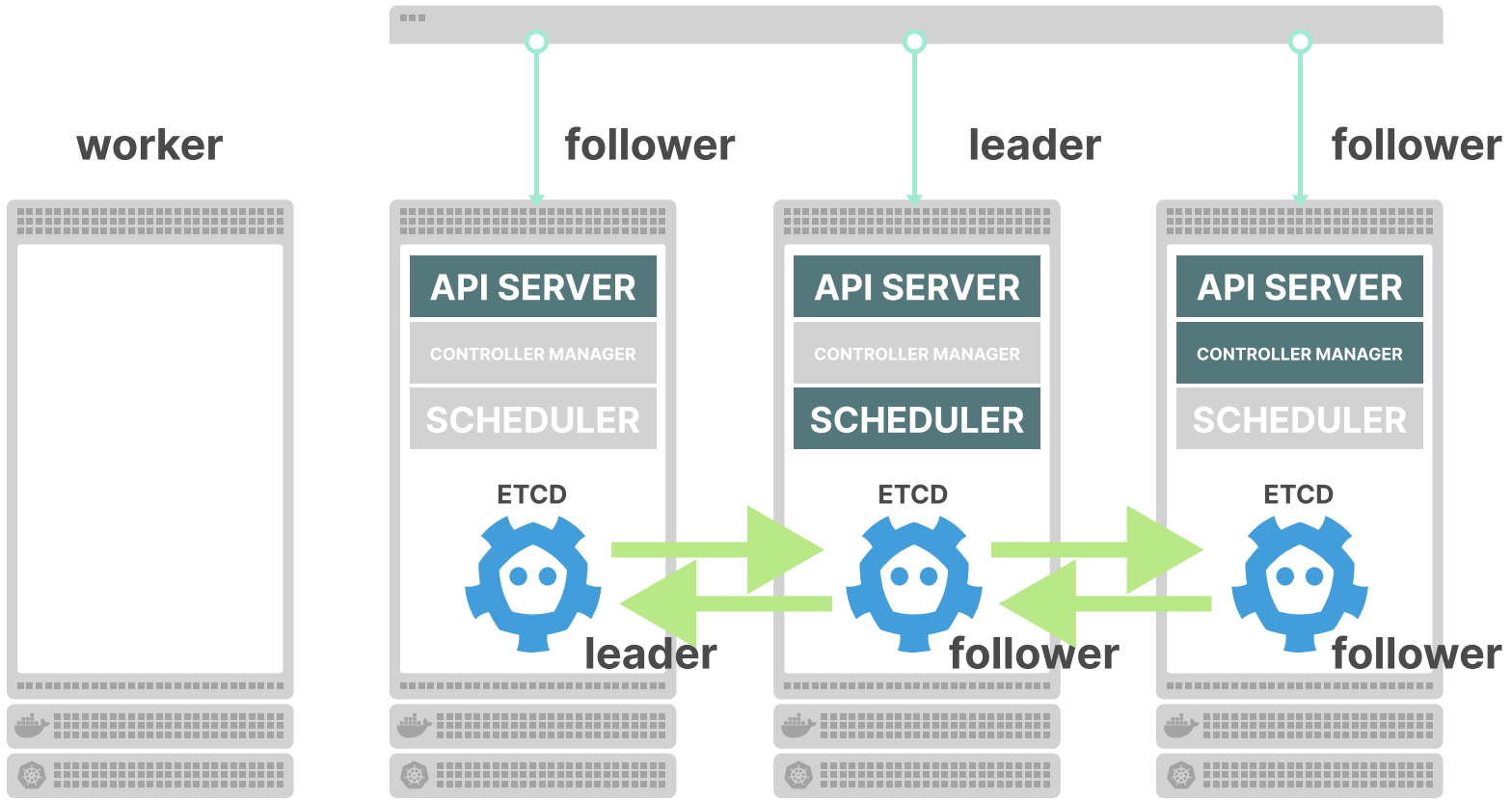


3 Pods 3 Pods 3 Pods

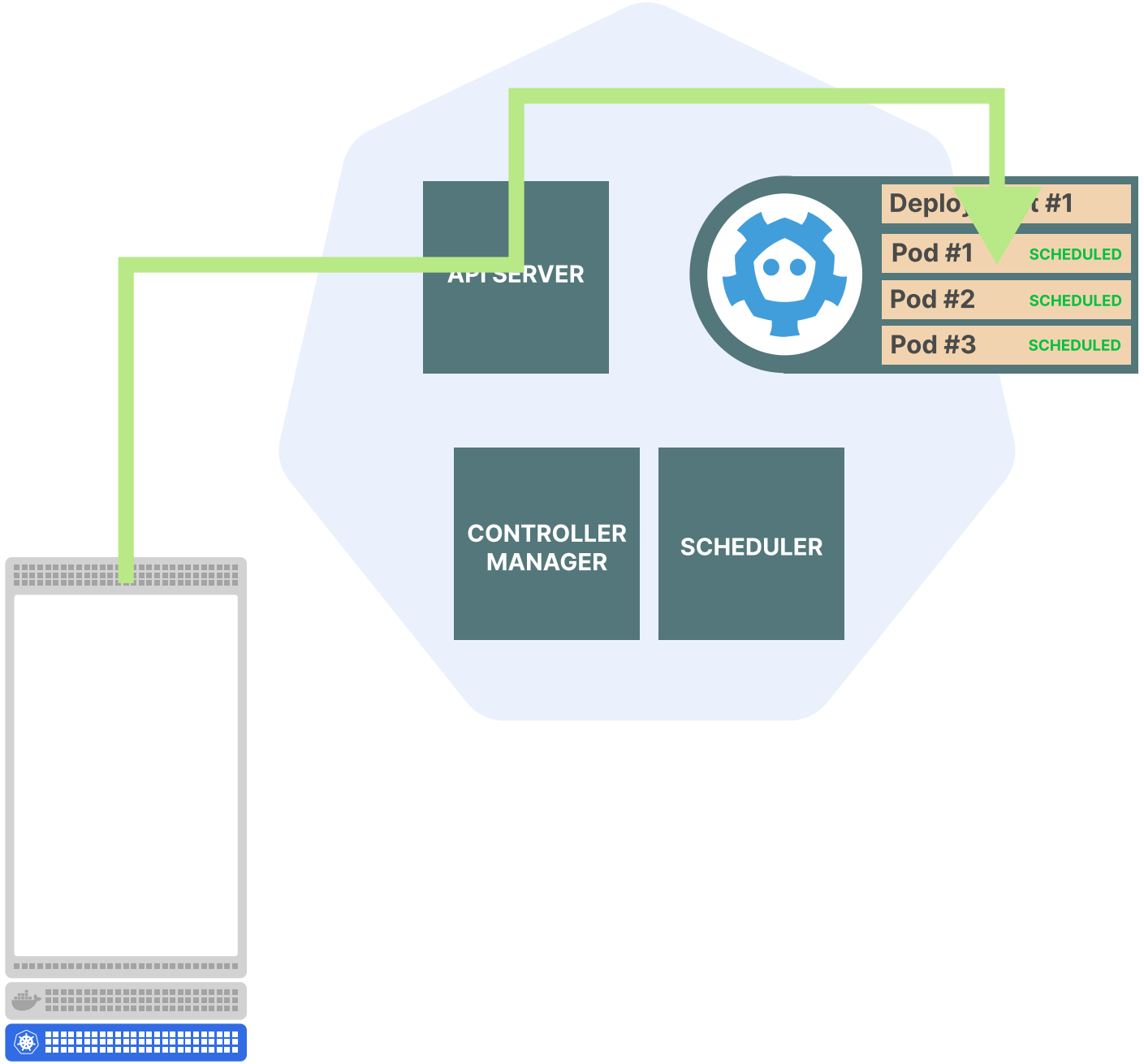
kubectl apply -f deployment.yaml



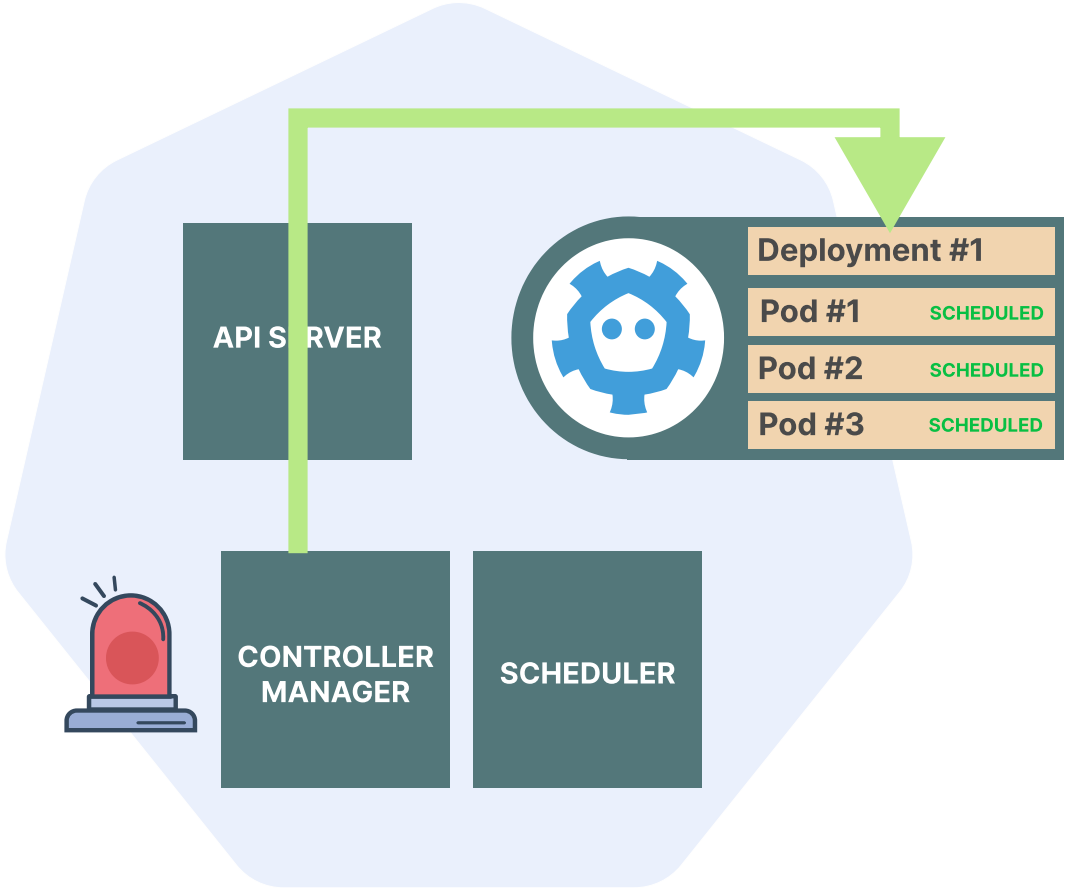
kubectl apply -f deployment.yaml

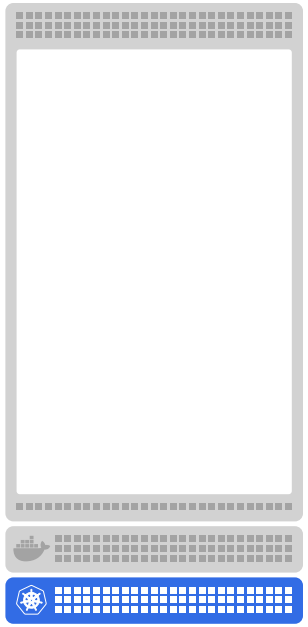


API design


APISERVER
Deplo #1
Pod #1
SCHEDULED
Pod #2
SCHEDULED
Pod #3
SCHEDULED
CONTROLLER MANAGER
SCHEDULER

kubelet


API SERVER
Deployment #1
Pod #1 SCHEDULED
Pod #2 SCHEDULED
Pod #3 SCHEDULED
CONTROLLER MANAGER
SCHEDULER



kubelet

API Server

Authentication

Authorization

Policies

API Server

Authentication

Authorization

Policies

API Server

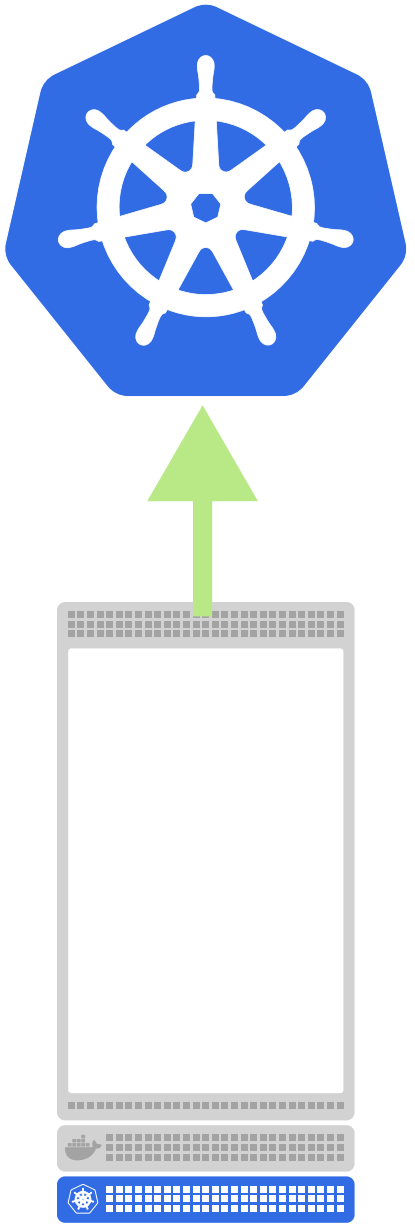
Authentication

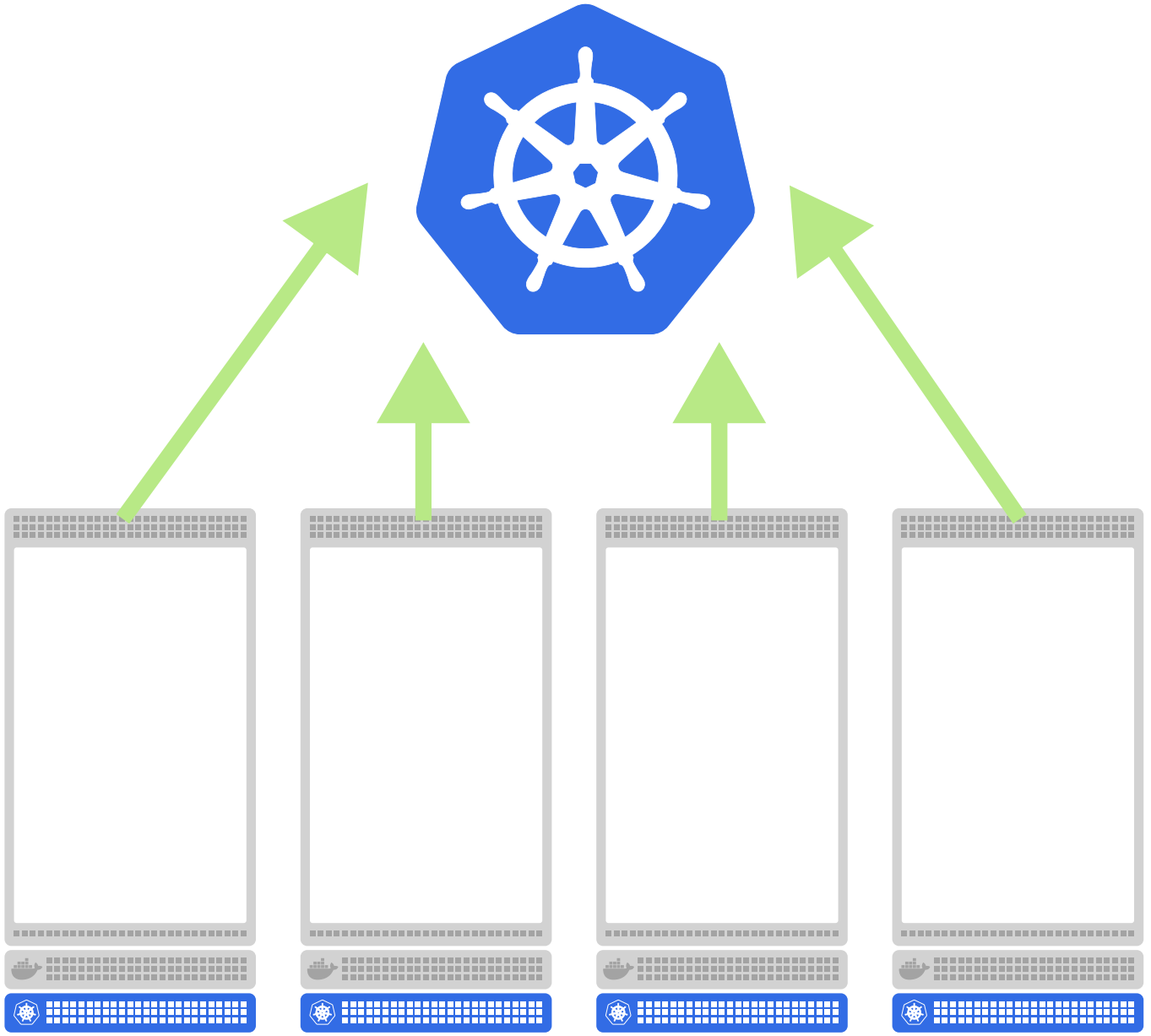
Authorization

Policies

API scalability

Is there anything for me?





Push vs Pull

Resilience

Bandwidth

API scalability

Push vs Pull

Resilience

Bandwidth

API scalability

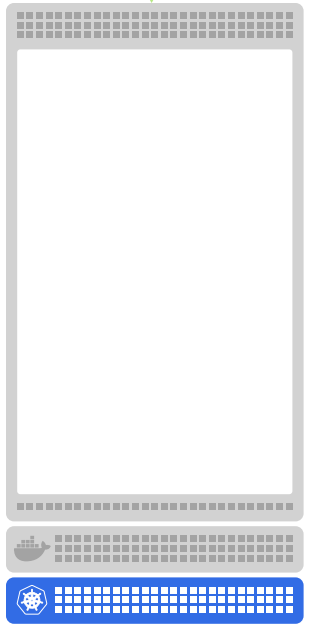
Push vs Pull

Resilience

Bandwidth



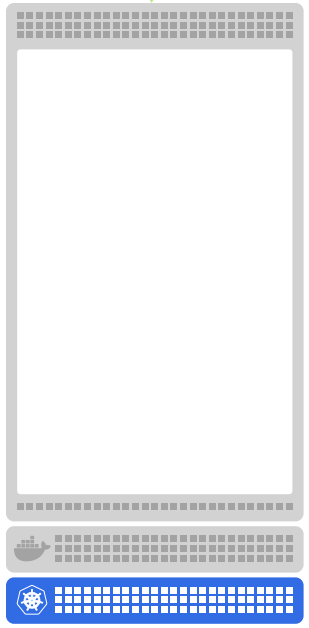




kubectl logs <pod-name>







kubectl logs <pod-name>

kubectl exec <pod-name>

Recap

Architecture

Recap — architecture

1. HA control plane
2. Failover for nodes & control plane
3. Control plane leader election
4. API server design
5. API scalability

Recap — architecture

1. HA control plane

2. Failover for nodes & control plane

3. Control plane leader election

4. API server design

5. API scalability

Recap — architecture

1. HA control plane

2. Failover for nodes & control plane

3. Control plane leader election

4. API server design

5. API scalability

Recap — architecture

1. HA control plane

2. Failover for nodes & control plane

3. Control plane leader election

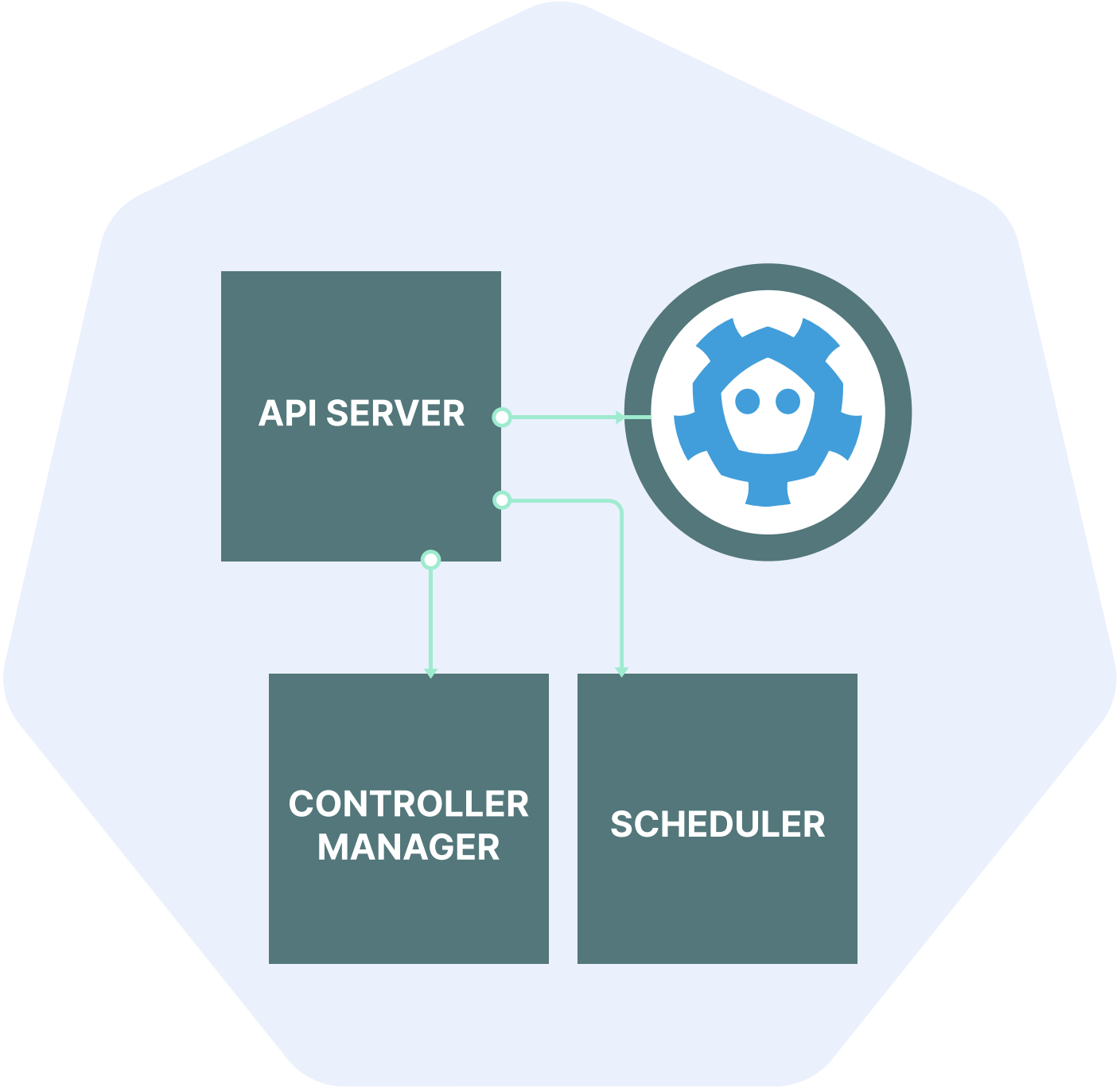
4. API server design

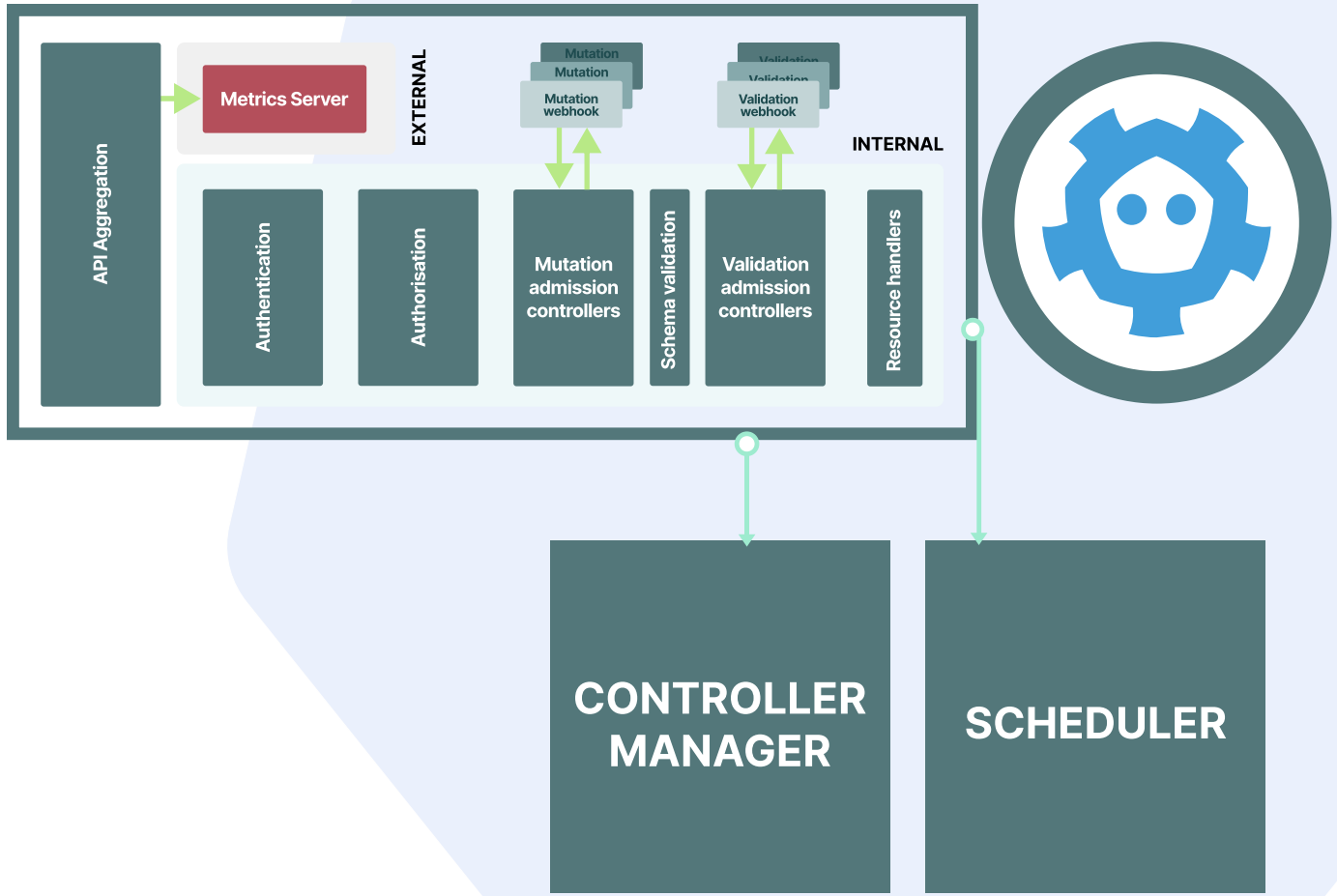
5. API scalability

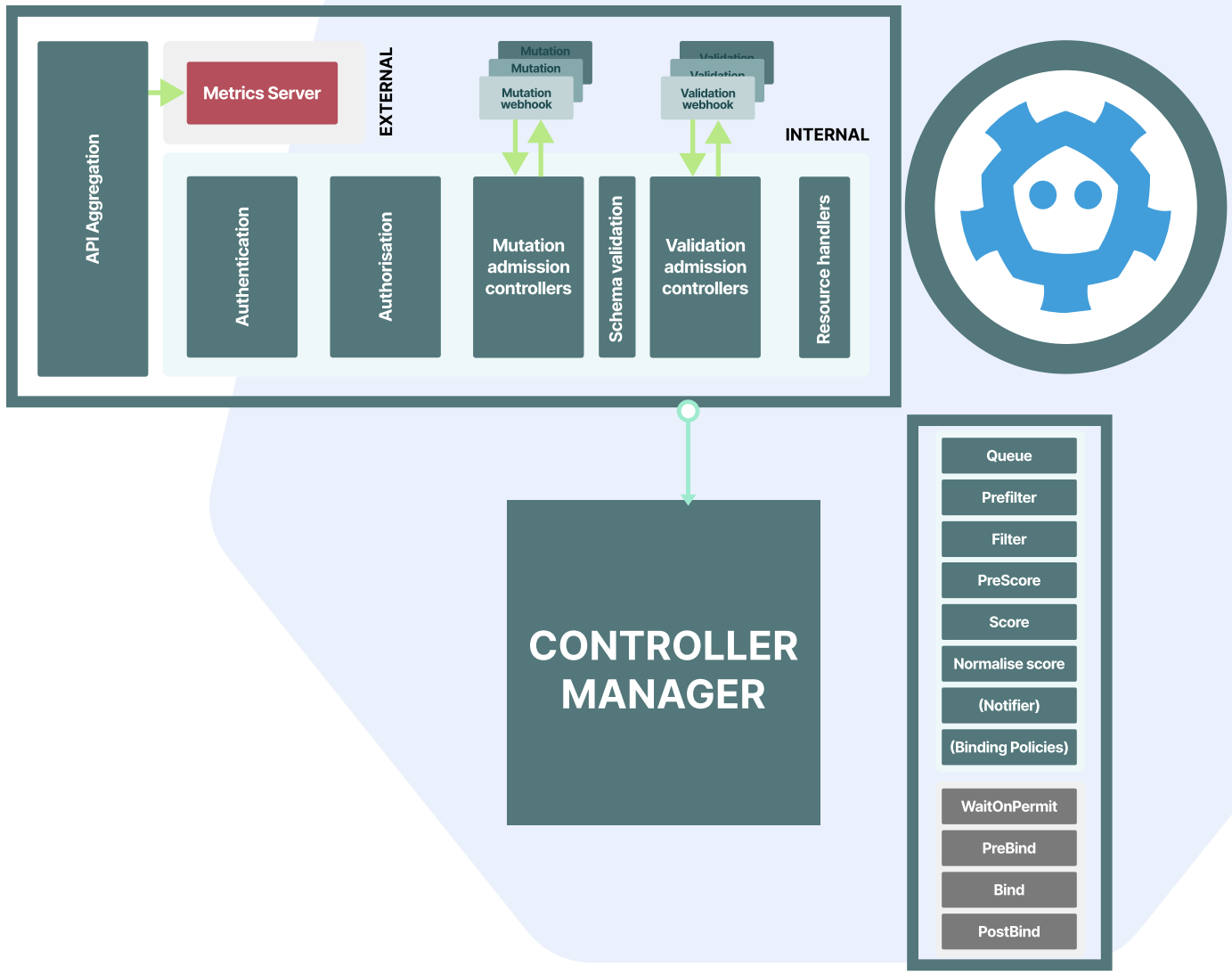
Recap — architecture

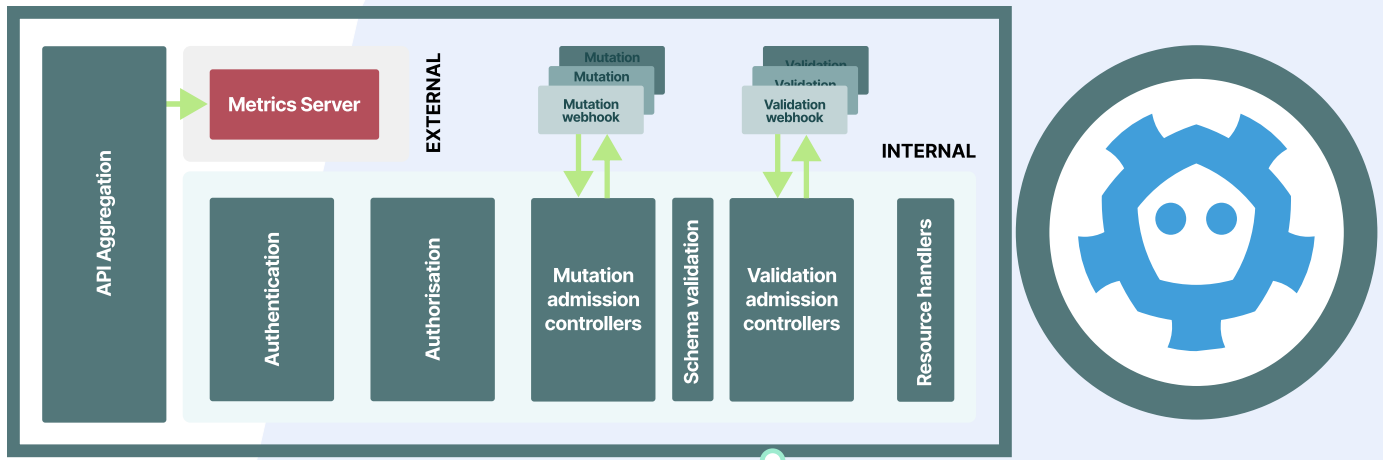
1. HA control plane
2. Failover for nodes & control plane
3. Control plane leader election
4. API server design
5. API scalability

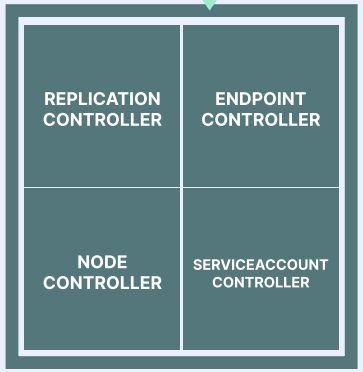
Control plane components

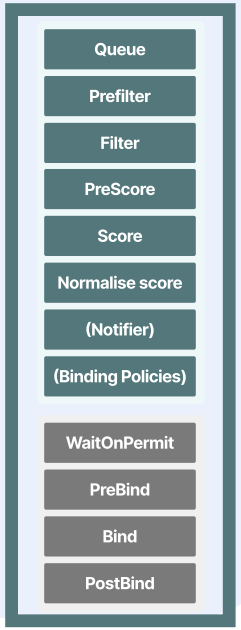

API SERVER
CONTROLLER MANAGER
SCHEDULER


API Aggregation
Metrics Server
EXTERNAL
Mutation
Mutation
Mutation webhook
Validation
Validation
Validation webhook
INTERNAL
Authentication
Authorisation
Mutation admission controllers
Schema validation
Validation admission controllers
Resource handlers
CONTROLLER MANAGER
SCHEDULER

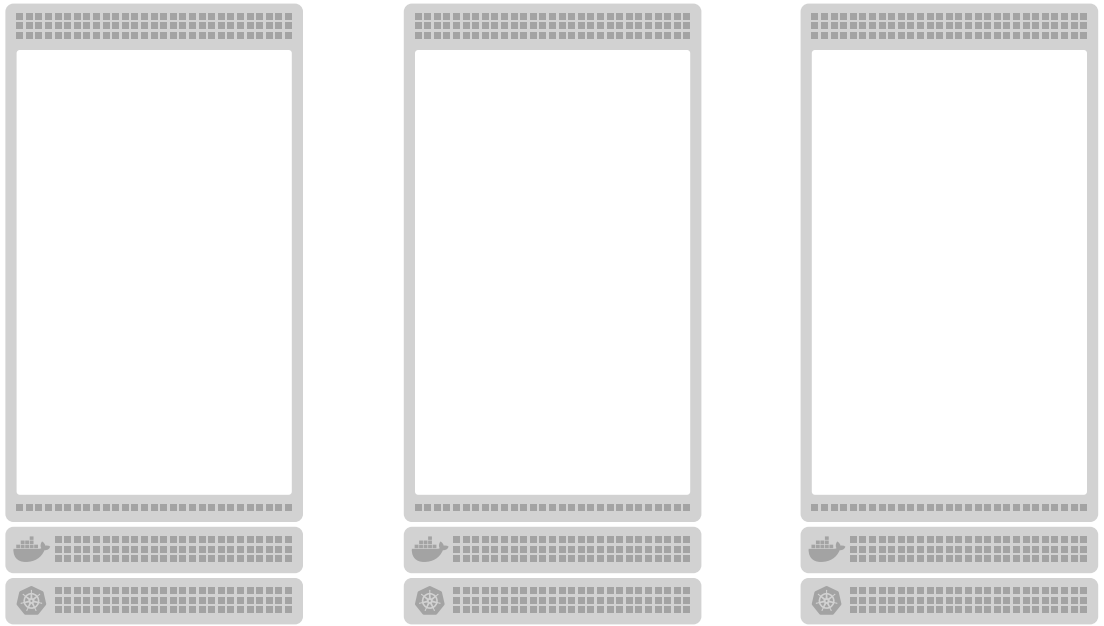

API Aggregation
Metrics Server
EXTERNAL
Mutation
Mutation
Mutation webhook
Validation
Validation
Validation webhook
INTERNAL
Authentication
Authorisation
Mutation admission controllers
Schema validation
Validation admission controllers
Resource handlers
CONTROLLER MANAGER
Queue
Prefilter
Filter
PreScore
Score
Normalise score
(Notifier)
(Binding Policies)
WaitOnPermit
PreBind
Bind
PostBind

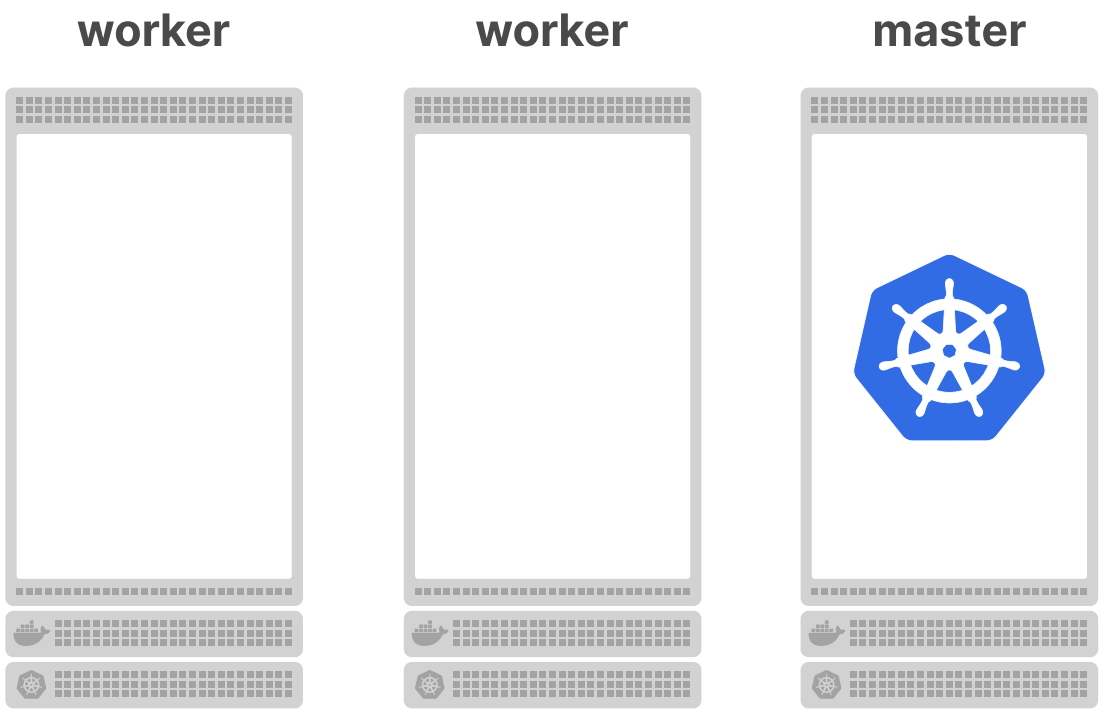

API Aggregation
Metrics Server
EXTERNAL
Mutation
Mutation
Mutation webhook
Validation
Validation
Validation webhook
INTERNAL
Authentication
Authorisation
Mutation admission controllers
Schema validation
Validation admission controllers
Resource handlers

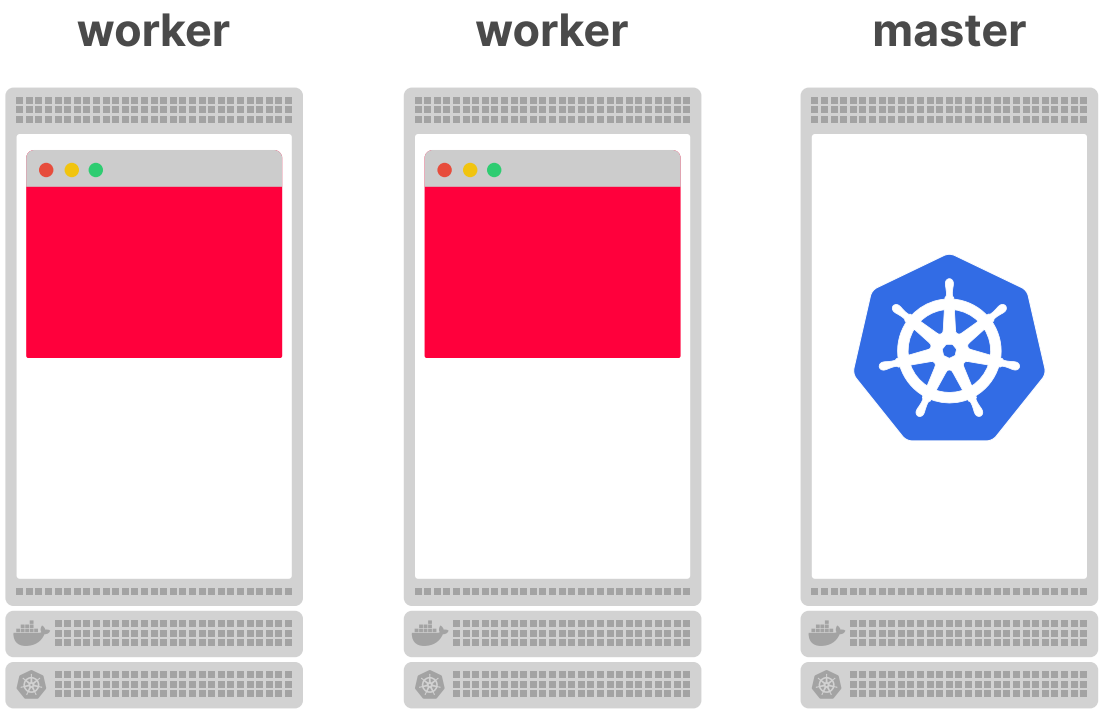

REPLICATION CONTROLLER
ENDPOINT CONTROLLER
NODE CONTROLLER
SERVICEACCOUNT CONTROLLER

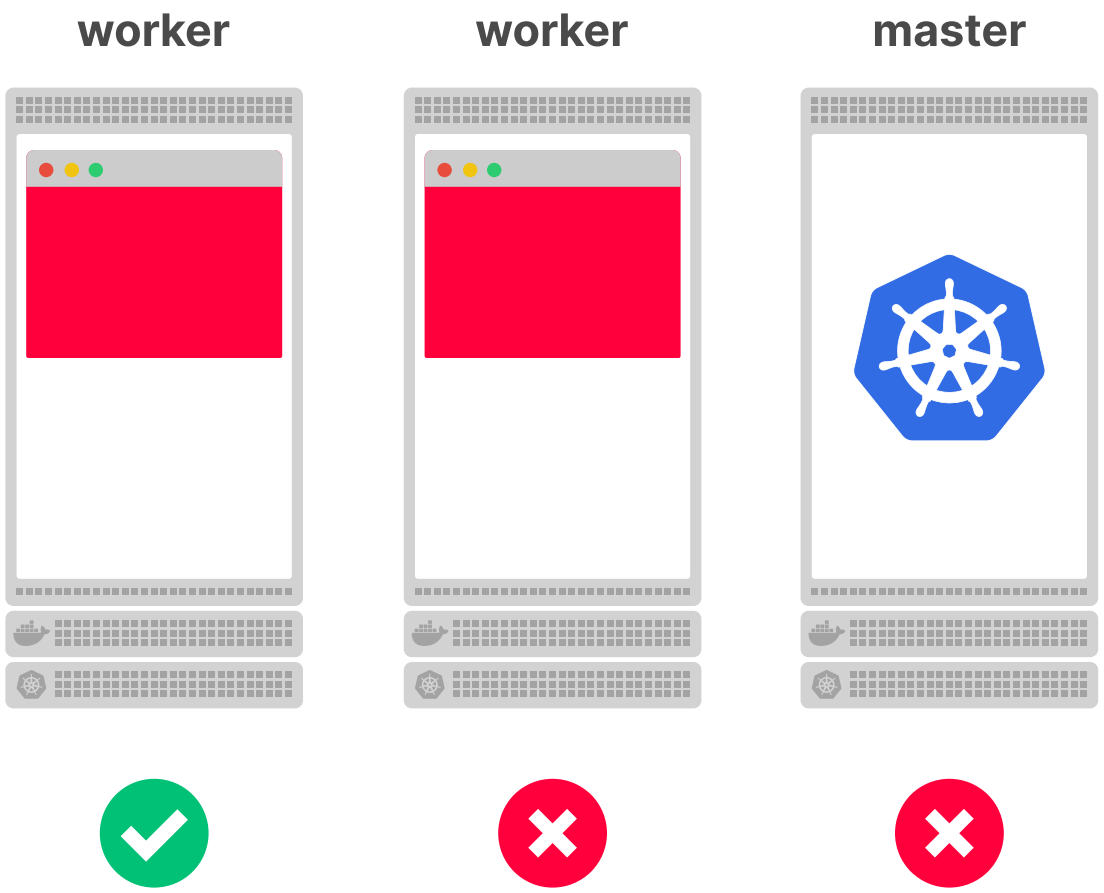

Queue
Prefilter
Filter
PreScore
Score
Normalise score
(Notifier)
(Binding Policies)
WaitOnPermit
PreBind
Bind
PostBind

Lab




worker
worker
master


worker
worker
master


worker
worker
master

Thanks.